

AK High-Dimensional Projection Structural Theory

Appendix HT

Higher Transfer and Unified Bridge Composition

Canonical Single-Appendix Edition

HC–PE–SS–RL–IW/Return Theory

A v20.0.0-dependent successor research appendix

Single-artifact rule. HC, PE, SS, RL, and IW/Return Theory are no longer distributed or typeset as separate appendices in this edition. Their complete retained mathematical content is normalized as internal Parts of Appendix HT. The document has one title, one table of contents, one continuous page sequence, one status algebra, and one self-contained XeLaTeX source. Former module codes are retained only as provenance and theorem-locator namespaces.

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Abstract

Appendix HT is the single integrated successor appendix for homological compression, finite persistence–Ext reconstruction, spectral-sequence transfer, derived inverse-limit control, and typed Return. The former HC, PE, SS, RL, and IW documents are internalized as Parts rather than appended as autonomous PDF artifacts. Their theorem strengths and non-claims remain sovereign, while HT supplies the common AK-v20 re-entry spine, interface certificates, defect stack, status precedence, and conditional composition theorem. Integration changes document architecture and management; it does not silently promote any local theorem, modify the frozen v20 Core, or discharge an external Return theorem.

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Part I

AK-v20 Connection and Unified HT Composition

1 Scope, sovereignty, and single-appendix internal integration

1.1 Purpose

The purpose of HT is not to normalize one appendix, to rename five documents, or to place a new layer above the AK Core by terminology. HT organizes a typed proof graph whose immutable central spine is the v20 Projection–Certificate–Return architecture:

$$X_{\text{ext}} \xrightarrow{\mathfrak{R}_{\text{purpose}}} F_X \in \text{FiltCh}(k) \text{ or } \text{Pers}_k^{\text{cons}} \xrightarrow{\text{Core}_{20}} Q_{\text{AK}} \xrightarrow{\text{Return}_{\text{source}}} Y_{\text{ext}}.$$

HC, PE, SS, RL, and IW/Return Theory attach to this spine only through declared interfaces. One demanding invocation pattern is

$$\text{HC} \longrightarrow \text{PE} \longrightarrow \text{SS} \longrightarrow \text{RL} \longrightarrow \text{IW/Return},$$

but this is not the universal architecture. HC may feed the Core directly; SS or RL may arise from an external source without PE; PE-6–PE-10 may remain entirely in the Technical Extension channel; and a module may be legitimately not_invoked. No arrow exists merely because adjacent modules occur in the same document.

Axiom 1.1 (Module sovereignty). Every theorem imported from HC, PE, SS, RL, or IW/Return Theory is consumed at exactly the strength stated in its internal Part. HT may compose certified arrows; it may not strengthen, reverse, or reinterpret a local theorem.

Axiom 1.2 (Frozen-baseline rule). The integrated HT document is a successor research artifact over AK-HPDST v20.0.0. It does not modify any frozen v20 definition, theorem, Claim UID, status rule, or artifact. A future release may integrate selected HT statements only by an explicit governance act.

Protocol 1.3 (Precedence). The normative order is:

1. the frozen v20.0.0 Main and Foundation baseline;
2. the local theorem statement inside the corresponding HT internal Part;
3. the HT interface statement that connects two modules;
4. the HT composite theorem.

If two readings appear to conflict, the weaker well-typed reading governs until a separate comparison theorem is proved.

1.2 Unified internal Parts

The present PDF includes, in full:

- HC: exact and filtered homological compression with persistence Return;

- PE: finite persistence–Ext reconstruction through A_∞ , deformation, and integral boundary Return;
- SS: spectral-sequence Return, including a nonzero-differential Hopf bridge;
- RL: derived inverse-limit defect and finite-to-infinite transfer theory;
- IW/Return Theory: typed Return strengths, obligations, defects, and certificates.

The integrated sections are immutable source modules. The HT composition layer is the new material preceding them. “Unified internal Parts” means complete retention of the mathematical body with module-local title and contents pages normalized away; it does not change the local evidence status or inherited open obligations of any module.

1.3 Inherited module-status boundary

Module	Inherited local role	HT reading
HC	v20-dependent successor bridge	Exact filtered compression and persistence Return only under its displayed zero-shift and certificate hypotheses.
PE	internally closed candidate	PE-1–PE-10 are consumed at their typed finite-quiver or enhancement strengths; independent audit and release integration remain pending.
SS	first-cycle successor bridge corpus	The worked Cech, Leray, and Serre–Hopf results retain their source and extension boundaries; corpus inclusion does not complete any missing line-addressed source registry.
RL	successor research draft / internal extension	Classical $\lim^1$ inputs, inverse-system comparison, and non-toy applications retain their displayed source-binding and audit requirements.
IW/Return	axiomatic successor Return theory	It supplies the final status and strength grammar; it does not turn a missing source theorem or comparison artifact into Return.

2 AK v20.0.0 connection and verifier-side re-entry

2.1 Frozen AK certificate spine

Definition 2.1 (Frozen AK projection record). For a source object X_{ext} , a frozen AK projection record contains a purpose-faithful realization

$$\mathfrak{R}_{\text{purpose}}(X_{\text{ext}}) = F_X \in \text{FiltCh}(k) \quad \text{or} \quad \text{Pers}_k^{\text{cons}},$$

together with, whenever invoked,

$$\mathbf{B}_{n,W}(F_X) := \mathcal{W}_W P_n(F_X), \quad \text{Eval}_{n,W,\tau}(F_X) := T_\tau^{\text{bar}}(\mathbf{B}_{n,W}(F_X)).$$

The record also fixes the source semantics, coefficient passage, variance, monitored degrees, right-open windows, thresholds, endpoint convention, and immutable artifacts. Window restriction precedes hard deletion.

Definition 2.2 (AK verifier-side spine). An invoked AK verifier-side spine consists of the applicable subset of:

1. the positive gate $\mathbf{B}\text{-Gate}^+$ and constructibility record;
2. the windowed pre-threshold and repaired post-threshold profiles;

3. a stage-correct detector with separate soundness and zero-completeness evidence;
4. an admissible representative $C_{\tau, W} F_X$ at its declared equivalence strength;
5. a standard degree- n eligible realization, edge extractor, and H^{-n} identification when a higher Ext discharge is invoked;
6. an actual tower comparison morphism $\phi_{i, W, \tau}$ and separate terminal, transient, and full-interior defect records when Type IV is invoked;
7. a metric comparison supported by the two-sided closed threshold band, a declared scalarization, and the post-threshold zero-separation condition when a metric inference is used;
8. a theorem-backed finite-to-infinite reduction when the target quantifies over an infinite tail or apex;
9. claim-use preflight, manifest consistency, artifact identity, and replay records;
10. a separately proved source-domain Return theorem for every external conclusion.

No item is inferred from the presence of HT terminology or from co-location in the integrated PDF.

Axiom 2.3 (AK Core sovereignty). HT may transport an existing AK verifier-side record or may construct a candidate input for one. It may not redefine $\text{Eval}_{n, W, \tau}$, replace window-first evaluation, weaken detector completeness, manufacture a representative, synthesize a natural higher edge, reverse a tower without a variance theorem, or interpret Core pass as an external theorem.

Definition 2.4 (HT-to-AK re-entry certificate). An HT output re-enters the AK verifier only through a certificate

$$\rho_{M \rightarrow \text{AK}} : O_M \rightsquigarrow O_{\text{AK}},$$

where O_M is the exact native output of an invoked module and O_{AK} is the exact Core-readable object or record consumed downstream. The symbol $\rightsquigarrow$ denotes the native comparison type: it may be an equality, isomorphism, filtered map, quasi-isomorphism, natural transformation, zigzag, extraction theorem, or record conversion, and is not presumed to be an ordinary morphism. The certificate declares its direction, category, coefficients, filtration, variance, monitored scope, comparison strength, source theorem, artifact identity, and failure route.

Theorem 2.5 (Conservative AK–HT compatibility). *Assume every invoked HT module is used at its local strength and every crossing of the AK Core boundary is supplied by an HT-to-AK or AK-to-HT interface certificate. Then adjoining the HT proof graph does not change any frozen v20 Core theorem or status rule. It only produces a conditional successor proof path whose conclusions are limited by the weakest consumed local theorem, re-entry certificate, reduction, and Return theorem.*

Proof. The frozen Core evaluates only its declared objects and verifier-side records. A sovereign HT module has no Core authority until a typed re-entry certificate identifies its output with a consumed AK object at a proved strength. Conversely, a Core verdict does not acquire external semantics without the separately declared Return theorem. Hence the composite path is conservative: it may transport certified information but cannot alter a Core definition, fill a missing non-scalar obligation, or promote a local status. $\square$

2.2 Module-to-AK connection map

Layer	AK connection	Licensed use	Forbidden shortcut
HC	finite proof compression and representative discipline	Exact zero-shift contraction may transport the full persistence module and finite barcode into an AK-readable presentation.	Ordinary homotopy equivalence is not persistence Return and HC is not the global collapse functor.

Layer	AK connection	Licensed use	Forbidden shortcut
PE-1	standard higher-edge calibration boundary	Complete-stencil transport-edge vanishing may calibrate a declared Ext edge.	No natural PH–Ext bridge or reverse implication without an independent realization and zero-reflection theorem.
PE-2–5	finite-quiver successor extension	Object, morphism, exact, and derived Return may be consumed after an exact finite type- <i>A</i> interface.	These theorems do not silently enlarge the Core higher-edge package.
PE-6–10	Appendix-N-style enhancement channel	A_∞ , Morita, deformation, descent, and integral boundary data may organize successor proofs through explicit extraction.	An enhancement is not Core-readable and proves no persistence theorem without a purpose-faithful Interface.
SS	spectral-sequence extension and Problem-Interface Return	Source-bound pages, differentials, convergence, and extensions may support a typed external Return.	Spectral-sequence degeneration is not AK collapse; E_∞ is not the target without extension analysis.
RL	complement to Appendix D tower discipline	Inverse systems and $\lim^1$ may discharge a separately typed inverse-limit edge and apex comparison.	A direct tower is not reversed by notation; finite-stage agreement is not infinite proof.
IW/Return	Chapter 8 Return successor grammar	Native comparison, target strength, equivalence relation, defects, and replay determine the final Return status.	Core pass, zero defect, or weaker Return is not automatic exact external Return.

Classification 2.6 (Status-axis separation). The following axes remain distinct:

- AK clause status: pass, reject, undefined, or justified not_invoked;
- HT module invocation: invoked or justified not_invoked;
- HT/Return primary status: REJECT, UNDEFINED, OBSTRUCTED, PARTIAL - RETURN, or RETURN;
- a source-certified replacement datum may separately emit REGULARIZED - RETURN.

AK pass is not external RETURN. A module marked not_invoked creates no missing obligation only when the scope exclusion is immutable and the requested conclusion does not consume that module.

Definition 2.7 (AK-admissible HT datum). An HT datum is AK-admissible when it additionally contains:

1. an immutable source-semantics and purpose profile;
2. a purpose-faithful realization or an explicit declaration that the claim is internal and does not invoke the AK Core;
3. the complete invoked verifier-side spine;
4. typed re-entry certificates for every crossing between HT and AK layers;
5. the selected proof-route DAG, with no double counting of alternative routes;

6. the final Return theorem and target-strength record when the conclusion is external.

Theorem 2.8 (AK–HT Projection–Transfer–Return theorem). *Let $\mathfrak{H}$ be an AK-admissible HT datum. Suppose the purpose-faithful realization, every invoked AK verifier-side clause, every local HT theorem, every AK/HT re-entry certificate, every inter-module interface, every required finite-to-infinite reduction, and the final source-domain Return theorem are proved at their displayed strengths. Then the composite proof DAG certifies the external target exactly at the requested Return strength Σ .*

If no external Return theorem is supplied, the strongest possible conclusion is the declared AK-internal or HT-internal result; it is not an external theorem.

Proof. The realization identifies the source feature with a Core-readable shadow. The frozen verifier-side spine certifies only that shadow. HT modules transport or refine certified objects only through their native local theorems and typed interfaces. Reduction controls every infinite target. The final Return theorem is the unique arrow assigning the audited target-side meaning. Composition is licensed by the supplied comparison and coherence cells; the weakest-strength rule prevents promotion. Therefore the external conclusion follows at Σ and at no stronger undeclared level. $\square$

Non-claim 2.9 (No new v20 claim). The preceding connection theorem is a successor conditional composition theorem. It does not insert HT into the frozen v20 Claim Register, mutate G16, or declare HC, PE, SS, RL, or IW to be canonical Core appendices.

3 The five sovereign transfer layers

3.1 HC: exact compression and protected persistence

HC starts with a finite based chain complex, optionally filtered by a finite totally ordered index, and constructs contractions, elementary cancellations, replay certificates, and exact persistence Return. Its strongest first-cycle statement used by HT is conditional on a zero-shift filtered contraction:

$$(T, \partial) \xrightleftharpoons[P]{I} (C, d), \quad PI = \text{id}_T, \quad \text{id}_C - IP = dH + Hd,$$

where I, P, H preserve every filtration sublevel. Then

$$\text{Pers}_n(T) \cong \text{Pers}_n(C)$$

naturally, and finite-type interval barcodes agree exactly.

Warning 3.1 (HC boundary). Ordinary chain-homotopy equivalence preserves terminal homology but need not preserve persistent homology. HT may pass from HC to PE or re-enter the AK persistence layer only through the exact zero-shift persistence Return or through a separately declared controlled-shift theorem. The HC compressed complex is not automatically the frozen representative $C_{\tau, W}F$.

3.2 PE: reconstruction and enhanced derived structure

PE receives a finite type- A persistence presentation or another explicitly eligible finite quiver presentation. It separates five finite-quiver Return levels from five nonhereditary enhancement/deformation levels:

1. object and interval-multiplicity Return;
2. morphism Return through Ext probes;
3. exact/Yoneda Return;
4. bounded-derived cone and triangle Return;

5. marked A_∞ , Morita, deformation, global descent, wall, and integral boundary Return.

HT does not infer the last group from the first. The PE-6–PE-10 layer is a separately source-bound enhancement programme. It occupies the successor Technical Extension channel and has no AK Core authority until an explicit one-parameter extraction or transport theorem supplies the exact verifier-side input.

3.3 SS: pagewise transfer and abutment

SS treats a source-bound spectral sequence

$$E_r^{p,q} \Rightarrow H^{p+q}$$

with actual pages, differentials, convergence data, filtrations on the abutment, and extension analysis. HT distinguishes:

- concentration Return;
- edge-isomorphism or five-term partial Return;
- nonzero-differential Return;
- unresolved extension or convergence defect.

A spectral sequence that degenerates is not thereby an AK canonical collapse, and an E_∞ page is not automatically an unfiltered object or ring. SS data become AK-facing evidence only through a source-bound Interface and the exact extraction or Return theorem consumed by the claim.

3.4 RL: inverse systems and derived-limit defect

RL complements, but does not replace, the frozen Appendix-D directed-tower calculus. It treats countable inverse systems

$$A_0 \longleftarrow A_1 \longleftarrow A_2 \longleftarrow \dots$$

and separates the categorical inverse limit, the Roos difference map, the ambient derived-limit object $\varprojlim^1 A_n$, the claim-relative connecting image, and the declared apex defect. Vanishing of one layer never discharges the others.

3.5 IW/Return Theory: native target comparison

IW/Return Theory is the final typed target layer. It supplies:

- a source object and observed presentation;
- an apex constructor when defined;
- a native comparison artifact to the target;
- a requested strength Σ and equivalence relation $\mathcal{E}$;
- a defect ledger and replay certificate;
- a deterministic primary status.

The strength structure is a preorder of registered forgetful maps, not one universal linear hierarchy.

4 HT primitive data and status calculus

Definition 4.1 (HT source-to-target datum). An HT datum is a tuple

$$\mathfrak{H} = (\mathcal{S}, X, G, \mathcal{I}, \mathcal{C}, \mathcal{T}, Y, \Sigma, \mathcal{E}, \mathcal{A}),$$

where:

1. $\mathcal{S}$ is the source category and $X \in \mathcal{S}$;
2. $G = (V, E)$ is a finite directed acyclic graph with $V \subseteq \{\text{HC, PE, SS, RL, IW}\}$; vertices are invoked modules and edges are requested transfers;
3. $\mathcal{I}$ is the collection of inter-module interface certificates;
4. $\mathcal{C}$ is the collection of local theorem certificates;
5. $\mathcal{T}$ is the target category and $Y \in \mathcal{T}$;
6. Σ is the requested Return strength;
7. $\mathcal{E}$ is the declared target equivalence relation;
8. $\mathcal{A}$ is the immutable artifact and source-binding package.

A module not in V is not invoked and contributes no implied theorem. An absent edge in E contributes no implied interface.

Definition 4.2 (Interface certificate). For every directed edge $e = (M \rightarrow N) \in E$, an interface certificate $\mathcal{I}_e$ contains:

1. the exact source object produced by M ;
2. the exact target object consumed by N ;
3. an equality, isomorphism, quasi-isomorphism, filtered map, natural transformation, zigzag, base-change cell, or other native comparison artifact;
4. compatibility with coefficients, grading, filtration, variance, marking, topology, tower direction, and artifact identity;
5. the permitted direction and strength of use;
6. a proof or source-bound theorem;
7. a failure route.

Classification 4.3 (Primary HT statuses). At a fixed datum and requested strength, the HT verifier emits exactly one of:

REJECT	a supplied assertion, source binding, type claim, compatibility equation, or artifact identity is false;
UNDEFINED	a required object, theorem scope, interface certificate, apex, or comparison artifact is absent;
OBSTRUCTED	all required objects and maps exist, but a declared algebraic or stability defect is nonzero;
PARTIAL-RETURN	a weaker registered strength is certified while the requested strength is not;
RETURN	every obligation at the requested strength is discharged.

A source-certified replacement target is a new Return datum. It may carry the distinct label

REGULARIZED-RETURN.

This is not a silent status mutation of the original target.

Protocol 4.4 (Ordered decision rule). The verifier applies the preceding status clauses in the displayed order. In particular, a fabricated map is rejected, a missing map is undefined, and only an existing valid map with a nonzero declared defect is obstructed.

Remark 4.5 (Local pass versus Return). A successful AK Core clause or complete HT subchain may be recorded as local pass evidence, but RETURN is emitted only for the fixed Return datum and requested strength. Co-locating a Core certificate and an external theorem statement does not supply the Return arrow between them.

5 Inter-module interfaces

5.1 HC to PE

Definition 5.1 (HC–PE interface). An HC–PE interface consists of:

1. a zero-shift filtered contraction, or another HC theorem, producing an exact isomorphism of persistence modules;
2. a finite critical grid $\Lambda = \{\lambda_0 < \dots < \lambda_N\}$ and declared endpoint convention;
3. a finite-dimensional type- A quiver representation M_C extracted from the returned persistence module;
4. an actual isomorphism $M_C \cong M_T$ induced by the HC natural isomorphism;
5. the PE orientation, handedness, and sentinel conventions.

Theorem 5.2 (Exact HC–PE transfer). *Let C and T be finite filtered chain complexes connected by a zero-shift filtered contraction. Fix a degree n and finite critical grid satisfying the PE finite type- A regime. Then the HC natural isomorphism*

$$\text{Pers}_n(T) \cong \text{Pers}_n(C)$$

induces an isomorphism of the corresponding finite quiver representations. Consequently every PE invariant or Return theorem that depends only on that representation and its declared marking has identical input on C and T .

Proof. The HC contraction restricts to every filtration sublevel, hence induces a natural isomorphism on homology functors over the finite poset Λ . A finite persistence module over a linearly ordered finite set is precisely a representation of the corresponding type- A quiver, after fixing the orientation convention. The component-wise homology isomorphisms commute with all structure maps, so they form a quiver-representation isomorphism. PE consumes the isomorphism class together with its declared marking and conventions; therefore its representation-level inputs agree. No chain-level, geometric, or enhancement-level conclusion is inferred unless PE separately proves it. $\square$

Corollary 5.3 (Compression invariance of finite PE object Return). *Under the theorem’s hypotheses, PE interval multiplicities, cumulative rank data, and finite Ext-signature inversions agree before and after HC compression.*

5.2 PE to SS

Definition 5.4 (PE–SS interface). A PE–SS interface is supplied only when PE returns or transports an actual filtered complex, filtered dg object, or filtered derived representative for which a source-bound spectral sequence exists. It includes the filtration, boundedness or convergence regime, coefficient policy, and an identification of the PE object with the spectral-sequence input.

Warning 5.5 (No spectral sequence from an abstract derived object alone). An object of a derived category, an Ext algebra, or an A_∞ quasi-isomorphism class does not by itself determine one canonical spectral sequence. The filtration and source construction are additional data.

Theorem 5.6 (Filtered PE–SS transfer). *Suppose a PE certificate provides a filtered complex (C, F) up to a filtered quasi-isomorphism that is compatible with an explicitly chosen source-bound spectral-sequence construction. Then the induced spectral sequences are isomorphic from the theorem-licensed page onward, and their abutments agree at the strength supplied by the convergence theorem.*

Proof. This is the standard comparison principle for the selected filtered-complex spectral sequence, applied only after the interface certificate identifies the filtration, the map, and the page from which the comparison theorem is valid. HT records the resulting pagewise isomorphism and does not promote it beyond the supplied convergence and extension analysis. $\square$

5.3 SS to RL

Definition 5.7 (SS–RL interface). An SS–RL interface consists of a countable inverse system of filtered objects, spectral sequences, pages, abutment filtrations, or cohomology objects with actual transition morphisms. It also records whether inverse limit is taken pagewise, on complexes, on abutments, or after cohomology.

Warning 5.8 (Noncommutation boundary). Neither cohomology nor spectral-sequence formation is assumed to commute with inverse limit. A comparison map and its derived-limit defect must be supplied at the exact level consumed by the target claim.

Theorem 5.9 (SS–RL edge decomposition). *For a countable inverse system in the RL module regime, any source-bound comparison between the cohomology of a limiting complex and the inverse limit of finite-stage cohomology factors through three separately audited layers:*

1. *validity of the inverse system and its transition morphisms;*
2. *the derived-limit edge, including the relevant $\varprojlim^1$ term or claim-relative connecting image;*
3. *agreement of the categorical inverse-limit object with the declared apex.*

No spectral-sequence collapse or pagewise stabilization eliminates an undeclared layer.

5.4 RL to IW/Return Theory

Definition 5.10 (RL–Return interface). An RL–Return interface identifies the RL output object and apex-comparison artifact with the observation, reconstructed candidate, and target-comparison artifact of a Return datum. It declares the target category, equivalence relation, strength, and the exact role of every RL defect.

Theorem 5.11 (Limit-to-Return transfer). *Assume an RL certificate proves the required system validity, derived-edge condition, and apex agreement. Assume further that the resulting apex is connected to a declared target Y by a native Return comparison satisfying the IW/Return obligations at strength Σ . Then the composite datum has RETURN at strength Σ .*

Proof. RL supplies the exact candidate object and certifies the finite-to-infinite or inverse-limit passage used by the Return datum. The IW/Return certificate then verifies typing, source identity, completeness, comparison, relative uniqueness, stability when invoked, obstruction, and replay certification. Since the hypotheses discharge every required layer and no stronger conclusion is read from the RL certificate alone, the ordered status rule emits RETURN. $\square$

6 Unified AK–HT defect stack

Definition 6.1 (AK–HT defect ledger). For an invoked AK-admissible HT datum define

$$\mathfrak{D}_{\text{AKHT}} = (D_{\text{src}}, D_{\text{real}}, D_{\text{Core}}, D_{\text{HC}}, D_{\text{PE}}, D_{\text{SS}}, D_{\text{RL}}, D_{\text{IW}}, D_{\text{int}}, D_{\text{red}}, D_{\text{return}}, D_{\text{cert}}),$$

where:

- D_{src} records source identity, theorem scope, coefficient, variance, or semantic mismatch;
- D_{real} records failure or absence of the purpose-faithful realization and re-entry typing;
- D_{Core} records invoked gate, profile, detector, representative, higher-edge, Type IV, transient, metric, or Core-manifest failure;
- D_{HC} records failed contraction, filtration shift, forbidden cancellation, or HC replay defect;
- D_{PE} records reconstruction, marking, variance, Morita, deformation, wall, coefficient, or boundary defect;
- D_{SS} records unknown differential, convergence, filtration, or extension defect;
- D_{RL} records inverse-system, derived-limit, edge, or apex defect;
- D_{IW} records target typing, native comparison, strength, uniqueness, or stability defect;
- D_{int} records AK/HT or inter-module mismatch;
- D_{red} records failure of a required finite-to-infinite or descent theorem;
- D_{return} records the absence or failure of the final source-domain Return theorem;
- D_{cert} records claim-use, source binding, artifact, serialization, and replay failure.

The source, realization, missing-object, and certificate entries determine rejection or undefinedness before a genuine Type IV obstruction may be emitted. Type IV remains a defect of an existing valid comparison artifact.

Proposition 6.2 (No scalar compensation). *There is no default scalar function whose small value compensates for a missing or false component of $\mathfrak{D}_{\text{AKHT}}$. In particular, metric reserve cannot manufacture source semantics, a realization, detector completeness, a filtered map, higher edge, spectral sequence, inverse system, apex, reduction, claim permission, or Return comparison.*

Proposition 6.3 (Defect locality and dependency control). *A nonzero local defect blocks exactly the conclusions whose selected proof route consumes that layer. It does not automatically invalidate a weaker, disjoint, or legitimately noninvoked Return strength.*

Proof. The HT strength structure is a preorder with module-specific forgetful maps, and the AK verifier uses a typed dependency graph. A claim may bypass a noninvoked module or may consume only a weaker quotient of an invoked output. Therefore the selected dependency path, not the mere presence of a defect elsewhere in the integrated corpus, determines the blocked conclusions. $\square$

7 The unified composition theorem

Definition 7.1 (Composable HT proof DAG). A composable HT proof DAG is a finite directed acyclic graph whose invoked module vertices lie in $\{\text{HC}, \text{PE}, \text{SS}, \text{RL}, \text{IW}\}$. A linear list is a special case. The graph is equipped with:

1. a valid local theorem certificate for every invoked M_i ;
2. a valid interface certificate for every adjacent pair;
3. a commutative comparison diagram whenever two routes reach the same object;
4. a declared target strength and Return comparison;
5. a complete replay package.

Theorem 7.2 (HT unified bridge composition). *Let $\mathcal{H}$ be a composable HT proof DAG from a source object X to a target object Y at strength Σ . If every local theorem hypothesis, every interface certificate, every required commutative cell, every finite-to-infinite condition, and every Return obligation is discharged, then*

$$\text{Return}_{\Sigma}^{\text{HT}}(X, Y)$$

is certified and the primary status is RETURN.

Conversely, if the ordered audit encounters:

1. *a false supplied statement, the status is REJECT;*
2. *otherwise a missing required datum, the status is UNDEFINED;*
3. *otherwise a nonzero declared comparison defect, the status is OBSTRUCTED;*
4. *otherwise a certified weaker strength but not Σ , the status is PARTIAL-RETURN.*

Proof. Compose the local native comparison artifacts along the selected DAG in the target category or bicategorical localization declared by the datum. Edge composition is permitted by the interface certificates; route independence is supplied by the commutative comparison and coherence cells. HC secures the finite presentation when invoked. PE secures the representation, exact, derived, or enhanced transfer when invoked. SS secures the page-to-abutment passage when invoked. RL secures the limit and apex passage when invoked. IW/Return Theory evaluates the final native comparison at strength Σ . The replay package records the complete dependency path. The positive conclusion follows from the conjunction of these certified obligations. The negative clauses follow from the ordered status calculus and the prohibition on converting false, absent, or nonzero-defect data into a stronger status. $\square$

Corollary 7.3 (AK-facing specialization). *If the HT proof DAG crosses the frozen AK Core boundary, the conclusion of the unified bridge composition theorem is AK-facing only when the datum is AK-admissible. Without the full verifier-side spine and source-domain Return theorem, the same DAG certifies at most its internal categorical conclusion.*

Corollary 7.4 (Associativity relative to coherence). *Suppose three certified HT subchains are composable and the declared associator or comparison cells satisfy the required coherence. Then either parenthesization yields the same Return status and target equivalence class. Without the coherence data, global functoriality is not asserted.*

Theorem 7.5 (Non-collapse of module strengths). *The unified composition theorem does not imply any of the following without a separate promotion theorem:*

- *terminal homology Return $\Rightarrow$ persistence Return;*
- *persistence-module Return $\Rightarrow$ chain-level filtered Return;*
- *derived equivalence $\Rightarrow$ marked A_{∞} equivalence;*
- *E_{∞} identification $\Rightarrow$ unfiltered ring identification;*
- *$\varprojlim^1 = 0 \Rightarrow$ apex agreement;*
- *characteristic or determinant Return $\Rightarrow$ exact source reconstruction.*

8 Canonical invocation patterns

8.1 Pattern A: compressed finite persistence to derived Return

A valid chain is

$$(C, F) \xrightarrow{\text{HC}} (T, F) \xrightarrow{\text{finite grid}} M_T \in \text{Rep}(A_N) \xrightarrow{\text{PE}} D^b(\text{Rep}(A_N)).$$

The Return target may be the exact barcode, Ext signature, Yoneda extension class, cone, or triangle. SS and RL are not invoked unless additional filtered or tower structure is declared.

8.2 Pattern B: spectral-sequence differential Return

The Hopf-fibration SS module supplies a finite symbolic certificate in which a nonzero transgression is essential. HC, PE, and RL are not needed. IW/Return Theory records the target strength as integral cohomology-group and ring Return, subject to the source-bound characteristic-class input and extension analysis.

8.3 Pattern C: tower and derived-limit Return

A source tower may pass directly through RL to Return Theory. A finite prefix is insufficient. An eventual-isomorphism, surjective-tail, Mittag-Leffler, Roos-surjectivity, or another theorem-backed certificate is required, together with actual apex agreement.

8.4 Pattern D: full HC–PE–SS–RL–Return chain

The full chain is admissible when a filtered source is exactly compressed, reconstructed in a PE-eligible finite presentation, supplied with a chosen filtered spectral sequence, organized into an inverse tower, and compared to an external apex. The full chain is not the default; it is the most demanding invocation pattern.

9 Artifact and replay contract

Definition 9.1 (Single-source HT artifact). The normative management artifact is one Appendix HT PDF generated from one self-contained XeLaTeX source. HC, PE, SS, RL, and IW are internal Parts, not separately consumed appendices and not external PDF dependencies. Every theorem citation records the HT Part and local locator, while all page numbers, bookmarks, status declarations, and replay metadata belong to the single HT artifact.

Protocol 9.2 (Minimal operational use). A theorem user should cite one HT interface or composition theorem and the exact local theorem in the internal Part. Citation of “Appendix HT” alone is insufficient to identify the hypotheses or Return strength.

Warning 9.3 (Self-audit boundary). Repeated internal review may expose defects and improve the artifact, but it is not independent external verification. The single-Appendix first cycle remains subject to source audit, proof-assistant formalization of selected kernels, and future integration governance.

10 Integrated claim matrix

Layer	Native input	Principal output	HT prohibition
HC	finite based or filtered chain complex	contraction, compressed complex, exact homology/persistence Return	no persistent conclusion from unfiltered homotopy alone
PE	finite type- A persistence or declared enhanced model	object, morphism, exact, derived, A_∞ , deformation, boundary Return	no synthetic promotion to a natural external bridge
SS	source-bound spectral sequence	page, differential, edge, abutment, extension Return	no abutment object/ring without convergence and extension analysis
RL	actual countable inverse system and apex comparison	inverse limit, $\varprojlim^1$, edge and apex audit	no infinite conclusion from finite agreement

Layer	Native input	Principal output	HT prohibition
IW/Return	observed presentation, apex, native comparison, target strength	deterministic Return status and replay certificate	no promotion from weaker to stronger Return without theorem
HT	certified local outputs and interfaces	conditional composite Return	no identification of sovereign modules or compensation across missing gates

11 Closure theorem and AK-connected first-cycle status

Theorem 11.1 (AK-connected HT first-cycle integration closure). *The HC, PE, SS, RL, and IW/Return programmes admit one conservative AK-connected integration architecture with:*

1. *content-complete single-source integration of five sovereign mathematical layers without status promotion;*
2. *an explicit frozen AK Projection–Certificate–Return spine;*
3. *typed AK/HT re-entry certificates and module interfaces;*
4. *a DAG-based, noncompensating defect and status calculus;*
5. *conditional internal and AK-facing unified composition theorems;*
6. *a single management and replay artifact whose local theorem authority remains source-located.*

No claim is made that every external problem admits the required realization, that the full five-module chain is the preferred route, or that inherited module source-audit and external-application obligations are discharged by integration.

Proof. The preceding sections define the frozen AK spine, the sovereign module layers, each admissible interface, the status-axis separation, the expanded defect ledger, and the conditional DAG composition theorems. The internal Parts supply the complete retained mathematical content without altering inherited theorem strengths. The conservative compatibility theorem prevents any silent Core mutation or external upgrade. These data establish integration closure at the architecture and artifact levels, subject to the inherited local proof and source boundaries. □

Classification 11.2 (HT AK-connected integrated status). The reaudited integrated HT cycle has status

```
higher-transfer-AK-v20-connected-integration-constructed
+ frozen-Core-spine-and-verifier-reentry-explicit
+ HC-PE-SS-RL-IW-single-source-content-integrated
+ local-module-statuses-inherited-without-promotion
+ conditional-DAG-composition-and-Return-theorems-proved
+ noncompensation-and-no-Core-bypass-enforced
+ internal-cross-module-and-AK-connection-audit-passed
+ inherited-source-audit-obligations-preserved
+ external-independent-audit-pending
+ future-release-integration-pending
```

Non-claim 11.3. Appendix HT proves no Iwasawa main conjecture, Navier–Stokes regularity theorem, new general spectral-sequence theorem, universal persistence–Ext equivalence, universal finite-to-infinite theorem, unconditional external Return, or canonical v20 Claim Register promotion. Its new content is the typed AK-connected successor composition architecture.

Part II

Homological Compression and Exact Persistence Return

12 Role inside Appendix HT

This Part supplies exact finite chain compression and zero-shift filtered persistence Return. It is the preferred finite-presentation engine when a later PE or AK certificate consumes the same filtered object. Ordinary chain-homotopy equivalence is not promoted to persistence Return.

Remark 12.1 (Normalization rule). The following is the complete retained source text of the former HC artifact, normalized into this HT Part. Its original numbering is a provenance namespace only; it no longer denotes a separate Appendix file.

Retained module abstract

HC-1 constructed a chain contraction onto homology, HC-2 proved exact elementary cancellation, HC-3 iterated the cancellation with cumulative Return maps, and HC-4 promoted the reduction record to a verifier-bound certificate. HC-5 adds filtration as protected information. It defines zero-shift filtered contractions, proves that they restrict at every filtration level, and therefore Return the entire persistence module naturally rather than merely the terminal homology groups. Under the usual finite-dimensional finite-type hypotheses, the barcode is returned exactly, including birth and death parameters. The appendix proves an exact filtered unit-pivot cancellation criterion, shows that equality of pivot filtration levels is forced for zero-shift cancellation, supplies a replay and terminal certificate extension, and gives a worked example in which an unfiltered cancellation would erase a genuine finite bar. It also separates exact Return from controlled-shift approximations and from unsupported metric or multiparameter claims. The principal result is a typed Interface–Compression–Return chain for one-parameter persistent homology.

Remark 12.2 (Native vector transcription rule). The following body is reconstructed as native XeLaTeX text and mathematics from the authoritative module PDF. It is not Verbatim text, not a raster image, and not an imported PDF. Original title pages, module-local tables of contents, and module-local page numbers are removed; mathematical statements, proofs, tables, diagrams, references, and source-boundary declarations are retained. The former numbering is preserved as the module provenance namespace.

1 Programme position and objective

The first HC cycle is:

- HC-1 : global chain contraction and homology Return,
- HC-2 : one exact unit-pivot cancellation,
- HC-3 : iterated sparse compression and cumulative Return,
- HC-4 : certificate, serialization, and verifier contract,
- HC-5 : filtration-preserving compression and persistence Return.

The new protected datum is not only the chain homotopy type of the terminal complex, but the entire nested family of subcomplexes.

Warning 1.1 (Primary firewall). An ordinary chain homotopy equivalence may preserve total homology while changing persistent homology. HC-5 licenses persistent conclusions only when the declared filtered identities are verified. “Homology preserved” is not a substitute for “persistence module preserved.”

2 Bridge identity and governance

Field	Value
Bridge code	HC-PERS-5
Input domain	finite based filtered chain complexes over a field, with a finite totally ordered filtration index
Compression engine	zero-shift filtered contractions and zero-shift filtered unit-pivot cancellations
Protected object	one-parameter persistence module in every homological degree
Exact Return target	natural isomorphism of persistence modules
Secondary Return	exact interval barcode under finite-type decomposition hypotheses
Representative Return	cycle representatives with no increase of filtration level
Certificate modes	filtered replay, filtered terminal contraction, and hybrid checkpoints
Forbidden shortcut	cancellation of a pivot whose generators have unequal filtration values
Choice defect	bases, equal-level pivot order, representative choices, interval endpoint convention
Status target	return_proved + persistent_bridge_complete_candidate
Frozen-baseline rule	no theorem or identifier of v20.0.0 is modified

Protocol 2.1 (Typed separation). HC-5 distinguishes four claims:

- (i) chain homology Return;
- (ii) exact persistence-module Return;

- (iii) barcode Return under interval-decomposition hypotheses;
- (iv) approximate Return under a separately declared shift budget.

No claim may be promoted by silently replacing one of these with another.

3 Filtered chain complexes and persistence modules

Definition 3.1 (Finite filtered chain complex). Let $\Lambda = \{\lambda_0 < \dots < \lambda_N\}$ be a finite totally ordered set. A filtered chain complex over a field $\mathbb{k}$ is a chain complex (C_*, d) with subcomplexes

$$0 = F_{<\lambda_0} C_* \subseteq F_{\lambda_0} C_* \subseteq \dots \subseteq F_{\lambda_N} C_* = C_*$$

such that

$$d(F_\lambda C_n) \subseteq F_\lambda C_{n-1}.$$

For a based complex, each basis generator x has filtration value

$$\text{fil}(x) := \min\{\lambda : x \in F_\lambda C\}.$$

Definition 3.2 (Strict filtered map). A linear map $f : C \rightarrow D$ is filtration-preserving with zero shift if

$$f(F_\lambda C) \subseteq F_\lambda D$$

for every $\lambda \in \Lambda$. A zero-shift filtered chain map is a chain map satisfying this condition.

Definition 3.3 (Persistence module). For each degree n , define

$$\text{Pers}_n(C)(\lambda) := H_n(F_\lambda C).$$

For $\lambda \leq \mu$, the structure map

$$\text{Pers}_n(C)(\lambda \leq \mu) : H_n(F_\lambda C) \rightarrow H_n(F_\mu C)$$

is induced by inclusion. Thus $\text{Pers}_n(C)$ is a functor from the poset Λ to $\mathbb{k}$ -vector spaces.

Definition 3.4 (Strict filtered contraction). A zero-shift filtered contraction from C onto T consists of zero-shift filtered chain maps

$$(T, \partial) \xrightleftharpoons[P]{I} (C, d)$$

and a filtration-preserving with zero shift degree-+1 map $H : C \rightarrow C[1]$ such that

$$PI = \text{id}_T, \quad \text{id}_C - IP = dH + Hd.$$

Optional side conditions are

$$PH = 0, \quad HI = 0, \quad H^2 = 0.$$

4 Exact persistence Return

Lemma 4.1 (Sublevel restriction). A zero-shift filtered contraction restricts at every $\lambda \in \Lambda$ to a chain contraction

$$(F_\lambda T, \partial) \xrightleftharpoons[P_\lambda]{I_\lambda} (F_\lambda C, d), \quad H_\lambda.$$

Proof. Strict filteredness ensures that I, P, H preserve each sublevel. Restricting the global identities gives

$$P_\lambda I_\lambda = \text{id}_{F_\lambda T}, \quad \text{id}_{F_\lambda C} - I_\lambda P_\lambda = dH_\lambda + H_\lambda d.$$

Theorem 4.2 (Exact persistence Return). *Let C and T be finite filtered chain complexes connected by a zero-shift filtered contraction. Then for every degree n , the maps*

$$(I_\lambda)_* : H_n(F_\lambda T) \rightarrow H_n(F_\lambda C)$$

form a natural isomorphism

$$\text{Pers}_n(T) \cong \text{Pers}_n(C).$$

Its inverse is induced by P .

Proof. By the sublevel restriction lemma, $(I_\lambda)_*$ and $(P_\lambda)_*$ are inverse isomorphisms for every λ . For $\lambda \leq \mu$, zero-shift filteredness gives commutative squares of chain maps with the inclusion maps. Passing to homology proves naturality.

Corollary 4.3 (Rank-invariant Return). *For all $\lambda \leq \mu$ and every n ,*

$$\text{rank}(H_n(F_\lambda T) \rightarrow H_n(F_\mu T)) = \text{rank}(H_n(F_\lambda C) \rightarrow H_n(F_\mu C)).$$

Hence every statement determined by the one-parameter persistence module is preserved.

Corollary 4.4 (Interval-endpoint Return). *Under the finite-type barcode hypotheses, the interval multiset has the same birth and death endpoints in T and C . More generally, a compatible family of persistent elements is transported by the natural isomorphism with the same lifespan. No canonical interval label or decomposition basis is asserted.*

5 Barcode Return

Definition 5.1 (Finite-type barcode regime). Assume each $F_\lambda C_n$ is finite-dimensional and Λ is finite. Extend the persistence module constantly between successive filtration values, using a declared half-open endpoint convention. Then each degree is a pointwise finite-dimensional one-parameter persistence module of finite type.

Theorem 5.2 (Exact barcode Return). *Under the finite-type barcode regime, a zero-shift filtered contraction yields equality of interval multisets:*

$$\text{Bar}_n(T) = \text{Bar}_n(C)$$

for every n , including multiplicities and declared endpoints.

Proof. The exact persistence Return theorem gives an isomorphism of persistence modules. The finite-type interval decomposition is unique up to permutation of interval summands. Therefore the interval multisets agree.

Remark 5.3 (What is canonical). The barcode multiset is invariant. A specific interval basis, pairing of matrix columns, or chosen representative for an interval is generally not canonical unless a deterministic convention is added.

Non-claim 5.4 (Multiparameter warning). For filtrations indexed by $\mathbb{R}^m$ with $m > 1$, an interval barcode need not classify the persistence module. The module-isomorphism theorem remains meaningful for zero-shift filtered contractions, but the barcode conclusion is not asserted without a separately typed invariant.

6 Filtered representative Return

Theorem 6.1 (Filtration-controlled representative Return). *Let $z_T \in F_\lambda T_n$ be a cycle. Then*

$$z_C := Iz_T \in F_\lambda C_n$$

is a cycle representing the returned class. Conversely, for a cycle $z_C \in F_\lambda C_n$,

$$z_T := Pz_C \in F_\lambda T_n$$

and

$$z_C - Iz_T = d(Hz_C), \quad Hz_C \in F_\lambda C_{n+1}.$$

Thus the equivalence is witnessed without increasing filtration.

Proof. The chain-map identities give cycle preservation. For $dz_C = 0$,

$$z_C - IPz_C = (dH + Hd)z_C = dHz_C.$$

Strict filteredness gives the stated filtration bounds.

Corollary 6.2 (Purpose-relative Return). *Any purpose depending only on the persistence module, its rank invariant, its finite-type barcode, or transported represented classes is exactly preserved. Chain-level geometric quantities not encoded by these objects require separate interfaces.*

7 Exact filtered elementary cancellation

Let a based filtered complex decompose in adjacent degrees as

$$C_n = \mathbb{k}a \oplus \bar{C}_n, \quad C_{n-1} = \mathbb{k}b \oplus \bar{C}_{n-1},$$

with boundary block

$$d_n = \begin{pmatrix} \lambda & \alpha \\ \beta & \Delta \end{pmatrix}, \quad \lambda \in \mathbb{k}^\times.$$

HC-2 defines the reduced differential

$$d'_n = \Delta - \beta\lambda^{-1}\alpha.$$

Definition 7.1 (Strictly admissible persistent pivot). The pivot (a, b) is zero-shift admissible when

$$\lambda = [b : da] \Big/ = 0 \quad \text{and} \quad \text{fil}(a) = \text{fil}(b).$$

Lemma 7.2 (Equality of pivot levels is forced). *For a filtered boundary, a nonzero coefficient $[b : da]$ implies*

$$\text{fil}(b) \leq \text{fil}(a).$$

The elementary homotopy $h(b) = \lambda^{-1}a$ is zero-shift filtered only if

$$\text{fil}(a) \leq \text{fil}(b).$$

Hence zero-shift filtered cancellation requires

$$\text{fil}(a) = \text{fil}(b).$$

Theorem 7.3 (Exact filtered cancellation). *A zero-shift admissible persistent pivot admits the HC-2 elementary contraction with reduced differential*

$$d' = \Delta - \beta\lambda^{-1}\alpha,$$

and the resulting maps i, p, h are zero-shift filtered. Consequently the cancellation returns the complete persistence module and barcode exactly.

Proof. Write

$$d(a) = \lambda b + \beta, \quad d(x) = \alpha(x)b + \Delta(x) \quad (x \in \bar{C}_n).$$

The HC-2 contraction is given on the affected generators by

$$h(b) = \lambda^{-1}a, \quad i(x) = x - \lambda^{-1}\alpha(x)a, \quad p(b) = -\lambda^{-1}\beta, \quad p(a) = 0,$$

with ordinary inclusion or projection elsewhere. Because d preserves filtration,

$$\alpha(x) \neq 0 \implies \text{fil}(b) \leq \text{fil}(x), \quad [y : \beta] \neq 0 \implies \text{fil}(y) \leq \text{fil}(a).$$

Together with $\text{fil}(a) = \text{fil}(b)$, these inequalities show term by term that h , i , and p have shift zero. The reduced differential is $d' = pdi$, equivalently

$$d'_n = \Delta - \beta\lambda^{-1}\alpha,$$

and is filtration preserving because it is a composite of zero-shift filtered maps. HC-2 supplies

$$pi = \text{id}, \quad \text{id} - ip = dh + hd.$$

The exact persistence Return theorem therefore applies at every sublevel.

Warning 7.4 (Cross-level cancellation prohibition). If $\text{fil}(b) < \text{fil}(a)$, the pair may be contractible in the unfiltered complex, but cancelling it with zero shift would erase an interval whose birth and death occur at different levels. Such a pivot is forbidden in exact HC-5.

8 Iterated filtered compression

Theorem 8.1 (Iterated exact persistent compression). *Let*

$$C^{(0)} \rightarrow C^{(1)} \rightarrow \dots \rightarrow C^{(m)}$$

be a finite sequence of zero-shift admissible filtered cancellations. Compose the local contractions as in HC-3:

$$I_{s+1} = I_s i_s, \quad P_{s+1} = P_s P_s, \quad H_{s+1} = H_s + I_s h_s P_s.$$

Then I_m, P_m, H_m are zero-shift filtered and define a contraction of $C^{(0)}$ onto $C^{(m)}$. Therefore

$$\text{Pers}_n(C^{(m)}) \cong \text{Pers}_n(C^{(0)})$$

and, in the finite-type regime,

$$\text{Bar}_n(C^{(m)}) = \text{Bar}_n(C^{(0)}).$$

Proof. Composition and sums of zero-shift filtered maps are zero-shift filtered. HC-3 proves the contraction identities. Apply exact persistence and barcode Return.

Remark 8.2 (Persistent irreducibility). Termination with no equal-level nonzero pivot means only that this elementary cancellation engine has no further zero-shift pivot. It does not prove minimality among all filtered complexes, uniqueness, or optimal sparsity.

9 Worked bridge: preserving a finite bar

Let the filtration indices be

$$0 < 1 < 2 < 3.$$

Define a filtered chain complex over $\mathbb{k}$ with generators

$$C_0 = \mathbb{k}\langle u, v \rangle, \quad C_1 = \mathbb{k}\langle e, z \rangle, \quad C_2 = \mathbb{k}\langle A \rangle,$$

and filtration values

$$\text{fil}(u) = 0, \quad \text{fil}(v) = \text{fil}(e) = 1, \quad \text{fil}(z) = 2, \quad \text{fil}(A) = 3.$$

Set

$$d(e) = v - u, \quad d(z) = 0, \quad d(A) = z.$$

9.1 Persistence before compression

At level 0, only u is present, so $H_0 \cong \mathbb{k}$. At level 1, e connects v to u , so no new H_0 class survives. At level 2, the cycle z creates one H_1 class. At level 3, A kills that class. Therefore

$$\text{Bar}_0(C) = \{[0, \infty)\}, \quad \text{Bar}_1(C) = \{[2, 3)\}.$$

9.2 Admissible compression

The pivot coefficient of v in $d(e)$ is 1, and

$$\text{fil}(e) = \text{fil}(v) = 1.$$

Thus (e, v) is zero-shift admissible. Cancelling it leaves

$$T_0 = \mathbb{k}\langle u \rangle, \quad T_1 = \mathbb{k}\langle z \rangle, \quad T_2 = \mathbb{k}\langle A \rangle, \quad \partial A = z.$$

The returned barcode is still

$$\text{Bar}_0(T) = \{[0, \infty)\}, \quad \text{Bar}_1(T) = \{[2, 3)\}.$$

9.3 Forbidden unfiltered cancellation

The pair (A, z) is a valid unfiltered unit pivot, but

$$\text{fil}(z) = 2 < 3 = \text{fil}(A).$$

Cancelling it would remove the finite interval $[2, 3)$. Total homology at the terminal level would remain correct, but the persistence module would not. This explicitly demonstrates the HC-5 firewall.

10 Certificate extension from HC-4

Definition 10.1 (Filtered input manifest). Extend the HC-4 input manifest by recording:

- (i) the totally ordered filtration index set and endpoint convention;
- (ii) a filtration value for every generator;
- (iii) verification that every boundary entry obeys

$$[dy : x] \neq 0 \implies \text{fil}(x) \leq \text{fil}(y);$$

- (iv) the exact or approximate Return mode;
- (v) the declared persistence degrees and output invariants.

Definition 10.2 (Filtered replay record). Each HC-3 replay record is extended by:

- (i) pivot filtration values $(\text{fil}(a_s), \text{fil}(b_s))$;
- (ii) strict-admissibility assertion $\text{fil}(a_s) = \text{fil}(b_s)$;
- (iii) digest of the current filtration-labelled basis;
- (iv) optional local proof that every Schur-update term respects filtration;
- (v) updated filtration-labelled state digest.

Verifier contract 10.3 (Filtered verifier). The verifier must check, in order:

- F1. HC-4 parsing, digest, coefficient, and chain-condition phases;
- F2. total ordering and canonical encoding of filtration values;
- F3. boundary compatibility with filtration;
- F4. equality of filtration levels for every exact pivot;
- F5. exact Schur update and basis deletion;
- F6. zero-shift filteredness of cumulative I , P , H , or reconstruct it from verified zero-shift local steps;
- F7. terminal chain and filtration conditions;
- F8. verification of a zero-shift contraction from which the typed natural persistence-module isomorphism follows;
- F9. barcode output only when finite-type decomposition hypotheses are declared and verified.

Theorem 10.4 (Filtered certificate soundness). *If the filtered replay or terminal contraction verifier returns ACCEPT, then the certified terminal complex and input complex have naturally isomorphic persistence modules in every declared degree. If the finite-type barcode phase is accepted, their declared barcodes are identical.*

Proof. Accepted replay certifies a finite composition of zero-shift filtered elementary contractions. Accepted terminal mode directly certifies a zero-shift filtered contraction. The exact persistence Return and barcode Return theorems apply.

Protocol 10.5 (Failure-code extension). The HC-5 verifier adds at least:

Code	Meaning
HC5-U101	filtration index encoding unsupported or ambiguous
HC5-R201	boundary violates filtration monotonicity
HC5-R202	exact pivot has unequal filtration levels
HC5-R203	cumulative map is not zero-shift filtered
HC5-R204	declared sublevel contraction identity fails
HC5-U301	barcode requested outside verified finite-type regime
HC5-R401	representative exceeds declared filtration level
HC5-R501	returned barcode disagrees with verified module decomposition

11 Controlled-shift boundary

Exact HC-5 uses zero filtration shift. Some applications permit a controlled error.

Definition 11.1 (Shifted filtered map). For $\varepsilon \geq 0$, a map $f : C \rightarrow D$ has filtration shift at most ε when

$$f(F_t C) \subseteq F_{t+\varepsilon} D.$$

Protocol 11.2 (Approximate branch). A cross-level pivot may enter an approximate HC branch only if all induced maps and homotopies carry explicit verified shift bounds and a separate theorem converts those bounds into an interleaving statement. Exact barcode equality must then be replaced by a stability bound. HC-5 does not infer such a theorem from ordinary cancellation alone.

Non-claim 11.3 (No silent bottleneck bound). The inequality

$$|\text{fil}(a) - \text{fil}(b)| \leq \varepsilon$$

by itself does not certify that the complete compressed persistence module is ε -interleaved with the input. All cumulative maps, homotopies, and structure-map compatibilities must be checked.

12 Defect stack

Define

$$D_{\text{HC5}} = (D_{\text{src}}, D_{\text{real}}, D_{\text{chain}}, D_{\text{fil}}, D_{\text{pivot}}, D_{\text{hom}}, D_{\text{pers}}, D_{\text{bar}}, D_{\text{return}}, D_{\text{cert}}, D_{\text{choice}}).$$

The components mean:

Component	Meaning
D_{src}	external theorem and convention binding incomplete
D_{real}	filtered complex not proved to realize the external object
D_{chain}	$d^2 = 0$ or dimension typing fails
D_{fil}	boundary or maps violate filtration
D_{pivot}	pivot nonunit, absent, or cross-level in exact mode
D_{hom}	contraction identity fails
D_{pers}	naturality across sublevels not established
D_{bar}	interval-decomposition hypotheses or endpoint convention absent
D_{return}	external conclusion lacks a typed Return map
D_{cert}	replay, serialization, digest, or verifier failure
D_{choice}	noncanonical basis, pivot order, or representative not declared

Theorem 12.1 (Zero-defect persistent bridge). *For a finite filtered chain complex over a field, a verified finite sequence of equal-level unit cancellations, together with accepted HC-5 certificate data and a separately valid realization interface, gives*

$$D_{\text{chain}} = D_{\text{fil}} = D_{\text{pivot}} = D_{\text{hom}} = D_{\text{pers}} = D_{\text{return}} = D_{\text{cert}} = 0.$$

If the finite-type barcode hypotheses are verified, then also $D_{\text{bar}} = 0$. The source, realization, and choice components remain governed by their own bindings.

13 AK interface and Return licensing

The typed chain is

$$\text{external filtered object} \xrightarrow{\mathcal{R}} (C, F) \xrightarrow{\text{HC-compression}} (T, F) \xrightarrow{\text{Pers}(I)} \text{Pers}(C) \xrightarrow{\mathcal{Q}} \text{external conclusion}$$

where:

- (i) $\mathcal{R}$ is the realization theorem or certified construction;
- (ii) HC-5 produces a zero-shift filtered contraction and proof certificate;
- (iii) $\text{Pers}(I)$ is the exact persistence Return isomorphism;
- (iv) $\mathcal{Q}$ interprets the returned invariant externally.

Warning 13.1 (No realization by compression). HC-5 does not prove that a Vietoris–Rips, Čech, cellular, simplicial, or other filtered complex correctly models an external space or signal. It preserves the persistence module of the declared filtered complex. Realization accuracy remains a separate theorem.

Warning 13.2 (No identification with AK collapse). Equal-level algebraic cancellation is not identified with AK canonical collapse, Type IV diagnostics, detector calculus, or any v20 threshold operation. Compatibility is architectural, not definitional.

14 Completion ledger

Obligation	Status	Evidence
Filtered object typing	complete	finite ordered filtration and sub-complex condition
Strict filtered contraction	complete	explicit definition and sublevel restriction
Persistence-module Return	proved	natural isomorphism theorem
Barcode Return	proved conditionally	finite-type interval decomposition regime
Representative Return	proved	I, P, H with filtration control
Elementary cancellation	proved	equal-level unit pivot criterion
Iterated compression	proved	HC-3 cumulative formulas remain strict
Cross-level failure boundary	complete	finite-bar counterexample
Replay certificate extension	complete candidate	filtration-labelled HC-4 contract
Approximate shifted branch	scoped only	no unlicensed interleaving claim
External realization Return	interface-dependent	must be supplied per application
Independent audit	open	not performed in this artifact
Formal proof assistant verification	open	not performed
Successor Claim Register entry	open	identifier and digest assignment pending

15 Final theorem and status

Theorem 15.1 (HC-5 exact Return closure). *Within the declared finite one-parameter regime, a verified zero-shift filtered contraction provides:*

- (i) *exact homology Return at every filtration level;*
- (ii) *a natural isomorphism of persistence modules;*
- (iii) *exact preservation of all persistent ranks, births, and deaths;*
- (iv) *exact barcode Return under finite-type decomposition hypotheses;*
- (v) *filtration-controlled cycle and boundary witnesses;*
- (vi) *a deterministic certificate path from compressed computation to the input complex.*

Therefore this zero-shift filtered-contraction branch is complete as an internal Return bridge under its stated hypotheses; no completeness claim is made for all possible persistence-preserving reductions.

Protocol 15.2 (Status assignment). The appropriate current status is

return_proved + persistent_bridge_complete_candidate + GPT_reaudited .

It is not yet integration_ready: independent mathematical audit, exact source-statement binding, formal verifier implementation, artifact hashes, successor claim identifiers, and release governance remain external obligations.

16 Source-statement orientation and binding boundary

The internal Return theorem is an elementary consequence of restricting a zero-shift filtered contraction to every filtration level. The following external sources orient, but do not replace, future line-level source binding:

Source role	Oriented statement
Zomorodian–Carlsson	finite-type one-parameter persistence modules over a field admit the graded-module representation underlying barcode decomposition
Corbet–Kerber	detailed reconstruction and clarification of the persistence representation theorem
Dłotko–Wagner	filtration-compatible Morse matchings pair cells at equal filtration level and preserve persistence
Allili–Kaczynski–Landi	filtration-preserving reductions in multidimensional persistence, relevant to the declared multiparameter non-claim
Chazal–de Silva–Glisse–Oudot	persistence modules, interval decompositions, interleavings, and stability for the separately scoped approximate branch

Warning 16.1 (Binding status). This appendix does not yet assign theorem numbers, page ranges, source-file hashes, or verbatim source statements. Consequently D_{src} remains open even though the internal proofs are complete.

References

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- [5] R. Corbet and M. Kerber, *The Representation Theorem of Persistence Revisited and Generalized*, arXiv:1707.08864.
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Part III

Finite Persistence–Ext Reconstruction, Enhancement, and Boundary Return

13 Normative scope, release status, and six integrated repairs

Classification 13.1 (Document role and status).

Role: theorem-bearing successor appendix over frozen AK-HPDST v20.0.0.

Evidence status: mixed proved internal theorems, conditional theorems with explicit transport certificates, and source-bound classical inputs.

Release status: integrated mathematical repair candidate; not yet integrated into v20 or a future canonical release.

Cycle status: first major PE cycle internally closed as an internal-audit candidate; all repair sets through M8/G4 applied; independent audit pending.

Protocol 13.2 (MF-1 through MF-6). The integrated text enforces the following rules.

1. PE-1 is an internal transport-edge Ext encoding and calibration, not a realization of persistent homology itself.
2. PE-6–PE-10 are nonhereditary enhancement/deformation theorems and require a separate Interface before any persistence reading.
3. PE-7 descent is preliminary; PE-9 is the sole normative global descent theorem.
4. PE-8 governs local formal deformation, PE-9 global atlas and wall crossing, and PE-10 valiative boundary and certified contact Return.
5. PE-2–PE-5 use the single convention table in section 15.
6. Finite scripts and check counts are regression evidence only and never theorem premises.

Protocol 13.3 (Second-reaudit repairs M1–M4 and G1–G2). The retained v1.1 repair layer additionally enforces:

1. PE-7 consumes a dg/A_∞ quasi-equivalence, or an equivalent certificate of quasi-fully-faithfulness on the relevant morphism complexes plus H^0 -essential surjectivity.
2. PE-8 Morita transport uses an explicit finite zigzag of complete filtered L_∞ quasi-isomorphisms compatible with the declared completions; a bare B_∞ homotopy-class statement is not silently converted into one strict controller map.
3. PE-9 separates the zero and nonzero cubic marked models by formality/nonformality at A_∞ quasi-isomorphism strength, and uses the Coxeter polynomial only for the standard-heart derived boundary.
4. PE-10 makes no general claim that the raw first nonzero cochain coefficient is gauge invariant. General contact Return requires a gauge-normal-form or associated-graded Deligne-groupoid certificate; the sparse marked cubic family has its own direct diagonal-contact theorem.
5. PE-1 is named only as a complete-stencil transport-edge/Ext calibration.

6. Dedicated PE-2 and PE-8 source-binding records are part of the local release package.

Protocol 13.4 (Fourth-reaudit repairs N1–N4 and G1). This v1.2 text additionally enforces:

1. PE-7 states comparison results in their native type: strict dg functors give strict dg-algebra maps, general A_∞ quasi-equivalences give marked A_∞ quasi-isomorphisms, and bimodule/quasi-functor data give source-bound zigzags in the Morita homotopy category.
2. PE-8 and PE-9 work over a field of characteristic zero and consume complete, separated, pronilpotent filtered controllers over local Artin or complete local bases.
3. Under the convention $g_{uv} = h_v h_u^{-1}$, tree trivializations are unique up to one simultaneous right multiplication.
4. PE-10 extends a finite increasing filtration by $F_p C = 0$ below its first step and $F_p C = C$ above its last step, and defines the Rees complex as a genuine graded $k[u]$ -submodule of $C[u, u^{-1}]$.
5. The PE-10 local source binding distinguishes conditional general contact data from the proved sparse cubic diagonal contact theorem.

Protocol 13.5 (Sixth- and seventh-reaudit repairs M5–M8 and G3–G4). This v1.4 text additionally enforces:

1. The cubic family is first defined polynomially over $k[T]$; its T -adic completion over $k[[T]]$ is a separate local formal object, and only the polynomial family has fibers $T = \lambda$ for arbitrary $\lambda \in k$.
2. A PE-10 integral boundary datum explicitly contains a DVR, a chosen uniformizer, its fraction and residue fields, and either a coefficient map or an R -linear lift of every consumed cubic datum.
3. PE-9 wall statements declare the Zariski site on $\mathbb{A}_k^1$ and use geometric fibers; real and complex path statements separately use the Euclidean topology.
4. Family definedness is recorded by a locus U_{family} and is not misidentified with local triviality or an empty wall at a fixed-target Return strength.
5. PE-10 uses an integral split-projective Rees construction over $R[u]$ when the Rees record is merged with a DVR boundary package; the field-valued Rees theorem is retained only as a specialization.
6. The typeset body, Claim Register, Source Binding Registry, audit manifest, package manifest, and ZIP use one synchronized v1.4 status and include the local source bindings for PE-1 through PE-10.

Warning 13.6 (No independent-referee upgrade). This integrated edition normalizes and consolidates the same research corpus. It is not independent peer review. Compilation, source registration, finite replay, and internal consistency do not establish novelty or substitute for independent proof checking.

14 Frozen dependency boundary and Return-strength architecture

The immutable lower baseline is AK-HPDST v20.0.0. Successor results do not silently alter its window-first semantics, hard-threshold discipline, finite-proof rules, representative policy, higher-edge eligibility, Type IV typing, Return separation, or Claim Register authority.

The finite-quiver progression is

$$\Sigma_{\text{edge}} \prec \Sigma_{\text{obj}} \prec \Sigma_{\text{cat}} \prec \Sigma_{\text{exact}} \prec \Sigma_{\text{der}},$$

where the successive targets are internal edge vanishing, object/barcode Return, category Return, exact/Yoneda Return, and bounded derived/cone/triangle Return. The enhancement progression is

$$\Sigma_{m_3} \prec \Sigma_{\text{marked}} \prec \Sigma_{\text{Morita}} \prec \Sigma_{\text{local def}} \prec \Sigma_{\text{global atlas}} \prec \Sigma_{\text{boundary}}.$$

No stronger level follows from a weaker one without the theorem and certificate displayed in the corresponding chapter.

15 Unified finite-quiver convention table

Item	Frozen convention
Quiver	$Q_n : 1 \rightarrow 2 \rightarrow \cdots \rightarrow n$.
Persistence objects	Covariant representations of Q_n in finite-dimensional k -vector spaces; equivalently finite-grid one-parameter persistence modules.
Modules	Left modules unless a displayed equivalence explicitly inserts an opposite algebra.
Intervals	$I[a, b]$ is supported at $a \leq i \leq b$ with identity maps on internal arrows.
Sentinel	$\iota_n M$ appends vertex $n + 1$ with value zero and zero terminal arrow.
Probe generator	$G_{n+1} = \bigoplus_{1 \leq c \leq d \leq n+1} I[c, d]$.
Probe algebra	$R_{n+1} = \text{End}(G_{n+1})$; the displayed probe modules carry the handedness stated in each construction.
Variance	$M \mapsto \text{YExt}^1(M, G_{n+1})$ is contravariant in M ; induced morphism Return reverses arrows exactly as displayed.
Derived generator	$P_n = \bigoplus_{i=1}^n I[i, n]$ and $\Gamma_n = \text{End}(P_n)^{\text{op}}$, so $\text{Hom}(P_n, -)$ is a left Γ_n -module functor.
Equality strength	Coordinate matrices are never canonical without declared bases; object, functor, exact, or triangle isomorphism is the default invariant strength.

16 Integrated dependency DAG and minimal claim kernel

UID	Result	Depends on	Integrated role
PE-CAL-001	Complete-edge derived encoding and vanishing calibration	frozen v20 edge package	calibration only
PE-FQ-001	Ext-signature inversion	finite type- A intervals	object/barcode Return
PE-FQ-002	Sentinel Hilton–Rees embedding	PE-FQ-001 context	morphism/category Return
PE-FQ-003	Hom–Ext boundary and exactness reflection	PE-FQ-002	exact/Yoneda Return
PE-FQ-004	Projective-generator derived equivalence	PE-FQ-003 context	cone/triangle Return
PE-ED-001	Cubic m_3 relation Return	separate nonhereditary source	enhancement Return
PE-ED-002	Marked minimal-model and Morita invariance	PE-ED-001	choice-independent Return
PE-ED-003	Hochschild local formal moduli	PE-ED-002	local deformation Return
PE-ED-004	Homotopy-coherent atlas and wall crossing	PE-ED-003	global deformation Return
PE-ED-005	Derived specialization and regularized boundary	PE-ED-004 context	boundary Return

Part IV

Calibration and finite one-parameter persistence reconstruction

17 PE-1 Calibration: Internal Transport-Edge Ext Encoding

Classification 17.1 (Integrated role after MF-1). This chapter is calibration only. It encodes an inherited complete transport edge in a derived object and returns its vanishing predicate. It is not a realization of the full persistent-homology profile and is not a headline identity $\text{PH}_1 \cong \text{Ext}^1$.

17.1 The inherited complete-edge package and the exact open gap

Definition 17.2 (Two Ext lanes). PE keeps two distinct lanes.

1. **Lane A: derived coefficient Ext.** For $A \in D^b(k\text{-mod})$,

$$\text{Ext}^1(A, k) := \text{Hom}_{D^b(k\text{-mod})}(A, k[1]).$$

This is the lane used in this appendix.

2. **Lane B: persistence-category Ext.** For persistence representations M, N ,

$$\text{Ext}_{\text{Rep}_k(A_n)}^1(M, N)$$

measures extension classes between two persistence objects. It is reserved for a later PE stage.

The two lanes are not identified by common notation.

Proposition 17.3 (Inherited canonical field edge). *For every bounded $A \in D^b(k\text{-mod})$ and every $n \geq 0$, the frozen Main Theorem 4.20 gives a natural contravariant isomorphism*

$$\text{Ext}^n(A, k) \cong \text{Hom}_k(H^{-n}(A), k).$$

In particular, for $A \in D^{[-1,0]}(k\text{-mod})$,

$$\text{Ext}^1(A, k) = 0 \iff H^{-1}(A) = 0.$$

Proposition 17.4 (Inherited point-evaluation failure). *A single evaluation $P \mapsto P(t_*)$ is not zero-reflecting on any profile class allowing a transient interval disjoint from t_* . Therefore Ext vanishing formed from one such evaluation cannot imply profile vanishing in general.*

Proof. This is the frozen Main Proposition 4.60: for $P = k_{[0,L)}$ and $t_* > L$, one has $P(t_*) = 0$ although $P \neq 0$. $\square$

Definition 17.5 (Exact PE-1 gap). The inherited family-mesh theorem supplies a fixed functorial edge and proves zero-reflection. The remaining hypothesis in Foundation Corollary 3.66 is a natural-family, independently meaningful isomorphism

$$H^{-1}(A(P)) \xrightarrow{\sim} \text{Edge}_{\Sigma, \tau}(P)$$

with non-adaptivity and change-of-realization compatibility. PE-1 closes exactly this gap for the fixed-stencil transport edge.

17.2 Finite repaired profile class and complete stencil

Fix throughout a field k , a bounded right-open window $W = [a, b)$, and a threshold $\tau > 0$. Put

$$W_\tau := \{t \in W \mid t + \tau \in W\}.$$

Definition 17.6 (Family-mesh PE scope). Let $\mathcal{F}$ be a nonempty family of finite, left-closed right-open windowed barcodes with a valid uniform excess certificate $e_{\tau,W}(\mathcal{F}) > 0$. Let $\Sigma \subset W_\tau$ be finite and satisfy

$$\text{Mesh}_{W_\tau}(\Sigma) < e_{\tau,W}(\mathcal{F}).$$

Let $\mathfrak{P} = \mathfrak{P}_1^{\mathcal{F},\Sigma,\tau}$ be the category whose objects are the repaired degree-one profiles

$$P = T_\tau^{\text{bar}}(B), \quad B \in \mathcal{F},$$

together with the zero profile, and whose morphisms are actual persistence natural transformations on the fixed window. The family, window, threshold, endpoint convention, excess certificate, and stencil are fixed before any verdict.

Remark 17.7 (Nontriviality). A nontrivial PE witness additionally records some $B \in \mathcal{F}$ with $T_\tau^{\text{bar}}(B) \neq 0$. This prevents an empty or already-zero family from being advertised as a substantive zero-reflection result.

Theorem 17.8 (Inherited complete-stencil zero-reflection). *On $\mathfrak{P}$, the fixed-threshold edge*

$$\text{Edge}_{\Sigma,\tau}(P) := \bigoplus_{t \in \Sigma} \text{im}(P(t) \xrightarrow{\rho_t^P} P(t + \tau))$$

is functorial and satisfies

$$\text{Edge}_{\Sigma,\tau}(P) = 0 \iff P = 0.$$

Proof. This is Foundation Appendix C, Proposition 3.63 and Theorem 3.65. The mesh/excess inequality ensures that every retained interval contains an active start sampled by Σ ; its τ -shift structure map is nonzero, so a nonzero repaired profile has nonzero edge. The exact internal source identities and locators are frozen in the dedicated PE-1 source-statement binding shipped with this v1.4 package; the binding cites the immutable v20 Main and Foundation artifacts and does not promote PE-1 beyond calibration strength. $\square$

17.3 Transport-coimage realization

Definition 17.9 (Stencil arrow functor). For $P \in \mathfrak{P}$, define

$$\text{Arr}_{\Sigma,\tau}(P) := (\rho_t^P : P(t) \rightarrow P(t + \tau))_{t \in \Sigma} \in \text{Arr}(\mathbf{Vect}_k^{\text{fd}})^\Sigma.$$

For a persistence morphism $\eta : P \rightarrow Q$, naturality gives commutative squares

$$\begin{array}{ccc} P(t) @>\rho_t^P>> P(t + \tau) \\ @V\eta_tVV @VV\eta_{t+\tau}V \\ Q(t) @>\rho_t^Q>> Q(t + \tau). \end{array}$$

These squares define $\text{Arr}_{\Sigma,\tau}(\eta)$.

Lemma 17.10 (Functoriality of coimage). *The assignment*

$$(f : V \rightarrow W) \mapsto \text{coim}(f) := \text{coker}(\ker f \hookrightarrow V)$$

defines a functor $\text{coim} : \text{Arr}(\mathbf{Vect}_k^{\text{fd}}) \rightarrow \mathbf{Vect}_k^{\text{fd}}$. For every arrow f , the canonical coimage–image morphism

$$\chi_f : \text{coim}(f) \xrightarrow{\sim} \text{im}(f)$$

is an isomorphism, natural in commutative squares.

Proof. A commutative square $(a, b) : f \rightarrow f'$ sends $\ker f$ into $\ker f'$, so a descends uniquely to a map on cokernels. Identity and composition are inherited from the universal property of cokernels. In vector spaces every morphism has the canonical epi–mono factorization, and the induced coimage–image map is an isomorphism. Naturality follows from uniqueness in the factorization square. $\square$

Definition 17.11 (Transport coimage and PE realization). Define the finite-dimensional transport coimage

$$\mathcal{E}_{\Sigma, \tau}(P) := \bigoplus_{t \in \Sigma} \text{coim}(\rho_t^P).$$

For each t , let $i_t^P : \ker \rho_t^P \hookrightarrow P(t)$ be the actual kernel inclusion, regarded as a map of complexes concentrated in degree zero. Define the PE derived realization by the standard mapping-cone construction

$$\mathcal{A}_{\Sigma, \tau}(P) := \bigoplus_{t \in \Sigma} \text{Cone}(i_t^P)[1] \in D^{[-1, -1]}(k\text{-mod}) \subset D^{[-1, 0]}(k\text{-mod}).$$

A persistence morphism induces maps of kernels and hence maps of the two-term complexes; the standard cone construction and finite direct sum define $\mathcal{A}_{\Sigma, \tau}(\eta)$. Thus $\mathcal{A}_{\Sigma, \tau} : \mathfrak{P} \rightarrow D^b(k\text{-mod})$ is fixed at scope level, before any zero/nonzero verdict.

Theorem 17.12 (Natural transport-edge realization). *There is a natural isomorphism*

$$\epsilon_P : H^{-1}(\mathcal{A}_{\Sigma, \tau}(P)) \xrightarrow{\sim} \text{Edge}_{\Sigma, \tau}(P)$$

for all $P \in \mathfrak{P}$. Moreover:

1. $\mathcal{A}_{\Sigma, \tau}(0) = 0$;
2. an actual profile isomorphism $P \xrightarrow{\sim} Q$ induces an isomorphism $\mathcal{A}_{\Sigma, \tau}(P) \xrightarrow{\sim} \mathcal{A}_{\Sigma, \tau}(Q)$;
3. the construction depends on the actual structure maps, not only on dimensions or a zero verdict;
4. the realization is non-adaptive and functorial on the declared morphism category.

Proof. For each kernel inclusion i_t^P , the standard cone has $H^0(\text{Cone}(i_t^P)) = \text{coker}(i_t^P) = \text{coim}(\rho_t^P)$ and no other cohomology. After shifting by $[1]$ and taking finite direct sums,

$$H^{-1}(\mathcal{A}_{\Sigma, \tau}(P)) \cong \mathcal{E}_{\Sigma, \tau}(P), \quad H^j(\mathcal{A}_{\Sigma, \tau}(P)) = 0 \ (j \neq -1).$$

The direct sum of the natural coimage–image isomorphisms $\chi_{\rho_t^P}$ gives ϵ_P . Naturality follows from the preceding lemma and finite direct sum. The remaining statements are immediate from functoriality and the fact that zero structure maps have zero coimage. $\square$

Remark 17.13 (Exact strength of the non-synthetic claim). The construction is not promoted merely because a vector space was written in degree -1 . The independent content is the fixed source category, actual arrow functor, actual kernel inclusions, functorial mapping-cone realization, universal coimage construction, natural coimage–image theorem, and profile-change compatibility. This discharges the precise natural-family realization hypothesis of Foundation Corollary 3.66.

Nevertheless, the construction is deliberately conservative: it internalizes the complete transport edge and creates no information beyond that edge. It is not a realization of the full filtered chain complex, the whole persistence module, or an external mathematical object. The correct label is *natural transport-edge realization*, not universal PH–Ext realization.

17.4 Bidirectional vanishing Return

Theorem 17.14 (Complete-stencil transport-edge/Ext vanishing calibration). *For every $P \in \mathfrak{P}$,*

$$P = 0 \iff \text{Ext}_{D^b(k\text{-mod})}^1(\mathcal{A}_{\Sigma,\tau}(P), k) = 0.$$

Equivalently, for every filtered input whose repaired degree-one profile $P = \text{Eval}_{1,W,\tau}(F)$ belongs to $\mathfrak{P}$ and whose exact repaired-profile record is available,

$$\text{Eval}_{1,W,\tau}(F) = 0 \iff \text{Ext}^1(\mathcal{A}_{\Sigma,\tau}(\text{Eval}_{1,W,\tau}(F)), k) = 0.$$

This is an equivalence of vanishing predicates only.

Proof. The canonical field edge and the realization theorem give natural isomorphisms

$$\text{Ext}^1(\mathcal{A}_{\Sigma,\tau}(P), k) \cong \text{Hom}_k(H^{-1}(\mathcal{A}_{\Sigma,\tau}(P)), k) \cong \text{Hom}_k(\text{Edge}_{\Sigma,\tau}(P), k).$$

All vector spaces are finite-dimensional, so the last term vanishes if and only if $\text{Edge}_{\Sigma,\tau}(P) = 0$. Complete-stencil zero-reflection gives $\text{Edge}_{\Sigma,\tau}(P) = 0$ if and only if $P = 0$. $\square$

Corollary 17.15 (Complete edge dual Return). *There is a natural covariant isomorphism*

$$\text{Edge}_{\Sigma,\tau}(P) \xrightarrow{\sim} \text{Ext}^1(\mathcal{A}_{\Sigma,\tau}(P), k)^\vee.$$

Thus the Ext group Returns the finite complete-stencil edge up to canonical finite-dimensional duality.

Proof. Dualize the canonical isomorphism $\text{Ext}^1(\mathcal{A}_{\Sigma,\tau}(P), k) \cong \text{Edge}_{\Sigma,\tau}(P)^\vee$ and use the natural evaluation isomorphism $V \cong V^{\vee\vee}$ for finite-dimensional vector spaces. The first Ext functor is contravariant in P through $\mathcal{A}_{\Sigma,\tau}$; the second dual restores covariance. $\square$

Non-claim 17.16 (No profile Return). The preceding corollary does not reconstruct P . The edge functor is zero-reflecting on $\mathfrak{P}$ but need not be faithful, full, conservative for isomorphisms, or barcode-complete. Zero-reflection distinguishes zero from nonzero; it does not distinguish all nonzero profiles.

17.5 Return-theory interpretation and status procedure

Definition 17.17 (PE Return datum). For $P \in \mathfrak{P}$, the PE Return datum is

$$\mathcal{R}_{\text{PE}}(P) = (\mathfrak{P}, D^b(k\text{-mod}), \{0, 1\}, P, \mathcal{A}_{\Sigma,\tau}(P), \text{Ext}^1(-, k), \epsilon_P, =, \Sigma_{\text{van}}, \mathcal{C}_P),$$

where Σ_{van} is vanishing-predicate strength and $\mathcal{C}_P$ is the replay certificate of section 17. The comparison is the displayed chain of canonical isomorphisms and the inherited zero-reflection theorem.

Classification 17.18 (Deterministic PE status). At fixed scope and requested vanishing-predicate strength:

1. a false scope assertion, false mesh/excess claim, invalid source binding, or invalid structure-map record gives REJECT;
2. otherwise, a missing family, stencil, actual profile, actual maps, coverage theorem, realization record, or coefficient hypothesis gives UNDEFINED;
3. otherwise, the theorem evaluates the Ext group and emits RETURN for the corresponding zero/nonzero profile verdict;
4. a request for full profile or barcode recovery receives PARTIAL-RETURN, because only edge-data and vanishing-predicate strengths are proved;

5. OBSTRUCTED is used only for a separately invoked stronger comparison with an existing valid comparison artifact and a nonzero declared defect.

Warning 17.19 (Type IV firewall). Missing realization, incomplete stencil coverage, or inability to promote from vanishing equivalence to profile Return is not Type IV. Under Return Theory it is respectively UNDEFINED, REJECT, or PARTIAL-RETURN, unless an actual comparison artifact for a stronger target exists and has a nonzero kernel, cokernel, cone, or other declared defect.

17.6 HC compatibility

Theorem 17.20 (Repaired-profile isomorphism transport). *Let $P, Q \in \mathfrak{P}$ and let $\eta : P \xrightarrow{\sim} Q$ be an actual persistence-profile isomorphism on the fixed scope (W, τ, Σ) . Then*

$$\mathcal{A}_{\Sigma, \tau}(\eta) : \mathcal{A}_{\Sigma, \tau}(P) \xrightarrow{\sim} \mathcal{A}_{\Sigma, \tau}(Q)$$

and the induced contravariant map

$$\text{Ext}^1(\mathcal{A}_{\Sigma, \tau}(Q), k) \xrightarrow{\sim} \text{Ext}^1(\mathcal{A}_{\Sigma, \tau}(P), k)$$

is an isomorphism. Hence the PE vanishing verdict is invariant under the supplied repaired-profile isomorphism.

Proof. Each component square of an isomorphism induces an isomorphism on coimages. Finite direct sum and shift preserve isomorphisms. The Ext functor sends derived isomorphisms to isomorphisms. $\square$

Corollary 17.21 (Conditional HC-to-PE composition). *Suppose an HC zero-shift filtered contraction between filtered complexes C and C' supplies its exact persistence-module Return and, in addition, an actual repaired-profile isomorphism*

$$\eta_{\text{HC}} : \text{Eval}_{1, W, \tau}(C) \xrightarrow{\sim} \text{Eval}_{1, W, \tau}(C')$$

on the same PE scope. Then the PE Ext groups are isomorphic and the bidirectional vanishing verdict agrees for C and C' .

Remark 17.22 (Why the extra comparison is displayed). HC exact persistence Return does not silently make hard threshold deletion a global functor. The HC-to-PE composition therefore consumes an actual repaired-profile comparison or a separately proved admissible transport theorem. Barcode equality alone is not substituted for this morphism-level record.

17.7 Worked examples and adversarial counterexamples

17.7.1 Single-point failure

Let $P = k_{[0, 2)}$ and choose $t_* = 3$. Then $P(t_*) = 0$ although $P \neq 0$. Any realization based only on $P(t_*)$ has vanishing Ext while the profile is nonzero. This is the inherited reverse-failure calibration.

17.7.2 Complete-stencil success

Let $W = [0, 5)$, $\tau = 1$, and

$$P = k_{[0, 3)} \oplus k_{[2, 5)}.$$

Choose $\Sigma = \{0.5, 1.5, 2.5, 3.5\}$. On $W_\tau = [0, 4)$ the augmented mesh is 1, while the retained excess of each interval is 2, so the strict family-mesh condition is satisfied. The first interval contributes at 0.5 and 1.5, and the second contributes at 2.5 and 3.5. Hence

$$\text{Edge}_{\Sigma, 1}(P) \cong k^4, \quad \mathcal{A}_{\Sigma, \tau}(P) \simeq k^4[1], \quad \text{Ext}^1(\mathcal{A}_{\Sigma, \tau}(P), k) \cong (k^4)^\vee \neq 0.$$

For the zero profile all three objects are zero.

17.7.3 Threshold boundary

Let $B = k_{[0,1]}$ and $\tau = 1$. The frozen hard-deletion convention removes finite bars of length at most τ , so $T_1^{\text{bar}}(B) = 0$ and the PE Ext group is zero. For $B_\delta = k_{[0,1+\delta]}$ with $\delta > 0$, the bar is retained; a complete stencil containing an active start gives nonzero edge and nonzero Ext. No alternative endpoint convention is introduced.

17.7.4 Incomplete-stencil failure

Let $W = [0, 3)$, $\tau = 1$, $P = k_{[0,1.5]}$, and $\Sigma = \{1.75\}$. Then $P(1.75) = 0$, so the stencil edge is zero although $P \neq 0$. The mesh/excess completeness hypothesis fails. The correct status for a claimed complete certificate is REJECT; without a completeness claim the reverse inference is merely unavailable.

17.7.5 Edge Return is not profile Return

Let $W = [0, 6)$, $\tau = 1$, and $\Sigma = \{0.5, 1.5, 2.5, 3.5, 4.5\}$. Consider

$$P = k_{[0,6)}, \quad Q = k_{[0,4)} \oplus k_{[3,6)}.$$

The augmented mesh on $W_\tau = [0, 5)$ is 1. The smallest retained excess in the two-object family is 2, so the same stencil is complete for both profiles. They are nonzero and nonisomorphic. The long interval P contributes at all five stencil points. For Q , the first interval contributes at 0.5, 1.5, 2.5 and the second at 3.5, 4.5. Hence

$$\text{Edge}_{\Sigma,1}(P) \cong k^5 \cong \text{Edge}_{\Sigma,1}(Q),$$

and the corresponding PE realizations and Ext groups are isomorphic. Thus even complete zero-reflection plus full edge dual Return does not recover the barcode or profile object.

17.7.6 HC compatibility toy model

Let a filtered chain complex over k have $C_1 = ka \oplus kz$, $C_0 = kb$, differential $d(a) = b$, $d(z) = 0$, and all generators at the same filtration level. Cancelling the equal-level unit pivot (a, b) leaves the complex generated by z . The HC contraction preserves the degree-one persistence module exactly. Once the actual repaired-profile isomorphism is recorded on a fixed PE scope, the theorem of the preceding section gives identical PE realizations and Ext verdicts.

17.7.7 General-ring failure

Over $R = k[\varepsilon]/(\varepsilon^2)$ with coefficient module $k = R/(\varepsilon)$, the object $A = k[0]$ has $H^{-1}(A) = 0$ but $\text{Ext}_R^1(k, k) \cong k \neq 0$. Hence the field-coefficient semisimplicity used in PE-1 cannot be removed without an Ext-acyclic coefficient or cogenerator package.

17.8 Defect ledger and failure boundary

Definition 17.23 (Non-aggregated PE defect ledger). The PE record keeps the following components separate:

Field	Meaning
D_{scope}	Missing or false field/window/threshold/family/profile-class data. This is a typed status field, not an algebraic defect object.
D_{cov}	Missing or false mesh/excess or stencil-coverage evidence.

D_{arr}	Missing actual persistence structure maps or morphism squares.
D_{real}	Missing functorial coimage realization or natural edge-realization isomorphism.
D_{coeff}	Failure of the field/finite-dimensional coefficient hypotheses.
D_{HC}	When HC composition is invoked, kernel/cokernel or non-isomorphism defect of the actual repaired-profile comparison.
D_{strength}	Difference between requested profile/barcode Return and proved edge/vanishing Return. Normally this yields partial Return, not Type IV.
D_{stab}	Perturbation or grid-refinement stability, not invoked by the exact PE-1 theorem unless a separate comparison record is supplied.

No scalar sum is defined, and favorable numerical slack cannot repair any missing non-scalar field.

Non-claim 17.24 (Failure boundaries). PE-1 does not license any of the following inferences:

1. point-evaluation zero $\Rightarrow$ profile zero;
2. dimension-vector equality $\Rightarrow$ equality of structure maps;
3. edge-vector-space isomorphism $\Rightarrow$ persistence-profile isomorphism;
4. barcode equality $\Rightarrow$ equality of richer extension data in another category;
5. general-ring Ext zero $\Leftrightarrow$ negative cohomology zero;
6. one-parameter interval methods $\Rightarrow$ multiparameter classification;
7. exact zero-predicate equivalence $\Rightarrow$ positive-radius metric stability;
8. finite-window theorem $\Rightarrow$ uncontrolled infinite-window theorem.

18 PE-2: Finite Ext-Signature and Barcode/Object Return

18.1 Finite one-parameter persistence as a quiver representation

Definition 18.1 (Linearly oriented persistence quiver). Let

$$Q_n : 1 \longrightarrow 2 \longrightarrow \cdots \longrightarrow n$$

and let $A_n = kQ_n$ be its path algebra. Write

$$\mathbf{Pers}_n(k) := \mathbf{Rep}_k(Q_n) \simeq A_n\text{-mod}_{\text{fd}}.$$

An object is a diagram

$$M_1 \xrightarrow{f_1} M_2 \xrightarrow{f_2} \cdots \xrightarrow{f_{n-1}} M_n$$

of finite-dimensional k -vector spaces.

Definition 18.2 (Interval representation). For $1 \leq a \leq b \leq n$, the interval representation $\mathbb{I}[a, b]$ has value k at vertices $a, a+1, \dots, b$, value 0 elsewhere, identity maps between consecutive nonzero vertices, and zero maps otherwise.

Theorem 18.3 (Finite interval decomposition). *Every $M \in \mathbf{Pers}_n(k)$ has a decomposition*

$$M \cong \bigoplus_{1 \leq a \leq b \leq n} \mathbb{I}[a, b]^{\oplus m_{a,b}(M)}$$

with uniquely determined nonnegative integers $m_{a,b}(M)$.

Proof. This is the finite linearly oriented specialization of the interval-decomposition and Krull–Schmidt theorem already used by the frozen one-parameter persistence baseline. The present appendix consumes only the finite-dimensional finite-type case. $\square$

Definition 18.4 (Barcode and rank invariant). The finite barcode is the multiset

$$\text{Bar}(M) = \{[a, b] \text{ with multiplicity } m_{a,b}(M)\}.$$

For $1 \leq i \leq j \leq n$, the rank invariant is

$$r_{i,j}(M) := \text{rank}(M_i \rightarrow M_j) = \sum_{a \leq i \leq j \leq b} m_{a,b}(M).$$

18.2 Grid-complete interface from a repaired profile

Definition 18.5 (Grid-complete discretization record). Let P be a repaired post-threshold degree-one persistence profile on a bounded right-open window W . A grid-complete record consists of:

1. a finite ordered grid $\Lambda = (\lambda_1 < \cdots < \lambda_n)$ compatible with the declared endpoint convention;
2. the actual structure maps defining

$$\text{Disc}_\Lambda(P) := (P(\lambda_1) \rightarrow \cdots \rightarrow P(\lambda_n)) \in \mathbf{Pers}_n(k);$$

3. a reconstruction convention Rec_Λ on the cells between successive grid values;
4. an actual natural isomorphism

$$\theta_P : \text{Rec}_\Lambda \text{Disc}_\Lambda(P) \xrightarrow{\sim} P$$

on the declared window.

The isomorphism θ_P , not finiteness of the sample list, is the completeness evidence.

Proposition 18.6 (Discrete Return implies continuous Return under a grid-complete record). *If P and P' have grid-complete records on the same grid and*

$$\text{Disc}_\Lambda(P) \cong \text{Disc}_\Lambda(P'),$$

then $P \cong P'$ on the declared window.

Proof. Compose the two reconstruction isomorphisms with the reconstructed discrete isomorphism. $\square$

Warning 18.7 (Finite observation is not grid completeness). A list of values and maps is not automatically a complete representation of a continuous profile. Missing critical values, incompatible endpoint decorations, or an absent reconstruction isomorphism makes the continuous Return UNDEFINED, even if the quiver-side Ext computation is correct.

18.3 The terminal-sentinel embedding

Definition 18.8 (Terminal-zero sentinel). Define

$$\iota_n : \text{Pers}_n(k) \longrightarrow \text{Pers}_{n+1}(k)$$

by

$$(\iota_n M)_i = M_i \quad (1 \leq i \leq n), \quad (\iota_n M)_{n+1} = 0,$$

using the original arrows through vertex n and the zero arrow $M_n \rightarrow 0$ at the end.

Proposition 18.9 (Exact full faithfulness). *The sentinel functor ι_n is exact and fully faithful. Moreover,*

$$\iota_n \llbracket a, b \rrbracket = \llbracket a, b \rrbracket \in \text{Pers}_{n+1}(k)$$

for every $1 \leq a \leq b \leq n$.

Proof. Kernels, cokernels, and finite direct sums in a quiver representation category are computed vertexwise, and appending a zero space preserves them. A morphism between sentinel objects is uniquely its restriction to the first n vertices, so the Hom map is bijective. The interval assertion is immediate. $\square$

Lemma 18.10 (The sentinel removes the projective blind spot). *Every nonzero interval summand $\llbracket a, b \rrbracket$ of $\iota_n M$ has $b \leq n$ and is therefore nonprojective in $\text{Pers}_{n+1}(k)$. The indecomposable projectives of Q_{n+1} are exactly $\llbracket i, n+1 \rrbracket$.*

Proof. For the orientation $1 \rightarrow \dots \rightarrow n+1$, the projective at vertex i is supported on every vertex reachable from i , hence is $\llbracket i, n+1 \rrbracket$. A sentinel interval ends no later than n . $\square$

Remark 18.11 (Why the extra zero vertex is structural). The sentinel is not padding for serialization. It moves the original terminal bars away from the projective boundary, where Yoneda Ext^1 would otherwise vanish against every probe.

18.4 Projective resolution and the interval Ext formula

Definition 18.12 (Projective interval). In $\text{Pers}_{n+1}(k)$ write

$$P_i := \llbracket i, n+1 \rrbracket.$$

Lemma 18.13 (Canonical projective resolution of a nonterminal interval). *For $1 \leq a \leq b \leq n$, there is a short exact sequence*

$$0 \longrightarrow P_{b+1} \xrightarrow{j_{a,b}} P_a \longrightarrow \llbracket a, b \rrbracket \longrightarrow 0.$$

Proof. The representation P_a is supported on $a, \dots, n+1$. Its subrepresentation supported on $b+1, \dots, n+1$ is canonically P_{b+1} . The quotient is supported on $a, \dots, b$ with identity arrows, hence is $\mathbb{I}[a, b]$. $\square$

Lemma 18.14 (Evaluation of Hom from a projective). *For every $N \in \mathbf{Pers}_{n+1}(k)$,*

$$\mathrm{Hom}(P_i, N) \cong N_i$$

naturally in N . Under these identifications, precomposition with $j_{a,b}$ is the structure map

$$N_a \longrightarrow N_{b+1}.$$

Proof. A morphism $P_i \rightarrow N$ is uniquely determined by the image of the generator at vertex i ; all later components are forced by commutativity. The second assertion follows by tracing the inclusion along the unique path $a \rightarrow b+1$. $\square$

Theorem 18.15 (Interval Yoneda-Ext formula). *For $1 \leq a \leq b \leq n$ and $1 \leq c \leq d \leq n+1$,*

$$\mathrm{YExt}_{A_{n+1}}^1(\mathbb{I}[a, b], \mathbb{I}[c, d]) \cong \begin{cases} k, & a < c \leq b+1 \leq d, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. Apply $\mathrm{Hom}(-, \mathbb{I}[c, d])$ to the projective resolution. Since P_a and P_{b+1} are projective, the relevant exact sequence is

$$\mathbb{I}[c, d]_a \longrightarrow \mathbb{I}[c, d]_{b+1} \longrightarrow \mathrm{YExt}^1(\mathbb{I}[a, b], \mathbb{I}[c, d]) \longrightarrow 0.$$

The target is k precisely when $c \leq b+1 \leq d$. If both a and $b+1$ lie in $[c, d]$, the first map is the identity and the cokernel is zero. If the target is nonzero and the source is zero, then necessarily $a < c$, and the cokernel is k . These are exactly the displayed conditions. $\square$

Corollary 18.16 (Finite additivity). *For finite direct sums in either argument, YExt^1 is additive. Hence the dimension of $\mathrm{YExt}^1(\iota_n M, \mathbb{I}[c, d])$ is the sum of the interval contributions in Theorem 18.15.*

18.5 The finite Ext signature

Definition 18.17 (Terminal-sentinel Ext signature). *For $M \in \mathbf{Pers}_n(k)$ define*

$$\mathfrak{E}_n(M) = (E_{c,d}(M))_{1 \leq c \leq d \leq n+1},$$

where

$$E_{c,d}(M) := \dim_k \mathrm{YExt}_{A_{n+1}}^1(\iota_n M, \mathbb{I}[c, d]).$$

For boundary arithmetic set

$$E_{c,d}(M) = 0 \quad \text{when } c > d, \quad E_{1,d}(M) = 0.$$

The latter identity also follows from the Ext formula.

Proposition 18.18 (Cumulative-transform formula). *If*

$$M \cong \bigoplus_{1 \leq a \leq b \leq n} \mathbb{I}[a, b]^{\oplus m_{a,b}},$$

then

$$E_{c,d}(M) = \sum_{a=1}^{c-1} \sum_{q=c}^d m_{a,q-1}$$

for $1 \leq c \leq d \leq n+1$.

Proof. By additivity and Theorem 18.15, an interval $\llbracket a, b \rrbracket$ contributes one exactly when

$$a < c \leq b + 1 \leq d.$$

Writing $q = b + 1$ gives the displayed double sum. □

Remark 18.19 (Finite proof-compression size). The signature has $(n+1)(n+2)/2$ integer entries, of which the row $c = 1$ is identically zero. It is finite and exact for the declared n ; this statement is not a universal optimality theorem.

18.6 Möbius inversion and exact barcode Return

Lemma 18.20 (First difference in the probe endpoint). *For $1 \leq c \leq q \leq n + 1$, define*

$$F_{c,q}(M) := E_{c,q}(M) - E_{c,q-1}(M).$$

Then

$$F_{c,q}(M) = \sum_{a=1}^{c-1} m_{a,q-1}(M).$$

Proof. Subtract the cumulative formula at $d = q - 1$ from the formula at $d = q$. All terms cancel except those with terminal index q . □

Theorem 18.21 (Exact interval-multiplicity inversion). *For every $1 \leq a \leq b \leq n$,*

$$\boxed{m_{a,b}(M) = E_{a+1,b+1}(M) - E_{a+1,b}(M) - E_{a,b+1}(M) + E_{a,b}(M).}$$

Proof. Apply the preceding lemma with $q = b + 1$:

$$F_{a+1,b+1} = \sum_{x=1}^a m_{x,b}, \quad F_{a,b+1} = \sum_{x=1}^{a-1} m_{x,b}.$$

Their difference is $m_{a,b}$. Expanding both F terms gives the formula. □

Corollary 18.22 (Exact barcode Return). *The signature $\mathfrak{E}_n(M)$ determines every barcode multiplicity and therefore determines $\text{Bar}(M)$ exactly.*

Corollary 18.23 (Exact finite-quiver object Return). *For $M, N \in \mathbf{Pers}_n(k)$,*

$$\mathfrak{E}_n(M) = \mathfrak{E}_n(N) \iff M \cong N.$$

Proof. Equality of signatures gives equality of every multiplicity by Theorem 18.21; uniqueness of interval decomposition gives $M \cong N$. The converse follows from invariance of Yoneda Ext under isomorphism. □

Corollary 18.24 (Rank-invariant Return). *The signature determines every rank*

$$r_{i,j}(M) = \sum_{a \leq i \leq j \leq b} m_{a,b}(M).$$

Hence it returns the dimension vector and all composite-map ranks.

Theorem 18.25 (Continuous repaired-profile Return under grid completeness). *Let P and P' be repaired degree-one profiles with grid-complete records on the same finite grid Λ . If*

$$\mathfrak{E}_n(\text{Disc}_\Lambda P) = \mathfrak{E}_n(\text{Disc}_\Lambda P'),$$

then $P \cong P'$ on the declared window. In particular, the Ext signature returns the exact finite barcode of P relative to the declared endpoint convention.

Proof. By Theorem 18.23, the discrete quiver representations are isomorphic. Apply the grid-complete reconstruction proposition. □

18.7 Return strength and categorical interpretation

Classification 18.26 (What PE-2 returns). PE-2 returns:

1. all interval endpoints on the declared grid;
2. every interval multiplicity;
3. the rank invariant;
4. the isomorphism class of the finite quiver representation;
5. under grid completeness, the isomorphism class of the repaired profile on the window.

PE-2 does not Return:

1. a preferred interval decomposition basis;
2. a canonical isomorphism from a reconstructed direct sum to the original object;
3. arbitrary morphisms between source representations;
4. chain-level representatives not encoded by the persistence object;
5. positive-radius perturbation stability.

Proposition 18.27 (Object Return is stronger than PE-1 edge Return). *There exist nonisomorphic finite persistence objects with equal PE-1 complete-edge vector spaces, whereas PE-2 distinguishes every pair of nonisomorphic objects on the fixed finite grid.*

Proof. PE-1 already supplies an explicit pair of distinct profiles with isomorphic complete stencil-edge spaces. PE-2 distinguishes them because equality of PE-2 signatures is equivalent to object isomorphism. $\square$

Warning 18.28 (Dimension signature is not morphism-level Yoneda Return). The matrix $\mathfrak{E}_n(M)$ remembers dimensions of a finite family of Ext groups, not the Ext functor with all induced maps between probes. The present theorem reconstructs objects up to isomorphism only. A morphism-level or fully faithful Return requires a stronger enriched probe record.

18.8 Worked examples

18.8.1 One interval

Let $n = 4$ and $M = \mathbb{I}[2, 3]$. In Q_5 , the interval Ext formula gives

$$E_{c,d}(M) = 1 \iff 2 < c \leq 4 \leq d.$$

Thus the nonzero signature entries are

$$E_{3,4} = E_{3,5} = E_{4,4} = E_{4,5} = 1.$$

The inversion at $(a, b) = (2, 3)$ gives

$$m_{2,3} = E_{3,4} - E_{3,3} - E_{2,4} + E_{2,3} = 1.$$

Every other multiplicity is zero.

18.8.2 A mixed barcode

Let $n = 4$ and

$$M = \mathbb{I}[1, 2] \oplus \mathbb{I}[2, 4] \oplus \mathbb{I}[2, 4] \oplus \mathbb{I}[4, 4].$$

The exact barcode is therefore

$$m_{1,2} = 1, \quad m_{2,4} = 2, \quad m_{4,4} = 1.$$

Computing the Ext signature and applying Theorem 18.21 returns precisely these three nonzero multiplicities. In particular, the two copies of the terminal original bar $[2, 4]$ are detected because the sentinel moves them away from the projective boundary in Q_5 .

18.8.3 Simple probes are insufficient

Let $n = 3$ and compare

$$M = \mathbb{I}[1, 2], \quad N = \mathbb{I}[2, 2].$$

For a simple probe $S_c = \mathbb{I}[c, c]$, the interval formula reduces to

$$\dim \text{YExt}^1(\iota_n \mathbb{I}[a, b], S_c) = 1 \iff c = b + 1.$$

Both M and N have the same simple-probe signature: one unit at $c = 3$ and zero elsewhere. Their births differ, so they are nonisomorphic. Interval probes with variable left and right endpoints are therefore essential to the proved inversion.

18.8.4 The sentinel is necessary

In the original category $\text{Pers}_n(k)$, every interval $\mathbb{I}[a, n]$ is projective. Hence

$$\text{YExt}_{A_n}^1(\mathbb{I}[a, n], X) = 0$$

for every probe X . Consequently the zero object and a nonzero terminal bar have identical all-probe Ext signatures if the source remains at the projective boundary. Appending the terminal zero vertex changes $\mathbb{I}[a, n]$ into a nonprojective interval of Q_{n+1} and removes this blind spot.

18.9 HC composition and v20 compatibility

Proposition 18.29 (HC-to-PE-2 composition). *Let C and T be filtered chain complexes connected by an HC zero-shift filtered contraction. Fix a monitored degree, window, threshold, and grid for which the repaired profiles have grid-complete records. Then the HC natural persistence isomorphism induces*

$$\text{Disc}_\Lambda(P(T)) \cong \text{Disc}_\Lambda(P(C)),$$

and therefore

$$\mathfrak{E}_n(\text{Disc}_\Lambda P(T)) = \mathfrak{E}_n(\text{Disc}_\Lambda P(C)).$$

Both routes Return the same finite barcode and persistence-object isomorphism class.

Proof. HC returns the full persistence module by an actual natural isomorphism at zero filtration shift. Restriction to the fixed repaired scope and grid gives an isomorphism in $\text{Pers}_n(k)$. Yoneda Ext dimensions are invariant under source isomorphism, and the PE-2 inversion returns the common barcode. $\square$

Warning 18.30 (No shortcut from terminal homology). An unfiltered chain-homology equivalence, an equality of terminal homology groups, or a numerical rank match does not imply equality of the PE-2 signature. Composition is licensed only from an actual persistence-level isomorphism on the same declared scope.

Protocol 18.31 (v20 evaluation order). For a v20 repaired profile, the permitted order is

source $\rightarrow$ window $\rightarrow$ pre-threshold profile $\rightarrow$ hard deletion
 $\rightarrow$ grid-complete discretization $\rightarrow$ sentinel $\rightarrow$ Ext signature $\rightarrow$ inversion.

PE-2 does not commute discretization, windowing, and hard deletion without a separately proved comparison theorem.

18.10 Adversarial boundaries

Proposition 18.32 (Boundary A: no-sentinel failure is exact). *Without the sentinel, no signature built solely from $\text{YExt}_{A_n}^1(M, -)$ can distinguish projective summands of M .*

Proof. Yoneda $\text{Ext}^1(P, -) = 0$ for every projective P . Thus adding or removing a projective direct summand leaves the entire functor $\text{YExt}^1(M, -)$ unchanged up to direct-sum zero contribution. $\square$

Proposition 18.33 (Boundary B: simple-probe failure is exact). *The family of simple probes records terminal multiplicities but does not determine interval births.*

Proof. The simple-probe formula depends only on $b + 1$. The worked counterexample gives two nonisomorphic intervals with the same signature. $\square$

Proposition 18.34 (Boundary C: no circularity). *The Ext signature and inversion do not consume the source barcode multiplicities as inputs.*

Proof. The executable route starts from the actual finite quiver matrices, constructs the sentinel representation and projective resolutions, computes Ext dimensions, and then applies finite differences. The multiplicities occur only in the proof of correctness and in the returned output. $\square$

Proposition 18.35 (Boundary D: exactness is discrete, not metric). *Exact equality of PE-2 signatures is not asserted to be stable under a positive bottleneck perturbation.*

Proof. A small perturbation can move an endpoint across a fixed grid cell and change an integer interval multiplicity. No threshold-gap or grid-transport theorem has been supplied for the Ext signature. $\square$

Proposition 18.36 (Boundary E: one-parameter finite type is essential). *The inversion theorem does not extend automatically to a branching quiver or a multiparameter persistence category.*

Proof. The proof uses the interval classification of linearly oriented A_n , its one-step projective resolutions, and the rectangular cumulative order in Theorem 18.18. These structures fail or change in more general representation types. $\square$

18.11 Defect ledger and status procedure

Field	Meaning
D_{scope}	Missing field, degree, window, threshold, endpoint, finite-type, or A_n scope.
D_{grid}	Missing grid completeness, reconstruction convention, or actual reconstruction isomorphism.
D_{diag}	Missing or ill-typed actual quiver vector spaces and arrows.
D_{sent}	Missing terminal-zero embedding or use of the wrong ambient Q_{n+1} category.
D_{probe}	Incomplete interval probe family or missing probe identities.
D_{ext}	Failed projective-resolution, Ext-space, or dimension computation.

D_{inv}	Negative or inconsistent inverted multiplicities, arithmetic mismatch, or failed reconstruction check.
D_{HC}	Missing actual persistence isomorphism when HC composition is invoked.
D_{strength}	Requested canonical basis, morphism Return, metric stability, infinite Return, or multiparameter Return beyond the proved level.
D_{artifact}	Missing immutable source, matrix, computation, or certificate identity.

Definition 18.37 (Non-aggregated PE-2 defect ledger). No scalar sum is defined. Numerical reserve cannot compensate for a missing non-scalar record.

Protocol 18.38 (Return-theory status). At a frozen datum and requested strength:

1. false typing, false matrix data, or a false computation gives REJECT;
2. missing scope, grid, sentinel, probe, Ext, or reconstruction evidence gives UNDEFINED;
3. a valid requested comparison artifact with nonzero declared defect may give OBSTRUCTED;
4. complete PE-2 evidence gives RETURN at finite-quiver object-isomorphism strength;
5. requesting canonical decomposition, morphism-level Return, metric stability, or a stronger domain gives PARTIAL-RETURN when the PE-2 object result is still valid.

A missing sentinel is undefined, not Type IV.

19 PE-3: Fully Faithful Ext-Probe Morphism Return

19.1 Ambient persistence category and sentinel subcategory

Definition 19.1 (Ambient finite persistence category). Fix a field k . Let

$$\mathcal{C}_{n+1} := \text{Rep}_k(Q_{n+1}) \simeq kQ_{n+1}\text{-mod}_{\text{fd}}.$$

This is a finite-length k -linear abelian category with enough projectives. Every object is a finite direct sum of interval representations $\mathbb{I}[a, b]$.

Definition 19.2 (Terminal-zero sentinel subcategory). Let $\mathcal{S}_n \subset \mathcal{C}_{n+1}$ be the full subcategory of objects M satisfying $M_{n+1} = 0$. The terminal-sentinel functor

$$\iota_n : \text{Rep}_k(Q_n) \longrightarrow \mathcal{S}_n, \quad M \longmapsto (M_1 \rightarrow \cdots \rightarrow M_n \rightarrow 0),$$

is an exact equivalence onto $\mathcal{S}_n$.

Proposition 19.3 (Extension closure). *The subcategory $\mathcal{S}_n$ is closed under kernels, cokernels, finite direct sums, and extensions inside $\mathcal{C}_{n+1}$.*

Proof. All these constructions are computed vertexwise. If every object in a kernel, cokernel, or short exact diagram has zero at vertex $n + 1$ except possibly the object being constructed, exactness at that vertex forces the constructed space to be zero as well. $\square$

Definition 19.4 (Ambient projectives). For $1 \leq i \leq n + 1$, write

$$P_i := \mathbb{I}[i, n + 1].$$

The indecomposable projectives of $\mathcal{C}_{n+1}$ are exactly the P_i , and every projective is a finite direct sum of them.

19.2 The Ext-probe functor

Definition 19.5 (Covariant Ext-probe functor). For $M \in \mathcal{S}_n$, define

$$\mathbb{E}_M := \text{YExt}_{\mathcal{C}_{n+1}}^1(M, -) : \mathcal{C}_{n+1} \longrightarrow \text{Vect}_k.$$

For a probe morphism $u : X \rightarrow Y$, the map

$$\mathbb{E}_M(u) : \text{YExt}^1(M, X) \longrightarrow \text{YExt}^1(M, Y)$$

is the Yoneda pushout map in the second variable.

Definition 19.6 (Contravariant source action). For $f : M \rightarrow N$ in $\mathcal{S}_n$, pullback of extensions in the first variable gives a natural transformation

$$\mathbb{E}(f) := \text{YExt}^1(f, -) : \mathbb{E}_N \Longrightarrow \mathbb{E}_M.$$

Thus

$$\mathbb{E} : \mathcal{S}_n^{\text{op}} \longrightarrow \text{Fun}_k(\mathcal{C}_{n+1}, \text{Vect}_k)$$

is a k -linear functor.

Proposition 19.7 (Identity and composition). *For $f : M \rightarrow N$ and $g : N \rightarrow L$,*

$$\mathbb{E}(\text{id}_M) = \text{id}_{\mathbb{E}_M}, \quad \mathbb{E}(g \circ f) = \mathbb{E}(f) \circ \mathbb{E}(g).$$

Proof. Yoneda pullback along an identity is the identity operation on extension classes. Pullback along a composite is the iterated pullback in reversed order. $\square$

Remark 19.8 (PE-2 as decategorification). PE-2 retains only

$$\dim_k \mathbb{E}_M(\mathbb{I}[c, d]).$$

PE-3 retains the functor values together with all linear maps $\mathbb{E}_M(u)$. The PE-2 signature is therefore the dimension-vector shadow of the PE-3 probe object.

19.3 Finite interval probe category

Definition 19.9 (Interval probe family). Let $\mathcal{J}_{n+1}$ be the full subcategory of $\mathcal{C}_{n+1}$ whose objects are the interval indecomposables

$$\mathbb{I}[c, d], \quad 1 \leq c \leq d \leq n+1.$$

There are exactly $(n+1)(n+2)/2$ probe objects.

Lemma 19.10 (Hom formula between probes). For intervals $X = \mathbb{I}[c, d]$ and $Y = \mathbb{I}[e, f]$,

$$\dim_k \operatorname{Hom}_{\mathcal{C}_{n+1}}(X, Y) = \begin{cases} 1, & e \leq c \leq f \leq d, \\ 0, & \text{otherwise.} \end{cases}$$

When nonzero, choose the canonical map that is the identity scalar on the overlap $[c, f]$ and zero elsewhere.

Proof. A nonzero component at the source birth c requires the target already to be alive, hence $e \leq c \leq f$. Commutativity propagates the scalar through the overlap. At the source death boundary, nonzero commutativity requires the target to die no later than the source, hence $f \leq d$. These conditions are sufficient and leave one scalar parameter. $\square$

Definition 19.11 (Finite probe diagram). The finite probe diagram of M is

$$\mathbb{E}_M|_{\mathcal{J}_{n+1}} : \mathcal{J}_{n+1} \longrightarrow \mathbf{Vect}_k,$$

recorded by:

1. every vector space $\operatorname{YExt}^1(M, \mathbb{I}[c, d])$;
2. for every nonzero canonical probe morphism $u : X \rightarrow Y$, the matrix of $\operatorname{YExt}^1(M, u)$;
3. the composition table of canonical probe morphisms, including zero composites.

Proposition 19.12 (Finite determination of natural transformations). For additive functors $F, H : \mathcal{C}_{n+1} \rightarrow \mathbf{Vect}_k$, restriction to interval probes induces a bijection between natural transformations $F \Rightarrow H$ and families of maps on interval probes satisfying naturality for every canonical probe morphism.

Proof. Every object of $\mathcal{C}_{n+1}$ is a finite direct sum of intervals and every morphism is a matrix of interval morphisms. Additivity forces the component on a direct sum to be the corresponding block direct sum. Naturality for all matrix morphisms is equivalent to naturality for the canonical basis maps between interval summands. $\square$

19.4 Hilton–Rees correspondence in the finite-quiver regime

Definition 19.13 (Projective-factorization ideal). For $A, B \in \mathcal{C}_{n+1}$, let $\mathcal{P}(A, B)$ be the subspace of morphisms $A \rightarrow B$ factoring through an ambient projective object. Set

$$\underline{\operatorname{Hom}}_{\mathcal{C}_{n+1}}(A, B) := \operatorname{Hom}_{\mathcal{C}_{n+1}}(A, B) / \mathcal{P}(A, B).$$

Theorem 19.14 (Hilton–Rees correspondence). For $A, B \in \mathcal{C}_{n+1}$, pullback in the first variable induces a natural isomorphism

$$\boxed{\underline{\operatorname{Hom}}_{\mathcal{C}_{n+1}}(B, A) \xrightarrow{\sim} \operatorname{Nat}(\mathbb{E}_A, \mathbb{E}_B)}$$

by

$$[f] \longmapsto \operatorname{YExt}^1(f, -).$$

Proof. Choose a projective presentation

$$0 \longrightarrow \Omega A \xrightarrow{i} P_A \xrightarrow{p} A \longrightarrow 0.$$

Applying $\text{Hom}_{\mathcal{C}_{n+1}}(-, X)$ gives a projective presentation of the additive functor $\mathbb{E}_A$:

$$\text{Hom}(P_A, -) \longrightarrow \text{Hom}(\Omega A, -) \longrightarrow \mathbb{E}_A \longrightarrow 0.$$

Apply $\text{Nat}(-, \mathbb{E}_B)$. Representable functors are projective and Yoneda gives

$$\text{Nat}(\text{Hom}(Z, -), \mathbb{E}_B) \cong \mathbb{E}_B(Z) = \text{YExt}^1(B, Z).$$

Therefore

$$\text{Nat}(\mathbb{E}_A, \mathbb{E}_B) \cong \ker(\text{YExt}^1(B, \Omega A) \rightarrow \text{YExt}^1(B, P_A)).$$

Applying $\text{Hom}(B, -)$ to the projective presentation gives the exact segment

$$\text{Hom}(B, P_A) \longrightarrow \text{Hom}(B, A) \longrightarrow \text{YExt}^1(B, \Omega A) \longrightarrow \text{YExt}^1(B, P_A).$$

Hence the kernel above is the cokernel of $\text{Hom}(B, P_A) \rightarrow \text{Hom}(B, A)$. Its image is exactly the set of maps $B \rightarrow A$ factoring through a projective: every such factorization lifts through P_A because the intermediate object is projective. Thus the cokernel is $\underline{\text{Hom}}(B, A)$. Tracing the construction identifies the correspondence with $[f] \mapsto \text{YExt}^1(f, -)$. $\square$

Remark 19.15 (Source status). The general theorem is classical and is source-bound to Hilton–Rees (1961). The proof above is included to freeze the variance, quotient convention, and finite-quiver specialization used by PE-3.

19.5 The sentinel eliminates the stable quotient

Lemma 19.16 (No maps from a sentinel interval to an ambient projective). *For $1 \leq a \leq b \leq n$ and every $1 \leq i \leq n+1$,*

$$\text{Hom}_{\mathcal{C}_{n+1}}(\mathbb{I}[a, b], P_i) = 0.$$

Proof. By theorem 19.10, a nonzero map to $P_i = \mathbb{I}[i, n+1]$ would require

$$i \leq a \leq n+1 \leq b,$$

which is impossible because $b \leq n$. $\square$

Proposition 19.17 (Projective-factorization annihilation). *For every $M \in \mathcal{S}_n$ and every ambient projective P ,*

$$\text{Hom}_{\mathcal{C}_{n+1}}(M, P) = 0.$$

Consequently, for $M, N \in \mathcal{S}_n$,

$$\mathcal{P}(M, N) = 0, \quad \underline{\text{Hom}}_{\mathcal{C}_{n+1}}(M, N) = \text{Hom}_{\mathcal{C}_{n+1}}(M, N).$$

Proof. Decompose M into sentinel intervals and P into the indecomposable projectives P_i . The first assertion follows from theorem 19.16 and additivity. A factorization $M \rightarrow P \rightarrow N$ then has zero first leg. $\square$

Warning 19.18 (Why this upgrade is special). Hilton–Rees generally returns only the projectively stable category. The upgrade to ordinary Hom is not automatic from “no projective direct summands.” It uses the stronger fact

$$\text{Hom}(M, P) = 0$$

for every sentinel source and every ambient projective P .

19.6 Fully faithful morphism Return

Theorem 19.19 (PE-3 natural-transformation Return). *For all $M, N \in \mathcal{S}_n$, pullback gives a natural k -linear isomorphism*

$$\boxed{\mathrm{Hom}_{\mathcal{S}_n}(M, N) \xrightarrow{\sim} \mathrm{Nat}(\mathbb{E}_N, \mathbb{E}_M)}$$

by

$$f \mapsto \mathrm{YExt}^1(f, -).$$

Proof. Apply theorem 19.14 with $A = N$ and $B = M$, then use theorem 19.17. $\square$

Corollary 19.20 (Contravariant full faithfulness). *The functor*

$$\mathbb{E} : \mathcal{S}_n^{\mathrm{op}} \longrightarrow \mathrm{Fun}_k(\mathcal{C}_{n+1}, \mathrm{Vect}_k)$$

is fully faithful.

Corollary 19.21 (Object reflection). *For $M, N \in \mathcal{S}_n$,*

$$\mathbb{E}_M \cong \mathbb{E}_N \iff M \cong N.$$

Proof. A natural isomorphism and its inverse correspond by full faithfulness to morphisms $N \rightarrow M$ and $M \rightarrow N$ whose composites map to the identity transformations. Faithfulness forces the source composites to be identities. $\square$

Corollary 19.22 (Composition Return). *The Ext-probe image returns identities and composition exactly. If $f : M \rightarrow N$ and $g : N \rightarrow L$, then the returned transformations satisfy*

$$\mathbb{E}(g \circ f) = \mathbb{E}(f) \circ \mathbb{E}(g).$$

Thus the source category is recovered up to equivalence with the essential image of $\mathbb{E}$.

19.7 Single finite probe module

Definition 19.23 (Additive interval generator). Let

$$G_{n+1} := \bigoplus_{1 \leq c \leq d \leq n+1} \mathbb{I}[c, d].$$

Every object of $\mathcal{C}_{n+1}$ is isomorphic to a direct summand of a finite direct sum of copies of G_{n+1} ; hence

$$\mathcal{C}_{n+1} = \mathrm{add}(G_{n+1}).$$

Set

$$R_{n+1} := \mathrm{End}_{\mathcal{C}_{n+1}}(G_{n+1}).$$

Definition 19.24 (Finite Ext-probe module). For $M \in \mathcal{S}_n$, define

$$\mathcal{P}_n(M) := \mathrm{YExt}_{\mathcal{C}_{n+1}}^1(M, G_{n+1}).$$

Covariance in the second variable gives a left R_{n+1} -module structure

$$r \cdot \xi := \mathrm{YExt}^1(M, r)(\xi), \quad r \in R_{n+1}.$$

Proposition 19.25 (Evaluation on an additive generator). *For additive functors $F, H : \mathcal{C}_{n+1} \rightarrow \mathrm{Vect}_k$, evaluation at G_{n+1} induces a natural isomorphism*

$$\mathrm{Nat}(F, H) \xrightarrow{\sim} \mathrm{Hom}_{R_{n+1}}(F(G_{n+1}), H(G_{n+1})).$$

Proof. A natural transformation gives an R_{n+1} -linear map at G_{n+1} by naturality with every endomorphism of G_{n+1} . Conversely, an R_{n+1} -linear map extends uniquely to finite direct sums of G_{n+1} . For a direct summand represented by an idempotent $e \in \text{End}(G_{n+1}^{\oplus r})$, R -linearity forces the extension to preserve the image of $F(e)$ and $H(e)$, so it descends uniquely to the summand. Since every object lies in $\text{add}(G_{n+1})$, this produces the unique natural transformation. $\square$

Theorem 19.26 (Finite module morphism Return). *For $M, N \in \mathcal{S}_n$, there is a natural isomorphism*

$$\text{Hom}_{\mathcal{S}_n}(M, N) \xrightarrow{\sim} \text{Hom}_{R_{n+1}}(\mathcal{P}_n(N), \mathcal{P}_n(M))$$

compatible with identities and reversed composition.

Proof. Combine theorem 19.19 and theorem 19.25 for $F = \mathbb{E}_N$ and $H = \mathbb{E}_M$. $\square$

Corollary 19.27 (Finite contravariant embedding). *The functor*

$$\mathcal{P}_n : \mathcal{S}_n^{\text{op}} \longrightarrow R_{n+1}\text{-mod}_{\text{fd}}$$

is fully faithful.

Corollary 19.28 (Probe-module object Return). *For $M, N \in \mathcal{S}_n$,*

$$\mathcal{P}_n(M) \cong \mathcal{P}_n(N) \iff M \cong N.$$

Thus the R_{n+1} -module structure alone returns the source object up to isomorphism.

Remark 19.29 (Relation to the PE-2 inversion). The idempotent summand of $\mathcal{P}_n(M)$ corresponding to $\llbracket c, d \rrbracket$ is exactly $\text{YExt}^1(M, \llbracket c, d \rrbracket)$. Its dimension is $E_{c,d}(M)$ from PE-2. Therefore the PE-2 Möbius inversion remains a compact object-level decoder, while PE-3 retains the additional R_{n+1} -action needed for morphisms.

19.8 Finite replay representation

Definition 19.30 (Probe algebra record). A finite probe algebra record contains:

1. the ordered interval list $\{\llbracket c, d \rrbracket\}$;
2. one basis vector for every nonzero interval Hom space from theorem 19.10;
3. the complete multiplication table, including zero composites;
4. the resulting finite-dimensional algebra R_{n+1} with primitive interval idempotents.

Definition 19.31 (Probe module record). For $M \in \mathcal{S}_n$, a probe module record contains:

1. the PE-2 component spaces $\text{YExt}^1(M, \llbracket c, d \rrbracket)$;
2. the matrix induced by each canonical interval morphism;
3. the assembled R_{n+1} -module $\mathcal{P}_n(M)$;
4. consistency checks for identities, multiplication, and zero composites.

Proposition 19.32 (Finite linear-system decoder). *Given valid probe module records for M and N , the space $\text{Hom}_{\mathcal{S}_n}(M, N)$ is returned by solving the finite system*

$$T(r \cdot x) = r \cdot T(x) \quad (r \in \mathcal{B}_{R_{n+1}})$$

for linear maps

$$T : \mathcal{P}_n(N) \rightarrow \mathcal{P}_n(M),$$

where $\mathcal{B}_{R_{n+1}}$ is the declared finite basis of the probe algebra.

Proof. The solution space is exactly $\text{Hom}_{R_{n+1}}(\mathcal{P}_n(N), \mathcal{P}_n(M))$. Apply theorem 19.26. $\square$

Protocol 19.33 (Returning an explicit source morphism). To reconstruct a source arrow from a probe-module homomorphism:

1. use PE-2 to return source and target isomorphism classes;
2. fix declared source and target quiver models, or declared isomorphisms to lexicographically ordered interval-sum models;
3. compute a basis of $\text{Hom}_{\mathcal{S}_n}(M, N)$ from the quiver commutativity equations;
4. compute the induced probe-module maps of this basis;
5. solve the unique coefficient system given by full faithfulness;
6. transport the resulting arrow back through the declared model isomorphisms.

The categorical arrow is unique. Its coordinate matrix depends on the declared model and bases.

19.9 Worked examples

19.9.1 Naturality removes false independent scalars

Take $n = 2$, so the ambient category is $\text{Rep}_k(Q_3)$, and let

$$M = \mathbb{I}[1, 1] \in \mathcal{S}_n.$$

The Ext-probe functor is nonzero on

$$X = \mathbb{I}[2, 3], \quad Y = \mathbb{I}[2, 2],$$

and both values are one-dimensional. There is a canonical nonzero quotient map

$$q : X \longrightarrow Y.$$

Using the projective resolution

$$0 \rightarrow P_2 \rightarrow P_1 \rightarrow M \rightarrow 0,$$

one has

$$\mathbb{E}_M(Z) \cong \text{coker}(Z_1 \rightarrow Z_2).$$

Therefore $\mathbb{E}_M(q) : k \rightarrow k$ is the identity. An endomorphism of the two component vector spaces would allow two unrelated scalars (α_X, α_Y) , but naturality forces

$$\alpha_Y = \alpha_X.$$

Thus

$$\text{End}_{R_{n+1}}(\mathcal{P}_2(M)) \cong k \cong \text{End}_{\mathcal{S}_n}(M),$$

whereas component dimensions alone do not encode this equality constraint.

19.9.2 Arrow reversal is exact

Still for $n = 2$, let

$$M = \mathbb{I}[1, 1], \quad N = \mathbb{I}[1, 2].$$

The interval Hom formula gives

$$\text{Hom}_{\mathcal{S}_n}(M, N) = 0, \quad \text{Hom}_{\mathcal{S}_n}(N, M) \cong k.$$

PE-3 returns

$$\mathrm{Hom}_{R_{n+1}}(\mathcal{P}_2(N), \mathcal{P}_2(M)) = 0,$$

while

$$\mathrm{Hom}_{R_{n+1}}(\mathcal{P}_2(M), \mathcal{P}_2(N)) \cong k.$$

This is the required contravariant orientation, not an ambiguity.

19.9.3 Scalar-distinct arrows

For any indecomposable sentinel interval M , every scalar $\lambda \in k$ gives an arrow $\lambda \mathrm{id}_M$. PE-2 sees the same object signature for all these arrows because it records no arrow data. PE-3 sends them to distinct module endomorphisms

$$\mathcal{P}_n(\lambda \mathrm{id}_M) = \lambda \mathrm{id}_{\mathcal{P}_n(M)}.$$

Faithfulness therefore separates zero, identity, and every other scalar arrow.

19.10 Return strength and categorical interpretation

Classification 19.34 (PE progression). The PE strengths now separate as follows:

Stage	Proved Return
PE-1	zero predicate and complete fixed-stencil edge data under natural realization and zero-reflection hypotheses;
PE-2	complete finite barcode, rank invariant, and object isomorphism class from Ext dimensions;
PE-3	finite source objects, all Hom spaces, identities, and composition from Ext-probe modules; equivalently a fully faithful contravariant category embedding.

Theorem 19.35 (Finite persistence category Return). *The category $\mathrm{Rep}_k(Q_n)$ is equivalent to the essential image of*

$$\mathcal{P}_n : \mathrm{Rep}_k(Q_n)^{\mathrm{op}} \longrightarrow R_{n+1}\text{-mod}_{\mathrm{fd}},$$

after identifying $\mathrm{Rep}_k(Q_n)$ with the sentinel subcategory $\mathcal{S}_n$.

Proof. The sentinel is an equivalence onto $\mathcal{S}_n$ and $\mathcal{P}_n$ is fully faithful. Any fully faithful functor is an equivalence from its source onto its essential image. $\square$

Warning 19.36 (Contravariant Return is not information loss). The reversal of arrows is intrinsic to pullback in the first Ext variable. It does not weaken full faithfulness. The returned category is the opposite category of the source, together with a declared involutive interpretation of arrow direction.

Non-claim 19.37 (No canonical coordinates). A fully faithful categorical embedding returns an arrow as a unique categorical morphism relative to identified source and target objects. It does not canonically choose bases for vertex spaces, a barcode decomposition isomorphism, or a matrix representative invariant under all automorphisms.

19.11 Exactness and extension-calculus boundary

Warning 19.38 (Full faithfulness is not exactness). The contravariant functor $\mathcal{P}_n$ is not asserted to send every short exact sequence in $\mathcal{S}_n$ to a short exact sequence of probe modules, nor to reflect exactness merely from the three object images.

Proposition 19.39 (Why the Ext-only image does not automatically return exact sequences). *For a short exact sequence*

$$0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$$

and a probe X , the first-variable long exact sequence contains

$$\mathrm{Hom}(L, X) \longrightarrow \mathrm{YExt}^1(N, X) \longrightarrow \mathrm{YExt}^1(M, X) \longrightarrow \mathrm{YExt}^1(L, X) \longrightarrow 0,$$

because $\mathcal{C}_{n+1}$ is hereditary. The omitted Hom boundary can affect exactness at $\mathrm{YExt}^1(N, X)$. Therefore the three Ext-probe modules alone do not supply a general exact-sequence criterion.

Non-claim 19.40 (Extension classes remain a separate stage). PE-3 returns every source morphism, so an already supplied short exact diagram can be replayed and checked. It does not yet classify all Yoneda extension classes between returned sentinel objects solely as internal exact sequences of R_{n+1} -modules. That is a distinct extension-calculus problem.

19.12 Continuous-profile interface and HC composition

Definition 19.41 (Grid-complete PE-3 source record). A repaired continuous persistence profile P enters PE-3 only through a PE-2 grid-complete record

$$\theta_P : \mathrm{Rec}_\Lambda \mathrm{Disc}_\Lambda(P) \xrightarrow{\sim} P$$

on a bounded declared window. A source morphism $f : P \rightarrow P'$ additionally requires actual sampled maps forming a quiver morphism

$$\mathrm{Disc}_\Lambda(f) : \mathrm{Disc}_\Lambda(P) \rightarrow \mathrm{Disc}_\Lambda(P')$$

and a commutative reconstruction square with θ_P and $\theta_{P'}$.

Theorem 19.42 (Continuous finite-window morphism Return). *Suppose P, P' and $f : P \rightarrow P'$ have compatible grid-complete records on the same declared window. Then the PE-3 probe-module transformation returns the discrete quiver morphism $\mathrm{Disc}_\Lambda(f)$ exactly and, through the reconstruction squares, returns f on the declared continuous profile scope.*

Proof. PE-3 full faithfulness returns the discrete arrow. Compatibility of the reconstruction isomorphisms transports it uniquely to the continuous profile morphism. $\square$

Protocol 19.43 (HC composition). An HC artifact may feed PE-3 only when it supplies an actual zero-shift persistence-module isomorphism on the same grid-complete scope. Then the induced PE-3 probe modules are isomorphic and all returned Hom spaces are transported by conjugation. Chain-homology preservation without persistence naturality is insufficient.

19.13 Adversarial boundaries

Proposition 19.44 (Dimension-only morphism failure). *The PE-2 dimension signature cannot return source morphisms.*

Proof. For any nonzero indecomposable M , the distinct arrows 0 and id_M have the same source and target dimension signatures. More generally, every scalar endomorphism has identical PE-2 object data. Hence no function of the two dimension vectors alone distinguishes these arrows. $\square$

Proposition 19.45 (Omitting probe morphisms enlarges false Hom spaces). *Recording only the component Ext spaces and not the induced maps can strictly enlarge the apparent transformation space.*

Proof. In the worked example $M = \mathbb{I}[1, 1]$ for Q_3 , two one-dimensional Ext components would permit independent scalar endomorphisms. The actual probe morphism induces an identity and forces the scalars to agree, reducing the endomorphism space from the false k^2 to the correct k . $\square$

Proposition 19.46 (Sentinel necessity for full faithfulness). *Without the terminal-zero restriction, the functor $A \mapsto \text{YExt}^1(A, -)$ is generally only fully faithful on the projectively stable quotient, not on the ordinary module category.*

Proof. Hilton–Rees identifies natural transformations with Hom modulo projective factorizations. A projective source object has zero Ext functor but may have nonzero identity morphism. The sentinel removes this failure by forcing every source Hom into every ambient projective to vanish. $\square$

Warning 19.47 (General quiver boundary). For another quiver or another full subcategory, absence of projective direct summands does not imply annihilation of the projective-factorization ideal. A PE-3 analogue requires a separate proof that $\text{Hom}(M, P) = 0$ for all source objects M and ambient projectives P , or it returns only the stable category.

Warning 19.48 (Coefficient boundary). The interval classification, finite additive generator, one-dimensional Hom formulas, and hereditary projective resolutions used here depend on the finite type- A representation category over a field. They are not silently transported to a general coefficient ring.

Warning 19.49 (Metric boundary). The integer dimensions and exact action matrices of a probe module may change discontinuously when an endpoint crosses a grid cell or threshold. No bottleneck or interleaving stability follows from full faithfulness.

19.14 Defect ledger and status procedure

Definition 19.50 (PE-3 defect ledger). A PE-3 run separates:

1. D_{scope} : monitored degree, window, threshold, stage, field, and endpoint convention;
2. D_{grid} : missing or false grid-complete reconstruction;
3. D_{sent} : failure of the terminal-zero source condition;
4. D_{probe} : incomplete interval list, Hom basis, or multiplication table;
5. D_{Ext} : missing or invalid Ext vector spaces or induced maps;
6. D_{module} : failed R_{n+1} -action identities;
7. D_{HR} : missing Hilton–Rees source binding or incorrect variance;
8. D_{proj} : nonzero projective-factorization ideal on the invoked source class;
9. D_{decode} : failed module-homomorphism or source-arrow reconstruction;
10. D_{strength} : requested exactness, metric, infinite, or multiparameter strength exceeds PE-3.

These defects are typed and noncompensating.

Protocol 19.51 (Primary status). At a fixed requested strength:

1. a false supplied assertion, invalid map, inconsistent algebra action, or failed replay gives REJECT;
2. a missing source, grid, sentinel, probe, Ext, source-binding, or reconstruction object gives UNDEFINED;
3. a valid requested comparison artifact with a separately declared nonzero defect may give OBSTRUCTED;
4. complete PE-3 evidence gives RETURN at exact finite-quiver category strength;
5. requesting exact-sequence classification, canonical coordinates, metric stability, infinite type, or multiparameter strength gives PARTIAL -RETURN when the PE-3 category result remains valid.

A missing sentinel or absent probe action is undefined, not Type IV.

20 PE-4: Exact-Structure and Yoneda-Class Return

20.1 Ambient category, sentinel category, and finite generators

Definition 20.1 (Ambient and sentinel categories). Fix a field k and let

$$Q_{n+1} : 1 \longrightarrow 2 \longrightarrow \cdots \longrightarrow n+1, \quad \mathcal{C}_{n+1} := \text{Rep}_k(Q_{n+1}).$$

Let $\mathcal{S}_n \subset \mathcal{C}_{n+1}$ be the full subcategory of objects M with $M_{n+1} = 0$. The sentinel functor

$$\iota_n : \text{Rep}_k(Q_n) \longrightarrow \mathcal{S}_n, \quad M \longmapsto (M_1 \rightarrow \cdots \rightarrow M_n \rightarrow 0),$$

is an exact equivalence.

Proposition 20.2 (Extension closure and hereditary regime). *The subcategory $\mathcal{S}_n$ is abelian, extension-closed in $\mathcal{C}_{n+1}$, and hereditary. In particular,*

$$\text{YExt}_{\mathcal{S}_n}^1(N, L) \cong \text{YExt}_{\mathcal{C}_{n+1}}^1(N, L)$$

for $L, N \in \mathcal{S}_n$, and $\text{YExt}_{\mathcal{S}_n}^r(-, -) = 0$ for $r \geq 2$.

Proof. Kernels, cokernels, and extensions in a quiver-representation category are computed vertexwise. At the terminal vertex, every kernel, cokernel, or middle term built from sentinel objects is therefore zero. Thus $\mathcal{S}_n$ is an exact abelian full subcategory equivalent to $\text{Rep}_k(Q_n)$. The path algebra of a finite acyclic quiver is hereditary, so the higher Ext groups vanish above degree one. $\square$

Definition 20.3 (Ambient additive generator and probe algebra). Set

$$G_{n+1} := \bigoplus_{1 \leq c \leq d \leq n+1} \mathbb{I}[c, d], \quad R_{n+1} := \text{End}_{\mathcal{C}_{n+1}}(G_{n+1}).$$

Every object of $\mathcal{C}_{n+1}$ lies in $\text{add}(G_{n+1})$.

Definition 20.4 (Finite Hom and Ext probe modules). For $X \in \mathcal{S}_n$, define

$$\mathcal{H}_n(X) := \text{Hom}_{\mathcal{C}_{n+1}}(X, G_{n+1}), \quad \mathcal{P}_n(X) := \text{YExt}_{\mathcal{C}_{n+1}}^1(X, G_{n+1}).$$

Both are finite-dimensional left R_{n+1} -modules under postcomposition and pushout:

$$r \cdot f := r \circ f, \quad r \cdot \eta := r_* \eta.$$

Remark 20.5 (Relation to PE-3). The module $\mathcal{P}_n(X)$ is exactly the PE-3 Ext-probe module. PE-4 adds $\mathcal{H}_n(X)$ only where exact-sequence and extension-class information requires it. Object and ordinary-morphism Return remain governed by PE-3.

20.2 Yoneda extensions and connecting transformations

Definition 20.6 (Endpoint-fixed extension). For $L, N \in \mathcal{S}_n$, an extension of N by L is a short exact sequence

$$\xi : \quad 0 \longrightarrow L \xrightarrow{i} E \xrightarrow{p} N \longrightarrow 0.$$

Two extensions are equivalent when there is an isomorphism of middle terms commuting with the identity maps on L and N . Their equivalence classes form the abelian group $\text{YExt}_{\mathcal{S}_n}^1(N, L)$ under Baer sum.

Definition 20.7 (Connecting transformation). For an extension class $\xi \in \text{YExt}_{\mathcal{S}_n}^1(N, L)$ and $X \in \mathcal{C}_{n+1}$, define

$$\delta_\xi^X : \text{Hom}(L, X) \longrightarrow \text{YExt}^1(N, X), \quad f \longmapsto f_* \xi,$$

where $f_* \xi$ is the pushout extension along $f : L \rightarrow X$.

Proposition 20.8 (Naturality of the connecting transformation). *The maps δ_ξ^X form a natural transformation*

$$\delta_\xi : \text{Hom}(L, -) \Longrightarrow \text{YExt}^1(N, -).$$

They are additive in f and linear in ξ .

Proof. For $u : X \rightarrow Y$, functoriality of pushout gives

$$u_*(f_*\xi) = (u \circ f)_*\xi,$$

which is precisely naturality. Additivity follows from the additive structure of the abelian category and the Baer group law. $\square$

Theorem 20.9 (Yoneda boundary classification). *For all $L, N \in \mathcal{S}_n$, the assignment*

$$\xi \longmapsto \delta_\xi$$

induces a natural k -linear isomorphism

$$\text{YExt}_{\mathcal{S}_n}^1(N, L) \xrightarrow{\sim} \text{Nat}(\text{Hom}_{\mathcal{C}_{n+1}}(L, -), \text{YExt}_{\mathcal{C}_{n+1}}^1(N, -)).$$

Proof. Given a natural transformation η , set

$$\Xi(\eta) := \eta_L(\text{id}_L) \in \text{YExt}^1(N, L).$$

For $f : L \rightarrow X$, naturality applied to f gives

$$\eta_X(f) = \text{YExt}^1(N, f)(\eta_L(\text{id}_L)) = f_*\Xi(\eta).$$

Hence $\eta = \delta_{\Xi(\eta)}$. Conversely,

$$\delta_\xi^L(\text{id}_L) = (\text{id}_L)_*\xi = \xi.$$

Thus the two constructions are inverse. Linearity and naturality in the endpoints follow from functoriality of pushout and pullback. $\square$

Remark 20.10 (No Hilton–Rees quotient appears here). The theorem concerns natural transformations from a representable Hom functor to an Ext functor. It is a direct Yoneda-lemma statement and does not quotient ordinary morphisms by projective factorizations. Hilton–Rees remains the PE-3 theorem governing natural transformations between two Ext functors.

20.3 Finite Hom–Ext boundary classification

Definition 20.11 (Finite boundary module map). For $\xi \in \text{YExt}^1(N, L)$, define

$$\partial_\xi := \delta_\xi^{G_{n+1}} : \mathcal{H}_n(L) \longrightarrow \mathcal{P}_n(N), \quad f \longmapsto f_*\xi.$$

Lemma 20.12 (R_{n+1} -linearity). *The map ∂_ξ is a homomorphism of left R_{n+1} -modules.*

Proof. For $r \in R_{n+1}$ and $f : L \rightarrow G_{n+1}$,

$$\partial_\xi(r \circ f) = (r \circ f)_*\xi = r_*(f_*\xi) = r \cdot \partial_\xi(f).$$

$\square$

Theorem 20.13 (Finite boundary-classification theorem). *For all $L, N \in \mathcal{S}_n$, evaluation at G_{n+1} induces a natural k -linear isomorphism*

$$\Theta_{N,L} : \text{YExt}_{\mathcal{S}_n}^1(N, L) \xrightarrow{\sim} \text{Hom}_{R_{n+1}}(\mathcal{H}_n(L), \mathcal{P}_n(N)), \quad \xi \longmapsto \partial_\xi.$$

Proof. By the Yoneda boundary-classification theorem,

$$\mathbf{YExt}^1(N, L) \cong \mathbf{Nat}(\mathbf{Hom}(L, -), \mathbf{YExt}^1(N, -)).$$

Because $\mathcal{C}_{n+1} = \text{add}(G_{n+1})$, PE-3's additive-generator evaluation theorem gives

$$\mathbf{Nat}(F, H) \cong \mathbf{Hom}_{R_{n+1}}(F(G_{n+1}), H(G_{n+1}))$$

for all additive functors $F, H : \mathcal{C}_{n+1} \rightarrow \mathbf{Vect}_k$. Apply this with

$$F = \mathbf{Hom}(L, -), \quad H = \mathbf{YExt}^1(N, -).$$

The resulting map is exactly $\xi \mapsto \partial_\xi$. □

Corollary 20.14 (Finite parameter-space equality). *The extension-parameter dimension is exactly*

$$\dim_k \mathbf{YExt}^1(N, L) = \dim_k \mathbf{Hom}_{R_{n+1}}(\mathcal{H}_n(L), \mathcal{P}_n(N)).$$

No extension degree of freedom is lost or added by the finite boundary representation.

Corollary 20.15 (Split detection). *For an extension class ξ ,*

$$\xi = 0 \iff \partial_\xi = 0 \iff \xi \text{ is represented by a split short exact sequence.}$$

Proof. The first equivalence follows from injectivity of $\Theta_{N,L}$. The zero element of the Yoneda extension group is represented by the split extension. □

20.4 Baer sum and endpoint functoriality

Theorem 20.16 (Baer-sum Return). *For $\xi, \eta \in \mathbf{YExt}^1(N, L)$,*

$$\partial_{\xi \boxplus \eta} = \partial_\xi + \partial_\eta.$$

Thus the complete abelian-group structure of $\mathbf{YExt}^1(N, L)$ is returned by addition in the finite module-Hom space.

Proof. The isomorphism $\Theta_{N,L}$ is k -linear, and Yoneda addition agrees with the Baer group law. □

Theorem 20.17 (Pushout and pullback formulas). *Let $a : L \rightarrow L'$ and $b : N' \rightarrow N$. Then*

$$\partial_{a_*\xi} = \partial_\xi \circ \mathcal{H}_n(a),$$

where

$$\mathcal{H}_n(a) : \mathcal{H}_n(L') \rightarrow \mathcal{H}_n(L)$$

is precomposition by a , and

$$\partial_{b^*\xi} = \mathcal{P}_n(b) \circ \partial_\xi,$$

where

$$\mathcal{P}_n(b) : \mathcal{P}_n(N) \rightarrow \mathcal{P}_n(N')$$

is pullback in the first Ext variable.

Proof. For $g : L' \rightarrow G_{n+1}$,

$$\partial_{a_*\xi}(g) = g_*(a_*\xi) = (g \circ a)_*\xi.$$

For $f : L \rightarrow G_{n+1}$,

$$\partial_{b^*\xi}(f) = f_*(b^*\xi) = b^*(f_*\xi).$$

These are the displayed formulas. □

Corollary 20.18 (Bifunctor Return). *The family of finite module-Hom spaces and the preceding compositions returns the complete bifunctor*

$$\mathbf{YExt}_{\mathcal{S}_n}^1(-, -) : \mathcal{S}_n^{\text{op}} \times \mathcal{S}_n \longrightarrow \mathbf{Vect}_k$$

at degree one.

20.5 Finite exactness reflection

Definition 20.19 (Injective cogenerator). Set

$$J_n := \bigoplus_{j=1}^n \mathbb{I}[1, j] \in \mathcal{S}_n.$$

The summands $\mathbb{I}[1, j]$ are the indecomposable injectives of $\mathcal{S}_n \simeq \text{Rep}_k(Q_n)$, so J_n is an injective cogenerator.

Proposition 20.20 (Exact faithful dual test functor). *The contravariant functor*

$$\mathcal{J}_n := \text{Hom}_{\mathcal{S}_n}(-, J_n) : \mathcal{S}_n^{\text{op}} \rightarrow \text{Vect}_k$$

is exact and faithful.

Proof. Injectivity of J_n makes $\text{Hom}(-, J_n)$ exact. Since J_n contains the injective envelope of every simple object, every nonzero object admits a nonzero map to J_n , and every nonzero morphism is detected after composition with a suitable map into J_n . Hence the functor is faithful. $\square$

Theorem 20.21 (Finite exactness-reflection criterion). *Let*

$$L \xrightarrow{i} E \xrightarrow{p} N$$

be morphisms in $\mathcal{S}_n$ with $p \circ i = 0$. Then

$$0 \longrightarrow L \xrightarrow{i} E \xrightarrow{p} N \longrightarrow 0$$

is short exact if and only if the finite vector-space sequence

$$0 \longrightarrow \text{Hom}(N, J_n) \xrightarrow{p^*} \text{Hom}(E, J_n) \xrightarrow{i^*} \text{Hom}(L, J_n) \longrightarrow 0$$

is exact.

Proof. If the source sequence is short exact, exactness follows from injectivity of J_n . Conversely, $\text{Hom}(-, J_n)$ is an exact faithful functor. An exact faithful functor between abelian categories reflects zero homology objects: if the transformed complex is exact, each source homology object is sent to zero; faithfulness forces its identity to be zero, hence the homology object itself is zero. Therefore the source complex is exact. $\square$

Warning 20.22 (Ranks alone are insufficient). The criterion consumes the actual maps p^* and i^* and their kernels and images. Equality of dimensions, Euler-characteristic balance, or rank totals without the map incidence data does not certify exactness.

20.6 The hereditary six-term probe sequence

Theorem 20.23 (Finite six-term exact sequence). *Let*

$$\xi : 0 \rightarrow L \xrightarrow{i} E \xrightarrow{p} N \rightarrow 0$$

be short exact in $\mathcal{S}_n$. Then the canonical boundary map gives an exact sequence of finite left R_{n+1} -modules

$$0 \rightarrow \mathcal{H}_n(N) \xrightarrow{\mathcal{H}_n(p)} \mathcal{H}_n(E) \xrightarrow{\mathcal{H}_n(i)} \mathcal{H}_n(L) \xrightarrow{\partial_\xi} \mathcal{P}_n(N) \xrightarrow{\mathcal{P}_n(p)} \mathcal{P}_n(E) \xrightarrow{\mathcal{P}_n(i)} \mathcal{P}_n(L) \rightarrow 0.$$

Proof. Apply the first-variable Hom–Ext long exact sequence to the short exact sequence and evaluate at G_{n+1} . Since $\mathcal{C}_{n+1}$ is hereditary, the next term $\text{YExt}^2(N, G_{n+1})$ vanishes. Naturality with every endomorphism of G_{n+1} makes all maps R_{n+1} -linear. $\square$

Proposition 20.24 (Cogenerator summand recovers the exactness test). *The injective cogenerator J_n is a direct summand of G_{n+1} . Applying its idempotent to the first four terms of the six-term sequence yields*

$$0 \rightarrow \text{Hom}(N, J_n) \rightarrow \text{Hom}(E, J_n) \rightarrow \text{Hom}(L, J_n) \rightarrow 0,$$

because $\text{YExt}^1(N, J_n) = 0$.

Remark 20.25 (Two-stage noncircular replay). The verifier first applies theorem 20.21 without assuming an extension class. Only after exactness passes does it construct ∂_ξ by actual pushout and verify theorem 20.23. Thus the existence of a long exact sequence is not used to prove the short exact sequence whose existence would generate it.

20.7 Reconstruction from a finite boundary map

Theorem 20.26 (Extension reconstruction from finite data). *Let $L, N \in \mathcal{S}_n$ and let*

$$\beta : \mathcal{H}_n(L) \longrightarrow \mathcal{P}_n(N)$$

be an R_{n+1} -linear map. Then there exists an endpoint-fixed short exact sequence

$$0 \rightarrow L \rightarrow E_\beta \rightarrow N \rightarrow 0$$

whose boundary map is β . It is unique up to equivalence of extensions.

Proof. By theorem 20.13, there is a unique class

$$\xi_\beta := \Theta_{N,L}^{-1}(\beta) \in \text{YExt}^1(N, L).$$

The Yoneda classification of Ext^1 supplies a short exact representative of this class, and two representatives are equivalent exactly when they determine the same Ext class. $\square$

Protocol 20.27 (Constructive representative). A replay implementation may construct a representative as follows:

1. recover the natural transformation corresponding to β from additive-generator evaluation;
2. compute

$$\xi_\beta = \delta_L(\text{id}_L);$$

3. choose a declared projective presentation

$$0 \rightarrow \Omega N \xrightarrow{j} P_N \rightarrow N \rightarrow 0;$$

4. represent ξ_β by a morphism $a : \Omega N \rightarrow L$ modulo maps extending across P_N ;
5. form the pushout of j along a to obtain E_β .

The resulting extension equivalence class is independent of the chosen presentation and representative. The displayed matrices are not canonical unless bases and presentation conventions are frozen.

Corollary 20.28 (Middle term versus extension class). *The boundary map returns the middle object together with the endpoint embeddings and quotient map up to extension equivalence. Returning only the isomorphism class of the middle object is strictly weaker.*

20.8 Worked bridge: the basic non-split interval extension

Definition 20.29 (The A_2 endpoint pair). In $\text{Rep}_k(1 \rightarrow 2)$, let

$$N := \mathbb{I}[1, 1], \quad L := \mathbb{I}[2, 2].$$

Then

$$\dim_k \text{YExt}^1(N, L) = 1.$$

Proposition 20.30 (Non-split generator). *The interval representation $E := \mathbb{I}[1, 2]$ fits into a non-split exact sequence*

$$\xi : \quad 0 \longrightarrow \mathbb{I}[2, 2] \longrightarrow \mathbb{I}[1, 2] \longrightarrow \mathbb{I}[1, 1] \longrightarrow 0,$$

and ξ generates $\text{YExt}^1(N, L) \cong k$.

Proof. The inclusion is zero at vertex 1 and the identity at vertex 2; the quotient is $\mathbb{I}[1, 1]$. The middle representation is indecomposable, so the sequence does not split. Since the Ext space is one-dimensional, its nonzero class is a generator. $\square$

Proposition 20.31 (Boundary distinguishes split and non-split classes). *For the non-split class,*

$$\partial_\xi^L(\text{id}_L) = \xi \neq 0.$$

For the split extension, the boundary is zero. Thus PE-4 distinguishes the two endpoint-fixed extension classes even before a middle term is supplied.

Remark 20.32 (Same middle isomorphism type may not fix the class). For $\lambda \in k^\times$, the classes $\lambda\xi$ are distinct when endpoint identities are fixed, although their middle terms are all isomorphic to $\mathbb{I}[1, 2]$. The boundary map records the scalar class; object Return alone cannot.

20.9 Exact-category Return

Definition 20.33 (PE-4 decorated essential image). Let

$$\mathcal{D}_n := \text{EssIm}(\mathcal{P}_n) \subset R_{n+1}\text{-mod}_{\text{fd}}$$

as a category, with arrows interpreted through PE-3's fully faithful contravariant correspondence. A *PE-4 conflation record* for objects corresponding to L, E, N consists of:

1. the PE-3 returned arrows $i : L \rightarrow E$ and $p : E \rightarrow N$;
2. the equation $pi = 0$;
3. the injective-cogenerator exactness receipt of theorem 20.21;
4. the canonical finite boundary map ∂_ξ ;
5. the exact six-term module sequence of theorem 20.23;
6. immutable endpoint, map, module-action, and artifact identities.

Theorem 20.34 (Exact-structure Return theorem). *The PE-3 fully faithful functor, equipped with PE-4 conflation records, yields an exact anti-equivalence*

$$\boxed{\mathcal{S}_n \simeq \mathcal{D}_n^{\text{op}}}$$

in the following precise sense:

1. objects and morphisms are returned by PE-3;

2. a composable source pair is short exact exactly when its finite cogenerator receipt passes;
3. every endpoint-fixed extension class is returned uniquely by its finite Hom–Ext boundary map;
4. split extensions, Baer sum, pullback, and pushout are preserved exactly;
5. the complete degree-one Yoneda bifunctor and the source exact structure are reconstructible from finite data.

Proof. PE-3 supplies an anti-equivalence onto $\mathcal{D}_n$. The exactness-reflection theorem identifies exactly which returned composable pairs are source short exact. The finite boundary-classification theorem gives a bijection on extension classes for every pair of endpoints, and the Baer and endpoint formulas preserve the exact extension calculus. Therefore the source exact structure is completely transported and finitely recognizable on the decorated image. $\square$

Warning 20.35 (Decorated exact image, not ordinary module exactness). The theorem does not identify source conflations with all ambient short exact sequences of R_{n+1} -modules among the objects $\mathcal{P}_n(M)$. The PE-4 exact structure is carried by the returned arrows, the cogenerator receipt, and the Hom–Ext boundary. Forgetting these decorations loses the theorem.

20.10 Relation to the PE strength ladder

Stage	Returned information	Strict remaining loss
PE-1	zero/nonzero vanishing predicate through a natural Ext edge	nonzero profile shape, object, arrows, extensions
PE-2	complete finite barcode and object isomorphism class	arrows and exact-sequence classes
PE-3	objects, Hom spaces, individual arrows, identities, composition	exactness and Yoneda class calculus
PE-4	short exactness, endpoint-fixed Ext ¹ classes, Baer/pushout/pullback, exact structure	canonical coordinates, higher derived structure, stability, infinite and multiparameter regimes

Proposition 20.36 (Strict progress over PE-3). *PE-4 strictly increases Return strength.*

Proof. Fix L, N with $\text{YExt}^1(N, L) \neq 0$. PE-3 returns the endpoint objects and every supplied arrow, but without a supplied middle diagram it does not select an extension class. PE-4 identifies the full extension group with a finite module-Hom space and reconstructs every class. The A_2 split and non-split example witnesses the strict separation. $\square$

20.11 Continuous profiles, threshold discipline, and HC composition

Definition 20.37 (Grid-exact continuous record). A continuous constructible persistence sequence on a bounded window enters PE-4 only through a common finite critical grid Λ and reconstruction isomorphisms identifying each profile and each morphism with its A_n representation. The record is *grid-exact* when the discrete representation sequence is vertexwise exact.

Theorem 20.38 (Finite-window continuous exact Return). *Under a common grid-complete reconstruction, a sequence of constructible one-parameter persistence modules is pointwise short exact on the declared window if and only if its finite quiver discretization passes the PE-4 exactness certificate. Its endpoint-fixed extension class is then returned by the PE-4 boundary map.*

Proof. On a common constructibility grid, the persistence modules and maps are determined by their finite quiver representations, and kernels and cokernels are computed pointwise. Exactness and extension equivalence therefore transport through the reconstruction isomorphisms. $\square$

Warning 20.39 (Hard threshold deletion is not imported as exact). The v20 hard deletion operation is not globally exact or functorial. PE-4 may consume an actual short exact sequence of repaired-profile objects only when that sequence is separately constructed and certified in an exact-admissible regime. A short exact sequence before hard deletion is not silently transported through deletion.

Protocol 20.40 (HC composition). An HC artifact may feed PE-4 when it supplies:

1. actual zero-shift persistence isomorphisms for all three objects;
2. a commutative morphism of short exact diagrams, not merely three independent object equivalences;
3. compatibility of the returned boundary maps under endpoint pushout and pullback.

Under these conditions the exactness receipt and extension class transport by conjugation. Homology preservation or objectwise barcode equality alone is insufficient.

20.12 Failure boundaries and adversarial examples

Proposition 20.41 (Ext-probe triples alone are not exact certificates). *The three modules*

$$\mathcal{P}_n(N), \quad \mathcal{P}_n(E), \quad \mathcal{P}_n(L)$$

without the Hom modules and connecting boundary do not determine the first-variable six-term sequence.

Proof. The long exact sequence contains

$$\mathrm{Hom}(L, G_{n+1}) \xrightarrow{\partial_\xi} \mathrm{Ext}^1(N, G_{n+1}).$$

Its image is the kernel of $\mathcal{P}_n(N) \rightarrow \mathcal{P}_n(E)$. Omitting the source Hom module and the boundary removes exactly the datum controlling that kernel. $\square$

Proposition 20.42 (Middle-object Return is not extension-class Return). *There can be distinct endpoint-fixed extension classes with isomorphic middle objects.*

Proof. In the one-dimensional A_2 example, every nonzero scalar multiple of the generator has middle object isomorphic to $\mathbb{I}[1, 2]$, while the classes are distinct relative to identity maps on the endpoints. $\square$

Proposition 20.43 (Dimension balance is not exactness). *A complex may have the dimension equality*

$$\dim E = \dim L + \dim N$$

vertexwise without the declared maps identifying L with $\ker p$ and N with $\mathrm{coker} i$.

Proof. Dimension equality constrains only cardinalities. Exactness additionally requires incidence of kernels and images. The cogenerator criterion uses the actual induced maps and therefore detects this missing structure. $\square$

Warning 20.44 (Stable-category boundary). Outside the sentinel subcategory, PE-3 returns morphisms only modulo projective factorizations unless additional lifting data are supplied. PE-4 cannot promote a stable arrow to an ordinary short exact diagram. Exact-structure Return outside the sentinel regime therefore requires a separate projective-lift certificate.

Warning 20.45 (Higher-degree boundary). Type- A persistence categories are hereditary; hence $\mathrm{Ext}^r = 0$ for $r \geq 2$. PE-4 is a complete degree-one exact-structure theorem in this regime. It is not a higher-extension or derived-category theorem. Nonhereditary or derived successors require new data and cannot be inferred by iteration.

20.13 Defect ledger and verifier status

Definition 20.46 (PE-4 defect stack). A PE-4 run records the following noncompensating defects:

1. D_{type} : source objects or maps are not typed in the same sentinel category;
2. D_{comp} : the declared arrows fail $pi = 0$;
3. D_{exact} : the cogenerator sequence is not exact;
4. D_{boundary} : the canonical pushout boundary is missing or not R_{n+1} -linear;
5. D_{six} : the finite six-term sequence fails exactness;
6. D_{source} : Yoneda or source binding is absent, stale, or mismatched;
7. D_{identity} : endpoint, module-action, or artifact identities do not match;
8. D_{return} : requested Return strength exceeds endpoint-fixed Ext^1 exact structure.

No numerical slack repairs any of these defects.

Classification 20.47 (Primary status procedure). For a frozen run:

1. false typing, false map equations, or invalid source bindings give REJECT;
2. missing objects, maps, module actions, or artifacts give UNDEFINED;
3. a valid composable pair failing the exactness test gives NOT_A_CONFLATION, not Type IV;
4. a valid exact sequence with incomplete boundary replay gives PARTIAL -RETURN;
5. all displayed obligations discharged gives EXACT -STRUCTURE -RETURN.

A nonzero extension class is not an obstruction; it is returned mathematical data.

21 PE-5: Bounded Derived, Cone, and Triangle Return

21.1 The hereditary persistence-derived regime

Definition 21.1 (Source abelian and derived categories). Fix a field k and the linearly oriented quiver

$$Q_n : 1 \longrightarrow 2 \longrightarrow \cdots \longrightarrow n.$$

Set

$$\mathcal{A}_n := \mathbf{Rep}_k(Q_n), \quad \mathcal{D}_n := D^b(\mathcal{A}_n).$$

Objects of $\mathcal{A}_n$ are finite one-parameter persistence representations on the declared finite mesh. Objects of $\mathcal{D}_n$ are bounded complexes of such representations, localized at quasi-isomorphisms.

Proposition 21.2 (Finite hereditary source). *The category $\mathcal{A}_n$ is a finite-length, Krull–Schmidt, hereditary abelian category with enough projectives and injectives. Its indecomposable objects are the intervals $\mathbb{I}[c, d]$, $1 \leq c \leq d \leq n$, and*

$$\mathrm{Ext}_{\mathcal{A}_n}^r(-, -) = 0 \quad (r \geq 2).$$

Proof. The interval classification and hereditary property are inherited from the finite line-quiver regime proved and consumed throughout PE-2–PE-4. Equivalently, $\mathcal{A}_n$ is the category of finite-dimensional modules over the path algebra kQ_n , whose projective resolutions have length at most one. $\square$

Theorem 21.3 (Hereditary cohomology splitting). *Every object $X \in \mathcal{D}_n$ admits an isomorphism*

$$X \cong \bigoplus_{q \in \mathbb{Z}} H^q(X)[-q],$$

with only finitely many nonzero summands.

Proof. The source is hereditary, so $\mathrm{Ext}_{\mathcal{A}_n}^2(B, A) = 0$ for all A, B . The classical hereditary-splitting theorem for bounded derived categories then gives the displayed isomorphism. The isomorphism is generally not canonical. $\square$

Warning 21.4 (Splitting is not canonical). The cohomology objects determine the isomorphism class of a derived object in the present hereditary regime, but they do not supply a canonical splitting map, a canonical chain representative, or a complete description of derived morphisms.

Definition 21.5 (Standard heart). The standard t -structure on $\mathcal{D}_n$ has heart equivalent to $\mathcal{A}_n$. We identify an object $M \in \mathcal{A}_n$ with the complex concentrated in degree zero.

Proposition 21.6 (Derived Hom between shifted heart objects). *For $A, B \in \mathcal{A}_n$ and integers q, r ,*

$$\mathrm{Hom}_{\mathcal{D}_n}(A[-q], B[-r]) \cong \mathrm{Ext}_{\mathcal{A}_n}^{q-r}(A, B).$$

Consequently this group is nonzero only when $q - r \in \{0, 1\}$.

Proof. By the shift identity,

$$\mathrm{Hom}_{\mathcal{D}_n}(A[-q], B[-r]) \cong \mathrm{Hom}_{\mathcal{D}_n}(A, B[q - r]).$$

The latter is $\mathrm{Ext}_{\mathcal{A}_n}^{q-r}(A, B)$ for nonnegative degree and zero for negative degree; hereditary vanishing removes all degrees at least two. $\square$

21.2 Projective interval generator and exact chain realization

Definition 21.7 (Projective interval generator). For $1 \leq i \leq n$, let

$$P_i := \llbracket i, n \rrbracket.$$

These are the indecomposable projectives of $\mathcal{A}_n$. Define

$$P_n := \bigoplus_{i=1}^n P_i, \quad \Gamma_n := \text{End}_{\mathcal{A}_n}(P_n)^{\text{op}}.$$

Proposition 21.8 (Finite projective-probe equivalence). *The functor*

$$\Phi_n := \text{Hom}_{\mathcal{A}_n}(P_n, -) : \mathcal{A}_n \longrightarrow \Gamma_n\text{-mod}_{\text{fd}}$$

is an exact equivalence. A quasi-inverse is

$$\Psi_n := P_n \otimes_{\Gamma_n} -.$$

After fixing the vertex-idempotent decomposition, $\Gamma_n \cong kQ_n$.

Proof. The object P_n is a finite projective generator. Projectivity makes Φ_n exact. If $M \neq 0$, some vertex space M_i is nonzero and $\text{Hom}(P_i, M) \cong M_i$, so Φ_n is faithful. The standard evaluation and coevaluation maps for the progenerator give mutually inverse natural isomorphisms

$$P_n \otimes_{\Gamma_n} \text{Hom}_{\mathcal{A}_n}(P_n, M) \cong M, \quad \text{Hom}_{\mathcal{A}_n}(P_n, P_n \otimes_{\Gamma_n} V) \cong V.$$

Thus Φ_n is an exact equivalence. Since P_n is the regular projective module decomposed by primitive idempotents, $\text{End}(P_n)^{\text{op}}$ identifies with the original path algebra after the declared choice of vertex labels. $\square$

Remark 21.9 (What the probe actually records). The component $\text{Hom}(P_i, M)$ is canonically the vertex space M_i . Precomposition by the canonical maps among the P_i returns the quiver structure maps. Thus the projective interval probe is not a dimension-only signature; it returns the complete persistence representation.

Definition 21.10 (Degreewise chain realization). For a bounded complex $X^\bullet$ in $\mathcal{A}_n$, define

$$\Phi_n^\bullet(X^\bullet)^q := \Phi_n(X^q), \quad d_{\Phi_n X}^q := \Phi_n(d_X^q).$$

For a chain map $f^\bullet$, define $\Phi_n^\bullet(f)^\bullet := \Phi_n(f^\bullet)$.

Proposition 21.11 (Exact chain and homotopy equivalence). *The degreewise functors $\Phi_n^\bullet$ and $\Psi_n^\bullet$ give inverse equivalences of bounded complex categories and bounded homotopy categories. They preserve and reflect:*

1. chain homotopies;
2. cohomology objects;
3. acyclic complexes;
4. quasi-isomorphisms.

Moreover,

$$H^q(\Phi_n^\bullet X) \cong \Phi_n(H^q(X)).$$

Proof. All assertions are computed degreewise from the exact equivalence in theorem 21.8. Exactness commutes with kernels and cokernels and hence with cohomology. A chain homotopy is a family of source morphisms satisfying a finite linear identity, which is preserved and reflected by full faithfulness. A map is a quasi-isomorphism exactly when its cohomology maps are isomorphisms, and equivalences reflect isomorphisms. $\square$

21.3 Global derived Return theorem

Theorem 21.12 (Projective-interval derived equivalence). *The degreewise projective-probe equivalence localizes to a k -linear triangle equivalence*

$$\Phi_n := \mathbf{RHom}_{\mathcal{A}_n}(P_n, -) : \mathcal{D}_n \xrightarrow{\sim} D^b(\Gamma_n\text{-mod}_{\text{fd}}).$$

Because P_n is projective, Φ_n is computed by the literal bounded Hom complex $\Phi_n^\bullet$. A quasi-inverse is the degree-wise derived tensor functor, which here is represented by $\Psi_n^\bullet$.

Proof. By theorem 21.11, $\Phi_n^\bullet$ is an equivalence of bounded complex categories that preserves and reflects quasi-isomorphisms. Therefore it induces an equivalence after localization. The same degreewise equivalence preserves shifts and standard mapping-cone triangles, so the localized functor is exact in the triangulated sense. Projectivity of P_n identifies the derived Hom with ordinary Hom on every bounded complex. $\square$

Corollary 21.13 (Derived object and morphism Return). *For all $X, Y \in \mathcal{D}_n$ and $r \in \mathbb{Z}$, the canonical map is an isomorphism*

$$\text{Hom}_{\mathcal{D}_n}(X, Y[r]) \xrightarrow{\sim} \text{Hom}_{D^b(\Gamma_n)}(\Phi_n X, \Phi_n Y[r]).$$

In particular, $\Phi_n X \cong \Phi_n Y$ if and only if $X \cong Y$.

Corollary 21.14 (Cohomology persistence Return). *For every $X \in \mathcal{D}_n$ and every q ,*

$$H^q(\Phi_n X) \cong \Phi_n(H^q(X)).$$

Applying the quasi-inverse returns the complete persistence representation $H^q(X)$, including its interval barcode under the finite type- A decomposition.

Classification 21.15 (Return strength). The theorem returns:

1. the isomorphism class of every bounded persistence complex;
2. every morphism in the derived category;
3. every shifted Hom and Ext group;
4. composition and identities;
5. the standard shift functor;
6. mapping-cone and distinguished-triangle isomorphism classes;
7. the standard heart and its PE-4 exact structure.

It does not return a unique chain representative or a basis-dependent matrix unless those choices are part of the certificate.

21.4 Strict mapping-cone and distinguished-triangle Return

Definition 21.16 (Selected chain-level cone certificate). A strict cone certificate for a derived morphism $f : X \rightarrow Y$ contains:

1. bounded chain representatives $X^\bullet, Y^\bullet$;
2. an actual chain map $\tilde{f} : X^\bullet \rightarrow Y^\bullet$ representing f ;
3. the standard cochain mapping cone

$$\text{Cone}(\tilde{f})^q := Y^q \oplus X^{q+1}, \quad d_{\text{Cone}}^q = \begin{pmatrix} d_Y^q & \tilde{f}^{q+1} \\ 0 & -d_X^{q+1} \end{pmatrix};$$

4. the standard triangle

$$X^\bullet \xrightarrow{\tilde{f}} Y^\bullet \longrightarrow \text{Cone}(\tilde{f}) \longrightarrow X^\bullet[1];$$

5. immutable identities for the representatives and all maps.

Theorem 21.17 (Strict cone compatibility). *For every selected chain map $\tilde{f}$,*

$$\boxed{\Phi_n^\bullet(\text{Cone}(\tilde{f})) = \text{Cone}(\Phi_n^\bullet(\tilde{f}))}$$

as literal complexes with their standard cone maps.

Proof. The degree-wise functor is additive and sends $-d_X$ to $-\Phi_n(d_X)$. Applying it to the displayed block differential gives exactly the block differential of the target mapping cone. $\square$

Corollary 21.18 (Distinguished-triangle preservation and reflection). *A triangle T in $\mathcal{D}_n$ is distinguished if and only if $\Phi_n(T)$ is distinguished in $D^b(\Gamma_n\text{-mod}_{\text{fd}})$.*

Proof. An exact equivalence preserves distinguished triangles. Its exact quasi-inverse reflects them. $\square$

Warning 21.19 (Cone uniqueness boundary). A cone of a fixed morphism in a triangulated category is unique only up to generally nonunique triangle isomorphism. PE-5 returns the cone isomorphism class globally. A strict cone object is returned only relative to the selected chain representative in the certificate.

Warning 21.20 (A module diagram is not automatically a triangle). Full faithfulness of a probe functor does not turn every composable four-term module diagram into a distinguished triangle. Distinguishedness requires either the strict mapping-cone certificate above or an independently source-bound distinguished-triangle record.

21.5 Finite shifted-interval probe on fixed amplitude

Definition 21.21 (Cohomological amplitude subcategory). For integers $a \leq b$, define the full subcategory

$$\mathcal{D}_n^{[a,b]} := \{X \in \mathcal{D}_n : H^q(X) = 0 \text{ for } q \notin [a, b]\}.$$

Definition 21.22 (Finite shifted-interval generator and algebra). Set

$$\mathbb{G}_{a,b} := \bigoplus_{q=a}^b \bigoplus_{1 \leq c \leq d \leq n} \mathbb{I}[c, d][-q], \quad B_{a,b} := \text{End}_{\mathcal{D}_n}(\mathbb{G}_{a,b}).$$

Define

$$\mathbb{Y}_{a,b}(X) := \text{Hom}_{\mathcal{D}_n}(\mathbb{G}_{a,b}, X),$$

viewed as a finitely generated right $B_{a,b}$ -module by precomposition.

Theorem 21.23 (Finite-amplitude generation). *One has*

$$\boxed{\mathcal{D}_n^{[a,b]} = \text{add}(\mathbb{G}_{a,b})}.$$

Proof. By theorem 21.3, each $X \in \mathcal{D}_n^{[a,b]}$ is isomorphic to

$$\bigoplus_{q=a}^b H^q(X)[-q].$$

Each finite persistence object $H^q(X)$ is a finite direct sum of interval modules. Hence $X \in \text{add}(\mathbb{G}_{a,b})$. The reverse inclusion is immediate from the cohomological degrees of the summands. $\square$

Theorem 21.24 (Finite shifted-Yoneda Return). *Restricted Yoneda is an additive equivalence*

$$\boxed{\mathbb{Y}_{a,b} : \mathcal{D}_n^{[a,b]} \xrightarrow{\sim} \text{proj-}B_{a,b}}.$$

For all $X, Y \in \mathcal{D}_n^{[a,b]}$,

$$\text{Hom}_{\mathcal{D}_n}(X, Y) \cong \text{Hom}_{B_{a,b}}(\mathbb{Y}_{a,b}X, \mathbb{Y}_{a,b}Y).$$

Proof. For any additive category and any object G , the functor $\text{Hom}(G, -)$ is fully faithful on $\text{add}(G)$ and its essential image is the category of finitely generated projective right $\text{End}(G)$ -modules. Apply this with $G = \mathbb{G}_{a,b}$ and use theorem 21.23. $\square$

Proposition 21.25 (Hom–Ext block structure of the finite probe algebra). *For shifted interval summands of $\mathbb{G}_{a,b}$,*

$$\text{Hom}_{\mathcal{D}_n}(\mathbb{I}[c, d][-q], \mathbb{I}[e, f][-r]) \cong \begin{cases} \text{Hom}_{\mathcal{A}_n}(\mathbb{I}[c, d], \mathbb{I}[e, f]), & q = r, \\ \text{Ext}_{\mathcal{A}_n}^1(\mathbb{I}[c, d], \mathbb{I}[e, f]), & q = r + 1, \\ 0, & \text{otherwise.} \end{cases}$$

Hence $B_{a,b}$ is finite-dimensional and records only actual Hom and degree-one Ext incidence, together with their composition.

Proof. Combine theorem 21.6 with hereditary vanishing and the finiteness of the shifted interval list. $\square$

Remark 21.26 (Derived barcode versus derived probe module). The graded interval multiplicities

$$\mathfrak{B}^{\text{der}}(X) := \{(q, [c, d], m_{q,c,d})\}$$

determine the object isomorphism class by hereditary splitting. They do not determine a supplied derived morphism. The $B_{a,b}$ -module action retains the Hom and Ext incidence needed to return morphisms and composition.

21.6 Finite triangle window and amplitude expansion

Proposition 21.27 (Cone amplitude bound). *If $X, Y \in \mathcal{D}_n^{[a,b]}$ and $f : X \rightarrow Y$, then*

$$\text{Cone}(f) \in \mathcal{D}_n^{[a-1,b]}.$$

Proof. The long exact cohomology sequence of

$$X \rightarrow Y \rightarrow \text{Cone}(f) \rightarrow X[1]$$

shows that $H^q(\text{Cone}(f))$ can be nonzero only when either $H^q(Y)$ or $H^{q+1}(X)$ is nonzero. This gives the interval $[a-1, b]$. $\square$

Definition 21.28 (Expanded triangle generator). Define

$$\mathbb{G}_{a,b}^{\Delta} := \mathbb{G}_{a-1,b}, \quad B_{a,b}^{\Delta} := \text{End}_{\mathcal{D}_n}(\mathbb{G}_{a-1,b}).$$

For a triangle with first two objects in $\mathcal{D}_n^{[a,b]}$, all four objects

$$X, \quad Y, \quad \text{Cone}(f), \quad X[1]$$

lie in $\text{add}(\mathbb{G}_{a-1,b})$.

Theorem 21.29 (Finite triangle-diagram Return). *Let*

$$T = (X \xrightarrow{f} Y \xrightarrow{g} C \xrightarrow{h} X[1])$$

be a source-bound distinguished triangle with $X, Y \in \mathcal{D}_n^{[a,b]}$. Applying

$$\mathbb{Y}_{a-1,b} = \text{Hom}_{\mathcal{D}_n}(\mathbb{G}_{a-1,b}, -)$$

returns the complete four-object, three-morphism triangle diagram up to unique source morphism correspondence. Together with the distinguishedness certificate, the module record returns the triangle isomorphism class.

Proof. By theorem 21.27, every object of the triangle is in $\text{add}(\mathbb{G}_{a-1,b})$. The restricted Yoneda equivalence is fully faithful there, so it returns all objects and arrows. Distinguishedness is supplied by the strict cone or source-bound triangle certificate and is reflected by the global triangle equivalence in theorem 21.18. $\square$

Non-claim 21.30 (Amplitude category is not triangulated). The category $\mathcal{D}_n^{[a,b]}$ is generally not closed under shifts or cones. The finite module category $\text{proj-}B_{a,b}$ is therefore an additive Return target, not by itself a triangulated target. Cone Return consumes the expanded range $[a-1, b]$ and a distinguishedness certificate.

21.7 Worked bridge: invisible cohomology maps, different cones

Definition 21.31 (A_2 source objects). Let $Q_2 : 1 \rightarrow 2$, let

$$S_1 := \mathbb{I}[1, 1], \quad S_2 := \mathbb{I}[2, 2], \quad E := \mathbb{I}[1, 2].$$

There is a nonsplit exact sequence

$$0 \longrightarrow S_2 \longrightarrow E \longrightarrow S_1 \longrightarrow 0,$$

representing a nonzero class

$$\xi \in \text{Ext}_{\mathcal{A}_n}^1(S_1, S_2).$$

Proposition 21.32 (Two cohomologically invisible morphisms). *Set*

$$X := S_1[-1], \quad Y := S_2.$$

The group

$$\text{Hom}_{\mathcal{D}_n}(X, Y) \cong \text{Ext}_{\mathcal{A}_n}^1(S_1, S_2) \cong k$$

contains the zero morphism f_0 and a nonzero morphism f_ξ . Both induce the zero map on every cohomology object.

Proof. The Hom–Ext identification is theorem 21.6. Since X has cohomology only in degree 1 and Y only in degree 0, every same-degree cohomology map $H^q(X) \rightarrow H^q(Y)$ is zero for either morphism. $\square$

Theorem 21.33 (Cone separation). *The two cones are not isomorphic:*

$$\text{Cone}(f_0) \cong S_2 \oplus S_1, \quad \text{Cone}(f_\xi) \cong E.$$

Thus cohomology-level maps do not determine the cone, while the PE-5 derived probe returns the distinction.

Proof. The cone of the zero map $S_1[-1] \rightarrow S_2$ is $S_2 \oplus S_1$. The nonzero class ξ is represented by the exact sequence above and yields the distinguished triangle

$$S_1[-1] \xrightarrow{f_\xi} S_2 \longrightarrow E \longrightarrow S_1.$$

Hence its cone is E . The module E is indecomposable, whereas $S_1 \oplus S_2$ is decomposable, so the cones are not isomorphic. $\square$

Corollary 21.34 (Necessity of the Ext block). *A Return architecture that stores only cohomology objects and the induced cohomology maps identifies f_0 and f_ξ incorrectly. The adjacent-degree Ext^1 block of theorem 21.25, or an equivalent chain-level record, is necessary for cone Return.*

21.8 Compatibility with PE-1–PE-4

Proposition 21.35 (PE-4 extensions as distinguished triangles). *A short exact sequence in the heart*

$$0 \rightarrow L \rightarrow E \rightarrow N \rightarrow 0$$

with Yoneda class $\xi \in \text{Ext}_{\mathcal{A}_n}^1(N, L)$ determines the distinguished triangle

$$L \longrightarrow E \longrightarrow N \xrightarrow{\xi} L[1].$$

Under Φ_n , the exact sequence, connecting morphism, and triangle are returned at the same strength.

Proof. This is the standard exact-sequence-to-triangle construction. The triangle equivalence preserves the connecting morphism and distinguishedness. PE-4 returns the heart-level extension class; PE-5 returns the same class as the final morphism of the corresponding distinguished triangle. $\square$

Proposition 21.36 (PE-3 and PE-2 restrictions). *Restricting PE-5 to the standard heart returns the same persistence objects and source morphisms as PE-2 and PE-3, although the probe algebras and variance conventions differ. Taking only the interval multiplicities of H^0 recovers the PE-2 object-level barcode Return.*

Warning 21.37 (No silent replacement). The global projective-generator realization is optimized for strict derived replay. The PE-2–PE-4 interval and Ext probe packages are optimized for explicit obstruction and extension semantics. PE-5 does not retroactively replace their finite signatures, source bindings, or certificates.

21.9 Failure boundaries and adversarial re-audit

Warning 21.38 (Audit A: cohomology determines objects but not maps). Hereditary splitting makes an object isomorphic to the direct sum of its shifted cohomology. It does not imply that a morphism is determined by its induced cohomology maps. The worked A_2 bridge is an explicit counterexample.

Warning 21.39 (Audit B: no canonical splitting). The isomorphism $X \cong \bigoplus H^q(X)[-q]$ is generally non-canonical. Any certificate using a particular splitting must record the choice and may claim only choice-relative coordinates.

Warning 21.40 (Audit C: no abstract functorial cone). The strict cone equality in theorem 21.17 is relative to selected complexes and a selected chain map. It does not construct a globally functorial cone on the abstract derived category.

Warning 21.41 (Audit D: nonhereditary boundary). If Ext^2 is nonzero, bounded complexes need not split into shifted cohomology objects. The finite shifted-interval algebra $B_{a,b}$ then omits higher extension and possible higher-composition data. A dg, differential graded category, or A_∞ enhancement may be required.

Warning 21.42 (Audit E: finite amplitude is claim-relative). A fixed $[a, b]$ probe package certifies only objects in that amplitude. A cone consumes the expanded lower bound $a - 1$. Iterated cones require an explicitly updated amplitude ledger; finite observation is not an unbounded derived certificate.

Warning 21.43 (Audit F: threshold operations remain separate). Hard barcode deletion and window restriction are not inferred to be exact or triangulated from PE-5. Any use of a repaired persistence profile inside a derived object must retain the frozen v20 stage, representative, and threshold-gap obligations.

Warning 21.44 (Audit G: no compression theorem). The equivalence $\Gamma_n \cong kQ_n$ shows lossless reconstruction, not a universal reduction in size. The finite shifted generator may be much larger than a minimal representation. Compression ratios remain separate claims.

Warning 21.45 (Audit H: triangle image is transported structure). The essential image of the finite amplitude probe is not declared triangulated inside the ordinary module category. Its accepted triangle structure is transported from the source and requires a triangle certificate.

Warning 21.46 (Audit I: source-derived truth is not external truth). A returned derived persistence object or triangle is still an AK-internal or successor-level result. An external arithmetic, analytic, geometric, or physical conclusion requires a separately proved realization and Return theorem.

Part V

Nonhereditary enhancement, deformation, global descent, and boundary Return

Warning 21.47 (Cross-layer firewall). The transition from PE-5 to PE-6 is a change of source and research layer, not a theorem converting an ordinary persistence object into the cubic algebra. Every external application must supply its own source-to-enhancement Interface, coefficient passage, purpose-faithfulness theorem, and Return theorem.

22 PE-6: Nonhereditary A_∞ Relation Return

Warning 22.1 (Layer firewall required by MF-2). From this chapter onward the source is a nonhereditary enhancement/deformation problem. PE-6–PE-10 are not theorems about ordinary one-parameter persistence unless a separate purpose-faithful Interface theorem maps a persistence problem into the displayed nonhereditary source and proves the required Return arrows.

22.1 The cubic-relation nonhereditary source

Definition 22.2 (Source algebra and categories). Fix a field k and define

$$Q : \quad x \xrightarrow{\alpha} y \xrightarrow{\beta} z \xrightarrow{\gamma} t, \quad B := kQ / \langle \gamma\beta\alpha \rangle.$$

Let

$$\mathcal{A}_B := \text{mod}_{\text{fd}}\text{-}B, \quad \mathcal{D}_B := D^b(\mathcal{A}_B),$$

using the right-module and variance convention fixed by the source binding. Let S_x, S_y, S_z, S_t denote the four simple modules and set

$$S := S_x \oplus S_y \oplus S_z \oplus S_t.$$

Proposition 22.3 (Finite length and generation). *The category $\mathcal{A}_B$ is a finite-length abelian category with simple objects S_x, S_y, S_z, S_t . Every object of $\mathcal{A}_B$ has a finite filtration with these simple subquotients, and*

$$\text{tria}(S_x, S_y, S_z, S_t) = \mathcal{D}_B.$$

Proof. The algebra B is finite-dimensional because Q is finite and acyclic. Hence every finite-dimensional module has finite length. A composition series gives the heart-level filtration. Every module is obtained from the simples by finite extensions, and every bounded complex is obtained from its cohomology objects by the standard truncation triangles. Thus the simples triangulated-generate $\mathcal{D}_B$. $\square$

Proposition 22.4 (Genuine higher extension). *The source is not hereditary. In the source-bound Ext convention,*

$$\text{Ext}_B^2(S_t, S_x) \cong k,$$

and the corresponding generator is denoted e . In particular, the PE-5 hereditary splitting theorem is unavailable.

Proof. This is the explicit cubic-relation calculation in the bound Keller example. The relation $\gamma\beta\alpha$ produces a degree-two relation class between the terminal and initial simples. The complete minimal Ext table is recorded in theorem 23.5; its nonzero degree-two component proves nonheredity. $\square$

Warning 22.5 (No inference from quiver shape alone). The underlying directed graph is of type A_4 , but the relation changes the homological algebra. Acyclicity of the graph does not imply heredity after quotienting by relations.

23 The dg Ext object and its minimal A_∞ model

Definition 23.1 (Chosen dg Ext algebra). Choose a bounded projective resolution $P^\bullet \rightarrow S$ in the module convention above and define

$$\mathcal{E}_{\text{dg}} := \text{End}_B^\bullet(P^\bullet)$$

with the standard commutator differential and composition product. Then

$$H^*(\mathcal{E}_{\text{dg}}) \cong \text{Ext}_B^*(S, S) =: E.$$

The decomposition of S gives an augmentation over the semisimple ring

$$R := ke_x \oplus ke_y \oplus ke_z \oplus ke_t.$$

Theorem 23.2 (Minimal-model transfer). *The graded R – R -bimodule E admits a strictly unital minimal A_∞ structure*

$$\mathcal{E}_\infty = (E, m_1 = 0, m_2, m_3, \dots)$$

such that m_2 is induced by the dg multiplication and there is an A_∞ quasi-isomorphism

$$\mathcal{E}_\infty \longrightarrow \mathcal{E}_{\text{dg}}$$

lifting the identity on cohomology. The minimal model is unique up to nonunique strictly unital A_∞ isomorphism.

Proof. Apply the Kadeishvili minimal-model theorem over the semisimple augmentation ring R . Strict unitality is available for the Ext category of the simple modules of a finite-dimensional basic algebra. The uniqueness statement is exactly at A_∞ -isomorphism strength; it does not select coordinates or a unique transfer contraction. $\square$

Definition 23.3 (Return-relevant equivalence strength). The invariant observation object of PE-6 is the strictly unital A_∞ quasi-isomorphism class of $\mathcal{E}_\infty$. A displayed basis and operation table are a selected representative of that class. Strict table equality is available only after the certificate freezes the simple ordering, basis, signs, transfer data, and normalization.

Warning 23.4 (Cohomology algebra versus enhancement). The associative graded algebra (E, m_2) is only the cohomology algebra of $\mathcal{E}_{\text{dg}}$. Replacing $\mathcal{E}_{\text{dg}}$ by (E, m_2) is valid precisely when the dg algebra is formal. PE-6 proves that the selected source is a smallest nonformal example, so this replacement loses theorem-bearing data.

23.1 Explicit higher Ext table and cubic relation class

Theorem 23.5 (Sparse minimal Ext table). *There is a normalized strictly unital minimal model with reduced positive-degree part*

$$E^1 = ka \oplus kb \oplus kc, \quad E^2 = ke, \quad E^r = 0 \quad (r \geq 3),$$

where, in the source-bound object convention,

$$a : y \rightarrow x, \quad b : z \rightarrow y, \quad c : t \rightarrow z, \quad e : t \rightarrow x.$$

The positive-degree binary products vanish, including

$$m_2(a, b) = 0, \quad m_2(b, c) = 0,$$

and all reduced higher products vanish except

$$\boxed{m_3(a, b, c) = e}.$$

The strict units at x, y, z, t act in the usual way.

Proof. This is the explicit minimal model computed in Keller’s cubic-relation example. The arrows a, b, c have cohomological degree one, e has degree two, and m_3 has degree $2 - 3 = -1$, so the displayed operation has degree

$$1 + 1 + 1 - 1 = 2.$$

Rescaling the one-dimensional target ke fixes the nonzero scalar to 1. $\square$

Corollary 23.6 (Nontrivial higher composition). *The relation class e is not obtained as an ordinary Yoneda product of the adjacent degree-one generators. It is obtained as their ternary A_∞ product.*

Proof. The relevant binary products vanish by theorem 23.5, whereas $m_3(a, b, c) = e \neq 0$. $\square$

Proposition 23.7 (Reduced Stasheff identities). *On the augmentation ideal generated by a, b, c, e , the operation table of theorem 23.5 satisfies the A_∞ identities: every possible nesting of the unique nonzero m_3 into another m_3 vanishes because the inner operation outputs e and m_3 is zero whenever one input is e .*

Proof. Minimality gives $m_1 = 0$, and all reduced m_2 and m_r for $r \geq 4$ vanish. The only potentially nonzero higher Stasheff sum is the arity-five sum of nested m_3 terms. Every inner nonzero term is e , while no nonzero m_3 accepts e as an input. Hence every summand vanishes individually. Strict-unit identities are standard and separate from the reduced calculation. $\square$

Remark 23.8 (Massey interpretation). The nonzero m_3 is a chosen lift of the corresponding triple Massey product. PE-6 consumes the source-bound A_∞ operation, not an untyped assertion that a Massey product “should” be nonzero.

23.2 Maurer–Cartan Return of the source relation

Definition 23.9 (Heart-level twisted datum). Let

$$V := V_x \otimes x \oplus V_y \otimes y \oplus V_z \otimes z \oplus V_t \otimes t$$

with finite-dimensional coefficient spaces. A degree-one upper-triangular twisting element is specified by maps

$$V_x \xleftarrow{V_a} V_y \xleftarrow{V_b} V_z \xleftarrow{V_c} V_t$$

and is denoted

$$\delta := V_a \otimes a + V_b \otimes b + V_c \otimes c.$$

Definition 23.10 (A_∞ Maurer–Cartan residual). For a strictly upper-triangular δ , define the finite residual

$$\text{MC}_{\mathcal{E}_\infty}(\delta) := \sum_{r \geq 1} (-1)^{r(r-1)/2} m_r(\delta, \dots, \delta).$$

The sum is finite. The pair (V, δ) is a twisted object exactly when

$$\text{MC}_{\mathcal{E}_\infty}(\delta) = 0.$$

Theorem 23.11 (Cubic relation Return). *For the table in theorem 23.5,*

$$\text{MC}_{\mathcal{E}_\infty}(\delta) = 0 \iff \boxed{V_a V_b V_c = 0}.$$

Thus the cubic source relation is exactly the finite A_∞ Maurer–Cartan equation.

Proof. Minimality removes the m_1 term. The positive-degree m_2 products of adjacent arrows vanish, and every m_r with $r \geq 4$ vanishes. The only remaining reduced term is the ternary term. Expanding $m_3(\delta, \delta, \delta)$, source and target idempotents force the unique composable tensor (a, b, c) , whose coefficient is $V_a V_b V_c$ and whose Ext component is e . Since $e \neq 0$, the residual vanishes exactly when the matrix composite vanishes. $\square$

Corollary 23.12 (Finite relation certificate). *A heart-level object certificate consists of the four dimensions, the three matrices V_a, V_b, V_c , the frozen m_3 table, and the matrix identity $V_a V_b V_c = 0$. No infinite observation or unstated relation oracle is required.*

Warning 23.13 (The source conclusion is not an input). The certificate may not mark $V_a V_b V_c = 0$ merely because the datum was declared to be a B -module. The verifier must multiply the supplied matrices or consume an independently checked symbolic proof.

23.3 Module-heart object, morphism, and exact-structure Return

Definition 23.14 (Returned heart). Let $\mathcal{H}_\infty$ be the full subcategory of $H^0 \text{tw}(\mathcal{E}_\infty)$ whose objects are finite filtered direct sums of the nonshifted simple objects x, y, z, t with twisting elements of the form above.

Theorem 23.15 (Heart reconstruction). *There is an equivalence of categories*

$$\boxed{\mathcal{H}_\infty \xrightarrow{\sim} \mathcal{A}_B.}$$

Under this equivalence, (V, δ) is sent to the source representation with vector spaces V_x, V_y, V_z, V_t , structure maps V_a, V_b, V_c , and relation $V_a V_b V_c = 0$.

Proof. Keller's filtered-module reconstruction theorem applies to the A_∞ Ext category of the four simples. Every finite-dimensional B -module has a finite simple filtration, so the filtered-module target is all of $\mathcal{A}_B$. The explicit twisted-object calculation is theorem 23.11, which recovers exactly the defining relation. $\square$

Proposition 23.16 (Morphism Return). *Let (V, δ) and (V', δ') be objects of $\mathcal{H}_\infty$. Their degree-zero morphisms in $H^0 \text{tw}(\mathcal{E}_\infty)$ are exactly the quadruples*

$$f_i : V_i \longrightarrow V'_i \quad (i = x, y, z, t)$$

satisfying

$$\boxed{f_x V_a = V'_a f_y, \quad f_y V_b = V'_b f_z, \quad f_z V_c = V'_c f_t.}$$

Consequently PE-6 returns every source morphism, identity, and composition.

Proof. The twisted differential on a degree-zero map is obtained by inserting f among the source and target twisting elements. In the present degrees, the zero-cycle equations reduce exactly to the three displayed commutative-square equations; negative-degree homotopies vanish in this nonshifted heart. These are precisely the module-homomorphism equations. $\square$

Corollary 23.17 (Exact-structure Return). *The equivalence in theorem 23.15 transports the abelian exact structure of $\mathcal{A}_B$ to $\mathcal{H}_\infty$. Hence kernels, cokernels, short exact sequences, Yoneda Ext^1 classes, pushouts, pullbacks, and Baer sums are returned at equivalence strength.*

Proof. An equivalence of categories preserves all existing finite limits and colimits up to unique isomorphism. Transport the abelian structure along the equivalence. This is a returned exact structure, not a claim that an arbitrary subcategory of an arbitrary triangulated category is abelian without the stated heart identification. $\square$

Warning 23.18 (No coordinate canonicity). The source module and its morphisms are returned up to categorical equivalence. Matrices are relative to selected bases of the coefficient spaces; changing bases changes the recorded matrices without changing the returned object.

23.4 Nonhereditary derived and twisted-complex Return

Theorem 23.19 (Derived A_∞ Return). *The twisted-object category of the minimal Ext enhancement returns the entire bounded derived category:*

$$\boxed{\Theta_B : H^0 \text{tw}(\mathcal{E}_\infty) \xrightarrow{\sim} \mathcal{D}_B.}$$

The equivalence takes the four generating objects to the four simple B -modules.

Proof. The A_∞ Yoneda embedding factors through twisted objects and induces an equivalence from $H^0 \text{tw}(\mathcal{E}_\infty)$ to the triangulated category generated by the representable $\mathcal{E}_\infty$ -modules. The standard A_∞ Ext functor sends these generators to S_x, S_y, S_z, S_t and induces an equivalence onto their triangulated closure in $\mathcal{D}_B$. By finite-length generation, that closure is all of $\mathcal{D}_B$. $\square$

Corollary 23.20 (Returned derived structure). *PE-6 returns, at triangle-equivalence strength:*

1. *all bounded derived objects;*
2. *all derived morphism spaces and compositions;*
3. *shifts;*
4. *twisted-complex cones;*
5. *distinguished triangles;*
6. *all higher extension groups represented as shifted morphisms.*

Proof. These structures are preserved and reflected by a triangle equivalence. $\square$

Proposition 23.21 (dg and minimal-model compatibility). *The A_∞ quasi-isomorphism $\mathcal{E}_\infty \rightarrow \mathcal{E}_{\text{dg}}$ induces equivalent perfect or twisted derived module theories. Thus the compact m_r table is a lossless minimal presentation of the chosen dg Ext enhancement at quasi-equivalence strength.*

Proof. An A_∞ quasi-isomorphism is a homotopy equivalence in the A_∞ sense and induces an equivalence of the corresponding derived module categories. The minimal model therefore preserves the generated triangulated category. “Lossless” here means equivalence of the returned category, not literal equality of cochain complexes. $\square$

Warning 23.22 (Twisted cones remain model-relative). The category $H^0 \text{tw}(\mathcal{E}_\infty)$ supplies an algebraic enhancement in which cones are represented by twisted objects. It does not turn the abstract cone of an arbitrary triangulated category into a globally canonical coordinate object. Strict replay remains relative to the chosen enhancement and operation table.

24 Why graded Ext and m_2 are insufficient

Definition 24.1 (m_3 -truncated observation). Let

$$\mathcal{E}_\infty^{(2)} := (E, m_1 = 0, m_2, m_r = 0 \text{ for } r \geq 3)$$

be the formal truncation retaining the same graded vector spaces, idempotents, and Yoneda product while deleting the higher composition.

Theorem 24.2 (Same graded Ext, different relation semantics). *The heart-level Maurer–Cartan equation for $\mathcal{E}_\infty^{(2)}$ imposes no condition on the triple (V_a, V_b, V_c) . Hence $\mathcal{E}_\infty^{(2)}$ does not return $\mathcal{A}_B$.*

Proof. The differential m_1 is zero. The only potentially composable positive-degree binary products are $m_2(a, b)$ and $m_2(b, c)$, and both are zero. By definition, all higher operations of the truncation vanish. Therefore every triple of matrices satisfies the truncated Maurer–Cartan equation, whether or not $V_a V_b V_c = 0$. $\square$

Corollary 24.3 (Explicit false positive). *Take*

$$V_x = V_y = V_z = V_t = k, \quad V_a = V_b = V_c = \text{id}_k.$$

The m_3 -truncated observation accepts this datum, but

$$V_a V_b V_c = \text{id}_k \neq 0,$$

so it is not a B -module. The deletion of m_3 produces a certified counterexample to graded-Ext-only Return.

Proposition 24.4 (An isolated Ext^2 class is still insufficient). *Knowing that $E^2 = ke$ and knowing all graded dimensions does not determine which ordered triple of degree-one extensions produces e . The map*

$$m_3 : E^1 \otimes_R E^1 \otimes_R E^1 \longrightarrow E^2$$

is additional typed incidence data.

Proof. The formal truncation and the cubic model have the same graded vector space and the same m_2 , including the same degree-two basis vector e , but different m_3 . By theorems 23.11 and 24.2, their Maurer–Cartan object classes differ. Therefore the isolated existence of e does not determine the relation semantics. $\square$

Corollary 24.5 (Nonformality witness). *The chosen dg Ext algebra is not formal. Otherwise it would be A_∞ quasi-isomorphic to the truncation with all m_r , $r \geq 3$, zero, contradicting the relation-reconstruction distinction above and the source-bound nonformality calculation.*

Warning 24.6 (Higher operation is not a scalar obstruction). The datum m_3 is a multilinear operation with source, target, degree, unit, and gauge semantics. Replacing it by one scalar “nonformality score” does not preserve the Return theorem.

24.1 Compatibility with PE-1–PE-5

Proposition 24.7 (PE-4 exact classes inside PE-6). *The heart equivalence returns ordinary short exact sequences and their Yoneda classes exactly as in PE-4, now inside a source whose Ext^2 need not vanish. Splicing two degree-one extensions is represented by m_2 when the ordinary Yoneda product is defined and nonzero; the cubic relation class in this example instead requires the higher operation m_3 .*

Proposition 24.8 (PE-5 hereditary boundary is respected). *If the source is replaced by the ordinary line-quiver persistence category of PE-5, then $\text{Ext}^2 = 0$ and the bounded derived category is already returned by the projective interval generator. PE-6 does not reinterpret a shifted PE-5 morphism as a new module-theoretic higher extension.*

Warning 24.9 (No silent replacement of PE-5). PE-5 is stronger and simpler on its hereditary scope because it has a strict projective-generator model. PE-6 is required only when a nonhereditary source contains relation data invisible to m_2 . The two certificates are complementary, not competing versions of one theorem.

Proposition 24.10 (PE-1–PE-3 restrictions). *After choosing a finite module or derived object, ordinary object, Hom, and degree-one Ext information may be extracted and compared with PE-1–PE-3 style probes. Such decategorifications do not reconstruct the m_3 operation and therefore cannot be promoted back to PE-6 strength without the enhancement certificate.*

24.2 Failure boundaries and adversarial re-audit

Warning 24.11 (Audit A: convention and sign sensitivity). The orientation of Ext arrows, the order of inputs to m_3 , and the sign of e depend on module, suspension, and composition conventions. The invariant statement is the source-bound nonzero ternary operation whose Maurer–Cartan equation returns the cubic relation. A certificate must freeze conventions before replay.

Warning 24.12 (Audit B: minimal model is not canonical). Kadeishvili transfer gives a minimal model unique only up to nonunique A_∞ isomorphism. PE-6 returns the quasi-isomorphism class and the induced category. A concrete table is choice-relative unless its contraction and basis records are supplied.

Warning 24.13 (Audit C: $m_3 \neq 0$ is not a universal invariant in raw coordinates). An individual coordinate of m_3 can change under an A_∞ change of model. The nonformality class and the reconstructed category are invariant at the declared equivalence strength. No coordinate is promoted beyond its frozen model.

Warning 24.14 (Audit D: heart identification is required). The category $H^0 \text{tw}(\mathcal{E}_\infty)$ is triangulated. The abelian module heart is the specifically identified nonshifted filtered-simple subcategory $\mathcal{H}_\infty$ with transported exact structure. An arbitrary full subcategory of twisted objects is not automatically a heart.

Warning 24.15 (Audit E: relation Return is source-relative). The equation $V_a V_b V_c = 0$ returns the defining relation of the selected algebra. It does not prove that an externally supplied physical, arithmetic, or geometric triple composite vanishes. That requires a separately proved source-to- B realization.

Warning 24.16 (Audit F: finite table does not imply finite representation type). The A_∞ operation table is finite, but the source module category may contain arbitrarily large vector spaces and may have complicated representation theory. Finite theorem data are not a uniform finite list of all objects.

Warning 24.17 (Audit G: no automatic truncation of higher operations). The present example has only m_2 and m_3 in a chosen minimal model. A different source may require $m_4, m_5, \dots$ or an infinite locally finite operation family. PE-6 supplies no theorem that a finite relation length always yields a globally finite enhancement outside the bound hypotheses.

Warning 24.18 (Audit H: Type IV terminology). A missing m_3 table, missing transfer, or absent source binding is UNDEFINED. A false claim that the formal truncation returns $\mathcal{A}_B$ is REJECT. Neither is a Type IV obstruction, because no valid comparison artifact with a nonzero declared defect has yet been supplied.

Warning 24.19 (Audit I: thresholds remain separate). Nothing in the dg or A_∞ equivalence makes hard threshold deletion exact, functorial, or triangulated. Any repaired persistence input retains the frozen window-first, stage, representative, and threshold-gap obligations.

25 PE-7: Marked Minimal-Model and Morita Invariance

Classification 25.1 (Descent responsibility after MF-3). The PE-7 tree/cycle discussion is preliminary and non-normative in this integrated edition. The sole normative global descent, monodromy, and cyclic-coherence results are stated in PE-9.

25.1 Marked enhanced Return and its strength levels

Definition 25.2 (Marked strictly unital model). Let

$$R = \bigoplus_{i=1}^s ke_i$$

be a fixed semisimple idempotent ring. A marked strictly unital A_∞ model is a strictly unital R -augmented A_∞ category or algebra together with:

1. an ordered object/idempotent list $(e_1, \dots, e_s)$;
2. graded source and target spaces for every operation;
3. the requested equivalence strength;
4. a source identity and artifact binding.

A morphism is marked when it preserves the idempotents, up to a declared permutation.

Definition 25.3 (Five Return strengths). The PE-7 strengths are:

1. Σ_{table} : literal equality of all frozen operation tensors in selected bases;
2. Σ_{mark} : marked strictly unital A_∞ isomorphism or quasi-isomorphism class;
3. Σ_{Mor} : equivalence of the returned pretriangulated/perfect category;
4. Σ_{heart} : equivalence of a declared transported finite-length heart with its marked simples;
5. Σ_{pres} : a standard module-heart and basic quiver-with-relations presentation with frozen generator and relation coordinates.

There are certified forgetful arrows

$$\Sigma_{\text{table}} \longrightarrow \Sigma_{\text{mark}} \longrightarrow \Sigma_{\text{Mor}}.$$

The heart and presentation strengths require additional data and are not automatic nodes on one unconditional linear chain.

Warning 25.4 (No strength collapse). An equivalence at Σ_{Mor} does not identify standard t -structures. An equivalence at Σ_{heart} does not choose bases of arrows or relations. A literal operation table is stronger as a replay artifact but less invariant than the marked A_∞ class.

25.2 Minimal-model invariance

Theorem 25.5 (Marked minimal-model uniqueness). *Let $\mathcal{C}$ be a strictly unital dg category over a field with fixed semisimple object ring R . Let $\mathcal{E}_{\min}$ and $\mathcal{E}'_{\min}$ be two strictly unital minimal A_∞ models of $H^*(\mathcal{C})$ obtained from possibly different resolutions, contractions, representatives, and transfer data, each marked by the same ordered idempotents. Then there is a marked strictly unital A_∞ isomorphism*

$$U : \mathcal{E}_{\min} \xrightarrow{\sim} \mathcal{E}'_{\min}$$

whose linear term induces the identity on the marked cohomology objects after the declared basis comparison.

Proof. The minimal-model theorem supplies strictly unital A_∞ quasi-isomorphisms from both minimal models to $\mathcal{C}$. The localization of strictly unital A_∞ categories at quasi-isomorphisms identifies the two models. Since both are minimal and the linear term is an isomorphism on cohomology, the quasi-isomorphism admits an A_∞ inverse and is an A_∞ isomorphism. The marking is preserved by fixing the semisimple idempotents and recording any allowed permutation. $\square$

Theorem 25.6 (Twisted-category invariance). *A marked strictly unital A_∞ quasi-equivalence*

$$U : \mathcal{E} \longrightarrow \mathcal{E}'$$

extends to a quasi-equivalence of pretriangulated twisted-object categories

$$\mathrm{tw}(U) : \mathrm{tw}(\mathcal{E}) \longrightarrow \mathrm{tw}(\mathcal{E}').$$

Consequently,

$$\boxed{H^0 \mathrm{tw}(\mathcal{E}) \simeq H^0 \mathrm{tw}(\mathcal{E}')} \quad$$

as triangulated categories, and the equivalence preserves the marked generators.

Proof. The twisted-object construction is the A_∞ pretriangulated hull. It sends quasi-equivalences to quasi-equivalences. Passing to H^0 gives the asserted triangle equivalence. Marked generators map according to the linear term and the frozen idempotent comparison. $\square$

Corollary 25.7 (Coordinate-independent PE-6 Return). *All minimal models of the PE-6 dg Ext category Return the same bounded derived category at Σ_{Mor} . A selected sparse table is one certificate representative, not the invariant target itself.*

25.3 Concrete cubic coefficient descent

Definition 25.8 (Cubic marked family). Let $\mathcal{E}_\kappa$ have marked degree-one generators a, b, c , a marked degree-two generator e , vanishing reduced positive-degree m_2 , and

$$m_3(a, b, c) = \kappa e, \quad \kappa \in k.$$

All other reduced higher products vanish, as in the PE-6 cubic source.

Theorem 25.9 (Diagonal gauge law). *For nonzero scalars $\lambda_a, \lambda_b, \lambda_c, \lambda_e$, set*

$$a' = \lambda_a a, \quad b' = \lambda_b b, \quad c' = \lambda_c c, \quad e' = \lambda_e e.$$

Then the same marked operation has coefficient

$$\boxed{\kappa' = \frac{\lambda_a \lambda_b \lambda_c}{\lambda_e} \kappa}$$

in the primed basis. Hence:

1. every $\kappa \neq 0$ is diagonally isomorphic to the normalized model $\mathcal{E}_1$;
2. $\kappa = 0$ cannot be changed to $\kappa \neq 0$ by an invertible marked basis change;
3. the coordinate-free invariant is the zero/nonzero orbit, not the scalar κ .

Proof. Multilinearity gives

$$m_3(a', b', c') = \lambda_a \lambda_b \lambda_c \kappa e = \frac{\lambda_a \lambda_b \lambda_c}{\lambda_e} \kappa e'.$$

The orbit statements follow because all scaling factors are invertible. $\square$

Proposition 25.10 (Maurer–Cartan zero-locus invariance). *Let twisting coefficients transform contragrediently,*

$$V'_a = \lambda_a^{-1} V_a, \quad V'_b = \lambda_b^{-1} V_b, \quad V'_c = \lambda_c^{-1} V_c.$$

Then

$$\kappa' V'_a V'_b V'_c = \lambda_e^{-1} \kappa V_a V_b V_c.$$

Therefore the Maurer–Cartan equations

$$\kappa V_a V_b V_c = 0 \quad \text{and} \quad \kappa' V'_a V'_b V'_c = 0$$

have exactly corresponding solutions.

Proof. Substitute the coefficient and matrix transformation laws from theorem 25.9. The two residuals differ by the invertible scalar λ_e^{-1} . $\square$

Corollary 25.11 (Presentation-scaling descent). *Rescaling the source arrows α, β, γ multiplies the cubic path $\gamma\beta\alpha$ by a unit and leaves the generated relation ideal unchanged. Thus literal relation coefficients and literal m_3 coefficients are presentation data; the relation ideal, enhanced category, and returned heart are invariant.*

Warning 25.12 (General gauge transformations). A general A_∞ isomorphism may redistribute information among several higher tensors. The diagonal law proves the explicit PE-6 normalization, but the general invariant remains the marked A_∞ quasi-isomorphism class, not the support pattern of one chosen table.

25.4 Enhanced derived-Morita transport

Definition 25.13 (Native enhanced derived-equivalence datum). Let A and B be finite-dimensional k -algebras, with declared dg or strictly unital A_∞ enhancements $\mathcal{D}_A$ and $\mathcal{D}_B$ of their bounded derived categories. A native comparison datum consists of the ordered simple generator S_A , its marked image T , and exactly one of the following typed certificates:

- (a) a strict dg quasi-equivalence $F_{\text{dg}} : \mathcal{D}_A \rightarrow \mathcal{D}_B$;
- (b) a strictly unital A_∞ quasi-equivalence $F_\infty : \mathcal{D}_A \rightarrow \mathcal{D}_B$, whose linear term is a quasi-isomorphism on every relevant morphism complex and whose H^0 functor is essentially surjective;
- (c) a dg quasi-functor represented by a right quasi-representable bimodule, together with a source-bound zigzag of quasi-equivalences or Morita equivalences comparing the marked endomorphism objects.

An abstract triangle equivalence, or a quasi-functor known only to induce an equivalence on H^0 , does not supply any of these certificates.

Theorem 25.14 (Native-type marked derived-endomorphism invariance). *For a native enhanced derived-equivalence datum:*

- (a) *in the strict dg case, restriction to S_A gives a strict dg-algebra quasi-isomorphism*

$$\boxed{\mathbf{R}\mathrm{Hom}_A(S_A, S_A) \xrightarrow{\sim} \mathbf{R}\mathrm{Hom}_B(T, T);}$$

- (b) *in the A_∞ case, the full collection of Taylor components restricts to a marked A_∞ quasi-isomorphism between the two derived endomorphism algebras; its linear term is a quasi-isomorphism of complexes, but it is not asserted to preserve multiplication strictly;*
- (c) *in the bimodule/quasi-functor case, the comparison is the declared zigzag in the dg/A_∞ Morita homotopy category rather than one invented strict algebra map.*

In all three cases the corresponding marked minimal models are A_∞ quasi-isomorphic, and their twisted-object Return categories are triangle equivalent.

Proof. A strict dg functor preserves composition and differential on the nose, so quasi-fully-faithfulness on the marked endomorphism complex gives (a). For a general A_∞ functor, the A_∞ functor identities show that the Taylor components form an A_∞ morphism; quasi-fully-faithfulness makes its linear term a quasi-isomorphism, proving (b) without a false strictness upgrade. Case (c) is precisely the native comparison supplied by the representing bimodule and its source-bound zigzag. Minimal-model invariance and the twisted-object quasi-equivalence then follow from theorems 25.5 and 25.6. $\square$

Corollary 25.15 (Derived-Morita Return). *The marked enhanced model of T satisfies*

$$H^0 \operatorname{tw}(\operatorname{Ext}_B^*(T, T)_\infty) \simeq D^b(\operatorname{mod}_{\operatorname{fd}} - B) \simeq D^b(\operatorname{mod}_{\operatorname{fd}} - A).$$

Thus PE-6 Return is invariant under enhanced derived equivalence when the simple collection is transported rather than replaced silently by the standard simples of B .

Proof. Let F denote the triangle equivalence induced by the selected native comparison datum. The image $T = F(S_A)$ triangulated-generates because S_A does and F is an equivalence. Apply the dg/ A_∞ Morita reconstruction and theorem 25.6. $\square$

Warning 25.16 (Morita equivalence is weaker than source identity). The theorem Returns the enhanced derived category and the marked transported generator. It does not identify A with B , their standard simple objects, or their chosen quiver presentations.

25.5 Heart descent and the standard-heart gate

Definition 25.17 (Transported heart). Let

$$F : D^b(\operatorname{mod}_{\operatorname{fd}} - A) \xrightarrow{\sim} D^b(\operatorname{mod}_{\operatorname{fd}} - B)$$

be the triangle equivalence induced by the selected native comparison datum of theorem 25.14. The transported heart is

$$\mathcal{H}_F := F(\operatorname{mod}_{\operatorname{fd}} - A).$$

Its marked simple objects are $T_i = F(S_i^A)$.

Proposition 25.18 (Transported-heart Return). *The subcategory $\mathcal{H}_F$ is the heart of the transported bounded t -structure and*

$$F|_{\operatorname{mod}_{\operatorname{fd}} - A} : \operatorname{mod}_{\operatorname{fd}} - A \xrightarrow{\sim} \mathcal{H}_F$$

is an exact equivalence of finite-length abelian categories. The objects T_i are precisely the simple objects of $\mathcal{H}_F$.

Proof. Transport the standard truncation subcategories along the triangle equivalence. The t -structure axioms and boundedness are preserved. The restriction to the heart is an equivalence; exactness follows because short exact sequences in a heart are the triangles whose three terms lie in that heart. Equivalences preserve simple objects. $\square$

Definition 25.19 (Standard-heart descent gate). The standard-heart gate consists of:

1. a proof that F is t -exact for the standard t -structures;
2. an identification $\mathcal{H}_F = \operatorname{mod}_{\operatorname{fd}} - B$;
3. a marked permutation carrying T_i to the standard simples of B ;
4. for presentation strength, a basic projective generator, primitive idempotents, arrow bases, relation bases, and normalization.

Theorem 25.20 (Conditional standard-heart and presentation descent). *If the standard-heart gate is discharged, then*

$$\text{mod}_{\text{fd}} - A \simeq \text{mod}_{\text{fd}} - B$$

as abelian categories. Consequently A and B are Morita equivalent; if both are basic and the simple marking is fixed, their basic algebras are isomorphic up to the declared idempotent permutation. A quiver-with-relations presentation is then Returnable only relative to the additional generator and relation-coordinate record.

Proof. By t -exactness, the restriction of F is an exact equivalence between the standard module hearts. Classical Morita theory gives Morita equivalence. A basic finite-dimensional algebra is determined up to isomorphism by its module category together with the marked indecomposable projectives. The final presentation still requires bases for radical layers and relation spaces, which are not categorical invariants. $\square$

Proposition 25.21 (Shift no-go). *For any nonzero finite-dimensional algebra A , the shift*

$$[1] : D^b(\text{mod}_{\text{fd}} - A) \xrightarrow{\sim} D^b(\text{mod}_{\text{fd}} - A)$$

is an enhanced triangle autoequivalence but is not t -exact for the standard heart. It sends

$$\text{mod}_{\text{fd}} - A \quad \text{to} \quad (\text{mod}_{\text{fd}} - A)[1] \neq \text{mod}_{\text{fd}} - A.$$

Hence enhanced derived equivalence alone cannot Return the standard heart or source presentation.

Proof. A nonzero module concentrated in cohomological degree zero is sent to degree -1 under the standard convention and therefore leaves the standard heart. The shift is nevertheless an exact autoequivalence of the dg/triangulated category. $\square$

25.6 Failure boundaries and adversarial reaudit

Warning 25.22 (Audit A: quasi-isomorphism versus table equality). A quasi-isomorphism of enhanced models does not imply equality of their displayed m_r tensors. The correct invariant target is Σ_{mark} or Σ_{Mor} unless coordinates are frozen.

Warning 25.23 (Audit B: derived equivalence versus Morita equivalence). A derived equivalence can carry the standard heart to a different heart. The shift no-go shows that standard-heart Return cannot be inferred from triangulated equivalence.

Warning 25.24 (Audit C: simple replacement). The transported collection $T = F(S_A)$ may consist of complexes rather than standard B -modules. Replacing T with the standard simples of B changes the observation object and requires a separate theorem.

Warning 25.25 (Audit D: relation coefficient). The scalar multiplying a relation or an m_3 tensor is basis-dependent. Only its orbit under the declared gauge group and the induced Maurer–Cartan category are invariant.

Warning 25.26 (Audit E: triangle equivalence without enhancement). PE-7 consumes an actual dg/A_∞ quasi-functor or theorem-backed standard enhancement. An abstract triangle functor does not by itself provide maps of derived endomorphism complexes.

Warning 25.27 (Audit F: local pairwise agreement). A list of pairwise quasi-equivalences is insufficient on cycles. The missing object is coherent comparison data, not a numerical defect that can be budgeted away.

Warning 25.28 (Audit G: strict descent obstruction status). An absent coherence cell gives UNDEFINED. A supplied cell that fails its equations gives REJECT. A Type IV-style OBSTRUCTED verdict is available only after a valid comparison artifact and a separately defined nonzero descent defect are supplied.

Warning 25.29 (Audit H: frozen threshold discipline). Nothing in A_∞ Morita invariance makes hard persistence threshold deletion exact, functorial, or triangulated. Any repaired persistence input retains all frozen window-first, representative, stage, and threshold-gap obligations.

26 PE-8: Local Hochschild Deformation and Obstruction Theory

Classification 26.1 (Responsibility boundary after MF-4). PE-8 governs local formal deformation: the Hochschild controller, Maurer–Cartan equation, tangent space, recursive obstruction classes, gauge, and Morita-stable local moduli. Global atlas and wall-crossing statements belong to PE-9; valuative boundary and contact-order statements belong to PE-10.

Protocol 26.2 (Standing characteristic-zero deformation regime). Throughout PE-8 and PE-9, k is a field of characteristic zero. The deformation base R is a local Artin commutative k -algebra or a complete local commutative k -algebra with maximal ideal $\mathfrak{m}_R$. Every controller carries a descending filtration that is complete, separated, compatible with all L_∞ brackets, and pronilpotent after tensoring with $\mathfrak{m}_R$. Every comparison morphism preserves the declared filtration and satisfies the hypotheses of the cited filtered Goldman–Millson and descent theorems. Positive-characteristic deformation theory is outside this theorem chain and requires a separately typed replacement.

26.1 The complete Hochschild deformation controller

Definition 26.3 (Marked bar-cochain Lie algebra). Let E be a graded R_0 -bimodule over a fixed finite semisimple idempotent ring

$$R_0 = \bigoplus_{i=1}^s ke_i,$$

and let $\bar{E}$ be its reduced augmentation ideal. Write $s\bar{E}$ for the suspension. The complete normalized Hochschild cochain space is

$$\mathfrak{g}_{\text{Hoch}}(E) := \prod_{r \geq 1} \text{Hom}_{R_0-R_0}((s\bar{E})^{\otimes r}, s\bar{E}),$$

with the graded commutator bracket of coderivations. The arity filtration is

$$F^p \mathfrak{g}_{\text{Hoch}}(E) = \prod_{r \geq p} \text{Hom}_{R_0-R_0}((s\bar{E})^{\otimes r}, s\bar{E}).$$

Proposition 26.4 (Completeness). *The arity filtration is descending, complete, and separated:*

$$\mathfrak{g}_{\text{Hoch}}(E) \cong \varprojlim_p \mathfrak{g}_{\text{Hoch}}(E) / F^p \mathfrak{g}_{\text{Hoch}}(E), \quad \bigcap_p F^p \mathfrak{g}_{\text{Hoch}}(E) = 0.$$

It is compatible with the bracket in the sense that

$$[F^p, F^q] \subseteq F^{p+q-1}.$$

Definition 26.5 (A_∞ structure as a Maurer–Cartan element). A marked strictly unital reduced A_∞ structure is a degree-one coderivation

$$b = b_1 + b_2 + b_3 + \cdots \in \mathfrak{g}_{\text{Hoch}}(E)^1$$

satisfying

$$[b, b] = 0.$$

Equivalently, $b^2 = 0$ on the reduced bar coalgebra. The twisted differential on cochains is

$$d_b(\phi) := [b, \phi].$$

Then

$$\mathfrak{g}_{\text{Hoch}}(b) := (\mathfrak{g}_{\text{Hoch}}(E), d_b, [-, -])$$

is a complete filtered dg Lie algebra.

Remark 26.6 (Degree convention). The suspension convention places deformation cochains in the dg-Lie degrees used below. Under the corresponding unshifted Hochschild convention,

$$H^1(\mathfrak{g}_{\text{Hoch}}(b)) \cong \text{HH}^2(E, b), \quad H^2(\mathfrak{g}_{\text{Hoch}}(b)) \cong \text{HH}^3(E, b).$$

The certificate must freeze the suspension, sign, and arity conventions before these identifications are consumed.

26.2 Formal perturbations, gauge, and the moduli groupoid

Definition 26.7 (Formal perturbation). Let $(R, \mathfrak{m}_R)$ be in the declared base regime. A formal perturbation of b is

$$\alpha \in F^1 \mathfrak{g}_{\text{Hoch}}(b)^1 \widehat{\otimes} \mathfrak{m}_R.$$

The perturbed operation is $b_R = b + \alpha$.

Theorem 26.8 (Maurer–Cartan deformation equation). *The perturbation $b_R = b + \alpha$ is an A_∞ structure if and only if*

$$d_b \alpha + \frac{1}{2}[\alpha, \alpha] = 0.$$

Proof. Since $[b, b] = 0$,

$$[b + \alpha, b + \alpha] = 2[b, \alpha] + [\alpha, \alpha].$$

The left side vanishes exactly when the displayed equation holds. $\square$

Definition 26.9 (Gauge or homotopy equivalence). The deformation object is not the raw set of Maurer–Cartan elements. It is the Deligne–Getzler–Hinich Maurer–Cartan ∞ -groupoid

$$\text{MC}_\bullet(\mathfrak{g}_{\text{Hoch}}(b) \widehat{\otimes} \mathfrak{m}_R).$$

Its connected components are deformation classes modulo filtered Maurer–Cartan homotopy. In a nilpotent dg-Lie model over characteristic zero, this agrees with the usual exponential gauge action of degree-zero elements.

Warning 26.10 (A set is not the complete moduli object). Automorphisms and higher homotopies may be nontrivial. Passing immediately to the set of gauge orbits forgets stabilizers and higher coherence. PE-8 Return at full deformation strength therefore records the Maurer–Cartan ∞ -groupoid, with π_0 only as a declared decategorification.

Theorem 26.11 (Gauge-equivalent perturbations have the same marked Return). *A Maurer–Cartan path between α and β yields a filtered marked A_∞ equivalence between $b + \alpha$ and $b + \beta$. Consequently their twisted-object categories are quasi-equivalent and have the same enhanced Return at the strength licensed by PE-7.*

Warning 26.12 (Converse boundary). Gauge inequivalence is a difference at marked deformation strength. It does not by itself prove that the associated pretriangulated categories are not Morita equivalent. A Morita-level separation requires an independent Morita invariant or a no-equivalence theorem.

26.3 Infinitesimal classes and recursive obstructions

Theorem 26.13 (First-order classification). *Let $R = k[\varepsilon]/(\varepsilon^2)$ and $\alpha = \varepsilon\phi$. Then α is Maurer–Cartan if and only if*

$$d_b \phi = 0.$$

Two such first-order perturbations are gauge equivalent if and only if their difference is a d_b -coboundary. Hence the tangent space to the deformation functor is

$$H^1(\mathfrak{g}_{\text{Hoch}}(b)) \cong \text{HH}^2(E, b).$$

Proof. Modulo ε^2 , the quadratic bracket term vanishes, so theorem 26.8 reduces to $d_b \phi = 0$. An infinitesimal degree-zero gauge parameter changes ϕ by $d_b \xi$. $\square$

Definition 26.14 (Order- N deformation). An order- N deformation over $k[t]/(t^{N+1})$ is

$$\alpha^{(N)} = \sum_{r=1}^N t^r \alpha_r$$

satisfying the Maurer–Cartan equation modulo t^{N+1} .

Theorem 26.15 (Recursive obstruction equation). *Suppose $\alpha^{(N-1)}$ is an order- $(N-1)$ deformation. Define*

$$\Omega_N := \frac{1}{2} \sum_{i+j=N} [\alpha_i, \alpha_j] \in \mathfrak{g}_{\text{Hoch}}(b)^2.$$

Then:

1. $d_b \Omega_N = 0$;
2. *an extension to order N exists if and only if*

$$[\Omega_N] = 0 \in H^2(\mathfrak{g}_{\text{Hoch}}(b));$$

3. *when the class vanishes, α_N may be chosen to satisfy*

$$d_b \alpha_N = -\Omega_N.$$

Under the unshifted convention, the obstruction lies in HH^3 .

Proof. The coefficient of t^N in the Maurer–Cartan equation is

$$d_b \alpha_N + \Omega_N.$$

The dg-Lie Jacobi identity and the lower-order Maurer–Cartan equations imply $d_b \Omega_N = 0$. The coefficient can be killed exactly when Ω_N is a coboundary. $\square$

Corollary 26.16 (Formal rigidity). *If*

$$H^1(\mathfrak{g}_{\text{Hoch}}(b)) = 0,$$

then every formal deformation in the declared complete pronilpotent regime is gauge equivalent to the trivial deformation. This is formal rigidity at the marked deformation strength.

Proof. Kill the first nonzero coefficient inductively by a degree-zero gauge parameter. Completeness permits passage to the inverse limit. $\square$

Corollary 26.17 (Unobstructed extension). *If*

$$H^2(\mathfrak{g}_{\text{Hoch}}(b)) = 0,$$

then every first-order deformation extends inductively to a formal deformation. The extension need not be unique; ambiguity is controlled by lower cohomology and gauge.

Warning 26.18 (No single-group overreading). The statements above do not imply:

1. $H^2 = 0$ gives uniqueness;
2. $H^1 \neq 0$ gives an actual nontrivial formal deformation;
3. one vanished obstruction eliminates all future obstructions;
4. a formal deformation converges analytically.

26.4 A finite obstructed dg-Lie calibration

Definition 26.19 (Toy obstruction controller). Let $\mathfrak{l}$ be the nilpotent dg Lie algebra with $d = 0$, degree-one basis u, v , degree-two basis w , and only nonzero bracket

$$[u, v] = [v, u] = w.$$

The symmetry is the graded symmetry appropriate to degree-one elements.

Proposition 26.20 (Obstructed and unobstructed directions). *For the first-order direction $\phi = u + v$,*

$$\frac{1}{2}[\phi, \phi] = w \neq 0,$$

so the direction is obstructed at second order. For $\phi = u$,

$$\frac{1}{2}[\phi, \phi] = 0,$$

so no second-order obstruction appears.

Remark 26.21 (Calibration status). This finite dg-Lie algebra is a proof-audit calibration for the recursive formula. It is not asserted to be the complete Hochschild controller of the PE-6 source.

26.5 The PE-6 cubic direction as an exact formal deformation

Definition 26.22 (Sparse cubic cochain). In the PE-6 marked minimal model, let a, b, c be the degree-one generators and e the degree-two relation generator. Define the reduced ternary cochain μ by

$$\mu(a, b, c) = e$$

and zero on every other typed input.

Proposition 26.23 (Cocycle and square-zero properties). *Let b_0 be the formal truncation retaining the fixed units, idempotents, graded space, and binary operation but no positive-arity higher operation. Then*

$$[b_0, \mu] = 0, \quad [\mu, \mu] = 0.$$

Proof. The first equality is the source-bound A_∞ identity linear in the unique ternary component. In the frozen sparse table, every reduced binary product on the relevant adjacent positive-degree generators vanishes. For the second equality, every nested nonzero inner μ outputs e , while μ is zero whenever any input is e . Thus every arity-five insertion vanishes individually. $\square$

Theorem 26.24 (Polynomial universal cubic family and formal completion). *Let T be an indeterminate. The finite operation table*

$$b_{\text{univ}} = b_0 + T\mu$$

defines an A_∞ structure over the polynomial ring $k[T]$. Its T -adic completion is the formal Maurer–Cartan family

$$\hat{b}_T = b_0 + T\mu \quad \text{over } k[[T]].$$

No higher correction terms are required in either regime.

Proof. By theorem 26.23, the Stasheff/Maurer–Cartan curvature is the polynomial

$$T[b_0, \mu] + \frac{T^2}{2}[\mu, \mu] = 0.$$

Hence all identities hold in $k[T]$. Base change along $k[T] \rightarrow k[[T]]$ gives the formal completion. The completion statement is local at $T = 0$ and is not used to define nonzero scalar fibers. $\square$

Corollary 26.25 (Finite heart-level polynomial relation wall). *For each $\lambda \in k$, base change along the genuine evaluation homomorphism*

$$k[[T]] \longrightarrow k, \quad T \longmapsto \lambda$$

defines a marked nonshifted simple-filtered heart $\mathcal{H}_\lambda$. For an object with maps

$$V_x \xleftarrow{V_a} V_y \xleftarrow{V_b} V_z \xleftarrow{V_c} V_t,$$

the object equation is

$$\boxed{\lambda V_a V_b V_c = 0.}$$

Thus:

1. *at $\lambda = 0$, every representation of the linearly oriented A_4 quiver is accepted;*
2. *at $\lambda \neq 0$, the accepted objects are exactly the modules over*

$$kQ/\langle \gamma\beta\alpha \rangle.$$

The $k[[T]]$ family is only the formal completion of this polynomial family at 0; no map $k[[T]] \rightarrow k$ with $T \mapsto \lambda \neq 0$ is invoked.

Proposition 26.26 (Coefficient orbit). *Under the marked diagonal changes of PE-7, all nonzero coefficients λ belong to one marked isomorphism orbit, while $\lambda = 0$ is a separate orbit. Hence the heart-level family has two coordinate-independent strata:*

$$\{0\}, \quad k^\times.$$

Warning 26.27 (Specialization is not an oracle). The polynomial family is an A_∞ operation family on a fixed graded model, and the formal family is only its completion at $T = 0$. PE-8 does not identify the ambient twisted category of the zero fiber with the standard derived category of the path algebra without a separate enhancement theorem. The statement below compares the standard bounded derived categories of the two marked hearts, which is exactly the declared heart-strength target.

26.6 Derived-Morita invariance of deformation moduli

Theorem 26.28 (Hochschild B_∞ invariance). *Let A and B be dg algebras or small dg categories related by an enhanced derived equivalence induced by a derived tensor bimodule. Then their Hochschild cochain complexes represent the same object in the homotopy category of B_∞ algebras. This source theorem supplies a higher-structure equivalence or zigzag; it does not, by itself, choose one strict map between the particular completed filtered controllers used by a PE certificate.*

Definition 26.29 (Filtered controller transport certificate). A filtered controller transport certificate from A to B is a finite zigzag

$$\mathfrak{g}_{\text{Hoch}}(A) = \mathfrak{h}_0 \xleftarrow{\simeq} \mathfrak{h}_1 \xrightarrow{\simeq} \cdots \xleftarrow{\simeq} \mathfrak{h}_{r-1} \xrightarrow{\simeq} \mathfrak{h}_r = \mathfrak{g}_{\text{Hoch}}(B)$$

of complete descendingly filtered L_∞ algebras and filtration-preserving L_∞ quasi-isomorphisms. The record identifies the arity or deformation-ideal filtrations, proves completeness and pronilpotence after tensoring with $\mathfrak{m}_R$, and binds the zigzag to the B_∞ Morita equivalence of theorem 26.28. Compatibility with the chosen completion is a separate non-scalar obligation.

Theorem 26.30 (Morita-stable Maurer–Cartan descent). *Given a filtered controller transport certificate satisfying the hypotheses of the filtered Goldman–Millson theorem at every edge, there is a zigzag of weak equivalences*

$$\boxed{\text{MC}_\bullet(\mathfrak{g}_{\text{Hoch}}(A) \widehat{\otimes} \mathfrak{m}_R) \simeq \text{MC}_\bullet(\mathfrak{g}_{\text{Hoch}}(B) \widehat{\otimes} \mathfrak{m}_R).}$$

Hence the formal deformation ∞ -groupoid is an enhanced-Morita Return invariant relative to the certified filtered zigzag.

Proof. Apply the filtered Goldman–Millson theorem to each filtration-compatible L_∞ quasi-isomorphism in the certified zigzag. Composition in the homotopy category of Kan complexes yields the displayed weak equivalence. The proof does not replace the zigzag by an unsupported strict controller map. $\square$

Corollary 26.31 (Tangent and obstruction transport). *The induced maps identify the tangent and obstruction spaces*

$$H^1(\mathfrak{g}_{\text{Hoch}}(A)) \cong H^1(\mathfrak{g}_{\text{Hoch}}(B)), \quad H^2(\mathfrak{g}_{\text{Hoch}}(A)) \cong H^2(\mathfrak{g}_{\text{Hoch}}(B)),$$

together with the higher brackets and obstruction maps carried by the transferred L_∞ structure.

Warning 26.32 (Cohomology groups alone are insufficient). An isomorphism

$$\text{HH}^*(A) \cong \text{HH}^*(B)$$

as graded vector spaces does not identify the Maurer–Cartan moduli problem. The bracket, braces, higher operations, filtration, and actual quasi-isomorphism are non-scalar obligations.

Warning 26.33 (Heart descent remains conditional). Morita-stable deformation Return does not select a standard t -structure. To descend a deformation to a standard module heart or literal quiver presentation, PE-7 still requires a transported-heart record, t -exactness when standard-heart equality is claimed, marked simples or projectives, and presentation coordinates.

26.7 Typed deformation certificate and verdict calculus

Definition 26.34 (PE-8 deformation certificate). A replayable certificate contains:

1. the marked source enhancement and immutable artifact identity;
2. the coefficient field and characteristic policy;
3. the local Artin or complete base $(R, \mathfrak{m}_R)$;
4. the complete filtration and convergence proof;
5. the normalized Hochschild cochain model and sign convention;
6. the baseline Maurer–Cartan element b ;
7. every perturbation coefficient α_r through the claimed order;
8. the computed curvature residual

$$d_b \alpha + \frac{1}{2}[\alpha, \alpha];$$

9. the gauge or Maurer–Cartan path artifact when equivalence is claimed;
10. the tangent and obstruction cocycles, their cohomology representatives, and any null-homotopy;
11. the enhanced-Morita comparison and filtered L_∞ transport when descent is invoked;
12. the heart, t -structure, and presentation gates when a stronger Return is requested;
13. the exact primary Return status and deformation-lifecycle status.

Classification 26.35 (Deformation lifecycle). The deformation-specific lifecycle is one of:

1. `not_invoked`;
2. `undefined`;
3. `rejected_curvature`;
4. `first_order_class`;
5. `gauge_trivial`;

6. `extended_to_order_N`;
7. `obstructed_at_order_N`;
8. `formally_integrated`;
9. `morita_descended`;
10. `heart_descended`;
11. `presentation_descended`.

These are not replacements for the AK Return Theory primary statuses `UNDEFINED`, `REJECT`, `OBSTRUCTED`, `PARTIAL-RETURN`, and `RETURN`. The two axes are recorded separately.

Protocol 26.36 (Status precedence). For a fixed requested deformation strength:

1. a false source, filtration, or curvature assertion gives `REJECT`;
2. a missing base, completion, controller, or comparison artifact gives `UNDEFINED`;
3. a valid nonzero obstruction class gives `OBSTRUCTED`;
4. a weaker certified order or weaker Return strength gives `PARTIAL-RETURN`;
5. only complete discharge at the declared strength gives `RETURN`.

A missing gauge path is not a nonzero obstruction. A nonzero obstruction is not a failed source binding.

26.8 Compatibility with PE-1–PE-7 and frozen v20

Proposition 26.37 (Compatibility chain). *PE-8 composes with earlier PE layers as follows:*

1. *PE-1 zero-reflection may be evaluated separately on each specialized persistence profile;*
2. *PE-2 and PE-3 object/morphism signatures are decategorified shadows of a specialized heart and do not reconstruct the deformation controller;*
3. *PE-4 exact-structure Return may be consumed only after the relevant heart has been descended;*
4. *PE-5 cone Return is available on a fixed enhanced specialization;*
5. *PE-6 supplies the concrete nonzero m_3 direction;*
6. *PE-7 supplies coordinate-independent marked and Morita strength separation;*
7. *PE-8 governs variation, obstruction, and moduli descent between such enhanced structures.*

Warning 26.38 (No threshold-exactness import). Nothing in Hochschild deformation theory makes hard persistence threshold deletion exact, globally functorial, or compatible with a deformed heart. Any use of the v20 repaired evaluation still obeys window-first order, threshold-gap transport, detector completeness, and typed realization.

Warning 26.39 (No finite-sample integration). Checking finitely many coefficients, parameter values, or finite-field specializations does not prove a formal or analytic deformation exists. Formal integration is certified by the recursive Maurer–Cartan equations in the complete base.

26.9 Adversarial reaudit

Warning 26.40 (Audit A: is every cocycle a deformation?). No. A cocycle is only a first-order direction. Its square and later brackets may create nonzero obstruction classes.

Warning 26.41 (Audit B: is every exact cocycle globally trivial?). Exactness kills the first-order term. Global triviality requires an inductive gauge construction and completeness. The certificate may not skip later coefficients.

Warning 26.42 (Audit C: does gauge inequivalence prove categorical inequivalence?). No. Gauge classes refine a fixed marked deformation problem. Distinct gauge classes can still become equivalent after forgetting to Morita strength.

Warning 26.43 (Audit D: does Morita invariance preserve the standard heart?). No. The deformation ∞ -groupoid of the enhancement is Morita invariant, but a standard heart requires a separate t -structure and marking descent.

Warning 26.44 (Audit E: can one use HH^2 and HH^3 dimensions only?). No. Tangent vectors, obstruction maps, brackets, stabilizers, and higher homotopies are lost by dimension data.

Warning 26.45 (Audit F: is the cubic coefficient itself invariant?). No. PE-7 proves that every nonzero coefficient is diagonally normalizable. The invariant information is the marked orbit and the resulting Maurer–Cartan semantics.

Warning 26.46 (Audit G: is the Coxeter calculation overused?). The calculation distinguishes the standard derived categories of the two returned hearts. It is not promoted to a comparison of every ambient fixed-graded-model twisted category.

Warning 26.47 (Audit H: does a formal family converge analytically?). No. PE-8 proves formal deformation statements over complete algebraic bases. Analytic convergence requires separate estimates and a different Return theorem.

Warning 26.48 (Audit I: does finite-field replay prove characteristic-zero gauge theory?). No. The finite-field script checks sparse operation identities, matrix wall counts, and integer Coxeter calculations. It does not validate exponential gauge or Goldman–Millson hypotheses.

27 PE-9: Global Maurer–Cartan Atlas and Wall-Crossing Return

Classification 27.1 (Normative global responsibility). This chapter supersedes the preliminary descent discussion of PE-7 and is the unique normative location for global atlas descent, monodromy, local-system versus constant Return, and strength-relative wall crossing.

27.1 Local charts and the global atlas object

Definition 27.2 (Finite deformation atlas). A finite deformation atlas is a tuple

$$\mathfrak{A} = (K, \{\mathfrak{g}_\sigma\}_{\sigma \in K}, r, \eta, \mathcal{A}),$$

where:

1. K is a finite simplicial complex or finite Čech nerve;
2. each nonempty simplex σ carries a complete filtered L_∞ algebra $\mathfrak{g}_\sigma$;
3. each face inclusion supplies an actual filtered L_∞ restriction morphism r with declared variance;
4. η contains the higher homotopies making the diagram cosimplicial or homotopy coherent;
5. $\mathcal{A}$ contains source identities, filtration data, totalization choices, and immutable artifacts.

A collection of quasi-isomorphism classes without actual transition morphisms is not an atlas.

Definition 27.3 (Global controller). For a cosimplicial complete filtered controller $\mathfrak{g}^\bullet$ associated to $\mathfrak{A}$, define the global controller at the consumed source strength by the Thom–Whitney or source-specified totalization

$$\mathfrak{g}_{\text{glob}} := \text{Tot}_{\text{TW}}(\mathfrak{g}^\bullet).$$

The notation does not assert that an arbitrary strict inverse limit computes the required derived global sections.

Theorem 27.4 (Source-bound global Maurer–Cartan descent). *Under the hypotheses of the consumed Hinich–Getzler–Bandiera descent theorem, there are natural weak equivalences of Kan complexes*

$$\text{MC}_\bullet(\mathfrak{g}_{\text{glob}}) \simeq \text{Tot}(\text{MC}_\bullet(\mathfrak{g}^\bullet)) \simeq \text{holim}_{[p] \in \Delta} \text{MC}_\bullet(\mathfrak{g}^p).$$

Consequently, the global formal deformation problem is determined by the homotopy-coherent atlas, not by its vertex set alone.

Proof. This is the exact permitted source statement: Hinich proves descent of Deligne groupoids through totalization in the dg-Lie regime, Getzler supplies the higher Maurer–Cartan Kan complex for nilpotent L_∞ algebras, and Bandiera proves compatibility with totalization and homotopy limits. PE-9 consumes the theorem only under the declared completeness and coherence hypotheses. $\square$

Corollary 27.5 (Hierarchy of gluing data). *A point of the right-hand homotopy limit consists, schematically, of:*

1. a local Maurer–Cartan element α_i on each vertex chart;
2. a gauge or 1-simplex comparing restrictions on each double overlap;
3. a 2-simplex witnessing compatibility of the two composite comparisons on each triple overlap;
4. higher simplices satisfying all higher overlap coherences.

Suppressing any invoked level changes the target from a homotopy limit to an unsupported truncation.

Warning 27.6 (Pairwise agreement is not descent). Even if every pair of overlapping charts is related by a filtered quasi-isomorphism, a global object does not follow until the triple and higher compatibility data required by theorem 27.5 are supplied. Pairwise equivalence is objectwise information; descent is coherent diagram information.

27.2 Strict graph atlases and monodromy

Definition 27.7 (Strict marked graph atlas). Let $\Gamma = (V, E)$ be a finite connected graph and G a group of marked coordinate automorphisms. A strict graph atlas assigns to each oriented edge $u \rightarrow v$ an element

$$g_{uv} \in G, \quad g_{vu} = g_{uv}^{-1}.$$

A global marked trivialization is a family $h_v \in G$ such that

$$\boxed{g_{uv} = h_v h_u^{-1}}$$

for every oriented edge.

Definition 27.8 (Cycle holonomy). For an oriented closed edge path

$$\gamma = (v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_m = v_0),$$

define

$$\text{Mon}_\gamma(g) := g_{v_{m-1}v_m} \cdots g_{v_0v_1} \in G.$$

For nonabelian G , changing the root conjugates the holonomy. The identity/nonidentity verdict is root-independent.

Theorem 27.9 (Spanning-tree descent criterion). *A strict marked graph atlas admits a global marked trivialization if and only if*

$$\text{Mon}_\gamma(g) = 1_G$$

for every cycle γ . Equivalently, it suffices to check the fundamental cycles relative to one spanning tree.

Proof. If $g_{uv} = h_v h_u^{-1}$, the factors telescope around every cycle. Conversely, choose a root and a spanning tree. Define h_v by multiplying transitions along the unique tree path from the root to v . Every tree edge then has the required form. A non-tree edge closes one fundamental cycle, and trivial holonomy is exactly the equation $g_{uv} = h_v h_u^{-1}$ for that edge. $\square$

Corollary 27.10 (Tree atlas closure). *If Γ is a tree, every strict edge assignment admits a global marked trivialization. The trivialization is unique up to simultaneous right multiplication of all h_v by one element of G .*

Proof. Existence follows from theorem 27.9. If h and h' are two trivializations and r is a root, put $c = h_r^{-1} h'_r$. Induction along the unique tree paths gives $h'_v = h_v c$ for every vertex v . Conversely, common right multiplication preserves each transition because $(h_v c)(h_u c)^{-1} = h_v h_u^{-1}$. $\square$

Proposition 27.11 (Triangle overlap defect). *For an abelian marked transition group, three pairwise transitions g_{01}, g_{12}, g_{02} on a triangle have defect*

$$c_{012} := g_{02}^{-1} g_{12} g_{01}.$$

They arise from vertex markings if and only if $c_{012} = 1$. Thus the first nontrivial strict overlap defect already occurs on a triple overlap.

Classification 27.12 (Monodromy status discipline). At a requested globally constant marked strength:

1. a missing transition, path composite, or loop record gives UNDEFINED;
2. a false supplied cocycle equation gives REJECT;
3. an actual valid transition system with nonidentity holonomy gives OBSTRUCTED;
4. trivial holonomy plus all remaining obligations gives RETURN.

At local-system or homotopy-limit strength, nontrivial monodromy may be part of a successful Return object rather than an obstruction.

Warning 27.13 (Type IV firewall). Gauge monodromy is not automatically the frozen v20 Type IV defect. It becomes a PE-9 descent obstruction only after the atlas, loop, requested constant target, and actual comparison data exist. Missing coherence is UNDEFINED, not nonzero obstruction.

27.3 Homotopy-coherent monodromy and Return strengths

Definition 27.14 (Atlas Return strengths). PE-9 inherits the standing characteristic-zero complete filtered regime of theorem 26.2.

PE-9 uses the following partially ordered strengths:

1. Σ_{local} : independently certified local charts;
2. Σ_{holim} : one global homotopy-limit Maurer–Cartan ∞ -groupoid;
3. Σ_{locsys} : a Return local system or stack with recorded monodromy;
4. Σ_{mark} : a fixed marked A_∞ quasi-isomorphism class, inherited from PE-7;
5. Σ_{const} : a globally constant marked target with coherently trivialized monodromy;
6. Σ_{heart} : a coherently transported heart and exact structure;
7. $\Sigma_{\text{std-derived}}$: identification with one fixed standard heart up to bounded-derived equivalence;
8. Σ_{pres} : a fixed literal presentation and coordinate system.

The forgetful arrows are declared individually. In particular, Σ_{locsys} does not imply Σ_{const} .

Proposition 27.15 (Replacement invariance). *If every local controller in an atlas is replaced by a filtered L_∞ quasi-isomorphic controller and the replacement maps form a coherent natural transformation of atlas diagrams, then the global Maurer–Cartan homotopy-limit Return is weakly equivalent to the original one.*

Proof. Filtered Goldman–Millson invariance gives a weak equivalence on every local Maurer–Cartan Kan complex. Coherent objectwise weak equivalences induce an equivalence of the corresponding homotopy limits under the source theorem’s hypotheses. An incoherent list of replacements does not supply this conclusion. $\square$

Definition 27.16 (Homotopy monodromy). For a loop in the atlas nerve based at a chart i , compose the actual overlap equivalences and their coherence cells to obtain an automorphism of the returned fiber in the relevant homotopy category or ∞ -groupoid. Its equivalence class is the homotopy monodromy

$$\text{Mon}_\gamma^\Sigma \in \pi_0 \text{Aut}_\Sigma(X_i).$$

A stronger marked certificate may retain an actual representative and higher homotopies.

Theorem 27.17 (Constant descent versus local-system descent). *Assume Σ_{locsys} has been certified. A promotion to Σ_{const} requires a coherent trivialization of the monodromy representation. Nontrivial monodromy blocks constant marked descent but does not invalidate the local-system Return.*

Proof. A constant marked object has identity transport around every loop. Conversely, a coherent monodromy trivialization identifies all path transports with the identity relative to a chosen base fiber. The weaker local system retains the automorphisms instead of annihilating them. $\square$

27.4 Strength-relative walls and path Return

Definition 27.18 (Wall site and certified local triviality). A wall datum includes a parameter object P together with a declared topology or Grothendieck site τ_P . The word “neighborhood” always means an open neighborhood in τ_P . Let $\mathcal{X}_p$ be the Return object at a point or geometric point p when defined. An open $U \ni p$ is Σ -trivial if there is a replayable atlas over U and a comparison with one fixed target X_U that is a valid equivalence at strength Σ on every relevant fiber, with coherent overlap data. Changing τ_P , the class of points, the target, or the equivalence relation changes the wall datum and requires a new certificate.

Definition 27.19 (Strength-relative discriminant). For a fixed wall datum (P, τ_P, Σ) , define

$$\Delta_\Sigma(P, \tau_P) := \{p \in P : \text{no certified } \Sigma\text{-trivial } \tau_P\text{-neighborhood of } p \text{ exists}\}.$$

A wall is therefore relative to the site, target, equivalence relation, and certificate strength. Once this datum is fixed, we abbreviate $\Delta_\Sigma(P, \tau_P)$ to Δ_Σ . The same parameter may be regular for Σ_{holim} and singular for $\Sigma_{\text{std-derived}}$.

Definition 27.20 (Path Return). Path Return is invoked only when P additionally carries a declared topological-space structure and $\gamma : [0, 1] \rightarrow P$ is continuous for that topology. A Σ -path Return is a chain of local atlas comparisons covering γ together with all overlap coherences. Its output is a transport equivalence

$$T_\gamma^\Sigma : \mathcal{X}_{\gamma(0)} \longrightarrow \mathcal{X}_{\gamma(1)}$$

at the declared strength, or a typed downgrade when the path meets the wall pulled back to the declared topology.

Theorem 27.21 (Path transport away from the wall). *If the image of γ lies in one connected component of $P \setminus \Delta_\Sigma$ and the covering atlas has certified homotopy coherence, then T_γ^Σ exists. For a closed path, it returns the monodromy automorphism. Independence from the chosen path requires a separately certified trivial monodromy or simple-connectivity argument at strength Σ .*

Proof. Cover the compact image of γ by finitely many Σ -trivial neighborhoods and compose their overlap equivalences. Homotopy coherence identifies refinements and parenthesizations. Closed paths retain the resulting loop automorphism unless it is separately trivialized. $\square$

Warning 27.22 (Crossing is not continuation). If a path meets Δ_Σ , the family may still Return at a weaker strength. The correct outcome is not automatic failure but a strength comparison: RETURN at the weaker family or homotopy-limit level and PARTIAL-RETURN, UNDEFINED, or REJECT at the stronger constant target, depending on which artifacts exist and which assertions are false.

27.5 PE-8 cubic wall-crossing calibration

Definition 27.23 (Cubic parameter family). Retain the PE-6/PE-8 marked graded model with

$$\mu(a, b, c) = e, \quad b_\lambda = b_0 + \lambda\mu, \quad \lambda \in k.$$

The Maurer–Cartan calculation of PE-8 proves that this is the fiber family of the polynomial universal structure $b_{\text{univ}} = b_0 + T\mu$ over $k[T]$. For the algebraic wall theorem below, the wall datum is $P = \mathbb{A}_k^1$ with its Zariski topology and geometric fibers after base change to an algebraic closure. Real and complex path statements use the Euclidean topology separately.

Theorem 27.24 (Nonzero-stratum trivialization). *On the principal Zariski open $\mathbb{G}_m = D(T) \subset \mathbb{A}_k^1$, the universal diagonal rescaling over $k[T, T^{-1}]$ identifies every geometric fiber b_λ with b_1 at marked A_∞ strength. Hence*

$$\Delta_{\Sigma_{\text{mark}}}(\mathbb{A}_k^1, \text{Zar}) \cap \mathbb{G}_m = \emptyset.$$

The restricted family over $\mathbb{G}_m$ admits a constant marked Return after recording this algebraic normalization.

Proof. PE-7 proves that a nonzero cubic coefficient can be normalized to one by rescaling the output basis vector, with the corresponding inverse transformation on twisting coefficients. The construction is defined for every $\lambda \neq 0$. $\square$

Theorem 27.25 (Zero is a marked and standard-heart wall). *The zero model and the nonzero cubic models belong to distinct marked A_∞ quasi-isomorphism classes. Moreover, the returned standard hearts are*

$$A_0 = kQ, \quad A_1 = kQ / \langle \gamma\beta\alpha \rangle,$$

with Coxeter polynomials

$$\chi_{A_0}(z) = z^4 + z^3 + z^2 + z + 1, \quad \chi_{A_1}(z) = (z + 1)^2(z^2 - z + 1).$$

Consequently, for the Zariski geometric-fiber wall datum,

$$\Delta_{\Sigma_{\text{mark}}}(\mathbb{A}_k^1, \text{Zar}) = \Delta_{\Sigma_{\text{std--derived}}}(\mathbb{A}_k^1, \text{Zar}) = V(T) = \{0\}.$$

Here $\{0\}$ denotes the closed origin of $\mathbb{A}_k^1$.

Proof. The zero specialization has no higher multiplication beyond m_2 and is therefore formal. Every nonzero geometric fiber is the source-bound nonnormal Keller model after diagonal normalization; its nonzero m_3 is indispensable to the relation reconstruction, as recorded in PE-6. Formality is invariant under marked A_∞ quasi-isomorphism, so the zero and nonzero fibers cannot lie in the same marked quasi-isomorphism class. Separately, PE-8 computes the two Coxeter polynomials, and derived equivalence in the finite-dimensional finite-global-dimension regime preserves that polynomial. Every Zariski neighborhood of the origin contains nonzero geometric points, so no such neighborhood is trivial at either fixed-target strength. The open complement $D(T)$ is trivial by theorem 27.24; therefore the wall is exactly $V(T)$. $\square$

Proposition 27.26 (Family-defined locus is not a wall statement). *The polynomial Maurer–Cartan formula $b_{\text{univ}} = b_0 + T\mu$ is defined on all of $\mathbb{A}_k^1$. Hence its family-defined locus is*

$$U_{\text{family}} = \mathbb{A}_k^1.$$

This proposition records existence of the varying family only. It does not assert local triviality against a fixed fiber, does not define a new Σ_{family} , and does not imply an empty wall under theorem 27.19.

Corollary 27.27 (Typed wall-crossing verdict). *For $k = \mathbb{R}$ or $\mathbb{C}$, equip the set of k -points with the Euclidean topology and pull back the algebraic wall $V(T)$. Any continuous path contained in the punctured stratum $\mathbb{G}_m(k)$ transports the nonzero marked model. Over $\mathbb{C}$, two nonzero coefficients may be joined inside $\mathbb{G}_m(\mathbb{C})$; over $\mathbb{R}$, a continuous path from a negative coefficient to a positive coefficient necessarily crosses zero. Any path from zero to a nonzero parameter crosses the marked and standard-derived-heart wall. The strongest globally defined output on $\mathbb{A}_k^1$ is the varying polynomial family, not one constant standard heart.*

Warning 27.28 (Ambient Morita boundary). The Coxeter calculation separates the bounded derived categories of the two returned standard hearts. PE-9 does not silently upgrade this to a theorem that every ambient twisted category associated to the underlying graded model changes Morita class.

27.6 Global certificate and status algebra

Definition 27.29 (PE-9 global atlas certificate). A PE-9 certificate records:

1. the atlas nerve, chart identities, filtration, and coefficient regime;
2. every local controller and Maurer–Cartan chart;
3. actual restriction or transition L_∞ morphisms;
4. overlap 1-simplices, triple 2-simplices, and all invoked higher coherence;
5. totalization or homotopy-limit construction and source theorem locator;
6. replacement quasi-isomorphisms and naturality cells when charts are changed;
7. a spanning tree, fundamental cycles, loop composites, and monodromy representatives;
8. the requested Return strength and target equivalence relation;

9. parameter strata, Δ_Σ , path records, and wall-crossing downgrade rules;
10. heart, t -structure, standard-heart, and presentation gates when invoked;
11. source bindings, verification artifacts, immutable hashes, and one primary status.

Protocol 27.30 (Replay order). The verifier executes:

1. validate all charts, filtrations, markings, and source identities;
2. validate actual transition maps and their variance;
3. replay pairwise, triple, and higher coherence in nerve order;
4. construct or consume the totalized global controller;
5. apply the source-bound Maurer–Cartan homotopy-limit theorem;
6. compute strict or homotopy monodromy on a declared generating loop set;
7. compare the obtained atlas strength with the requested constant, heart, or presentation strength;
8. for parameter paths, verify the strength-relative discriminant and every crossing;
9. emit the primary status and all weaker successful Return strengths.

Classification 27.31 (Primary statuses). At a fixed datum and requested strength:

1. REJECT: a supplied transition, coherence equation, wall claim, source binding, or equivalence assertion is false;
2. UNDEFINED: a required chart, overlap map, higher cell, totalization, loop record, or target gate is absent;
3. OBSTRUCTED: the complete comparison exists but a declared constant-descent monodromy or coherence defect is nontrivial;
4. PARTIAL - RETURN: a global homotopy-limit, local-system, or family Return is certified while a stronger constant, heart, or presentation Return is unavailable;
5. RETURN: every obligation at the requested strength is discharged.

28 PE-10: Derived Specialization and Regularized Boundary Return

Classification 28.1 (Normative boundary responsibility). This chapter is the unique normative location for derived specialization, Tor and lattice defects, Rees/associated-graded regularization, valuative contact order, and regularized boundary Return.

28.1 Integral families and the three fibers

Definition 28.2 (Integral boundary datum and coefficient regime). An integral boundary datum is

$$\mathfrak{B}_R = (R, t, K, \kappa, \iota, \mathcal{X}_R, \mathcal{A}),$$

where R is a discrete valuation ring with chosen uniformizer t , $K = \text{Frac}(R)$, $\kappa = R/(t)$, and ι is either a declared coefficient homomorphism from the source coefficient ring into R or an explicit R -linear integral lift of every marked graded object, operation, and relation consumed by the family. The object $\mathcal{X}_R$ is a bounded perfect complex, finite dg/A_∞ algebra, or finite twisted category over R , and $\mathcal{A}$ records markings, source identity, the coefficient-lift proof, filtration, and immutable artifacts. Completeness of R is required only when a later complete-local or formal-gauge theorem explicitly consumes it; the universal-coefficient theorem below needs only the DVR hypotheses.

Definition 28.3 (Generic and special fibers). For a perfect complex C over R , define

$$C_\eta := C \otimes_R K, \quad C_0^{\mathbf{L}} := C \otimes_R^{\mathbf{L}} \kappa.$$

If a selected representative is termwise flat, its ordinary special complex is

$$C_0^{\text{naive}} := C \otimes_R \kappa,$$

and represents $C_0^{\mathbf{L}}$. Equality of representatives does not identify $H^*(C) \otimes \kappa$ with $H^*(C_0^{\mathbf{L}})$.

Definition 28.4 (Exact and regularized boundary strengths). The principal boundary strengths are:

1. Σ_{generic} : Return the generic fiber only;
2. Σ_{pair} : Return the generic and special fibers as a pair;
3. Σ_{flat} : Return cohomology by flat base change with no boundary correction;
4. $\Sigma_{\text{regularized}}$: Return the special derived fiber together with all declared correction and contact data;
5. Σ_{integral} : Return the actual integral family up to the declared regular equivalence.

The forgetful arrows point from integral to regularized to pair to generic. No reverse arrow is automatic.

28.2 Derived specialization and the Tor boundary defect

Theorem 28.5 (DVR universal-coefficient boundary sequence). Let $\mathfrak{B}_R$ be an integral boundary datum and let $C \in D_{\text{perf}}^{\mathbf{b}}(R)$. For every integer n , there is a natural short exact sequence

$$0 \longrightarrow H^n(C) \otimes_R \kappa \longrightarrow H^n(C_0^{\mathbf{L}}) \longrightarrow \text{Tor}_1^R(H^{n+1}(C), \kappa) \longrightarrow 0.$$

The sequence need not split canonically.

Proof. The derived tensor product spectral sequence has

$$E_2^{-p,q} = \text{Tor}_p^R(H^q(C), \kappa) \implies H^{q-p}(C \otimes_R^{\mathbf{L}} \kappa).$$

Since a DVR has global dimension one for the residue module, only $p = 0, 1$ occur. The two surviving filtration pieces in total degree n give the displayed exact sequence. $\square$

Definition 28.6 (Tor boundary defect). Define

$$\mathbf{D}_{\text{Tor}}^n(C) := \text{Tor}_1^R(H^{n+1}(C), \kappa).$$

The family is cohomologically flat at the boundary when every $\mathbf{D}_{\text{Tor}}^n(C)$ vanishes and the displayed base-change maps are the consumed comparison artifacts.

Corollary 28.7 (Finite DVR flatness criterion). *Assume every $H^n(C)$ is a finite R -module. Then the following are equivalent:*

1. every $\mathbf{D}_{\text{Tor}}^n(C)$ vanishes;
2. every $H^n(C)$ is t -torsion-free;
3. every $H^n(C)$ is finite free over R ;
4. cohomology commutes with specialization in every degree:

$$H^n(C_0^{\mathbf{L}}) \cong H^n(C) \otimes_R \kappa.$$

Warning 28.8 (Termwise flat is not cohomologically flat). A bounded complex of finite free R -modules computes its derived special fiber by ordinary tensor product, but its cohomology may still contain t -torsion. The missing distinction is exactly $\mathbf{D}_{\text{Tor}}$.

Proposition 28.9 (Minimal boundary-jump calibration). *For $m \geq 1$, let*

$$C_m = [R \xrightarrow{t^m} R]$$

be placed in cohomological degrees 0 and 1. Then:

1. $(C_m)_\eta$ is acyclic;
2. $H^0(C_m) = 0$ and $H^1(C_m) = R/(t^m)$;
3. $(C_m)_0^{\mathbf{L}} = [\kappa \xrightarrow{0} \kappa]$, so

$$H^0((C_m)_0^{\mathbf{L}}) \cong \kappa, \quad H^1((C_m)_0^{\mathbf{L}}) \cong \kappa;$$
4. $\mathbf{D}_{\text{Tor}}^0(C_m) \cong \kappa$ and the sequence of theorem 28.5 is exact;
5. the generic fiber, special fiber, and dimensions of all $\mathbf{D}_{\text{Tor}}^n$ are independent of m .

Thus derived specialization detects a boundary jump but not its contact order.

Proof. Multiplication by t^m is invertible over K and zero modulo t . The cohomology over R is immediate. Finally,

$$\text{Tor}_1^R(R/(t^m), \kappa) \cong \kappa.$$

□

28.3 Integral Rees degeneration and associated-graded Return

Definition 28.10 (Split-projective filtered complex). Let A be a commutative ring. A split-projective finite filtered complex (C, F) over A is a bounded complex of finite projective A -modules with an increasing finite filtration by subcomplexes

$$0 = F_{p_0-1}C \subset F_{p_0}C \subset \cdots \subset F_{p_1}C = C$$

such that, in every cochain degree, each inclusion $F_{p-1}C^n \hookrightarrow F_pC^n$ is split and every graded quotient $F_pC^n / F_{p-1}C^n$ is finite projective. Extend the filtration to all $p \in \mathbb{Z}$ by

$$F_pC = 0 \quad (p < p_0), \quad F_pC = C \quad (p \geq p_1).$$

Definition 28.11 (Integral Rees complex). Define

$$\mathrm{Rees}_{F,A}(C) := \bigoplus_{p \in \mathbb{Z}} F_p C u^p \subset C[u, u^{-1}],$$

where every element has finite support. Multiplication by u is well defined because $F_p C \subseteq F_{p+1} C$. Thus $\mathrm{Rees}_{F,A}(C)$ is a graded complex of $A[u]$ -modules.

Theorem 28.12 (Three exact integral Rees fibers). *For a split-projective finite filtered complex (C, F) over A :*

1. *after inverting u ,*

$$\mathrm{Rees}_{F,A}(C)[u^{-1}] \cong C \otimes_A A[u, u^{-1}];$$

2. *at $u = 1$,*

$$\mathrm{Rees}_{F,A}(C) \otimes_{A[u]} A[u]/(u-1) \cong C;$$

3. *at $u = 0$,*

$$\mathrm{Rees}_{F,A}(C) \otimes_{A[u]} A[u]/(u) \cong \mathrm{gr}_F C := \bigoplus_{p \in \mathbb{Z}} F_p C / F_{p-1} C;$$

4. *every cochain term of $\mathrm{Rees}_{F,A}(C)$ is finite projective, hence flat, over $A[u]$.*

Therefore ordinary and derived $u = 0, 1$ fibers agree at complex level.

Proof. Exhaustivity gives the localization isomorphism. Evaluation at $u = 1$ sends $x = \sum_p x_p u^p$ to $\sum_p x_p$. If the sum is zero, set

$$y_p = - \sum_{q \leq p} x_q.$$

Then $y_p \in F_p C$, only finitely many y_p are nonzero, and

$$x = (u-1) \sum_p y_p u^p.$$

Hence the evaluation kernel is $(u-1) \mathrm{Rees}_{F,A}(C)$. Modulo u , the degree- p quotient is $F_p C / F_{p-1} C$. Finally choose degreewise projective complements G_p^n with $F_p C^n = F_{p-1} C^n \oplus G_p^n$. Noncanonically,

$$\mathrm{Rees}_{F,A}(C)^n \cong \bigoplus_p G_p^n u^p A[u],$$

a finite projective $A[u]$ -module. The differential preserves the filtration, so all identifications are isomorphisms of complexes. $\square$

Theorem 28.13 (DVR-to-Rees base bridge). *Let $(R, t, K, \kappa, \iota, C_R, \mathcal{A})$ be an integral boundary datum whose perfect complex representative C_R carries a split-projective finite filtration F . For $s \in \{K, \kappa\}$, let F_s be the base-changed filtration on $C_s = C_R \otimes_R s$. Then there are canonical isomorphisms*

$$\mathrm{Rees}_{F,R}(C_R) \otimes_R s \cong \mathrm{Rees}_{F_s,s}(C_s), \quad \mathrm{gr}_F C_R \otimes_R s \cong \mathrm{gr}_{F_s} C_s.$$

Moreover the t -base changes $R \rightarrow K, \kappa$ commute with the u -base changes $u \mapsto 1, 0$ in the evident base-change squares. Thus the DVR specialization coordinate t and the filtration-degeneration coordinate u are distinct but compatibly recorded.

Proof. The filtration steps and their quotients are finite projective, so ordinary and derived base change agree and commute with the finite direct-sum decompositions used in the Rees complex. The displayed isomorphisms follow coefficientwise. Associativity of tensor product gives the commuting t/u base-change squares. $\square$

Definition 28.14 (Integral Rees-regularized boundary). For an invoked filtered integral representative (C_R, F) , the Rees record is

$$(\mathrm{gr}_F C_R, \mathrm{Rees}_{F,R}(C_R), \text{the three } u\text{-fiber maps, and the } t/u \text{ base-change squares}).$$

It Returns the R -linear associated graded and a filtration-degeneration family over $R[u]$. It does not identify the u -degeneration with the t -adic specialization and does not assert $\mathrm{gr}_F C_R \cong C_R$.

Warning 28.15 (Associated graded is not inverse Return). Different filtered complexes can have isomorphic associated graded complexes. The integral Rees object with its $R[u]$ -module structure is stronger than the zero fiber alone. Forgetting the Rees lattice discards extension data between filtration levels.

28.4 Gauge-class contact data and the sparse marked normal form

Warning 28.16 (Raw cochain order is not gauge invariant). For a general regular Maurer–Cartan arc

$$\alpha(t) = \sum_{j \geq 1} t^j \alpha_j,$$

a gauge parameter of lower order may insert a nonzero exact coefficient before the first nonzero cohomology class. Therefore the smallest j with $\alpha_j \neq 0$ is not, by itself, a gauge-class invariant and is not used as a general PE-10 theorem.

Definition 28.17 (Certified normal-form contact datum). A certified normal-form contact datum for a Maurer–Cartan gauge class consists of:

1. a declared gauge slice, associated-graded Deligne-groupoid model, or other normalization theorem;
2. a representative $\alpha^{\mathrm{nf}}(t)$ in that normal form;
3. a proof that any two accepted normal-form representatives have the same first nonzero filtration degree and initial cohomology orbit;
4. the resulting invariants

$$\mathrm{ord}_t^{\mathrm{nf}}[\alpha], \quad \mathrm{in}^{\mathrm{nf}}[\alpha].$$

Without this certificate, contact order is UNDEFINED at general gauge strength rather than inferred from a raw series.

Proposition 28.18 (Generic Return does not determine certified contact). *Even when two integral families become equivalent after extending scalars to $k((t))$, their integral lattices or certified normal forms may carry different contact data. Thus generic equivalence alone does not imply integral or regularized-contact Return.*

28.5 The cubic integral degeneration

Definition 28.19 (Cubic DVR family). Assume the integral boundary datum contains an R -linear marked lift $(b_{0,R}, \mu_R)$ of the PE-6 cubic data, obtained either by scalar extension through ι or by a separately certified integral model. To lighten notation write $b_0 = b_{0,R}$ and $\mu = \mu_R$. Let μ satisfy

$$\mu(a, b, c) = e, \quad [b_0, \mu] = 0, \quad [\mu, \mu] = 0.$$

For $m \geq 1$ and $u(t) \in R^\times$, define

$$b_{m,u} = b_0 + t^m u(t) \mu.$$

This is an actual A_∞ family because its Maurer–Cartan equation has no higher correction term.

Definition 28.20 (Sparse marked diagonal contact order). For the frozen one-dimensional cubic direction and its marked diagonal coordinate, define

$$\text{ord}_t^{\text{cub}}(b_{m,u}) = m, \quad \text{in}^{\text{cub}}(b_{m,u}) = [\mu].$$

The admissible integral diagonal marked transformations multiply the cubic coefficient by a unit of R and preserve the ordered idempotents and the one-dimensional cubic line.

Theorem 28.21 (Generic coincidence, special coincidence, and sparse integral separation). *For the declared sparse marked cubic family:*

1. every generic fiber $(b_{m,u})_K$ is marked A_∞ isomorphic to the normalized nonzero model b_1 ;
2. every special fiber $(b_{m,u})_\kappa$ equals b_0 ;
3. every admissible integral diagonal marked equivalence preserves $\text{ord}_t^{\text{cub}} = m$;
4. the unit $u(t)$ can be removed by an R -invertible diagonal marked rescaling, so m is the complete orbit invariant inside this frozen sparse family;
5. consequently $b_{m,u}$ and $b_{n,v}$ have equivalent generic fibers and identical special fibers but are not equivalent by an admissible integral diagonal marked transformation when $m \neq n$.

No claim is made here that m classifies arbitrary regular A_∞ gauge orbits outside this marked normal form.

Proof. Over K , $t^m u(t)$ is invertible, so the PE-7 diagonal coefficient normalization applies. Modulo t , the cubic term vanishes. An admissible integral diagonal marked transformation changes the coefficient $t^m u(t)$ only by multiplication by a unit of R , which preserves its t -adic valuation m . Conversely, for fixed m , multiplication of the output basis vector by the unit $u(t)$ is an integral marked automorphism and normalizes the coefficient to t^m . $\square$

Definition 28.22 (Normalized lattice defect). The generic normalization uses the K -basis vector

$$e' = t^m u(t)e.$$

Relative to the selected integral marked lattice Re , define

$$\boxed{D_{\text{lat}}(b_{m,u}) := Re / (t^m u(t)Re) \cong R / (t^m).}$$

Its length is

$$\text{length}_R D_{\text{lat}}(b_{m,u}) = m.$$

Proposition 28.23 (The lattice defect is the contact certificate). *The generic normalizing change of basis extends to an R -linear marked automorphism if and only if $D_{\text{lat}} = 0$. For $m \geq 1$, the normalization is defined only after inverting t , and the finite length of D_{lat} recovers the sparse marked diagonal contact order exactly.*

Corollary 28.24 (Pair Return is not regularized Return). *The arcs $b_{1,1}$ and $b_{2,1}$ have the same generic marked Return and the same special marked Return. They have different lattice defects and are not equivalent at integral or regularized-contact strength. Thus*

$$\Sigma_{\text{pair}} \not\Rightarrow \Sigma_{\text{regularized}}.$$

28.6 Coefficientwise categorical base change

Definition 28.25 (Finite twisted boundary category). Let $\mathcal{E}_R$ be a finite strictly unital R -linear A_∞ category whose graded morphism modules are finite free over R . Let $\text{tw}_{\text{fin}}(\mathcal{E}_R)$ denote the category of finite twisted objects with finite free coefficient modules and nilpotent twisting matrices.

Theorem 28.26 (Finite twisted base change). *For $s = K$ or $s = \kappa$, coefficientwise tensor product defines an A_∞ functor*

$$(-)_s : \mathrm{tw}_{\mathrm{fin}}(\mathcal{E}_R) \longrightarrow \mathrm{tw}_{\mathrm{fin}}(\mathcal{E}_R \otimes_R s)$$

which preserves shifts, finite sums, twisting equations, closed morphisms, and selected cones. For two finite twisted objects X, Y there is a natural identification of morphism complexes

$$\mathrm{hom}_{\mathrm{tw}(\mathcal{E}_R)}(X, Y) \otimes_R s \cong \mathrm{hom}_{\mathrm{tw}(\mathcal{E}_s)}(X_s, Y_s).$$

If the base-changed marked generators split-generate the target perfect category, then after idempotent completion the base-changed finite twisted category Returns that perfect fiber.

Proof. All structure maps and Maurer–Cartan expressions are finite sums of R -multilinear operations, so they commute with coefficientwise base change. Morphism complexes are finite free combinations of the source morphism modules. Shifts, sums, and selected cone matrices are defined by the same finite formulas. The final statement is the definition of split-generation followed by Karoubi completion. $\square$

Warning 28.27 (Heart descent remains separate). Theorem 28.26 Returns the finite twisted or perfect fiber. It does not identify a standard heart across the boundary. In the cubic calibration, PE-8 and PE-9 already prove that the zero and nonzero standard hearts have different Coxeter polynomials.

28.7 Regularized boundary Return

Definition 28.28 (Regularized boundary package). A PE-10 regularized boundary package is

$$\begin{aligned} \mathfrak{B}^{\mathrm{reg}} = & (\mathcal{X}_\eta, \mathcal{X}_0^{\mathrm{L}}, \mathrm{D}_{\mathrm{Tor}}, \mathrm{Rees}_{F,R}(C_R), \mathrm{gr}_F C_R, \mathrm{BC}_{t,u}, \\ & \mathrm{Contact}^{\mathrm{nf}}[\alpha] \text{ when certified, } \mathrm{Contact}^{\mathrm{cub}} \text{ in the sparse cubic calibration,} \\ & \mathrm{D}_{\mathrm{lat}}, \mathrm{D}_{\mathrm{wall}}, \mathcal{C}). \end{aligned}$$

with fields omitted only by a declared non-invocation rule. The Rees fields may be invoked only after selecting a split-projective filtered perfect representative (C_R, F) of the integral family and supplying theorem 28.13; $\mathrm{BC}_{t,u}$ denotes its commuting base-change squares. Here $\mathcal{C}$ is the replay certificate and $\mathrm{D}_{\mathrm{wall}}$ records any separately proved failure of a constant heart or category Return.

Theorem 28.29 (Regularized boundary Return theorem). *Suppose the integral family, derived fibers, source identities, and invoked correction records are valid. Then:*

1. *the generic and derived special fibers are Returned at their own native strengths;*
2. $\mathrm{D}_{\mathrm{Tor}}$ *exactly records the cohomological correction to flat specialization;*
3. *an invoked integral $R[u]$ -Rees record Returns the associated-graded filtered boundary and a filtration-degeneration path, while the certified t/u base-change squares relate it to the generic and special fibers without identifying the two parameters;*
4. *a certified normal-form contact datum Returns the valuative tangent only at the declared gauge-slice or associated-graded strength; in the sparse cubic calibration, the direct marked diagonal datum $m, [\mu]$ is available;*
5. $\mathrm{D}_{\mathrm{lat}}$ *records the finite failure of a chosen generic normalization to preserve the integral lattice;*
6. *nonzero correction data may block Σ_{flat} or a constant-heart Return while leaving $\Sigma_{\mathrm{regularized}}$ valid;*
7. *the regularized package does not, in general, reconstruct the complete integral family unless a separate classification theorem is supplied.*

Proof. Items (1)–(5) are the preceding theorems assembled with the explicit type bridges of theorem 28.13 and without identifying the DVR parameter t with the Rees parameter u . Item (6) follows because each stronger strength requires vanishing of its own declared correction, whereas the regularized target includes rather than suppresses that correction. Item (7) is witnessed generally by higher-order deformation coefficients and, at pair strength, already by the cubic families of different order. $\square$

Corollary 28.30 (Exact classification in the sparse cubic family). *Within the one-dimensional sparse marked cubic family $b_{m,u}$, the regularized contact datum m determines the integral marked orbit because every unit $u(t)$ is integrally normalizable. This is a special classification theorem and is not promoted to arbitrary deformation directions.*

28.8 Defect stack, certificate, and status algebra

Definition 28.31 (Boundary defect stack). The PE-10 defect stack keeps separate:

1. D_{Tor} : cohomological base-change defects;
2. D_{lat} : integral-lattice defects of a selected generic normalization;
3. D_{wall} : independently proved constant-heart or category wall defects;
4. missing filtration, Rees, marking, comparison, or source records, which are non-scalar obligations and not algebraic defect objects.

No sum of these quantities is defined by default.

Definition 28.32 (PE-10 boundary certificate). A PE-10 certificate records:

1. R, t, K, κ and all coefficient conventions;
2. the actual integral complex or A_∞ operation table;
3. generic, naive special, and derived special fiber constructions;
4. the Tor spectral-sequence or exact-sequence data consumed;
5. every filtration and the complete Rees object when invoked;
6. the Maurer–Cartan arc and, when invoked, the gauge-normal-form certificate or the sparse cubic marked diagonal contact datum;
7. marked bases, integral lattices, generic normalization, and lattice quotient;
8. finite twisted generators and base-change comparison maps;
9. the requested Return strength and every weaker successful strength;
10. source bindings, immutable artifacts, verification outputs, and primary status.

Protocol 28.33 (Replay order). The verifier executes:

1. validate the DVR, chosen uniformizer, fraction/residue fields, and the coefficient homomorphism or explicit R -linear integral lift;
2. construct the generic and derived special fibers;
3. replay the Tor boundary sequence and flatness claims;
4. replay the Rees fibers when a filtered regularization is invoked;
5. validate the invoked gauge-normal-form contact certificate, or replay the sparse cubic diagonal contact datum;
6. replay the generic normalization over K and test integrality over R ;

7. compute D_{lat} and every independent wall invariant;
8. compare the resulting package with the requested exact, flat, regularized, or integral strength;
9. emit one primary status and list all certified weaker Returns.

Classification 28.34 (Primary statuses). At a fixed datum and requested strength:

1. REJECT: a supplied fiber, flatness assertion, gauge equation, source binding, or defect computation is false;
2. UNDEFINED: the integral model, derived fiber, filtration, lattice, comparison, or theorem scope required by the requested strength is absent;
3. OBSTRUCTED: all objects and comparisons exist, but a nonzero declared Tor, lattice, or wall defect prevents the requested exact or constant Return;
4. REGULARIZED-RETURN: the exact target fails or changes, but the separately identified regularized boundary package is fully certified;
5. PARTIAL-RETURN: only generic, special, or pair strength is certified;
6. RETURN: every obligation at the requested strength is discharged.

A regularized target is a distinct Return datum; it is not a silent relabeling of the failed exact target.

Part VI

Consolidated governance, source binding, audit, and closure

29 Unified defect stack and deterministic status algebra

For a requested Return strength Σ , the integrated verifier separates at least the following defects:

$$D_{\text{type}}, D_{\text{source}}, D_{\text{real}}, D_{\text{complete}}, D_{\text{compare}}, D_{\text{exact}}, D_{\text{higher}}, D_{\text{coh}}, D_{\text{wall}}, D_{\text{Tor}}, D_{\text{lat}}, D_{\text{cert}}.$$

No scalar reserve compensates for a missing non-scalar obligation. The ordered status rule is:

1. a false supplied assertion or invalid source binding gives REJECT;
2. otherwise a missing required object, map, enhancement, coherence, or theorem scope gives UNDEFINED;
3. otherwise a valid comparison with a nonzero defect gives OBSTRUCTED;
4. otherwise all obligations at Σ give RETURN;
5. otherwise a registered weaker strength may give PARTIAL - RETURN;
6. a source-certified replacement target is a distinct Return datum and may give REGULARIZED - RETURN.

30 Unified replay certificate schema

An accepted integrated PE claim records:

1. immutable source object, category, coefficient, variance, and convention records;
2. requested Return strength and target equivalence relation;
3. all actual maps, functors, resolutions, probe generators, enhancements, transition data, or integral lattices consumed;
4. exact hypotheses and source-bound classical theorem locators;
5. finite scope, endpoint, amplitude, completion, nilpotence, flatness, and coherence policies as applicable;
6. comparison artifacts and the complete defect stack;
7. basis/normalization data only when coordinate replay is requested;
8. status decision, failure route, dependency ancestry, and immutable artifact identities;
9. regression scripts as non-proof attachments, never as theorem premises.

31 Consolidated source-binding regime

The theorem-bearing classical inputs are consumed only at their displayed strengths. They include: Gabriel's finite type- A classification and Krull–Schmidt interval decomposition, with the PE-2 Hom/Ext and inversion

formulas proved directly in this appendix; Hilton–Rees natural transformations of extension functors; Yoneda extensions, pushout/pullback, Baer sum, and exact-sequence calculus; projective generators and derived Morita equivalence; hereditary splitting and mapping cones; Kadeishvili minimal models and twisted-object reconstruction; dg/A_∞ Morita theory and transported hearts; Gerstenhaber–Hochschild tangent and obstruction theory; Keller’s higher-structure invariance of Hochschild cochains; an explicit completion-compatible filtered L_∞ zigzag followed by Dolgushev–Rogers Goldman–Millson invariance; homotopy-coherent descent; Coxeter-polynomial invariance at the standard-heart derived strength; derived specialization, Tor spectral sequences, split-projective integral Rees constructions over $R[u]$, and derived base change.

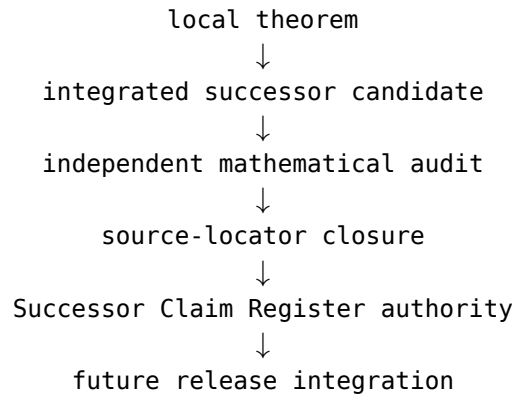
The exact local source-statement files and artifact hashes are listed in the accompanying Source Binding Registry. A bibliography entry or local statement binding records provenance and scope; it does not replace verification of the hypotheses in a consuming theorem.

32 Regression evidence boundary

All available PE verification scripts were replayed in the earlier local audits and passed their declared finite checks. Those checks validate explicit formulas, enumerate small counterexamples, and protect implementations against regression. In this integrated edition, check counts are deliberately omitted from theorem statements and moved to the external audit manifest. No theorem obtains its generality or proof status from finite enumeration.

33 Integrated promotion ladder and release gates

The promotion ladder is:



This document remains at the integrated successor-candidate stage after applying every internal repair identified through the seventh systemic reaudit and passing the focused eighth internal closure audit. Remaining mandatory gates are independent proof audit, exact bibliographic/statement reconciliation, formal or equivalently independent replay of the headline proofs, canonical artifact hashing under release authority, and an explicit decision on whether any result is suitable for a future v21 package.

Classification 33.1 (Integrated PE first-cycle structural closure declaration). Under the hypotheses and source-bound inputs displayed in the respective chapters:

1. PE-2–PE-5 form a coherent finite type-*A* reconstruction chain from Ext dimensions through object, morphism, exact, and bounded-derived Return;
2. PE-6–PE-10 form a coherent but separate enhancement/deformation chain from a genuine higher composition through marked/Morita invariance, local formal moduli, global homotopy-coherent descent, and valuative regularized boundary Return;

3. PE-1 is a compatible internal edge calibration but is not required by, and is weaker than, the finite-quiver reconstruction chain;
4. no cross-layer persistence interpretation is valid without a separate Interface theorem;
5. no stronger Return strength is inferred from a weaker one without its displayed promotion theorem and certificate.

Consequently the first major PE research cycle is internally closed as an internal-audit candidate with all repair sets through M8/G4 applied. Independent mathematical audit, source-authority reconciliation, and release integration remain open gates.

Classification 33.2 (Final status).

first-major-PE-cycle-internally-closed-candidate
 + all-internal-reaudit-repairs-through-M8-G4-applied
 + claim-source-manifest-package-synchronized
 + independent-audit-pending
 + integration-pending

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Part VII

Spectral-Sequence Transfer and Nonzero-Differential Return

34 Role inside Appendix HT

This Part supplies concentration, edge-exact, and genuinely nonzero-differential spectral-sequence Return patterns. Spectral-sequence degeneration remains terminologically and mathematically distinct from AK canonical collapse.

Remark 34.1 (Normalization rule). The following is the complete retained source text of the former SS artifact, normalized into this HT Part. Its original numbering is a provenance namespace only; it no longer denotes a separate Appendix file.

Retained module abstract

This document records the final GPT reaudit of the first construction cycle of the AK-HPDST spectral-sequence programme. The earlier finite-acyclic-cover Cech bridge exhibited one-row degeneration and a derived Return to global cohomology. The Leray bridge exhibited row and column edge-isomorphism regimes together with a five-term exact obstruction Return. The remaining structural gap was a worked bridge in which a genuinely nonzero higher differential is indispensable. That gap is filled here by the integral cohomological Serre spectral sequence of the Hopf fibration $S^1 \rightarrow S^3 \xrightarrow{p} S^2$. The transgression $d_2 : E_{0,1}^2 \rightarrow E_{2,0}^2$ is identified with the Euler/first-Chern class of the circle bundle and is an isomorphism. The resulting E_∞ page has exactly two surviving terms and returns the integral cohomology ring of S^3 . The document also records corrections and precision upgrades for the Leray bridge, provides a cross-bridge coverage audit, and records the governance decision that further generic SS expansion presently has lower marginal value than moving to the Homological Compression programme.

Remark 34.2 (Native vector transcription rule). The following body is reconstructed as native XeLaTeX text and mathematics from the authoritative module PDF. It is not Verbatim text, not a raster image, and not an imported PDF. Original title pages, module-local tables of contents, and module-local page numbers are removed; mathematical statements, proofs, tables, diagrams, references, and source-boundary declarations are retained. The former numbering is preserved as the module provenance namespace.

1 Scope and closure question

This is a successor research artifact over the frozen AK-HPDST v20.0.0 baseline. It neither modifies nor retroactively enlarges any v20 Core claim.

The closure question is whether the SS programme has demonstrated all three logically distinct behaviours needed for an AK-compatible spectral-sequence architecture:

(C1) concentration and complete isomorphism Return;

(C2) partial Return with an explicit residual obstruction;

(C3) a nonzero differential whose computation is essential to the final external conclusion.

The Čech bridge supplies (C1), the Leray bridge supplies (C1)–(C2), and the Serre–Hopf bridge below supplies (C3).

Warning 1.1 (Terminological firewall). A spectral sequence “collapsing” or “degenerating” is not AK canonical collapse. The terms remain disjoint unless a separate comparison theorem is proved.

2 Reaudit of the Leray bridge

The Leray construction is mathematically sound under its declared hypotheses, but its maximal edition requires the following precision upgrades.

Proposition 2.1 (Correct source binding). *For a morphism of ringed spaces $f : X \rightarrow Y$ and a bounded-below complex $\mathcal{F}^\bullet$ of $\mathcal{O}_X$ -modules, the source-bound Leray theorem is*

$$E_2^{p,q} = H^p(Y, R^q f_* \mathcal{F}^\bullet) \implies H^{p+q}(X, \mathcal{F}^\bullet).$$

For a sheaf $\mathcal{F}$ placed in degree zero this specializes to the version used in the Leray bridge.

Proof. This is the standard Grothendieck spectral sequence for $\Gamma(Y, -) \circ f_* = \Gamma(X, -)$, equivalently Stacks Project Tag 01F2. Tag 01EY is the containing section rather than the most precise lemma-level binding.

Proposition 2.2 (Affine implementation precision). *If $f : X \rightarrow Y$ is an affine morphism of schemes and $\mathcal{F}$ is quasi-coherent, then $R^q f_* \mathcal{F} = 0$ for $q > 0$, so*

$$H^n(Y, f_* \mathcal{F}) \xrightarrow{\cong} H^n(X, \mathcal{F}).$$

If Y is affine, both sides vanish for $n > 0$.

Proof. Use higher-direct-image vanishing for affine morphisms and affine-scheme vanishing for quasi-coherent sheaves. Suitable bindings are Stacks Project Tags 0AVV and 01XE.

Warning 2.3 (Correction of computational wording). The transgression $d_2^{0,1}$ is a page-level obstruction morphism, but it is not automatically a finite computational certificate. Finiteness requires separately declared finite-generation or finite-dimensional hypotheses and an effective representation of the groups and differential.

Proposition 2.4 (Five-term Return boundary). *The Leray five-term exact sequence*

$$0 \rightarrow H^1(Y, f_* \mathcal{F}) \rightarrow H^1(X, \mathcal{F}) \rightarrow H^0(Y, R^1 f_* \mathcal{F}) \xrightarrow{d_2^{0,1}} H^2(Y, f_* \mathcal{F}) \rightarrow H^2(X, \mathcal{F})$$

returns canonical subobjects, quotients, and an obstruction kernel, but does not canonically split $H^1(X, \mathcal{F})$.

3 Selection of the third bridge

The missing case must contain a nonzero differential that is known independently and that changes the abutment.

Candidate	Advantage	Risk	Decision
Hypercohomology	natural next algebraic bridge	often degenerates under easy examples; external target may become formal	later
Grothendieck	maximum categorical breadth	duplicates Leray architecture without a vivid nonzero differential	later
Hochschild–Serre	arithmetic and group-theoretic relevance	local coefficients and extension data increase initial load	later
Serre–Hopf	explicit nonzero transgression, finite page, complete abutment	topological rather than algebraic-geometric	selected

Protocol 3.1 (Third-bridge selection decision). Among the candidates listed above, the Hopf-fibration Serre spectral sequence is selected for first-cycle closure because its unique possible non-trivial differential is geometrically identified by a characteristic class, is an isomorphism, and determines the declared abutment without unresolved extension data. This is a research-priority decision, not a mathematical optimality theorem.

4 Source binding for the Serre bridge

The bridge imports the following classical results in the restricted Hopf regime.

- (S1) For a Serre fibration $F \rightarrow E \rightarrow B$ of spaces of CW type, with path-connected base and coefficient ring R , there is a cohomological Serre spectral sequence with the induced local coefficient systems

$$E_2^{p,q} = H^p\left(B; \mathcal{H}^q(F; R)\right) \Longrightarrow H^{p+q}(E; R),$$

with the usual finite-filtration interpretation in the present first-quadrant finite-support case.

- (S2) If B is simply connected, the monodromy action is trivial and the local coefficient systems are constant.
- (S3) For an oriented principal S^1 -bundle, the transgression of a chosen generator of $H^1(S^1; \mathbb{Z})$ is the Euler class, equivalently the first Chern class, up to sign convention.
- (S4) The Hopf principal circle bundle has first Chern class a generator of $H^2(S^2; \mathbb{Z})$.
- (S5) The cohomological Serre spectral sequence is multiplicative in this integral principal-bundle regime, and the differential satisfies the graded Leibniz rule.

A direct source binding for the Serre construction and its applications is Hatcher’s *Spectral Sequences in Algebraic Topology*, Chapter 1. The characteristic-class input is to be bound separately to a standard source on principal circle bundles and first Chern classes. The present artifact records theorem-level bindings but does not yet supply a line-addressed source registry. No new Serre theorem, characteristic-class theorem, or Hopf-invariant computation is claimed.

5 Typed realization

Definition 5.1 (Serre input record). A typed Serre record is

$$\mathbf{S} = (F \twoheadrightarrow E \xrightarrow{p} B, R, \mathcal{L}, \mathbf{P}),$$

where p is a declared Serre fibration, R is the coefficient ring, $\mathcal{L}$ is the induced local coefficient system $\mathcal{H}^q(F; R)$, and $\mathbf{P}$ is the external predicate to be returned.

Definition 5.2 (Hopf realization). For the Hopf fibration

$$S^1 \twoheadrightarrow S^3 \xrightarrow{p} S^2$$

with integral coefficients, the realization is the multiplicative first-quadrant spectral sequence

$$E_2^{p,q} = H^p(S^2; H^q(S^1; \mathbb{Z})) \Rightarrow H^{p+q}(S^3; \mathbb{Z}).$$

Because S^2 is simply connected, no twisted coefficient ambiguity remains.

6 The nonzero differential

Let $x \in H^2(S^2; \mathbb{Z}) \cong \mathbb{Z}$ and $y \in H^1(S^1; \mathbb{Z}) \cong \mathbb{Z}$ be generators. Then

$$E_2^{p,q} = 0$$

only at

$$(0, 0), \quad (2, 0), \quad (0, 1), \quad (2, 1),$$

and each nonzero group is isomorphic to $\mathbb{Z}$.

Theorem 6.1 (Hopf transgression). *With compatible choices of orientation,*

$$d_2(y) = x.$$

In particular,

$$d_2 : E_2^{0,1} \rightarrow E_2^{2,0}$$

is an isomorphism.

Proof. For an oriented circle bundle, the Serre transgression of the fiber generator is the Euler class, or equivalently the first Chern class, up to sign. The Hopf bundle has first Chern class $\pm x$, a generator of $H^2(S^2; \mathbb{Z})$. After choosing the orientation so that the sign is positive, $d_2(y) = x$.

Remark 6.2 (Why this is not circular). The differential is not inferred from already knowing $H^*(S^3; \mathbb{Z})$. It is source-bound to the independently known characteristic class of the Hopf bundle. The abutment calculation therefore genuinely uses the differential.

7 Page computation and Return

Proposition 7.1 (The E_3 page). *After applying d_2 , the terms $E_2^{0,1}$ and $E_2^{2,0}$ are killed, whereas $E_2^{0,0}$ and $E_2^{2,1}$ survive. No further differential can enter or leave the surviving terms.*

Proof. The asserted d_2 is an isomorphism between $E_2^{0,1}$ and $E_2^{2,0}$. The unit in $(0, 0)$ is permanent. The product class xy in $(2, 1)$ satisfies $d_2(xy) = d_2(x)y - xd_2(y) = -x^2 = 0$ because $H^4(S^2; \mathbb{Z}) = 0$; equivalently, its target position is zero. No possible source reaches $(2, 1)$ in the first quadrant. Degree reasons eliminate all d_r for $r \geq 3$.

Corollary 7.2 (Stable page).

$$E_3 = E_\infty,$$

with

$$E_\infty^{0,0} \cong \mathbb{Z}, \quad E_\infty^{2,1} \cong \mathbb{Z},$$

and all other terms zero.

Theorem 7.3 (Serre–Hopf Return). *The spectral sequence returns*

$$H^n(S^3; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & n = 0, 3, \\ 0, & \text{otherwise.} \end{cases}$$

Moreover, as a graded ring,

$$H^*(S^3; \mathbb{Z}) \cong \Lambda_{\mathbb{Z}}(z_3),$$

where $|z_3| = 3$.

Proof. For total degree zero, only $E_\infty^{0,0}$ occurs. For total degree three, only $E_\infty^{2,1}$ occurs. Every other total degree has no surviving graded piece. Hence there is no additive extension ambiguity. The degree-three generator squares to zero for dimensional reasons, so the ring is the exterior algebra on one degree-three generator.

Definition 7.4 (Differential Return certificate). For this bridge, the Return certificate is the tuple

$$c_{\text{Hopf}} = (E_2^{0,1} \cong \mathbb{Z}, E_2^{2,0} \cong \mathbb{Z}, c_1(p) = \pm 1, d_2(y) = \pm x, E_3 = E_\infty).$$

It is finite as a symbolic proof record. This does not assert a general complexity bound for computing characteristic classes or spectral-sequence differentials.

8 Failure boundary

Non-claim 8.1. No claim is made that every circle bundle has transgression an isomorphism. For Euler class mx , the differential is multiplication by m : for $m \neq 0$ its kernel is zero and its cokernel is $\mathbb{Z}/m$, while for $m = 0$ the full source survives.

Non-claim 8.2. No claim is made that every Serre spectral sequence has constant coefficients. Non-simply-connected bases may act nontrivially on the fiber cohomology.

Non-claim 8.3. No claim is made that E_∞ determines a ring without multiplicative and extension analysis. The Hopf example is safe because only one additive piece survives in each relevant total degree and dimensional reasons determine the product.

Non-claim 8.4. No claim is made that the characteristic-class identification is an AK theorem. It is external source mathematics bound into the bridge.

9 AK defect stack for the third bridge

Definition 9.1 (Serre bridge defect record). Define

$$D_{\text{Serre}} = (D_{\text{fib}}, D_{\text{loc}}, D_{\text{diff}}, D_{\text{conv}}, D_{\text{ext}}, D_{\text{return}}, D_{\text{nat}}),$$

where the entries record respectively failure of the fibration input, unresolved local coefficients, unknown relevant differentials, convergence failure, unresolved extensions, failure of the external Return, and naturality mismatch.

Theorem 9.2 (Hopf defect clearance). *For the integral Hopf record, every component of D_{Serre} affecting the declared predicate “determine $H^*(S^3; \mathbb{Z})$ ” is cleared.*

Proof. The Hopf map is a principal S^1 -bundle; the base is simply connected; the sole relevant differential is fixed by the generator Chern class; first-quadrant finite support gives convergence; one graded piece survives per relevant total degree; and the edge/abutment identification returns the stated cohomology.

10 Cross-bridge coverage audit

Obligation	Cech bridge	Leray bridge	Serre–Hopf bridge
External input	finite acyclic cover	morphism and higher direct images	fibration and local coefficients
Reduction mechanism	one-row concentration	row/column concentration or five-term truncation	nonzero transgression
Convergence	bounded double complex	first-quadrant Leray filtration	finite first-quadrant page
Extension handling	one piece per degree	one piece or exact retention	one piece per returned degree
Return strength	derived isomorphism and cohomology isomorphism	full isomorphism or exact obstruction Return	complete cohomology-ring Return
Actual nonzero d_r	not needed	represented strictly by $d_2^{0,1}$	explicitly computed and essential
Primary geometry	local-to-global	map-relative	fibration/characteristic class

Protocol 10.1 (SS first-cycle coverage audit). The three bridges jointly realize the minimum complete spectral-sequence test suite for the successor programme:

- (i) degeneration with full Return;
- (ii) non-degenerate low-degree exact Return with a retained obstruction;
- (iii) an explicitly nonzero differential producing the final Return.

Consequently, within the declared first-cycle test suite, no further generic SS bridge is presently required to demonstrate the distinction among realization, differential reduction, convergence, extension control, and Return. This is a programme-closure judgment rather than a theorem about all spectral sequences.

11 Compatibility with frozen v20

v20 obligation	SS closure treatment
Interface before conclusion	every bridge fixes its external object, coefficients, category, and purpose
Realization before Return	the filtered/double-complex or fibration realization is constructed first
No finite-to-infinite shortcut	convergence is separately source-bound in each regime
Extension discipline	one-piece filtrations or exact unsplit extensions are explicitly distinguished
Return sovereignty	external conclusions use only proved quasi-isomorphisms, edge maps, exact sequences, or abutment identifications
Undefined versus reject	missing hypotheses mean unproved/undefined, not a negative theorem

Collapse firewall	no spectral-sequence degeneration is equated with AK canonical collapse
Core sovereignty	no v20 canonical definition or theorem is modified

12 Corpus-level v20 reaudit note

A text-level comparison against the frozen Main and minimal Claim Register confirms that the present SS package respects the following explicit v20 policies: realization is not Return; external conclusions require a complete Interface–Core–Reduction–Return chain; missing realization, reduction, or Return leaves the external conclusion unlicensed; and known-theorem recovery is limited to the exact strength returned. The SS package does not identify its spectral-sequence reductions with AK canonical collapse and does not alter a frozen Core theorem. This check is a GPT corpus reaudit, not an independent third-party certification and not a line-addressed release audit.

13 Completion ledger

Item	Status	Evidence
General SS defect framework	complete	reaudited framework
Cech Return bridge	bridge complete	one-row and derived Return
Leray Return bridge	bridge complete after corrections	two isomorphism regimes plus five-term
Nonzero-differential bridge	bridge complete	Hopf d_2 computation
Extension audit	complete for declared examples	one-piece or exact retention
Failure boundary	complete	explicit non-claims
v20 compatibility	corpus-level GPT reaudit completed	Main and Claim Register policy comparison
Independent external audit	pending	outside GPT-only scope
Release registration	pending	successor governance step

14 Final closure theorem and next step

Protocol 14.1 (GPT closure decision for the SS construction phase). Within the declared classical regimes, the successor SS programme has reached the maximum useful first-cycle construction state available without independent external audit or selecting a new application domain. It contains:

Cech: finite local data $\rightarrow \mathbf{R}\Gamma$,

Leray: relative data $\rightarrow$ edge/exact Return,

and

Serre–Hopf: nonzero $d_2 \rightarrow H^*(S^3; \mathbb{Z})$.

The programme may therefore be frozen at

return_proved + bridge_complete + GPT_reaudited + integration_candidate.

It is not integration_ready until independent mathematical audit, exact source-registry records, hashes, claim identifiers, and release approval are completed.

Protocol 14.2 (Marginal-value decision). Further additions such as hypercohomology, Grothendieck, Hochschild–Serre, Adams, or Eilenberg–Moore spectral sequences would expand the library but would not repair a remaining foundational SS defect. The highest-value next mathematical programme is Homological Compression, using the completed SS package as an input and audit layer. This is a strategic prioritization, not a mathematical theorem.

Source note

The principal web-verifiable source bindings used in the reaudit are Stacks Project Tags 01F2 (Leray spectral sequence), 0AVV (affine morphisms and derived pushforward), 01XE (affine cohomology vanishing), and standard university-hosted texts on the Serre spectral sequence and characteristic classes, including Allen Hatcher's Spectral Sequences in Algebraic Topology. These references bind classical results only; they do not support a novelty claim for AK.

Part VIII

Derived-Limit Defect and Finite-to-Infinite Transfer

35 Role inside Appendix HT

This Part governs countable inverse systems, the Roos difference map, $\varprojlim^1$, claim-relative edge defects, and apex agreement. It complements rather than reverses the frozen v20 directed-tower calculus.

Remark 35.1 (Normalization rule). The following is the complete retained source text of the former RL artifact, normalized into this HT Part. Its original numbering is a provenance namespace only; it no longer denotes a separate Appendix file.

Retained module abstract

This research appendix develops a theorem-level extension of the finite-to-infinite discipline of AK-HPDST v20.0.0. Appendix D of the frozen baseline treats directed tower comparison artifacts and proves an exact reduction under eventual constancy and explicit apex agreement. The present appendix does not replace that result. It studies the complementary situation in which an external problem is represented by an inverse tower, completion, or cohomological limiting construction, and a derived inverse-limit term may obstruct passage from finite stages to the declared infinite object. The primary objective is to separate three logically distinct defects: failure inside the inverse system, failure of exactness at the inverse-limit edge, and failure of agreement with the declared apex. No external conclusion is asserted without a separately typed realization and Return theorem.

Remark 35.2 (Native vector transcription rule). The following body is reconstructed as native XeLaTeX text and mathematics from the authoritative module PDF. It is not Verbatim text, not a raster image, and not an imported PDF. Original title pages, module-local tables of contents, and module-local page numbers are removed; mathematical statements, proofs, tables, diagrams, references, and source-boundary declarations are retained. The former numbering is preserved as the module provenance namespace.

1 Governance, scope, and relation to the frozen baseline

1.1 Research status

This document is a successor research artifact built over AK-HPDST v20.0.0. It is not a Foundation Appendix of v20.0.0, does not modify any v20 canonical definition, and does not enlarge the v20 claim closure. Every theorem below has one of the statuses

classical_recovery AK_internal_extension
bridge_candidate conditional_application

A theorem may be considered for later v21 integration only after source binding, independent proof audit, compatibility audit against the frozen baseline, and at least one non-toy application. Section 9 supplies an external derived-category calibration in which a nonzero derived-limit term is indispensable. Section 10 first records the exact-sequence calibration. Section 11 then strengthens it by constructing, directly from the tower, a canonical edge-comparison artifact and an independent Roos-complex defect object. Their natural comparison yields a complete restricted realization–evaluation–apex–Return bridge for degreewise vanishing along countable towers. This successor bridge is not retroactively inserted into the frozen v20 Core.

1.2 What v20.0.0 already proves

The frozen Appendix D already provides:

- (i) typed directed tower-profile systems;
- (ii) a declared source profile and declared apex profile;

- (iii) an actual comparison morphism from the tower-side source to the apex;
- (iv) kernel and cokernel defect profiles and their terminal/transient diagnostics;
- (v) the rule that finite observation is not infinite proof;
- (vi) an exact finite-to-infinite theorem under eventual constancy, existence of the genuine colimit, and explicit apex agreement.

RL therefore does not re-prove eventual-constancy reduction as a new AK theorem. It uses that theorem as the zero-derived-defect calibration case.

1.3 The new gap addressed by RL

Many external theories use inverse systems rather than directed colimits. Typical schematic forms are

$$X_0 \leftarrow \cdots \leftarrow X_1 \leftarrow \cdots \leftarrow X_2 \leftarrow \cdots, \quad X_\infty \simeq \varprojlim_n X_n,$$

or cohomology of a limiting object compared with the inverse limit of finite-stage cohomology. In such settings, levelwise exactness need not remain exact after applying $\varprojlim$. A derived term $\varprojlim^1$ may survive.

The logical gap is therefore not merely

$$\text{finite stage} \not\Rightarrow \text{infinite stage},$$

but more precisely

$$\text{levelwise exact data} \not\Rightarrow \text{exact inverse-limit data}$$

unless the derived-limit obstruction is controlled.

1.4 Normative non-claims

This appendix does not assert any of the following without a separate theorem:

- (i) that an externally given object called X_∞ is the categorical inverse limit;
- (ii) that cohomology commutes with inverse limit;
- (iii) that $\varprojlim^1$ vanishes merely because finitely many levels agree;
- (iv) that the Mittag-Leffler condition holds merely because dimensions are bounded;
- (v) that vanishing of $\varprojlim^1$ proves apex agreement;
- (vi) that an inverse-limit isomorphism proves the external target predicate;
- (vii) that a direct tower from Appendix D may be silently reversed into an inverse tower;
- (viii) that a numerical plateau is eventual constancy or a finite-to-infinite certificate.

2 Typed inverse systems and apex semantics

2.1 Base regime

Assumption 2.1 (RL base category). Unless explicitly stated otherwise, R is a ring and all inverse systems lie in $R\text{-Mod}$. This restriction is deliberate: countable products, kernels, cokernels, inverse limits, and the standard countable description of $\varprojlim$ are then available without importing additional higher-categorical hypotheses.

Definition 2.2 (Countable inverse system). A countable inverse system is a diagram

$$A_0 \xleftarrow{u_0} A_1 \xleftarrow{u_1} A_2 \xleftarrow{u_2} \dots$$

in $R\text{-Mod}$. For $m \geq n$, write

$$u_{m,n} = u_n u_{n+1} \dots u_{m-1} : A_m \rightarrow A_n,$$

with $u_{n,n} = \text{id}_{A_n}$.

Definition 2.3 (Inverse-system record). An RL inverse-system record contains:

- (i) the objects A_n and actual transition morphisms u_n ;
- (ii) the ambient module category and coefficient ring;
- (iii) provenance identifiers for all objects and maps;
- (iv) the external meaning of each level;
- (v) the monitored degree or invariant, when the system is extracted from a graded object;
- (vi) the exact claim for which the system is used.

A family of isomorphism classes, ranks, dimensions, barcodes, or numerical invariants is not by itself an inverse-system record.

Definition 2.4 (Declared inverse-limit source). For a valid inverse-system record $A_\bullet$, the inverse-limit source is

$$L(A_\bullet) := \varprojlim_n A_n$$

with its canonical projections $p_n : L(A_\bullet) \rightarrow A_n$. Its status is `genuine_inverse_limit` only when this categorical construction is the source required by the consuming claim.

Definition 2.5 (Declared apex and apex comparison). A declared apex consists of an object A_∞ , a provenance and semantic-authority record, and an actual comparison morphism

$$\alpha_A : A_\infty \rightarrow \varprojlim_n A_n \quad \text{or} \quad \beta_A : \varprojlim_n A_n \rightarrow A_\infty,$$

with its direction fixed by the external construction. No apex agreement is inferred from notation.

Definition 2.6 (Apex defect). For a declared comparison map f , define the apex defect object

$$D_{\text{apex}}(f) := \ker(f) \oplus \text{coker}(f).$$

The verdict `apex_agreement` means that the relevant actual map f is an isomorphism, equivalently $D_{\text{apex}}(f) = 0$. This verdict is logically independent of derived-limit vanishing.

3 The countable derived-limit defect

3.1 The Roos difference map

Definition 3.1 (Roos difference map). For a countable inverse system $A_\bullet$, define

$$\Delta_A : \prod_{n \geq 0} A_n \rightarrow \prod_{n \geq 0} A_n, \quad \Delta_A((a_n)_{n \geq 0}) = (a_n - u_n(a_{n+1}))_{n \geq 0}.$$

Proposition 3.2 (Kernel description of the inverse limit). *For every countable inverse system in $R\text{-Mod}$, there is a canonical isomorphism*

$$\varprojlim_n A_n \cong \ker(\Delta_A).$$

Proof. An element of $\ker(\Delta_A)$ is exactly a sequence (a_n) satisfying $a_n = u_n(a_{n+1})$ for every n . This is the defining compatibility condition for the inverse limit.

Definition 3.3 (Roos first-edge defect). For a countable inverse system in the RL base regime, define the Roos first-edge object

$$D_{\lim}(A_\bullet) := \text{coker}(\Delta_A).$$

Theorem 3.6 below identifies this object canonically with the first right-derived inverse limit $R^1 \varprojlim A_n$. Until that identification is invoked, $D_{\lim}$ is only the explicitly constructed cokernel above. Its vanishing status is

$$\text{derived_limit_clear} \iff D_{\lim}(A_\bullet) = 0.$$

Remark 3.4 (Interpretation). The equation $D_{\lim}(A_\bullet) = 0$ means that every sequence in $\varprojlim_n A_n$ is a Roos difference. It is an exactness statement about the whole tail of the inverse system. It is not a statement about one level, finitely many levels, or the apex.

Theorem 3.5 (Classical countable Roos exact sequence). *For every countable inverse system in $R\text{-Mod}$, there is a natural exact sequence*

$$0 \rightarrow \varprojlim_n A_n \rightarrow \prod_{n \geq 0} A_n \xrightarrow{\Delta_A} \prod_{n \geq 0} A_n \rightarrow D_{\lim}(A_\bullet) \rightarrow 0.$$

Proof. The first identification is Proposition 3.2; the final object is Definition 3.3. Exactness at the two products is the kernel–image and image–cokernel exactness of Δ_A .

Theorem 3.6 (Identification with the derived inverse limit). *In the countable module regime, the Roos first-edge object is naturally isomorphic to the first right-derived inverse limit:*

$$D_{\lim}(A_\bullet) \cong R^1 \varprojlim_n A_n = \varprojlim_n^1 A_n,$$

and $R^q \varprojlim_n A_n = 0$ for $q \geq 2$.

Remark 3.7 (Source-binding status). Theorem 3.6, the Mittag–Leffler vanishing theorem, and the six-term exact sequence below are classical results. The release candidate must bind them to an exact primary or standard reference and record the indexing and variance conventions. The exact source bindings used in this draft are recorded in Section 7. The fixed Stacks Project tags bind the indexing, variance, completion, and Mittag–Leffler conventions used here. These bindings establish classical provenance; they do not remove the independent proof-audit requirement for a later release candidate.

3.2 System defect, edge defect, and apex defect

Definition 3.8 (Three-layer finite-to-infinite audit record). An RL finite-to-infinite audit record separates:

- (i) *system-validity status* S_{sys} : whether the declared transition system and the hypotheses required by the chosen transfer theorem are present and proved;
- (ii) *ambient derived-limit object* $D_{\text{lim}}(A_{\bullet}) = \lim^1 A_n$, which records the possible right-exactness defect available at the inverse-limit edge;
- (iii) *claim-relative edge defect* $D_{\text{edge}}(\partial) := \text{im}(\partial)$ for the actual connecting map used by the consuming claim;
- (iv) *apex defect* $D_{\text{apex}}(f)$ for the actual map relating the categorical inverse limit to the declared realized apex.

The system-validity status is not an algebraic defect object. The other three objects are never merged into one scalar without a separately proved aggregation theorem.

Warning 3.9 (Vanishing of one layer does not discharge the others). The implications

$$D_{\text{lim}}(A_{\bullet}) = 0 \Rightarrow D_{\text{apex}}(f) = 0, \quad D_{\text{apex}}(f) = 0 \Rightarrow D_{\text{lim}}(A_{\bullet}) = 0$$

are not available in general. Moreover, $D_{\text{lim}}(A_{\bullet}) \neq 0$ does not by itself obstruct every target claim: the claim-relative obstruction may still vanish when $D_{\text{edge}}(\partial) = \text{im}(\partial) = 0$. Likewise, levelwise isomorphisms over a finite initial segment determine none of these tail or apex objects.

4 Vanishing criteria and finite certificates

4.1 Mittag-Leffler control

Definition 4.1 (Mittag-Leffler condition). An inverse system $A_{\bullet}$ satisfies the Mittag-Leffler condition if, for every n , the descending family of submodules

$$\text{im}(A_m \xrightarrow{u_{m,n}} A_n), \quad m \geq n,$$

is eventually constant.

Theorem 4.2 (Classical Mittag-Leffler vanishing). *Let $A_{\bullet}$ be a countable inverse system of modules satisfying the Mittag-Leffler condition. Then*

$$\lim_{\leftarrow}^1 A_n = 0.$$

Remark 4.3 (Status and proof obligation). Theorem 4.2 is a classical-recovery theorem. Its later release version requires exact source binding and an independent proof audit. RL uses it only through its stated hypotheses; it never replaces a proof of the Mittag-Leffler condition.

Corollary 4.4 (Surjective tower criterion). *If every transition map $u_n : A_{n+1} \twoheadrightarrow A_n$ is surjective, then $A_{\bullet}$ is Mittag-Leffler and*

$$\lim_{\leftarrow}^1 A_n = 0.$$

Proof. For fixed n , every image $\text{im}(A_m \rightarrow A_n)$ equals A_n . Apply Theorem 4.2.

Corollary 4.5 (Eventually constant inverse system). *If there exists N such that every u_n is an isomorphism for $n \geq N$, then*

$$\lim_{\leftarrow}^1 A_n = 0, \quad \lim_{\leftarrow} A_n \cong A_N.$$

Remark 4.6 (Calibration against frozen Appendix D). Corollary 4.5 is the inverse-system analogue of the frozen Appendix D eventual-constancy reduction. It is not claimed to supersede the directed-colimit theorem there. The two results occupy opposite variance channels and must not be silently identified.

4.2 Certificate discipline

Definition 4.7 (RL vanishing certificate). An RL vanishing certificate for $D_{\lim}(A_\bullet) = 0$ contains exactly one theorem-backed route:

- (i) an eventual-isomorphism bound N and proof for the full quantified tail;
- (ii) levelwise surjectivity with proof for every transition in the declared tail;
- (iii) a proved Mittag–Leffler image-stabilization bound for each required index;
- (iv) a direct proof that the Roos map Δ_A is surjective;
- (v) another explicitly named theorem whose hypotheses are fully discharged.

A finite computation, repeated rank equality, or observed plateau is not a vanishing certificate.

Definition 4.8 (Finite certificate with quantified tail). A finite certificate with quantified tail consists of finite data $C_{\leq N}$ together with a theorem proving that the stated tail property holds for every $n \geq N$. The theorem, not the finite data alone, performs the finite-to-infinite promotion.

5 Exact transfer across inverse limits

5.1 The edge problem

Definition 5.1 (Levelwise short exact inverse system). A levelwise short exact inverse system is a commutative diagram of inverse systems

$$0 \rightarrow A_n \rightarrow B_n \rightarrow C_n \rightarrow 0 \quad (n \geq 0)$$

whose rows are short exact and whose transition maps commute with the row maps.

Theorem 5.2 (Derived-limit edge exact sequence). *For a levelwise short exact inverse system of modules, there is a natural exact sequence*

$$0 \rightarrow \lim_{\leftarrow} A_n \rightarrow \lim_{\leftarrow} B_n \rightarrow \lim_{\leftarrow} C_n \xrightarrow{\partial} \lim_{\leftarrow}^1 A_n \rightarrow \lim_{\leftarrow}^1 B_n \rightarrow \lim_{\leftarrow}^1 C_n \rightarrow 0.$$

Proof. Apply the Roos two-term complex to the three inverse systems. Exactness of products in $R\text{-Mod}$ gives a short exact sequence of two-term Roos complexes. The snake lemma yields the displayed natural six-term exact sequence; Theorem 3.6 identifies the three cokernels with $\lim_{\leftarrow}^1$.

Remark 5.3 (AK interpretation). The ambient object $\lim^1 A_n$ is the space in which an edge obstruction may live. For this particular short exact system, the actual claim-relative obstruction is

$$D_{\text{edge}}(\partial) := \text{im} \left(\partial : \varprojlim C_n \rightarrow \varprojlim^1 A_n \right).$$

Thus right exactness at $\varprojlim C_n$ holds exactly when $D_{\text{edge}}(\partial) = 0$. The stronger condition $\varprojlim^1 A_n = 0$ is a reusable sufficient certificate, not a necessary condition for every consuming claim. Both are distinct from any later comparison with an external apex.

Corollary 5.4 (Claim-relative exact transfer). *In the setting of Theorem 5.2, the sequence*

$$0 \rightarrow \varprojlim A_n \rightarrow \varprojlim B_n \rightarrow \varprojlim C_n \rightarrow 0$$

is exact if and only if $D_{\text{edge}}(\partial) = 0$.

Corollary 5.5 (Reusable sufficient clearance). *In the same setting, $\varprojlim^1 A_n = 0$ implies $D_{\text{edge}}(\partial) = 0$, and hence the inverse-limit sequence is short exact.*

5.2 AK finite-to-infinite transfer schema

Definition 5.6 (Certificate carrier and target predicate). A certificate carrier is a left-exact functorial construction Z from the realized finite-stage objects to modules, together with a declared target predicate of the form

$P_\infty : z_\infty \in Z_\infty$ lies in a specified kernel, image, zero class, or quotient class.

The finite certificates must determine a compatible element $(z_n)_n \in \varprojlim Z_n$, not merely unrelated levelwise verdicts.

Definition 5.7 (RL transfer package). An RL transfer package for an external target predicate $Q(X_\infty)$ contains:

- (i) an external inverse tower $X_\bullet$, a declared realized apex X_∞ , and actual transition morphisms;
- (ii) a typed realization producing the exact inverse systems of modules and certificate carriers consumed by the proof;
- (iii) finite-stage AK certificates that form compatible inverse-system data, together with a theorem producing the associated limit class or limit morphism;
- (iv) for every inverse-limit edge, either a proof that the claim-relative defect $D_{\text{edge}}(\partial)$ vanishes or a stronger reusable certificate such as $\varprojlim^1 A_n = 0$;
- (v) an actual apex comparison and proof of the exact target-relative agreement required by the claim;
- (vi) a proved predicate-transfer theorem transporting the compatible limit class to an AK-internal infinite certificate;
- (vii) a separately proved Return theorem from that infinite certificate to $Q(X_\infty)$.

No item may be replaced by the conclusion of a later item.

Proposition 5.8 (AK–RL finite-to-infinite composition rule). *Given a valid RL transfer package, assume:*

- (i) the finite-stage certificates produce compatible inverse-system data;
- (ii) every actual claim-relative edge defect used in constructing the limit certificate vanishes;
- (iii) the predicate-transfer theorem constructs the declared AK-internal infinite certificate from that compatible limit data;
- (iv) the required target-relative apex comparison is valid;
- (v) the separately typed Return theorem applies.

Then the external predicate $Q(X_\infty)$ follows.

Proof. Compatibility gives an element or morphism in the appropriate inverse limit. Vanishing of each actual edge defect supplies precisely the exactness needed to preserve the kernel, image, zero, or quotient predicate specified by the carrier. The predicate-transfer theorem therefore constructs the AK-internal infinite certificate rather than assuming it. Apex agreement transports that certificate to the declared external object, and the Return theorem yields $Q(X_\infty)$.

Remark 5.9 (Logical status of the composition rule). Proposition 5.8 is a formally typed composition principle, not a new existence theorem. Its mathematical content in any application lies in constructing the compatible limit data, proving the edge-clearance theorem, establishing apex agreement, and proving Return. The proposition must not be counted as an external theorem merely because these obligations are listed in one package.

Proposition 5.10 (No unlicensed implication between RL and frozen Type IV). *Without an additional comparison theorem, no implication may be inferred between an RL verdict and the frozen v20 Type IV verdict. More precisely:*

- (i) RL audits an inverse-limit or completion edge and its target-relative apex comparison;
- (ii) frozen Type IV audits the kernel and cokernel profiles of an actual directed-colimit comparison morphism

$$\phi_{i,W,\tau} : \text{ColimProfile}_{i,W,\tau}(F_\bullet) \rightarrow \text{ApexProfile}_{i,W,\tau}(F_\infty);$$

- (iii) when one external proof uses both channels, it must provide an explicit variance-changing construction or comparison square and prove its commutativity;
- (iv) passing one channel does not discharge any artifact, terminal-tameness, transient-defect, or apex obligation in the other.

This proposition is a typing and governance rule. It does not claim logical independence in the model-theoretic sense; such a claim would require explicit counterexamples or an independence theorem.

Proof. The two audits consume diagrams of opposite variance and different comparison maps. Hence no typed morphism relating their defect objects is available from the stated data alone. Any implication therefore requires an additional theorem constructing such a relation. The frozen non-compensation discipline then requires every independently invoked obligation to be discharged separately.

Definition 5.11 (v20 readiness-bundle extension record). For a future v21 integration, an invoked inverse-limit claim extends—and does not replace—the frozen finite-to-infinite readiness bundle by recording:

- (i) variance inverse;
- (ii) the actual inverse-system record and provenance;
- (iii) the chosen Roos/derived-limit theorem and source binding;
- (iv) every ambient D_{lim} object and actual D_{edge} map used;
- (v) the target-relative apex comparison;
- (vi) the predicate-transfer theorem;
- (vii) the external Return theorem;
- (viii) any separately invoked frozen Type IV artifact.

Until v21 promotion, this is a successor research record and has no normative force inside v20.

Remark 5.12 (What is and is not new). The classical algebra in this section is not claimed as new. The AK contribution is the non-circular composition discipline: finite certificates must first become compatible limit data; actual edge defects must then be cleared; only afterwards may an infinite AK certificate be constructed, compared with the apex, and returned externally.

6 Failure taxonomy

Definition 6.1 (RL failure classes). The RL layer uses the following separately recordable failure classes; several may occur simultaneously:

- (RL-F1) **Variance failure**: a directed system is used where an inverse system is required, or conversely.
- (RL-F2) **System failure**: transition morphisms or compatibility laws are missing.
- (RL-F3) **Tail failure**: a finite plateau is used without a quantified tail theorem.
- (RL-F4) **Mittag–Leffler failure**: image stabilization is asserted but not proved.
- (RL-F5) **Ambient derived-limit nonvanishing**: a relevant $\lim_{\leftarrow}^1$ term is nonzero; this is not automatically a claim failure.
- (RL-F6) **Claim-relative edge failure**: the actual $D_{\text{edge}}(\partial) = \text{im}(\partial)$ required by the claim is nonzero.
- (RL-F7) **Derived-limit undefinedness**: the ambient category or hypotheses do not support the invoked construction.
- (RL-F8) **Edge failure**: inverse-limit exactness is used without controlling the connecting term.
- (RL-F9) **Apex failure**: the categorical limit and declared apex are not proved to agree.
- (RL-F10) **Return failure**: an infinite AK certificate is obtained but no external Return theorem is proved.
- (RL-F11) **Scope inflation**: a target-relative transfer is reported as full information preservation.

7 Classical source-binding registry

7.1 Binding policy

A source binding identifies the exact classical result consumed by RL, the convention imported from that source, and the local theorem that uses it. A binding is not a proof by authority: the principal algebraic constructions are reproduced in RL, while the source fixes provenance and guards against silent changes of indexing or variance.

Binding ID	Fixed source	RL use
RL-SB-001	Stacks Project, Tag 0594, “Mittag–Leffler systems” [1]	Definition 4.1, Theorem 4.2, and exactness under countable Mittag–Leffler control.
RL-SB-002	Stacks Project, Tag 07KV, “ $R\lim_{\leftarrow}$ of abelian groups” [2]	Derived inverse-limit convention, Roos-type derived-limit formalism, and the interpretation of the first derived edge.
RL-SB-003	Stacks Project, Homological Algebra, inverse-systems chapter [3]	Cross-check of countable inverse-system exactness and derived-limit indexing.
RL-SB-004	Stacks Project, Tag 00M9, “Completion” [4]	Definition of I -adic completion $M^\wedge = \varprojlim M/I^n M$ and the meaning of I -adic completeness.
RL-SB-005	Stacks Project, Tag 07E7, “Topological rings and modules” [5]	Separation/completeness convention for the I -adic topology.

RL-SB-008: continuous ℓ -adic cohomology. Jannsen, *Continuous ℓ -adic cohomology* (1988), together with Stacks Tag 03QQ for the equivalence between ℓ -adic cohomology of a field and continuous Galois cohomology. Role: definition and finite-field calculation lane.

RL-SB-009: derived completion. Stacks Tags 091N and 0BKF. Role: derived-completion convention and principal-ideal model.

RL-SB-010: Selmer/Iwasawa architecture. Nekovář, *Selmer complexes*, Astérisque 310 (2006). Role: architecture reference only; no control theorem from that monograph is imported into this draft.

7.2 Convention lock

RL uses towers indexed by $n \geq 0$ with transition maps $A_{n+1} \rightarrow A_n$. For completion examples the finite stage is written $M/I^{n+1}M$ so that stage zero is M/IM . Replacing this with $M/I^n M$ is harmless after reindexing, but the two conventions may not be mixed inside one certificate or source record.

8 Calibration application: I -adic completion and zero Return

8.1 Purpose

This section supplies the first complete RL recovery path. It is not a new theorem about completion. Its purpose is to verify that the RL data model can pass through an actual infinite construction without assuming the infinite certificate in advance.

Assumption 8.1 (Completion calibration data). Let R be a ring, let $I \subseteq R$ be an ideal, and let M be an R -module. Define

$$A_n := M/I^{n+1}M$$

with the canonical quotient maps

$$u_n : A_{n+1} \twoheadrightarrow A_n.$$

Let

$$\widehat{M}_I := \varprojlim_n A_n$$

be the categorical I -adic completion and let

$$\eta_M : M \rightarrow \widehat{M}_I, \quad x \mapsto (x \bmod I^{n+1}M)_n$$

be the canonical comparison map.

Proposition 8.2 (System and derived-limit clearance). *The tower $A_\bullet$ in Assumption 8.1 is a valid countable inverse system with surjective transitions. Consequently it is Mittag-Leffler and*

$$\varprojlim_n^1 A_n = 0.$$

Proof. The quotient map $M/I^{n+2}M \rightarrow M/I^{n+1}M$ is well defined and surjective because $I^{n+2}M \subseteq I^{n+1}M$. The transition maps compose canonically, so the system-validity obligation passes. Corollary 4.4 gives the derived-limit clearance.

Proposition 8.3 (Apex agreement for the categorical completion). *When the declared apex is $\widehat{M}_I := \varprojlim_n M/I^{n+1}M$, the apex comparison is the identity and*

$$D_{\text{apex}}(\text{id}_{\widehat{M}_I}) = 0.$$

If instead the declared external apex is M , then the relevant comparison is $\eta_M : M \rightarrow \widehat{M}_I$, and apex agreement is exactly the additional assertion that M is I -adically complete.

Proof. The first statement is definitional. The second is the standard definition of I -adic completeness under source binding RL-SB-004.

Proposition 8.4 (Kernel of the completion comparison). *The canonical map η_M satisfies*

$$\ker(\eta_M) = \bigcap_{n \geq 1} I^n M.$$

Hence η_M reflects zero if and only if M is I -adically separated.

Proof. An element $x \in M$ lies in $\ker(\eta_M)$ precisely when its residue class is zero in every $M/I^{n+1}M$. This is equivalent to $x \in I^{n+1}M$ for every $n \geq 0$, hence to $x \in \bigcap_{n \geq 1} I^n M$.

Definition 8.5 (Finite-stage zero certificates). For $x \in M$, the stage- n zero certificate is

$$c_n(x) : x \bmod I^{n+1}M = 0 \text{ in } A_n.$$

The family $(c_n(x))_n$ is compatible because every transition sends zero to zero. A finite initial list $c_0(x), \dots, c_N(x)$ is not an infinite certificate unless a separate theorem proves $x \in I^{n+1}M$ for every $n > N$.

Theorem 8.6 (AK–RL completion zero-transfer theorem). *Assume the completion calibration data. Let $x \in M$. If*

(i) *the finite-stage certificates form the compatible all-stage family*

$$x \bmod I^{n+1}M = 0 \quad (n \geq 0),$$

(ii) *the declared external apex is M and the Return direction requires zero reflection,*

(iii) *M is I -adically separated,*

then $x = 0$ in M .

If, in addition, M is I -adically complete, then η_M is an isomorphism and the categorical completion and external apex agree fully.

Proof. The compatible all-stage family defines the zero element in $\widehat{M_I}$ and states that $\eta_M(x) = 0$. Proposition 8.4 and separatedness imply $x = 0$. Completeness is, by Proposition 8.3, precisely the additional surjectivity and full apex-agreement condition.

Remark 8.7 (AK typing and calibration limit). The example records four distinct facts:

(i) the quotient tower is an actual inverse system;

(ii) $\varprojlim 1$ vanishes by a theorem-backed surjective-tail certificate;

(iii) the compatible residues construct the completion-side certificate;

(iv) separatedness, not derived-limit vanishing, supplies the external zero Return.

However, the proof of Theorem 8.6 does not consume the vanishing of $\varprojlim 1$: no nontrivial inverse-limit exactness edge occurs in that zero-reflection argument. The $\varprojlim 1$ computation therefore calibrates an independent RL channel but is not essential to the stated Return theorem. Consequently this example validates field separation and certificate construction, but it is not yet an essential derived-limit application. A later non-toy bridge must use a genuine connecting morphism or Milnor-type exact edge essentially.

Warning 8.8 (What this example does not prove). A finite number of congruences $x \equiv 0 \pmod{I^{n+1}M}$ never proves $x = 0$. Surjectivity of the quotient tower proves $\varprojlim A_n = 0$, but it does not prove that $M \rightarrow \widehat{M_I}$ is injective or surjective. Injectivity requires separation; surjectivity requires completeness or another theorem. These obligations cannot compensate for one another.

Corollary 8.9 (p -adic specialization). *For $R = \mathbf{Z}$, $I = (p)$, and $M = \mathbf{Z}$, the tower*

$$\mathbf{Z}/p\mathbf{Z} \leftarrow \mathbf{Z}/p^2\mathbf{Z} \leftarrow \dots$$

has vanishing $\varprojlim 1$, inverse limit $\mathbf{Z}_p$, and an injective comparison

$$\mathbf{Z} \hookrightarrow \mathbf{Z}_p.$$

Therefore, an integer whose residue is zero modulo p^n for every $n \geq 1$ is zero.

Proof. The transition maps are surjective. The inverse limit is the standard definition of $\mathbf{Z}_p$. Finally $\varprojlim_{n \geq 1} p^n \mathbf{Z} = 0$, so Theorem 8.6 applies.

9 Essential external calibration: a Milnor edge with nonzero derived defect

9.1 Purpose and external status

This section leaves the module-level completion regime and calibrates RL against an inverse tower in the external derived category $D(\mathbf{Ab})$. The example is classical rather than new, but it is essential in the precise sense missing from Section 8: every finite stage has vanishing degree-one cohomology, while the derived inverse limit has nonzero degree-one cohomology, and that class is supplied entirely by $R^1\lim$.

Thus the example proves that the derived-limit channel is not optional bookkeeping for any future AK bridge using such a derived inverse limit. Deleting it produces a false finite-to-infinite conclusion in the external calibration problem. No claim is made here that the complexes K_n have already been realized as frozen AK objects or that an AK Eval verdict has been computed.

9.2 The tower

Definition 9.1 (Doubling tower in the derived category). For every $n \geq 0$, let

$$K_n := \mathbf{Z}[0] \in D(\mathbf{Ab}),$$

the complex with $\mathbf{Z}$ in cohomological degree zero and zero elsewhere. Let the transition morphism

$$f_n : K_{n+1} \rightarrow K_n$$

be multiplication by 2. Let

$$K := \varprojlim_n K_n$$

be the derived inverse limit in $D(\mathbf{Ab})$.

Proposition 9.2 (Finite-stage verdict). For every n ,

$$H^0(K_n) \cong \mathbf{Z}, \quad H^q(K_n) = 0 \quad (q \neq 0).$$

In particular, every finite stage has the valid degree-one certificate

$$H^1(K_n) = 0.$$

Proof. This is immediate from the definition of $\mathbf{Z}[0]$.

9.3 The nonzero derived-limit term

Proposition 9.3 (Ordinary inverse limit of the doubling tower). For the inverse system

$$\mathbf{Z} \xleftarrow{2} \mathbf{Z} \xleftarrow{2} \mathbf{Z} \xleftarrow{2} \dots,$$

one has

$$\varprojlim_n \mathbf{Z} = 0.$$

Proof. A compatible family (a_n) satisfies $a_n = 2a_{n+1}$ for every n . Hence a_0 is divisible by 2^m for all m , so $a_0 = 0$, and then every $a_n = 0$.

Theorem 9.4 (Explicit nonvanishing of the first derived limit). *For the same tower,*

$$\varprojlim_n^1 \mathbf{Z} \simeq \mathbf{Z}_2 / \mathbf{Z},$$

where $\mathbf{Z}$ is embedded diagonally in the ring $\mathbf{Z}_2$ of 2-adic integers. In particular,

$$\varprojlim_n^1 \mathbf{Z} / \mathbf{Z} = 0.$$

Proof. The Roos map is

$$\Delta : \prod_{n \geq 0} \mathbf{Z} \rightarrow \prod_{n \geq 0} \mathbf{Z}, \quad \Delta((a_n)) = (a_n - 2a_{n+1})_n.$$

Define

$$\Phi : \prod_{n \geq 0} \mathbf{Z} \rightarrow \mathbf{Z}_2, \quad \Phi((b_n)) := \sum_{n \geq 0} 2^n b_n.$$

The series converges 2-adically. For $b = \Delta(a)$, its partial sums telescope:

$$\sum_{n=0}^N 2^n (a_n - 2a_{n+1}) = a_0 - 2^{N+1} a_{N+1},$$

and the final term tends to zero in $\mathbf{Z}_2$. Hence $\Phi(\text{im } \Delta) \subseteq \mathbf{Z}$, so Φ induces

$$\Phi : \text{coker } \Delta \rightarrow \mathbf{Z}_2 / \mathbf{Z}.$$

Every 2-adic integer has a digit expansion $\sum_{n \geq 0} 2^n \epsilon_n$ with $\epsilon_n \in \{0, 1\}$, so Φ is surjective.

To prove injectivity, suppose $\Phi((b_n)) = m \in \mathbf{Z}$. Set $a_0 = m$ and recursively

$$a_{n+1} := \frac{a_n - b_n}{2}.$$

The congruence $m - \sum_{j=0}^n 2^j b_j \in 2^{n+1} \mathbf{Z}_2 \cap \mathbf{Z} = 2^{n+1} \mathbf{Z}$ shows that every a_{n+1} is an integer. Then $b_n = a_n - 2a_{n+1}$, so $(b_n) \in \text{im } \Delta$. Therefore Φ is an isomorphism. Theorem 3.6 identifies $\text{coker } \Delta$ with $\varprojlim_n^1 \mathbf{Z}$.

9.4 Milnor exact sequence and the infinite class

Theorem 9.5 (Milnor edge for the doubling tower). *The canonical Milnor short exact sequence gives*

$$0 \rightarrow \varprojlim_n^1 H^0(K_n) \rightarrow H^1(K) \rightarrow \varprojlim_n H^1(K_n) \rightarrow 0.$$

Consequently,

$$H^1(K) \cong \mathbf{Z}_2 / \mathbf{Z} = 0.$$

Moreover,

$$H^0(K) = 0.$$

Proof. By source binding RL-SB-007, for every inverse system in $D(\mathbf{Ab})$ and every p there is a canonical exact sequence

$$0 \rightarrow R^1 \varprojlim H^{p-1}(K_n) \rightarrow H^p(\mathbf{R} \varprojlim K_n) \rightarrow \varprojlim H^p(K_n) \rightarrow 0.$$

For $p = 1$, Proposition 9.2 gives $H^1(K_n) = 0$, while Theorem 9.4 gives

$$R^1 \varprojlim H^0(K_n) = \varprojlim_n^1 \mathbf{Z} \simeq \mathbf{Z}_2 / \mathbf{Z}.$$

Hence the left injection is an isomorphism onto $H^1(K)$. For $p = 0$, the $R^1 \varprojlim H^{-1}(K_n)$ term is zero and Proposition 9.3 gives $H^0(K) = 0$.

Corollary 9.6 (Finite-stage vanishing does not transfer). *The implication*

$$\left(H^1(K_n) = 0 \text{ for every } n \right) \implies H^1(\mathbf{R}\lim_{\leftarrow} K_n) = 0$$

is false.

Proof. The premise holds by Proposition 9.2, while the conclusion fails by Theorem 9.5.

9.5 AK–RL interpretation

Proposition 9.7 (Essential consumption of the RL defect channel). *For the target predicate*

$$Q(K) : \quad H^1(K) = 0,$$

the degree-one finite-stage vanishing statements are levelwise valid and trivially compatible as zero statements, but they are insufficient. The exact infinite certificate is controlled by the Milnor edge

$$0 \rightarrow \lim_{\leftarrow}^1 H^0(K_n) \rightarrow H^1(K) \rightarrow \lim_{\leftarrow} H^1(K_n) \rightarrow 0,$$

and the nonzero ambient derived-limit object maps isomorphically into the target group. Therefore the derived-limit term is essential to the external verdict. This establishes a necessary AK–RL audit obligation, not an AK-internal certificate by itself.

Proof. The right-hand inverse limit is zero because every $H^1(K_n)$ is zero. Theorem 9.5 identifies the entire target group with the left-hand derived-limit term. Omitting that term would incorrectly certify $Q(K)$.

Remark 9.8 (Initial-reader interpretation). Every finite floor of the tower has “nothing in degree one.” Nevertheless the infinite derived object contains a degree-one class. The class is not present on any individual floor; it records the impossibility of choosing the finite-stage data compatibly through the whole infinite tail. In RL language, this is exactly the information carried by $\lim_{\leftarrow}^1$.

Remark 9.9 (What has and has not been achieved). This is a genuine external derived-category calibration and an essential use of $\lim_{\leftarrow}^1$, not merely a parallel calculation. It is still a classical recovery example, not a new theorem, not an AK-internal theorem, and not yet an arithmetic bridge. It validates the necessity and typing of the RL architecture. A genuine AK bridge additionally requires a realization theorem into declared AK objects, an AK-internal predicate or certificate, an apex comparison, and a separately proved Return theorem.

10 Restricted AK–RL Milnor bridge

10.1 Exact-sequence calibration

Definition 10.1 (Milnor-admissible tower). Fix an integer p . A *Milnor-admissible tower in degree p* is a countable inverse tower

$$K_0 \leftarrow K_1 \leftarrow K_2 \leftarrow \dots$$

in $D(\mathbf{Ab})$ for which a derived limit exists and the canonical Milnor short exact sequence is available:

$$0 \rightarrow R^1 \lim_{\leftarrow} H^{p-1}(K_n) \xrightarrow{\iota_p} H^p(\mathbf{R}\lim_{\leftarrow} K_n) \xrightarrow{c_p} \lim_{\leftarrow} H^p(K_n) \rightarrow 0.$$

The external target predicate is

$$Q_p(K_\bullet) : \quad H^p(\mathbf{R}\lim_{\leftarrow} K_n) = 0.$$

Theorem 10.2 (Automatic Milnor admissibility in $D(\mathbf{Ab})$). *Every countable inverse tower of objects of $D(\mathbf{Ab})$ admits a derived limit and, for every $p \in \mathbf{Z}$, has the canonical short exact sequence*

$$0 \rightarrow R^1 \lim_{\leftarrow} H^{p-1}(K_n) \rightarrow H^p(\mathbf{R} \lim_{\leftarrow} K_n) \rightarrow \lim_{\leftarrow} H^p(K_n) \rightarrow 0.$$

Thus, within the declared $D(\mathbf{Ab})$ regime, “Milnor-admissible” is not an extra mathematical hypothesis; it is a typing reminder fixing the chosen derived-limit model and comparison map.

Proof. Countable products exist in $D(\mathbf{Ab})$, so the derived limit is defined by the standard distinguished triangle

$$\mathbf{R} \lim_{\leftarrow} K_n \rightarrow \prod_n K_n \rightarrow \prod_n K_n \rightarrow (\mathbf{R} \lim_{\leftarrow} K_n)[1].$$

Products are exact in $\mathbf{Ab}$, hence cohomology commutes with these countable products. The resulting long exact cohomology sequence identifies the relevant kernel and cokernel with $\lim$ and $R^1 \lim$ through the Roos difference map, and breaks into the displayed short exact sequences. This is precisely the construction bound to Stacks Project Tags 07KV and 08U5.

Definition 10.3 (Milnor tuple calibration). For a Milnor-admissible tower put

$$F_p := \lim_{\leftarrow} H^p(K_n), \quad E_p := R^1 \lim_{\leftarrow} H^{p-1}(K_n), \quad A_p := H^p(\mathbf{R} \lim_{\leftarrow} K_n).$$

The tuple

$$(E_p \xrightarrow{\iota_p} A_p \xrightarrow{c_p} F_p)$$

is called the *Milnor tuple calibration*. It is an externally supplied exact-sequence profile, not yet the independent AK realization constructed below.

Proposition 10.4 (Formal exact-sequence bridge). *For every Milnor-admissible tower,*

$$E_p = 0 \text{ and } F_p = 0 \iff A_p = 0.$$

Proof. This follows from injectivity of ι_p , surjectivity of c_p , and exactness at A_p .

Remark 10.5 (Status of the calibration). Proposition 10.4 is mathematically non-vacuous because it consumes the external Milnor exactness theorem. Nevertheless its carriers are read directly from that same exact sequence. It is therefore classified only as a formal exact-sequence calibration. The independent bridge begins in Section 11.

11 Canonical edge–Roos AK bridge

11.1 The actual comparison morphism

Definition 11.1 (Canonical cohomological edge map). For a Milnor-admissible tower define the actual comparison morphism

$$c_p(K_\bullet) : H^p(\mathbf{R} \lim_{\leftarrow} K_n) \rightarrow \lim_{\leftarrow} H^p(K_n)$$

to be the canonical map induced by the projections $\mathbf{R} \lim_{\leftarrow} K_n \rightarrow K_n$. Set

$$A_p(K_\bullet) := H^p(\mathbf{R} \lim_{\leftarrow} K_n), \quad F_p(K_\bullet) := \lim_{\leftarrow} H^p(K_n).$$

Definition 11.2 (Canonical edge-comparison artifact). The degree- p edge-comparison artifact is

$$C_p(K_\bullet) := \left(A_p(K_\bullet), F_p(K_\bullet), c_p(K_\bullet), D_p^-(K_\bullet), D_p^+(K_\bullet) \right),$$

where

$$D_p^-(K_\bullet) := \ker c_p(K_\bullet), \quad D_p^+(K_\bullet) := \operatorname{coker} c_p(K_\bullet).$$

This is the RL analogue of an actual-map kernel/cokernel audit. It is not identified with frozen Type IV, whose variance and source construction are different.

Theorem 11.3 (Milnor identification of the edge defects). *For a Milnor-admissible countable tower,*

$$D_p^-(K_\bullet) \cong R^1 \varprojlim H^{p-1}(K_n), \quad D_p^+(K_\bullet) = 0.$$

Proof. These are exactly the kernel and cokernel statements of the Milnor short exact sequence in Definition 10.1.

11.2 Independent Roos-complex realization

Definition 11.4 (Cohomological Roos complex). For a tower $K_\bullet$ and degree p , define the two-term cochain complex

$$\mathbf{R}_p(K_\bullet) := \left[\prod_{n \geq 0} H^{p-1}(K_n) \xrightarrow{\Delta_{p-1}} \prod_{n \geq 0} H^{p-1}(K_n) \right],$$

placed in cohomological degrees 0 and 1, where

$$\Delta_{p-1}((x_n)) = (x_n - H^{p-1}(f_n)(x_{n+1}))_n.$$

The complex is constructed directly from the transition maps of the cohomology tower. It does not use A_p , the Milnor injection, or the Milnor short exact sequence.

Definition 11.5 (Independent Roos edge defect). Define

$$D_p^{\text{Roos}}(K_\bullet) := H^1(\mathbf{R}_p(K_\bullet)) = \operatorname{coker}(\Delta_{p-1}).$$

This is the independently constructed AK–RL defect carrier.

Proposition 11.6 (Internal computation of the Roos artifact). *There are canonical identifications*

$$H^0(\mathbf{R}_p(K_\bullet)) \cong \varprojlim H^{p-1}(K_n), \quad H^1(\mathbf{R}_p(K_\bullet)) \cong R^1 \varprojlim H^{p-1}(K_n).$$

Proof. The first statement is the kernel description of the inverse limit. The second is the countable Roos identification of Theorem 3.6.

Theorem 11.7 (Independent defect-to-edge comparison). *For every Milnor-admissible tower there is a canonical isomorphism*

$$\kappa_p(K_\bullet) : D_p^{\text{Roos}}(K_\bullet) \xrightarrow{\sim} D_p^-(K_\bullet) = \ker c_p(K_\bullet).$$

Proof. Proposition 11.6 canonically identifies the source with $R^1 \varprojlim H^{p-1}(K_n)$. Theorem 11.3 canonically identifies that derived limit with the kernel of the actual comparison map c_p . Their composite is κ_p .

Remark 11.8 (Why this is stronger than the Milnor tuple). The defect object D_p^{Roos} is constructed before the apex group and before the edge exact sequence are used. Theorem 11.7 is therefore a comparison theorem between an independently generated AK–RL artifact and the externally meaningful kernel of the apex-to-underived-limit map. This removes the principal carrier-circularity of the earlier tuple-only bridge. It does not make the construction independent of classical derived-limit theory: the comparison isomorphism still consumes the standard Roos model and the Milnor edge theorem.

11.3 Naturality and invariance

Theorem 11.9 (Naturality of the canonical edge artifact). *Let $g_\bullet : K_\bullet \rightarrow L_\bullet$ be a morphism of Milnor-admissible towers. Then the square*

$$\begin{array}{ccc} H^p(\mathbf{R}\lim_{\leftarrow} K_n) & \xrightarrow{c_p(K_\bullet)} & \lim_{\leftarrow} H^p_y(K_n) \\ H^p(\mathbf{R}\lim_{\leftarrow} g_\bullet) \downarrow & & \downarrow \lim_{\leftarrow} H^p(g_n) \\ H^p(\mathbf{R}\lim_{\leftarrow} L_n) & \xrightarrow{c_p(L_\bullet)} & \lim_{\leftarrow} H^p(L_n) \end{array} \quad \downarrow$$

commutes. Consequently $g_\bullet$ induces maps on D_p^- and D_p^+ .

Proof. Both composites are induced by the same compatible family of maps from the derived limit of $K_\bullet$ to the stages of $L_\bullet$. Functoriality of cohomology and the universal property of the inverse limit give commutativity.

Theorem 11.10 (Naturality of the Roos comparison). *A tower morphism $g_\bullet$ induces a cochain map*

$$\mathbf{R}_p(g_\bullet) : \mathbf{R}_p(K_\bullet) \rightarrow \mathbf{R}_p(L_\bullet),$$

and the comparison isomorphisms satisfy

$$\kappa_p(L_\bullet) \circ H^1(\mathbf{R}_p(g_\bullet)) = D_p^-(g_\bullet) \circ \kappa_p(K_\bullet).$$

Proof. The componentwise maps on H^{p-1} commute with the tower transitions, hence with the Roos difference maps. Naturality of the derived-limit identification and of the Milnor exact sequence gives the displayed identity.

Corollary 11.11 (Levelwise quasi-isomorphism invariance). *If $g_\bullet : K_\bullet \rightarrow L_\bullet$ is levelwise a quasi-isomorphism, then it induces isomorphisms*

$$F_p(K_\bullet) \cong F_p(L_\bullet), \quad D_p^{\text{Roos}}(K_\bullet) \cong D_p^{\text{Roos}}(L_\bullet), \quad A_p(K_\bullet) \cong A_p(L_\bullet),$$

and identifies their edge-comparison artifacts.

Proof. Levelwise quasi-isomorphisms induce isomorphisms on all cohomology towers. They induce an isomorphism of Roos complexes and, in the derived category, an isomorphism of the corresponding derived limits. The conclusion follows from Theorems 11.9 and 11.10.

Proposition 11.12 (Tail invariance). *Deleting finitely many initial stages of a countable tower does not change*

$$F_p, \quad D_p^{\text{Roos}}, \quad A_p,$$

up to canonical isomorphism, and hence does not change the edge verdict.

Proof. Let $j_N : \mathbf{N}^{op} \rightarrow \mathbf{N}^{op}$ be the shift $n \mapsto n + N$. This functor is cofinal for inverse limits. The restriction morphism from the full tower to its N th tail therefore induces isomorphisms on $\lim$ and $R^1\lim$. By Theorem 10.2, the same restriction induces an isomorphism on the derived limits. Naturality of the Milnor edge sequence then identifies F_p , D_p^{Roos} , A_p , and the comparison map. This proof uses cofinality and naturality only; no unproved chain-level “finite contractible correction” is required.

Proposition 11.13 (Split pro-isomorphism invariance). *A morphism of pro-systems of abelian groups induces canonical maps on $\lim$ and $R^1\lim$, and a pro-isomorphism induces isomorphisms on both. For towers in a triangulated category, a pro-isomorphism can be used to produce an isomorphism between derived-limit objects, but that apex isomorphism need not be canonical.*

Proof. The first assertion follows by applying the explicit Roos product maps associated with a representative of the pro-morphism; independence of the representative and functoriality give canonical maps on the two cohomology objects. A pro-inverse makes them isomorphisms. The derived-limit assertion is obtained by extending the product maps to a morphism of defining distinguished triangles. The extension exists and is an isomorphism, but the triangulated-category completion is not unique in general, so no canonical apex identification is asserted.

Warning 11.14 (Pro-isomorphism audit policy). RL distinguishes canonical invariant-level transport from noncanonical apex transport. A release may identify $\lim$ and $R^1\lim$ across a specified pro-isomorphism, but it must not silently identify the derived-limit apex or its actual comparison map. The representative tower, the pro-map, and any chosen apex isomorphism must remain in the provenance record.

11.4 Sound sufficient-condition detectors

Definition 11.15 (Sound RL detector). A sound RL detector is a finite or explicitly checkable certificate whose success implies a stated defect verdict. Failure of the detector means only **not certified**; it does not imply that the defect is nonzero.

Theorem 11.16 (Certified defect-clearance detectors). *For the cohomology tower $H^{p-1}(K_n)$, each of the following is a sound sufficient certificate for $D_p^{\text{Roos}}(K_\bullet) = 0$:*

- (i) *all transition maps are surjective;*
- (ii) *the tower satisfies the Mittag-Leffler condition;*
- (iii) *the tower is eventually constant;*
- (iv) *every $H^{p-1}(K_n)$ is finite-dimensional over a fixed field;*
- (v) *an explicit right inverse to the Roos difference map Δ_{p-1} is supplied.*

Proof. Cases (i)–(iii) are the classical surjective, Mittag-Leffler, and eventual-constancy criteria. In case (iv), for each fixed n the descending chain of subspaces $\text{im}(H^{p-1}(K_m) \rightarrow H^{p-1}(K_n))$ stabilizes because $H^{p-1}(K_n)$ is finite-dimensional; hence the tower is Mittag-Leffler. In case (v), surjectivity of Δ_{p-1} gives zero cokernel by definition.

Proposition 11.17 (Finite-prefix impossibility). *For every $N \geq 0$ there exist two countable inverse systems of abelian groups that agree through stage N , including every displayed object and transition map in that prefix, but whose first derived inverse limits are respectively zero and nonzero. Hence no sound general procedure that reads only a fixed finite prefix can decide vanishing of $R^1\lim$.*

Proof. Let $B_\bullet$ be the zero tower. Let $A_n = 0$ for $n \leq N$, let $A_n = \mathbf{Z}$ for $n > N$, take the transition $A_{N+1} \rightarrow A_N$ to be zero, and take every later transition to be multiplication by 2. The two towers have identical prefixes through stage N . The zero tower has vanishing $R^1 \lim_{\leftarrow} A_n$. By tail invariance, $R^1 \lim_{\leftarrow} A_n$ is the same as for the doubling tower, hence is $\mathbf{Z}_2/\mathbf{Z} = 0$ by Theorem 9.4.

Remark 11.18 (No complete finite detector is claimed). Theorem 11.16 deliberately lists sufficient conditions only. Proposition 11.17 proves, rather than merely asserts, the obstruction to a general fixed-prefix decision rule. This preserves the frozen AK rule that finite observation is not infinite proof.

11.5 Strengthened independent Bridge theorem

Definition 11.19 (Independent edge–Roos evaluation). Define

$$\text{Eval}_p^{\text{ER}}(K_\bullet) := \left(D_p^{\text{Roos}}(K_\bullet), F_p(K_\bullet) \right).$$

The zero verdict means simultaneous vanishing of the independently constructed Roos defect and the underived inverse-limit carrier.

Theorem 11.20 (Restricted independent AK–RL edge–Roos Bridge). *For every Milnor-admissible tower in degree p ,*

$$\text{Eval}_p^{\text{ER}}(K_\bullet) = 0 \iff H^p(\mathbf{R} \lim_{\leftarrow} K_n) = 0.$$

Equivalently,

$$D_p^{\text{Roos}}(K_\bullet) = 0 \text{ and } \lim_{\leftarrow} H^p(K_n) = 0 \iff Q_p(K_\bullet).$$

Proof. Theorem 11.7 identifies D_p^{Roos} with $\ker c_p$, while Theorem 11.3 gives $\text{coker } c_p = 0$. Thus

$$0 \rightarrow D_p^{\text{Roos}} \rightarrow A_p \xrightarrow{c_p} F_p \rightarrow 0$$

is exact. The same two-sided zero argument as in Proposition 10.4 yields the equivalence.

Corollary 11.21 (Independent Return theorem). *A zero edge–Roos evaluation returns the external apex predicate:*

$$\text{Eval}_p^{\text{ER}}(K_\bullet) = 0 \implies H^p(\mathbf{R} \lim_{\leftarrow} K_n) = 0.$$

Conversely, apex vanishing forces both internal carriers to vanish.

Theorem 11.22 (Execution on the doubling tower). *For the doubling tower of Definition 9.1 and $P = 1$,*

$$F_1 = 0, \quad D_1^{\text{Roos}} \cong \mathbf{Z}_2/\mathbf{Z} = 0,$$

and hence

$$\text{Eval}_1^{\text{ER}}(K_\bullet) = 0.$$

The Bridge theorem therefore returns

$$H^1(\mathbf{R} \lim_{\leftarrow} K_n) = 0.$$

Proof. The finite carrier is zero by Proposition 9.2. The Roos defect is $\mathbf{Z}_2/\mathbf{Z}$ by Theorem 9.4 and Proposition 11.6. The conclusion follows from Theorem 11.20.

Remark 11.23 (Exact connection status after strengthening). The artifact $\mathbf{R}_p(K_\bullet)$ is constructed directly from the tower, while c_p is the actual external apex comparison map. Theorem 11.7 supplies the nontrivial comparison between them. Therefore the bridge is stronger than a tuple-level repackaging of the Milnor short exact sequence. Nevertheless, the Roos complex is a classical model of $R^1\text{lim}$, not a novel frozen-v20 detector, and the comparison theorem still relies on classical Milnor exactness. It remains a restricted successor bridge for sequential derived towers; it is not yet an encoding into the frozen v20 one-parameter persistence detector calculus.

12 Finite-field continuous e -tale realizations and calibration examples

12.1 Fixed site and derived-global-sections convention

Fix a finite field $k = \mathbf{F}_q$, a prime $\ell = \text{char}(k)$, and

$$X = \text{Spec}(k), \quad X'_{et}$$

with its small e -tale site. The absolute Galois group is

$$G_k = \text{Gal}(k/k) \cong \hat{\mathbf{Z}},$$

topologically generated by arithmetic Frobenius. The standard equivalence between abelian sheaves on X'_{et} and discrete continuous G_k -modules identifies

$$R\Gamma(X'_{et}, \mathcal{F})$$

with continuous Galois cohomology. For a countable inverse coefficient system $\mathcal{F}_\bullet$, define the continuous apex complex by

$$R\Gamma_{\text{cont}}(X, \mathcal{F}_\bullet) := \mathbf{R}\varprojlim_n R\Gamma(X'_{et}, \mathcal{F}_n).$$

This is the derived-global-sections model used below. It is a concrete specialization of the continuous-cohomology formalism, not a claim that every competing ℓ -adic formalism is definitionally identical without comparison.

12.2 Standard constant ℓ -adic system

For $n \geq 1$, let

$$\mathcal{F}_n = \mathbf{Z}/\ell^n \mathbf{Z}$$

with trivial G_k -action and transition maps the reductions

$$\rho_n : \mathbf{Z}/\ell^{n+1} \mathbf{Z} \twoheadrightarrow \mathbf{Z}/\ell^n \mathbf{Z}.$$

Proposition 12.1 (Finite-stage cohomology of the standard system). *For every $n \geq 1$,*

$$H^0(X'_{et}, \mathcal{F}_n) \cong \mathbf{Z}/\ell^n \mathbf{Z}, \quad H^1(X'_{et}, \mathcal{F}_n) \cong \mathbf{Z}/\ell^n \mathbf{Z},$$

and

$$H^i(X'_{et}, \mathcal{F}_n) = 0 \quad (i \geq 2).$$

Under these identifications, the transition maps on H^0 and H^1 are the usual reductions.

Proof. For a trivial finite discrete $G_k \cong \bar{\mathbf{Z}}$ -module A , one has $H^0(G_k, A) = A$ and

$$H^1(G_k, A) = \text{Hom}_{\text{cont}}(\bar{\mathbf{Z}}, A) \cong A,$$

by sending a continuous homomorphism to the image of a topological generator. The ℓ -cohomological dimension of a finite field is one, hence the higher ℓ -primary groups vanish. Naturality gives the transition maps.

Theorem 12.2 (Standard ℓ -adic continuous apex computation). *For the system $\mathcal{F}_\bullet$ above,*

$$H_{\text{cont}}^0(X, \mathcal{F}_\bullet) \simeq \mathbf{Z}_\ell, \quad H_{\text{cont}}^1(X, \mathcal{F}_\bullet) \simeq \mathbf{Z}_\ell,$$

and

$$H_{\text{cont}}^i(X, \mathcal{F}_\bullet) = 0 \quad (i \geq 2).$$

For $q = 0, 1$, the edge map

$$c_q^{\text{cont}} : H_{\text{cont}}^q(X, \mathcal{F}_\bullet) \rightarrow \varprojlim_n H^q(X'_{et}, \mathcal{F}_n)$$

is an isomorphism and the Roos defect vanishes.

Proof. The transition maps on H^0 and H^1 are surjective, hence both towers are Mittag-Leffler and their first derived inverse limits vanish. Their ordinary inverse limits are $\mathbf{Z}_\ell$. Apply the canonical Milnor short exact sequence in each degree. For $i \geq 2$, both the finite carrier and the preceding derived-limit carrier vanish.

Corollary 12.3 (AK Return for the standard system). *The degreewise AK–RL record*

$$\left(D_q^{\text{Roos}}, \varprojlim_n H^q(X'_{et}, \mathcal{F}_n), c_q^{\text{cont}} \right)$$

returns the exact external verdict

$$H_{\text{cont}}^q(X, \mathcal{F}_\bullet) \simeq \begin{cases} \mathbf{Z}_\ell, & q = 0, 1, \\ 0, & q \geq 2. \end{cases}$$

This is a completed zero-defect realization in the fixed finite-field domain. It verifies the interface on a standard coefficient tower, but because the edge defect vanishes it is not by itself the nonzero-defect witness required for promotion.

12.3 A non-Mittag-Leffler constant system

Let

$$\mathcal{G}_n = \mathbf{Z}$$

for all $n \geq 0$, with trivial G_k -action and transition maps

$$\mathcal{G}_{n+1} \xrightarrow{\times \ell} \mathcal{G}_n.$$

This is an inverse system of abelian sheaves on X'_{et} , although it is not the standard constructible ℓ -adic coefficient system above.

Proposition 12.4 (Finite stages are degree-one acyclic). *For every n ,*

$$H^0(X'_{et}, \mathcal{G}_n) = \mathbf{Z}, \quad H^1(X'_{et}, \mathcal{G}_n) = 0.$$

Proof. The degree-zero statement is immediate. Moreover

$$H^1(X'_{et}, \mathbf{Z}) \cong \text{Hom}_{\text{cont}}(G_k, \mathbf{Z}) = 0,$$

because the continuous image of the compact profinite group G_k in the discrete group $\mathbf{Z}$ is finite, while $\mathbf{Z}$ has no nonzero finite subgroup.

Theorem 12.5 (Essential Jannsen-style edge defect). *For the system $\mathcal{G}_\bullet$,*

$$\varprojlim_n H^1(X'_{et}, \mathcal{G}_n) = 0,$$

but

$$D_1^{\text{Roos}}(R\Gamma(X, \mathcal{G}_\bullet)) \cong \varprojlim_n^1 \left(\mathbf{Z} \xleftarrow{\times \ell} \mathbf{Z} \xleftarrow{\times \ell} \cdots \right) \cong \mathbf{Z}_\ell / \mathbf{Z}.$$

Consequently,

$$H_{\text{cont}}^1(X, \mathcal{G}_\bullet) \simeq \mathbf{Z}_\ell / \mathbf{Z} = 0.$$

Proof. The finite carrier is zero by Proposition 12.4. The Roos computation is the same digit-expansion computation as in Theorem 9.4, with 2 replaced by ℓ : the homomorphism

$$(b_n)_{n \geq 0} \mapsto \sum_{n \geq 0} \ell^n b_n$$

induces an isomorphism from the Roos cokernel to $\mathbf{Z}_\ell / \mathbf{Z}$. The degree-one Milnor sequence then reduces to an isomorphism between this Roos defect and the continuous apex group.

Corollary 12.6 (Nonzero site-instantiated calibration Return). *The external predicate*

$$Q_{\text{cont},1}(\mathcal{G}_\bullet) : H_{\text{cont}}^1(X, \mathcal{G}_\bullet) = 0$$

is false, although every finite-stage degree-one group vanishes. The AK–RL Return is determined entirely by the nonzero Roos defect.

Remark 12.7 (Exact status of the example). Theorem 12.5 is a genuine calculation on the $'_e$ site of a fixed scheme, with a fixed coefficient system and fixed transition maps. Its finite-stage complexes are derived global sections on X'_{et} and its apex is the derived inverse-limit cohomology of that sheaf tower. However, after the finite-stage calculation the relevant degree-zero tower is exactly the classical multiplication-by- ℓ tower on $\mathbf{Z}$. The example is therefore a site-instantiated essential calibration, not an independent arithmetic phenomenon and not a sufficient non-toy promotion witness. Moreover, $\mathbf{Z} \xleftarrow{\times \ell} \mathbf{Z}$ is not the standard constructible ℓ -adic coefficient system. The standard ℓ -adic system is treated separately in Theorem 12.2.

13 Derived reduction-limit Bridge and comparison with derived completion

13.1 General edge package

Let A be a ring, $I \subset A$ an ideal, and $M \in D(A)$. Define the derived reduction-limit object

$$K_n(M) := M \otimes_A^{\mathbf{L}} A/I^n, \quad L_I^{\text{red}}(M) := \mathbf{R}\varprojlim_n K_n(M).$$

A separate comparison theorem is required before $L_I^{\text{red}}(M)$ is identified with a preferred notion of derived I -adic completion for arbitrary (A, I, M) . In the principal-ideal example

below, the displayed reduction-limit model is the intended derived-completion model. For each q there is a canonical edge package

$$0 \rightarrow \varprojlim_{\leftarrow} {}^1 H^{q-1}(K_n(M)) \rightarrow H^q(L_I^{\text{red}}(M)) \rightarrow \varprojlim_{\leftarrow} H^q(K_n(M)) \rightarrow 0.$$

Thus define

$$D_{I,q}^{\text{comp}}(M) := H^1 \left(\text{Roos}(H^{q-1}(K_{\bullet}(M))) \right).$$

Proposition 13.1 (Completion Return schema). *For every countable derived-reduction tower for which the displayed derived inverse limit exists,*

$$H^q(L_I^{\text{red}}(M)) = 0 \iff D_{I,q}^{\text{comp}}(M) = 0 \text{ and } \varprojlim_{\leftarrow} H^q(K_n(M)) = 0.$$

Proof. This is Theorem 11.20 applied to the tower $K_n(M)$.

13.2 A concrete derived completion invisible to classical completion

Fix a prime p , take

$$A = \mathbf{Z}, \quad I = (p), \quad M = \mathbf{Q}_p/\mathbf{Z}_p$$

as a discrete $\mathbf{Z}$ -module in degree zero.

Lemma 13.2 (Finite derived reductions). *For $n \geq 1$,*

$$K_n(M) = M \otimes_{\mathbf{Z}}^{\mathbf{L}} \mathbf{Z}/p^n \mathbf{Z}$$

has

$$H^{-1}(K_n(M)) \cong M[p^n] \cong \mathbf{Z}/p^n \mathbf{Z}, \quad H^0(K_n(M)) \cong M/p^n M = 0,$$

and no other cohomology. Under the canonical transition $K_{n+1}(M) \rightarrow K_n(M)$, the degree -1 map is the usual reduction under the identifications above.

Proof. Use the two-term free resolution

$$[\mathbf{Z} \xrightarrow{p^n} \mathbf{Z}]$$

of $\mathbf{Z}/p^n \mathbf{Z}$. Tensoring with M gives $[M \xrightarrow{p^n} M]$ in degrees $-1, 0$. Since M is p -divisible, the cokernel is zero, while the kernel is $M[p^n]$. A chain map from the $(n+1)$ -resolution to the n -resolution is multiplication by p in degree -1 and the identity in degree zero; on the compatible torsion generators this induces reduction.

Theorem 13.3 (Derived reduction-limit computation for $\mathbf{Q}_p/\mathbf{Z}_p$). *There is a quasi-isomorphism*

$$L_{(p)}^{\text{red}}(\mathbf{Q}_p/\mathbf{Z}_p) \simeq \mathbf{Z}_p[1].$$

Equivalently,

$$\left(\begin{array}{c} \\ H^{-1} L_{(p)}^{\text{red}}(M) \end{array} \right) \cong \mathbf{Z}_p, \quad \left(\begin{array}{c} \\ H^q L_{(p)}^{\text{red}}(M) \end{array} \right) = 0 \quad (q \neq -1).$$

The classical completion is instead

$$\varprojlim_n M/p^n M = 0.$$

Proof. By Lemma 13.2, the degree -1 tower is the surjective system $\mathbf{Z}/p^n \mathbf{Z}$, whose inverse limit is $\mathbf{Z}_p$ and whose first derived limit vanishes. The degree-zero tower is zero. The Milnor edge sequences therefore give the stated cohomology. Since the apex has only one nonzero cohomology group, it is quasi-isomorphic to $\mathbf{Z}_p[1]$. Classical completion vanishes because $p^n M = M$.

Remark 13.4 (What is and is not a Roos defect here). The nonzero class in Theorem 13.3 is carried by the ordinary inverse limit of the finite-stage Tor_1 groups, not by a nonzero $\lim^1$. Thus derived tensoring is essential, while the Roos edge itself is clear. This example proves that the derived-completion Bridge must monitor both the finite carrier in every cohomological degree and the Roos defect; it would be incorrect to identify “derived” solely with “nonzero $\lim^1$ ”.

Corollary 13.5 (AK completion Return). *The ordinary-completion verdict “zero” does not Return the derived reduction-limit verdict. The typed AK record must retain the degree -1 finite carrier*

$$\lim_{\leftarrow n} \mathrm{Tor}_1^{\mathbf{Z}}(M, \mathbf{Z}/p^n) \simeq \mathbf{Z}_p.$$

With that carrier included, the Edge–Roos Return recovers the exact apex $\mathbf{Z}_p[1]$.

14 Iwasawa-cohomology interface

14.1 Fixed derived inverse-limit architecture

Let F be a number field, let p be a prime, and let

$$F = F_0 \subset F_1 \subset \cdots \subset F_\infty$$

be a fixed $\mathbf{Z}_p$ -extension with

$$\Gamma = \mathrm{Gal}(F_\infty/F), \quad \Lambda = \mathbf{Z}_p[[\Gamma]].$$

Fix a finite set S containing the archimedean places, the places above p , and all places at which the chosen representation is ramified. Let T be a finite free $\mathbf{Z}_p$ -module with continuous $G_{F,S}$ -action. Define

$$K_n(T) := R\Gamma(G_{F_n,S}, T)$$

and use corestriction to form the inverse tower. Assume that explicit cochain models and compatible complex-level corestriction morphisms have been chosen, so that $K_\bullet(T)$ is an actual inverse tower. Define the sequential RL tower apex

$$R\Gamma_{\mathrm{RL-Iw}}(F_\infty/F, T) := \mathbf{R}\lim_{\leftarrow n} K_n(T).$$

This object is first taken in the underlying derived category of abelian groups. It may be identified with a standard Iwasawa-cohomology complex only after a separately sourced comparison theorem, including the Λ -action, topology, and convention for twists or involutions.

Theorem 14.1 (Sequential Iwasawa edge package). *For every degree q there is a natural short exact sequence of underlying abelian groups*

$$0 \rightarrow \lim_{\leftarrow n}^1 H^{q-1}(G_{F_n,S}, T) \rightarrow H_{\mathrm{RL-Iw}}^q(F_\infty/F, T) \rightarrow \lim_{\leftarrow n} H^q(G_{F_n,S}, T) \rightarrow 0.$$

Consequently the Iwasawa Roos carrier

$$D_{\mathrm{Iw},q}(T) := H^1 \left(\mathrm{Roos}(H^{q-1}(G_{F_\bullet,S}, T)) \right)$$

is naturally isomorphic to the kernel of the sequential RL–Iwasawa edge map.

Proof. Apply the countable Milnor theorem and Theorem 11.7 to the corestriction tower $K_\bullet(T)$. The statement is made first on underlying abelian groups so that no unproved exactness assertion about a chosen topological Λ -module category is imported.

Corollary 14.2 (Iwasawa vanishing Return). *For each q ,*

$$H_{\mathbf{RL-Iw}}^q(F_\infty/F, T) = 0$$

if and only if

$$D_{\mathbf{Iw}, q}(T) = 0 \quad \text{and} \quad \varprojlim_n H^q(G_{F_n, S}, T) = 0.$$

Definition 14.3 (Selmer-complex extension record). A future Selmer extension of Theorem 14.1 must additionally bind:

- (i) the local complexes and local-condition morphisms at every $v \in S$;
- (ii) the global-to-local cone defining the finite-layer Selmer complex;
- (iii) compatibility of corestriction with the local conditions;
- (iv) the compact or discrete coefficient category and Pontryagin-dual convention;
- (v) the exact Λ -module finiteness or torsion hypothesis used by the Return theorem;
- (vi) the control morphism relating finite-layer Selmer cohomology to the Iwasawa apex.

Warning 14.4 (No control theorem is inferred). Theorem 14.1 constructs the sequential derived inverse-limit interface under the stated complex-level corestriction hypothesis. It does not by itself identify this tower apex with every standard convention for Iwasawa cohomology. It does not imply a Selmer control theorem or any cotorsion, pseudo-nullity, Leopoldt, Gross–Kuz'min, or main-conjecture statement. Each such conclusion requires a separately sourced comparison map and Return theorem.

15 Export to the Spectral Sequence successor appendix

15.1 Derived-limit spectral sequence

For an inverse system of complexes in a setting where the hyperderived-limit spectral sequence exists and converges to the stated abutment, it has the form

$$E_2^{i,j} = R^i \varprojlim_n H^j(K_n) \Rightarrow H^{i+j}(\mathbf{R} \varprojlim_n K_n).$$

Theorem 15.1 (RL is the two-column edge layer). *Assume the displayed spectral sequence exists and converges for the chosen countable inverse system of complexes. For the underlying countable inverse systems of abelian groups or modules,*

$$R^i \varprojlim_n = 0 \quad (i \geq 2).$$

Hence its E_2 page has only columns $i = 0, 1$, no nonzero differential can enter or leave either surviving term after the E_2 page, and the induced filtration on the abutment yields the canonical Milnor short exact sequence

$$0 \rightarrow R^1 \varprojlim_n H^{q-1}(K_n) \rightarrow H^q(\mathbf{R} \varprojlim_n K_n) \rightarrow \varprojlim_n H^q(K_n) \rightarrow 0.$$

Proof. The vanishing of higher derived inverse limits in the sequential module regime is Theorem 3.6. With only two adjacent columns, all differentials d_r for $r \geq 2$ have source or target outside the nonzero range. Under the stated convergence hypothesis, the two-step filtration of the abutment gives the short exact edge sequence. Independently of this spectral-sequence presentation, the same Milnor exact sequence is already established earlier from the countable derived-limit theorem.

Definition 15.2 (RL-to-SS export bundle). The RL appendix exports the following certified data to the Spectral Sequence successor appendix:

- (i) indexing convention $E_2^{i,j} = R^i \lim_{\leftarrow} H^j$;
- (ii) the actual edge map c_q ;
- (iii) the Roos model for the $E_2^{1,q-1}$ term;
- (iv) the proof that sequential systems have cohomological dimension one for $\lim_{\leftarrow}$;
- (v) naturality, tail invariance, and quasi-isomorphism invariance;
- (vi) the warning that general indexing categories may have higher $R^i \lim_{\leftarrow}$ and nontrivial differentials.

Proposition 15.3 (Interface compatibility). *The Edge–Roos Bridge verdict in degree q agrees with the vanishing verdict obtained from the two-column spectral-sequence filtration:*

$$H^q(\mathbf{R}\lim_{\leftarrow} K_n) = 0 \iff E_2^{0,q} = 0 \text{ and } E_2^{1,q-1} = 0.$$

Proof. By Theorem 15.1, the two E_2 terms are the quotient and subobject in the Milnor short exact sequence. The middle object vanishes exactly when both vanish.

Warning 15.4 (Boundary of the export). Proposition 15.3 is special to the two-column sequential regime. For a general indexing category, higher columns, extension data, convergence conditions, and possible differentials must be audited by the Spectral Sequence appendix itself.

16 Post-implementation self-audit

16.1 Implemented results and remaining comparison obligations

The present version implements the following four previously pending tasks, with the stated boundaries:

- (a) fixed finite-field continuous ℓ -adic calculations, consisting of a standard zero-defect ℓ -adic realization and a site-instantiated non-Mittag–Leffler calibration; neither is counted as the still-required independent non-toy promotion witness;
- (b) a derived-completion Bridge and the explicit computation $(\mathbf{Q}_p/\mathbf{Z}_p)_{(p)}^{\mathbf{L}} \simeq \mathbf{Z}_p[1]$;
- (c) a sequential RL–Iwasawa tower interface over a $\mathbf{Z}_p$ -extension, conditional on explicit compatible complex-level corestriction and still awaiting comparison with a standard Λ -linear Iwasawa complex;
- (d) a conditional compatibility export identifying sequential RL with the two-column edge layer whenever the relevant derived-limit spectral sequence exists and converges.

16.2 Adversarial checks

The following potential overclaims were tested and excluded:

- (i) The standard ℓ -adic example has zero Roos defect; it is not advertised as an essential $\varprojlim^1$ example.
- (ii) The nonzero $'_e$ tale example uses a nonstandard calibration coefficient system and is labelled accordingly.
- (iii) The derived completion of $\mathbf{Q}_p/\mathbf{Z}_p$ is nonzero because of the inverse limit of Tor_1 , not because of a nonzero Roos defect.
- (iv) The Iwasawa interface is stated on underlying abelian groups before any stronger topological Λ -linear claim.
- (v) No Selmer control theorem, Leopoldt theorem, or main-conjecture consequence is inferred.
- (vi) The spectral-sequence collapse is restricted to countable module towers.

16.3 Strongest defensible status

The strongest defensible status after this implementation is

sequential_edge–Roos_bridge/proved;
finite-field_continuous-etale_realizations/completed-as-calibrations;
principal-derived-reduction-limit_example/completed;
RL-Iwasawa_tower-interface/conditional-on-complex-corestriction;
RL-to-spectral-sequence_edge-export/conditional-on-convergence;
frozen-v20_persistence-realization/pending;
Selmer-control-and-Leopoldt-Return/pending.

16.4 System-level compatibility verdict

The RL appendix may be used by later appendices only under the following namespace and dependency rules:

- (roman*) the symbols $D_{\mathrm{mathrmlim}}$, $D_{\mathrm{mathrmedg}}$, and $D_{\mathrm{mathrmapex}}$ are RL-local typed objects and must not overwrite frozen v20 defect profiles or Type IV terminology;
- (roman*) the Spectral Sequence appendix may consume the RL edge bundle, but RL does not pre-empt its convergence, differential, extension, or multiplicative audits;
- (roman*) the Iwasawa appendix may consume the sequential tower interface, but it must provide the standard-complex comparison, Λ -linearity, topology, local conditions, and control morphisms independently;
- (roman*) the Main text may cite RL only as a successor inverse-limit theorem package and may not enlarge frozen-v20 claim closure before promotion;
- (roman*) no example in the present version discharges the promotion requirement of an independent non-toy application with a standard coefficient system and a substantive external Return predicate.

Warning 16.1 (Promotion witness remains open). The finite-field examples are mathematically correct and domain-instantiated, but the nonzero example reduces after cohomology to the classical multiplication tower on

$\mathbb{Z}$, while the standard

ℓ -adic example has zero edge defect. They therefore validate the interface but do not yet constitute the strongest non-toy witness required for v21 integration. A suitable witness should involve a standard arithmetic or geometric coefficient system for which the derived-limit edge changes a substantive external conclusion.

17 Remaining application lanes

The remaining lanes are no longer formal omissions inside RL itself; they are separate successor projects:

- (i) a frozen-v20 persistence realization, if a non-lossy target category can be constructed;
- (ii) a sourced Selmer-complex tower with compatible local conditions;
- (iii) an actual control morphism and its kernel/cokernel theorem;
- (iv) a Leopoldt or Gross–Kuz’min Return predicate;
- (v) extension from sequential towers to general indexing categories;
- (vi) multiplicative and duality compatibility.

18 Promotion gate for later v21 integration

RL is integration-ready only if all of the following hold:

- (i) the classical inverse-limit theorems have exact source bindings;
- (ii) every proof and variance convention has passed independent audit;
- (iii) the four-part audit record and Theorem 5.10 are cross-audited against frozen Type IV, transient-defect, and readiness-bundle semantics;
- (iv) the completion calibration in Section 8, the essential Milnor-edge calibration in Section 9, and the formal calibration in Section 10, the edge–Roos Bridge in Section 11, the post-implementation self-audit in Section 16, and the finite-field continuous ℓ -adic realizations in Section 12, the derived reduction-limit Bridge in Section 13, the Iwasawa interface in Section 14, and the spectral export in Section 15 remain valid under independent audit; at least one additional independent non-toy domain bridge, using a standard coefficient system and a substantive external Return predicate, must be completed; the finite-field calibrations in the present version do not alone discharge this condition;
- (v) no theorem relies on a finite numerical plateau as infinite evidence;
- (vi) the Main/Foundation placement and Claim Register consequences are fully specified.

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Part IX

Typed Return Theory and IW Strength Architecture

36 Role inside Appendix HT

This Part supplies the common Return datum, strength preorder, R1–R8 obligations, Type IV classification, deterministic primary statuses, replay certificate, and non-promotion rules used by the entire HT Appendix.

Remark 36.1 (Normalization rule). The following is the complete retained source text of the former IW artifact, normalized into this HT Part. Its original numbering is a provenance namespace only; it no longer denotes a separate Appendix file.

Retained module abstract

This document consolidates the Iwasawa successor sequence IW-1–IW-9 into a standalone AK Return Theory. Return is not identified with inversion, inverse limit, reconstruction, equality of characteristic ideals, or derived equivalence. It is a typed and certified passage from an observed presentation to a declared target at an explicitly chosen strength. The architecture separates source data, observation functor, candidate apex, comparison artifact, defect object, and replayable certificate. Eight core Return obligations govern typing, source identity, completeness, comparison, uniqueness relative to strength, stability, obstruction, and certification; two separate governance laws control non-collapse and composition. A status lattice distinguishes UNDEFINED, REJECT, OBSTRUCTED, partial Return, regularized Return, and exact Return. Type IV is unified as a family of defects of an existing comparison artifact and is no longer used as a synonym for divergence or missing data. The resulting theory explains IW-1–IW-9 as instances of one schema while preserving their different Return strengths: completion, arithmetic inverse-limit, analytic ideal, higher-Fitting, pseudo-isomorphism, derived-complex, determinant-line, and concrete zeta-determinant Return. A final adversarial reaudit removes three possible overclaims: Return is not globally functorial without declared coherence data; uniqueness is only relative to the target category and equivalence relation; and stronger Return levels do not follow automatically from weaker ones. The final status is `axiomatic_return_theory_closed_and_reaudited`.

Remark 36.2 (Native vector transcription rule). The following body is reconstructed as native XeLaTeX text and mathematics from the authoritative module PDF. It is not Verbatim text, not a raster image, and not an imported PDF. Original title pages, module-local tables of contents, and module-local page numbers are removed; mathematical statements, proofs, tables, diagrams, references, and source-boundary declarations are retained. The former numbering is preserved as the module provenance namespace.

1 Decision: why a standalone Return theory is justified

The IW sequence produced genuine structural advances through IW-7 and theorem-bound arithmetic closures in IW-8 and IW-9. Continuing by adding more external theorems would enlarge the application catalogue but would not clarify the AK-specific invariant. The reusable invariant is the Return architecture itself.

The central question is therefore:

Given a possibly lossy, finite, filtered, completed, analytic, or derived presentation of mathematical data, under which typed obligations may one certify that a declared target has been recovered at a stated strength?

Return Theory answers this question without pretending that every observation is invertible. It records exactly what is recovered, in which category, under which equivalence relation, with which defects annihilated, and by which replayable certificate.

2 Primitive data of a Return problem

Definition 2.1 (Return datum). A Return datum is a tuple

$$\mathbf{R} = (\mathcal{S}, \mathcal{O}, \mathcal{T}, X, Y, \text{Obs}, \alpha, \mathcal{E}, \Sigma, \mathcal{C}),$$

where:

(D1) $\mathcal{S}$ is the source category and $X \in \mathcal{S}$ is the declared source object;

(D2) $\mathcal{O}$ is the observation category and $\text{Obs}(X) \in \mathcal{O}$ is the observed presentation;

(D3) $\mathcal{T}$ is the target category and $Y \in \mathcal{T}$ is the declared Return target;

(D4) α is the declared comparison artifact relating the reconstructed candidate to Y ;

(D5) $\mathcal{E}$ is the target equivalence relation or localization, for example equality, isomorphism, quasi-isomorphism, pseudo-isomorphism, equality modulo codimension two, or equality of ideals;

(D6) Σ is the requested Return strength;

(D7) $\mathcal{C}$ is a certificate package containing source bindings, hypotheses, normalization, comparison data, defects, and replay instructions.

Definition 2.2 (Apex constructor). An apex constructor is a partial operation

$$\text{Apex} : \mathcal{O} \dots \rightarrow \mathcal{T}$$

that assigns a candidate target $\widehat{X} := \text{Apex}(\text{Obs}(X))$ when the required completeness, convergence, compatibility, or realization conditions are certified.

The comparison artifact is then typed as

$$\alpha : \widehat{X} \rightarrow Y,$$

or, when morphisms are not the correct language, as a pair of inclusions, a zig-zag of quasi-isomorphisms, an equality in a localization, or an isomorphism of determinant objects.

Warning 2.3 (No universal shape for α). An invented morphism between ideals is not permitted merely because an equality theorem exists. The comparison artifact must use the native type of the theorem: morphism, pair of inclusions, equivalence class, derived isomorphism, determinant trivialization, or another explicitly declared structure.

3 Return strengths and the non-collapse principle

Definition 3.1 (Strength preorder). Return strengths form a preorder $(\mathbf{S}, \preceq)$. The relation

$$\Sigma_1 \preceq \Sigma_2$$

means that a valid Σ_2 -Return canonically forgets to a valid Σ_1 -Return through a declared forgetful map. It does not mean that Σ_1 implies Σ_2 .

The IW programme supplies several certified weakening maps, but not one universal linear chain. The following relations are representative only when the indicated comparison functors have been registered:

complete inverse-system Return $\rightarrow$ finite-prefix consistency,

higher-Fitting profile Return $\rightarrow$ principal-ideal Return,

pseudo-isomorphism Return $\rightarrow$ characteristic-divisor Return,

zeta-trivialized determinant Return $\rightarrow$ determinant-line Return.

A derived-complex Return, a pseudo-isomorphism Return, and an arithmetic-growth Return are generally not comparable without additional functors and hypotheses. Thus the strength structure is a preorder of registered forgetful maps, not a universal total chain.

Axiom 3.2 (Governance law G1: Non-collapse). A certificate at strength Σ may not be promoted to a strictly stronger strength Σ^* unless an explicit promotion theorem or additional certificate is supplied.

Examples:

- equality of characteristic ideals does not imply exact module isomorphism;
- all finite prefixes do not imply a specified infinite object unless compatibility and completeness are certified;
- a derived equivalence of arithmetic complexes does not isolate an arbitrary cohomology module up to exact presentation;
- a determinant-line trivialization does not reconstruct every extension class.

4 The eight core Return obligations

Axiom 4.1 (R1: Typing). Every object, map, equivalence relation, normalization, and defect must be declared in a category in which the claimed operation is defined.

Axiom 4.2 (R2: Source identity). The theorem, computation, or certificate used to construct the apex and comparison must apply to the same objects, branches, coefficient rings, and normalizations as the Return datum.

Axiom 4.3 (R3: Completeness). A finite or partial observation may Return only the strength justified by its coverage. Infinite, completed, or global targets require a complete compatible system or a theorem-backed infinite specification.

Axiom 4.4 (R4: Comparison). Return requires an actual comparison artifact in the native type of the target claim. Mere resemblance, equal numerical samples, asymptotic agreement, or common notation is insufficient.

Axiom 4.5 (R5: Relative uniqueness). Uniqueness is asserted only in the declared target category modulo the declared equivalence relation $\mathcal{E}$. If two accepted targets are both compared to the same apex by certified $\mathcal{E}$ -equivalences, then they are unique up to $\mathcal{E}$.

Axiom 4.6 (R6: Stability). A stability claim must provide a perturbation class, a metric or topology, a tolerance, and a theorem showing that accepted Return status persists or degrades in a controlled way.

Axiom 4.7 (R7: Obstruction). A Type IV obstruction may be emitted only when the apex and comparison artifact exist and a declared terminal defect of that artifact is nonzero. Missing data or an absent apex is `UNDEFINED`, not `OBSTRUCTED`.

Axiom 4.8 (R8: Certification). Every accepted Return must carry a replayable certificate sufficient to verify R1–R7 and to reproduce the declared status without relying on unstated human judgment.

5 Status algebra

Classification 5.1 (Primary statuses). A verifier emits one of the following primary statuses.

<code>UNDEFINED</code>	A required object, branch, normalization, apex, theorem scope, or comparison artifact is absent.
<code>REJECT</code>	A stated premise, typing assertion, compatibility equation, source binding, or certificate claim is false.
<code>OBSTRUCTED</code>	The full comparison artifact exists, but a declared kernel, cokernel, quotient, cone, or localized defect is nonzero.
<code>PARTIAL-RETURN</code>	A weaker strength has been certified while a stronger requested level remains unproved.
<code>RETURN</code>	All obligations at the declared strength are discharged.
<code>REGULARIZED-RETURN</code>	A source-certified replacement target, such as a leading term, is returned under a new identifier.

Proposition 5.2 (Deterministic primary status at fixed datum and strength). *For a frozen Return datum R and requested strength Σ , the verifier emits exactly one primary status under the ordered decision rule:*

(P1) *a false supplied assertion or invalid source binding gives* `REJECT`;

(P2) *otherwise, a missing required object or theorem scope gives* `UNDEFINED`;

(P3) *otherwise, a defined nonzero terminal comparison or stability defect gives* `OBSTRUCTED`;

(P4) otherwise, if all obligations at Σ hold, emit RETURN ;

(P5) otherwise, if a weaker registered strength is certified, emit PARTIAL-RETURN ;

(P6) a source-certified replacement target is evaluated as a distinct Return datum and may emit $\text{REGULARIZED-RETURN}$.

Proof. The ordered rule partitions verifier states after treating a regularized target as a new datum. The first applicable clause is unique. In particular, partial Return is not simultaneously Return at the requested strength, and regularized Return is not a silent status change of the original target.

Remark 5.3 (Why precedence matters). A fabricated comparison map with a claimed nonzero cokernel is not an obstruction; it is a rejection of the certificate. An absent map is undefined. Only a valid map with nonzero defect is obstructed.

6 Type IV unification

Definition 6.1 (Return defect ledger). For a proposed Return datum, the full defect ledger is

$$\mathbf{D}_{\text{Ret}}(\mathbf{R}) = (D_{\text{src}}, D_{\text{apex}}, D_{\text{cmp}}, D_{\text{faith}}, D_{\text{stab}}, D_{\text{cert}}),$$

where:

(IV-A) D_{src} records source-theorem and object-identity defects;

(IV-B) D_{apex} records completeness, convergence, or realization defects;

(IV-C) D_{cmp} records kernel, cokernel, quotient, cone, or localized inclusion defects;

(IV-D) D_{faith} records information lost by the chosen equivalence relation or observation functor;

(IV-E) D_{stab} records perturbation-instability defects;

(IV-F) D_{cert} records serialization and replay defects.

Only D_{cmp} and, under a declared perturbation contract, D_{stab} are terminal Type IV obstruction objects. The source, apex, faithfulness, and certificate entries are governance diagnostics that determine rejection, undefinedness, or a promotion ceiling before a genuine obstruction is emitted.

Classification 6.2 (Four operational Type IV classes).

- Type IV-U** Undefined interface: no valid apex or comparison artifact. This is a boundary state, not a genuine obstruction.
- Type IV-C** Comparison obstruction: a valid comparison exists and its terminal defect is nonzero.
- Type IV-F** Faithfulness ceiling: Return succeeds only modulo a declared equivalence that forgets information relevant to a stronger target.
- Type IV-S** Stability obstruction: a valid base Return and a valid perturbation comparison exist, but the declared stability defect is nonzero.

Type IV-U is retained for operational continuity with v20 but is never reported as OBSTRUCTED . Type IV-F is the correct diagnosis for characteristic-ideal or pseudo-isomorphism Return when exact module data are requested: the weaker Return may be valid, while promotion is blocked by a known faithfulness ceiling.

Theorem 6.3 (Type IV correctness criterion). *At a fixed Return strength, OBSTRUCTED is sound if and only if:*

- (i) $R1$ – $R4$ hold;
- (ii) the declared apex and comparison artifact exist;

(iii) the terminal defect object is defined and nonzero;

(iv) the defect is invariant under the declared target equivalence.

Proof. The forward direction is R7 plus target invariance. Conversely, a valid existing comparison with a nonzero invariant terminal defect cannot satisfy the Return acceptance equation at that strength, so obstruction is the unique valid status after the precedence rule.

7 Consistency witness and non-independence notice

The core obligations are intended as a compatible verifier contract, not as a claim of logical independence in the model-theoretic sense.

Proposition 7.1 (A concrete consistency witness). *There exists a nontrivial Return model satisfying R1–R8 and G1–G2.*

Proof. Take $\mathcal{S} = \mathcal{O} = \mathcal{T} = \mathbf{FinSet}$, let $\mathbf{Obs} = \mathbf{Id}$, let the admissible apex constructor be $\mathbf{Apex} = \mathbf{Id}$, choose isomorphism $\mathbf{as}_{\mathcal{E}}$, and let a certificate consist of the finite multiplication tables of the identity maps together with exact equality checks. The comparison artifact is $\alpha = \mathbf{Id}_X$, all terminal defects vanish, perturbation claims are omitted unless the discrete metric is declared, and composition is ordinary composition of identities. This supplies a model of all obligations and governance laws; hence the system is jointly satisfiable.

Remark 7.2 (No axiom-independence claim). The document does not claim that R1–R8 are pairwise logically independent. For example, particular formalizations may encode parts of source identity inside typing or certification. The numbering records verifier responsibilities, not a minimal independent basis.

8 Categorical architecture

8.1 Observation and reconstruction

Let

$$\mathbf{Obs} : \mathcal{S} \rightarrow \mathcal{O}$$

be an observation functor. A Return constructor is not assumed to be a global inverse. It is a partially defined functor or pseudofunctor

$$\mathbf{Ret}_{\Sigma} : \mathcal{O}_{\Sigma}^{\text{adm}} \rightarrow \mathcal{T}_{\Sigma}$$

on an admissible subcategory selected by the certificate predicates.

Definition 8.1 (Return witness). A Return witness for X at strength Σ is an equivalence

$$\eta_X : X_{\Sigma} \xrightarrow{\sim \varepsilon} \mathbf{Ret}_{\Sigma}(\mathbf{Obs}(X))$$

in the declared target category, where X_{Σ} is the image of X under the target-forgetting or localization functor appropriate to Σ .

Warning 8.2 (No automatic adjunction). Return Theory does not postulate that $\mathbf{Ret}$ is left or right adjoint to $\mathbf{Obs}$. An adjunction, unit, counit, or monadic reconstruction may be registered only when proved in the concrete application.

8.2 Composition

Suppose

$$\mathcal{O}_0 \xrightarrow{R_1} \mathcal{O}_1 \xrightarrow{R_2} T$$

are two certified Return stages.

Axiom 8.3 (Governance law G2: Compositionality under certificate closure). The composite is an accepted Return only if object identities match, intermediate equivalences are coherent, and the combined certificate contains all hypotheses and defects of both stages. Status is propagated by the worst unresolved stage.

Thus a finite-tower Return followed by an analytic comparison is not accepted if either the inverse-limit certificate or the main-conjecture comparison is absent.

9 Core theorems of Return Theory

Theorem 9.1 (Relative uniqueness). *Let $Y_1, Y_2 \in T$ be accepted Return targets for the same apex $\widehat{X}$ at the same strength and equivalence relation. If*

$$\alpha_i : \widehat{X} \xrightarrow{\sim_\varepsilon} Y_i \quad (i = 1, 2),$$

then

$$Y_1 \sim_\varepsilon Y_2.$$

Proof. Compose α_2 with an inverse representative of α_1 in the localized target category.

Theorem 9.2 (Safe weakening). *If a Return at strength Σ_2 is accepted and a certified forgetful map*

$$F_{21} : \mathcal{T}_{\Sigma_2} \rightarrow T_{\Sigma_1}$$

exists with $\Sigma_1 \preceq \Sigma_2$, then Return at strength Σ_1 is accepted.

Proof. Apply F_{21} to the Return witness and transport the certificate through the registered coherence map.

Theorem 9.3 (No promotion without rigidity). *Assume the observation/forgetful map from strength Σ_2 to Σ_1 is not faithful on the admissible target class. Then no verifier using only Σ_1 -data can soundly promote every accepted Σ_1 -Return to Σ_2 -Return.*

Proof. Non-faithfulness gives non-equivalent objects with identical Σ_1 observations. Any uniform promotion rule would assign the same stronger target to both and therefore misclassify at least one. A rigidity theorem may remove this counterexample only by restricting the admissible class.

Corollary 9.4 (Characteristic ideal ceiling). *A characteristic ideal alone cannot soundly provide exact module Return in a category containing both*

$$\Lambda/(f^2) \quad \text{and} \quad \Lambda/(f) \oplus \Lambda/(f),$$

since they have the same zeroth Fitting ideal but need not be isomorphic.

Theorem 9.5 (Certificate monotonicity). *Adding verified evidence can move a datum from `UNDEFINED` to Return, or from `OBSTRUCTED` at a stronger level to partial Return at a weaker level. It cannot convert a false premise from `REJECT` to Return without replacing or correcting that premise.*

10 Replay certificate schema

Verifier contract 10.1 (Minimal Return certificate). A certificate must serialize at least:

- (C1) source and target identifiers, versions, and hashes;
- (C2) coefficient ring, category, topology, branch, and normalization;
- (C3) observation coverage and completeness declaration;
- (C4) apex constructor and theorem/source binding;
- (C5) comparison artifact in its native mathematical type;
- (C6) requested strength and target equivalence relation;
- (C7) defect objects and zero/nonzero witnesses;
- (C8) status precedence trace;
- (C9) replay commands or deterministic verification procedure;
- (C10) explicit non-claims and promotion barriers.

Protocol 10.2 (Acceptance algorithm). Given R :

- Step 1. validate all hashes, identifiers, and theorem scope;
- Step 2. type-check objects, maps, branches, and normalizations;
- Step 3. verify observation completeness required by Σ ;
- Step 4. construct or validate the apex;
- Step 5. validate the native comparison artifact;
- Step 6. compute or verify terminal defects;
- Step 7. apply the precedence rule;
- Step 8. emit the strongest certified status, together with all weaker safe Returns;
- Step 9. record every blocked promotion as Type IV-F or Type IV-U/C/S as appropriate.

11 IW-1–IW-9 as instances

Track	Apex	Return strength	Principal firewall
IW-1	$\lim_{\leftarrow -r} X/m^{r+1}X$	completed module isomorphism	finite generation, separatedness, completeness, transition compatibility
IW-2	arithmetic inverse limit	arithmetic module Return	control maps, norm compatibility, source reconstruction, vanishing defects
IW-3	analytic branch or regularized apex	coefficient/ideal Return	no apex $\Rightarrow$ undefined; actual nonzero terminal defect $\Rightarrow$ obstructed
IW-4	class-group tower, growth data, analytic ideal	inverse-limit, ideal Return	finite prefix prohibition; characteristic ideal non-promotion; two-inclusion comparison

Track	Apex	Return strength	Principal firewall
IW-5	executable branch certificate	replayable ideal Return	manifest separation, normalization, stagewise status lattice
IW-6	higher Fitting tower	refined ideal profile	stratumwise defect vector; presentation/exact-module promotion barrier
IW-7	all higher cyclotomic/elliptic ideals	pseudo-isomorphism class	codimension-two equivalence; pseudo-null and extension-data ceiling
IW-8	perfect arithmetic complex	derived and determinant Return	weak-Leopoldt scope; zeta trivialization separately theorem-bound
IW-9	explicit $\mathbf{Q}(i, \sqrt{2})/\mathbf{Q}(i)$ datum at $p = 5$	concrete zeta-determinant Return	admissible S, T , integral branch decomposition, theorem scope

Proposition 11.1 (IW coherence). *Each IW track is an instance of the same Return datum after choosing its target category, equivalence relation, apex constructor, and strength. No track forces all other tracks to share its comparison type.*

12 Stability and perturbation discipline

Definition 12.1 (Stable Return). A Return datum is (ε, δ) -stable if a declared perturbation $d_{\mathcal{O}}(o, o') \leq \varepsilon$ implies either:

$$d_{\mathcal{T}}(\text{Ret}(o), \text{Ret}(o')) \leq \delta,$$

or preservation of the declared equivalence class/status, according to the target category.

Warning 12.2 (No metric laundering). A symbol such as δ does not create a stability theorem. The observation metric, target metric or topology, admissible perturbations, and proof of continuity must be separately registered.

In arithmetic applications the appropriate stability may be discrete, p -adic, ideal-theoretic, or functorial rather than metric. Return Theory allows these variants but requires their native structure to be declared.

13 Regularized Return

Definition 13.1 (Replacement-target rule). If the raw target is undefined because of a pole, exceptional zero, or singular specialization, a regularized target may be introduced only under a new identifier and a source-certified replacement theorem.

Thus

$$L_p(\chi) = 0$$

does not by itself obstruct Return. A leading term, derivative, Bockstein image, or zeta element may be the correct apex, but it is a different target. Silent replacement is forbidden.

14 Adversarial reaudit

This section subjects the proposed theory to counterexample-driven review.

14.1 Audit A: is Return merely inversion?

No. Many valid instances have no inverse observation functor. Pseudo-isomorphism Return and determinant Return intentionally land in localized or compressed categories. The theory therefore defines Return by a certified witness at a declared strength, not by global invertibility.

14.2 Audit B: is uniqueness overstated?

The initial formulation risked suggesting absolute uniqueness. This is corrected: uniqueness is relative to $(\mathcal{T}, \mathcal{E}, \Sigma)$. Different presentations or stronger categories may distinguish targets identified at a weaker level.

14.3 Audit C: is functoriality automatic?

No. Objectwise reconstruction does not imply naturality. The final version requires explicit coherence data before declaring a Return functor, adjunction, or natural transformation.

14.4 Audit D: does zero defect prove source truth?

No. A zero terminal defect computed from a false or mismatched source certificate is rejected before obstruction analysis. Source identity and typing precede defect evaluation.

14.5 Audit E: can finite agreement imply infinite Return?

No. Arbitrarily long finite prefixes can agree while infinite completions differ or fail to exist. R3 requires a complete compatible system or an independently proved infinite specification.

14.6 Audit F: does derived Return dominate every module Return?

Not in the naive sense. A derived equivalence preserves the derived object but extracting a particular module with a preferred presentation may require a t -structure, cohomological identification, and additional rigidity. The strength preorder is therefore application-dependent and not globally total.

14.7 Audit G: is Type IV too broad?

The earlier IW sequence used Type IV across several layers. The final classification separates undefined interface, genuine comparison obstruction, faithfulness ceiling, and instability. Only Type IV-C and Type IV-S are genuine obstructions at their declared levels; Type IV-U is undefined and Type IV-F is a promotion ceiling. Consequently, the phrase “Type IV family is reserved for the operational classes, while the six-component object above is the broader Return defect ledger.”

14.8 Audit H: is IW-9 overused as universal evidence?

No. IW-9 is retained as a concrete theorem-bound example, not as evidence that every number field or branch satisfies eTNC or zeta-determinant Return.

14.9 Audit I: is the strength preorder accidentally linearized?

The earlier presentation displayed a representative chain that could be misread as a universal order, while later correctly stating that derived and module-level Returns can be incomparable. The chain has been replaced by individually registered weakening maps. No cross-family order is asserted without an explicit functor.

14.10 Audit J: are there exactly eight axioms?

R1–R8 are now called the eight core Return obligations. Non-collapse and compositionality are separately labelled governance laws G1–G2. This removes the previous counting inconsistency.

14.11 Audit K: does the Type IV defect object presuppose a valid comparison while recording an absent apex?

The six-component object is now the full Return defect ledger, defined for a proposed datum. Genuine Type IV terminal obstruction is restricted to valid comparison or stability defects. Source mismatch, absent apex, and certificate failure are handled earlier by rejection or undefinedness.

14.12 Audit L: is logical independence established?

No independence theorem was proved and none is now claimed. A concrete finite-set model is supplied only to witness joint consistency. Minimality or pairwise independence remains outside scope.

Theorem 14.1 (Reaudit verdict). *After corrections A–H and the second reaudit corrections I–L, the obligations and governance laws are jointly satisfiable and do not imply any of the prohibited promotions:*

- (i) *finite prefix to infinite object;*
- (ii) *characteristic ideal to exact module;*
- (iii) *pseudo-isomorphism to isomorphism;*
- (iv) *derived equivalence to a preferred exact presentation;*
- (v) *determinant trivialization to full arithmetic reconstruction;*
- (vi) *missing apex to obstruction.*

Proof. Each prohibited promotion is blocked by R3, G1, R5, or R7. Explicit counterexamples or non-faithfulness arguments establish that removing these barriers would be unsound.

15 Scope and non-claims

Non-claim 15.1. Return Theory does not prove external arithmetic theorems used by IW-4–IW-9. It binds, types, composes, and audits them.

Non-claim 15.2. The axioms do not assert that every observation admits an apex, every apex admits a comparison, or every valid comparison is an equivalence.

Non-claim 15.3. The theory is not a universal solution to inverse problems. It is a governance and proof architecture for stating exactly which inverse or reconstruction claim has been certified.

Non-claim 15.4. No general equivalence is claimed between Return Theory and adjunctions, monadicity, descent, reconstruction theorems, or realizability. These may instantiate Return Theory when separately proved.

16 Research consequences and next valid progression

The strongest next progression is not another automatic IW-number. It is one of the following independently valuable tasks:

- (N1) implement the certificate schema and verifier as an executable reference checker;
- (N2) prove a general categorical representation theorem for a restricted class of Return data;
- (N3) construct a realization bridge in which a concrete finite certificate Returns an externally meaningful theorem, not merely an internal object;
- (N4) identify a rigidity class where higher-Fitting or derived data promote to exact module Return;
- (N5) formalize R1–R8 in a proof assistant and replay IW-1 or IW-9 end to end.

These are genuine advances. Merely adding more examples of known theorems is an application expansion, not a further axiom-level evolution.

17 Final closure declaration

Theorem 17.1 (AK Return Theory v1.0 closure). *For every declared application satisfying R1–R8 and the applicable governance laws G1–G2, the AK verifier may soundly emit the strongest status among the registered strengths actually checked by its certificate, propagate certified weakenings, and classify each examined blocked stage as undefined, rejected, obstructed, partial, regularized, or limited by faithfulness without conflating these states.*

Proof. Typing and source identity establish semantic validity. Completeness and comparison establish an admissible apex and native comparison artifact. Relative uniqueness fixes the target only at the declared strength. Stability and obstruction control perturbations and genuine failures. Certification makes the reasoning replayable. The ordered status rule then yields one primary status at each frozen requested strength; it does not claim completeness over unregistered strengths or unexamined theorem scopes.

The final registered status is

axiomatic_return_theory_closed_and_reaudited ,
typeIV_taxonomy_unified ,
IW1_through_IW9_coherently_embedded ,
promotion_overclaims_blocked .

Part X

Unified Closure, Embedded Registry, and Canonical Single-Artifact Status

37 Canonical single-Appendix closure theorem

Theorem 37.1 (HT true single-Appendix integration closure). *The HC, PE, SS, RL, and IW/Return mathematical layers are represented in one Appendix HT with one XeLaTeX source, one primary title hierarchy, one continuous pagination, one AK-v20 re-entry architecture, one directed-interface schema, and one composite status calculus. HC, SS, RL, and IW are retained by page-faithful native vector XeLaTeX transcription; PE is retained as structured semantic XeLaTeX. The primary table of contents indexes the HT-level Parts and semantic sections; the canonical clickable statement locator below supplies statement-level navigation for the vector-transcribed Parts. No Verbatim extraction, raster page, imported PDF, subordinate module source, or external module PDF is required to compile the canonical edition.*

Proof. Part I defines the frozen AK spine and the DAG-valued HT datum. Parts II–IX contain the retained mathematical bodies of HC, PE, SS, RL, and IW. Module-local title pages, tables of contents, and page numbers are intentionally normalized away; their mathematical definitions, theorems, proofs, worked examples, defect ledgers, source boundaries, and references remain in the single source. The source uses no pdfpages, includepdf, input, or include dependency. The embedded provenance registry below fixes the authoritative inputs without making them compilation dependencies. $\square$

38 Embedded provenance and source-binding registry

Artifact role	SHA-256
AK v20 Main	c0ad406c023bc1a10c90ede494ea380b42dceeb630264998016fd841b93b166b
AK v20 Foundation	546499168884b7ff9e525309d9ce0ff9e78a6202e88f1cbb612699aceb1c06cb
AK v20 Problem Interface	f2adfc9191eb50a548753c830da26f05d6ed938ff6942592e84ed38bff2f298
HC provenance	35048cea239841eab07b3310ae82f8fdb35683cf640b3002aa6bd2bbedf1eec
PE native source	5a766936de035bb7a7970f7c2233fee030fd82b400d9813c9bf72f8d42ce887b
SS provenance	473114c0a23faeb0d87aecdeb44a87bb545a41c45a1fc99f3ba38a39ac1fb654
RL provenance	f68acda79b7037ccab3ba99cbcce6c5cdda181d4569f6db02a45447e79349b57

Artifact role	SHA-256
IW provenance	30cefbb191a974034221a264ef1c2e50ce688d31bbbb1950df6eb669118f2560

Warning 38.1 (Registry meaning). The hashes identify the frozen baseline and provenance sources consumed in constructing HT. They are audit anchors, not hidden compilation inputs. Mathematical authority remains limited by the local theorem statements and their declared source boundaries.

39 Unified theorem-locator namespace

The primary table of contents records the unified HT Part and semantic-section hierarchy. For the page-faithful vector-transcribed HC, SS, RL, and IW Parts, the following one-to-one clickable register is the canonical statement-level navigation layer. Every locator links directly to the statement hypertarget, and the displayed page number links to the same destination. PE retains its native semantic labels and cross-references. This navigation contract does not assert that coordinate-positioned module pages have been converted into ordinary LaTeX theorem environments.

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40 Final canonical status

Classification 40.1 (True single-Appendix status).

higher-transfer-canonical-single-appendix-closed-candidate + HC-PE-SS-RL-IW-mathematical-bodies-internalized + page-faithful-native-vector-transcription-for-HC-SS-RL-IW + structured-native-XeLaTeX-PE-retained + no-Verbatim-no-raster-no-imported-PDF-no-subsource-dependency + one-title-one-primary-TOC-one-pagination-one-DAG-schema + canonical-clickable-statement-locator-one-to-one + embedded-provenance-registry-and-AK-v20-verifier-reentry-explicit + local-theorem-strengths-inherited-without-promotion + internal-single-artifact-governance-and-navigation-gates-closed + independent-external-mathematical-audit-pending + future-release-integration-pending
